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Simulating 500 million years of evolution with a language model

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Abstract

More than three billion years of evolution have produced an image of biology encoded into the space of natural proteins. Here we show that language models trained on tokens generated by evolution can act as evolutionary simulators to generate functional proteins that are far away from known proteins. We present ESM3, a frontier multimodal generative language model that reasons over the sequence, structure, and function of proteins. ESM3 can follow complex prompts combining its modalities and is highly responsive to biological alignment. We have prompted ESM3 to generate fluorescent proteins with a chain of thought. Among the generations that we synthesized, we found a bright fluorescent protein at far distance (58% identity) from known fluorescent proteins. Similarly distant natural fluorescent proteins are separated by over five hundred million years of evolution.

Introduction

The proteins that exist today have developed into their present forms over the course of billions of years of natural evolution, passing through a vast evolutionary sieve. In parallel experiments conducted over geological time, nature creates random mutations and applies selection, filtering proteins by their myriad sequences, structures, and functions.

As a result, the patterns in the proteins we observe reflect the action of the deep hidden variables of the biology that have shaped their evolution across time. Gene sequencing surveys of Earth's natural diversity are cataloging the sequences (1–3) and structures (4, 5) of proteins, containing billions of sequences and hundreds of millions of structures that illuminate patterns of variation across life. A consensus is developing that underlying these sequences is a fundamental language of protein biology that can be understood using language models (6–11).

A number of language models of protein sequences have now been developed and evaluated (5–10, 12–17). It has been found that the representations that emerge within language models reflect the biological structure and function of proteins (6–8, 18), and are learned without any supervision on those properties (19, 20), improving with scale (5, 21). In the field of artificial intelligence, scaling laws have been found that predict the growth in capabilities with increasing scale, describing a frontier in compute, parameters, and data (22–24).

We present ESM3, a frontier multimodal generative model that reasons over the sequences, structures, and functions of proteins. ESM3 is trained as a generative masked language model over discrete tokens for each modality. Structural reasoning is achieved by encoding three-dimensional atomic structure as discrete tokens rather than with the complex architecture and diffusion in three-dimensional space employed in recent predictive (25) and generative models (26–28) of proteins. All-to-all modeling of discrete tokens is scalable, and allows ESM3 to be prompted with any combination of its modalities, enabling controllable generation of new proteins that respect combinations of prompts.

ESM3 at its largest scale was trained with 1.07×10^{24} floating point operations (FLOPs) on 2.78 billion proteins and 771 billion unique tokens, and has 98 billion parameters. Scaling ESM3 to this 98 billion parameter size results in improvements in the representation of sequence, structure, and function, as well as on generative evaluations. We observe that ESM3 is highly responsive to prompts, and finds creative solutions to complex combinations of prompts, including solutions for which we can find no matching structure in nature. Models at all scales can be aligned to better follow prompts, and larger models are far more responsive to alignment, showing greater capability to solve the hardest prompts after alignment.

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使用语言模型模拟五亿年的演化

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摘要

超过三十亿年的演化产生了生物学图像，编码在自然蛋白质的空间中。在这里，我们展示了通过演化生成的标记训练的语言模型能够作为演化模拟器，生成与已知蛋白质相距甚远的功能蛋白质。我们提出了ESM3，这是一种前沿的多模态生成语言模型，能够在蛋白质的序列、结构和功能上进行推理。ESM3可以遵循组合其模态的复杂提示，并对生物对齐高度响应。我们已向ESM3提示以生成具有思维链的荧光蛋白。在我们合成的生成物中，我们发现了一个与已知荧光蛋白相距甚远的明亮荧光蛋白（58% 相似性）。同样距离较远的自然荧光蛋白在五亿年以上的演化中是分开的。

引言

今天存在的蛋白质在数十亿年的自然进化过程中发展成其当前形态，经历了一个庞大的进化筛选过程。在地质时间上进行的平行实验中，自然创造了随机突变并应用选择，通过各种各样的序列、结构和功能过滤蛋白质。

因此，我们观察到的蛋白质中的模式反映了塑造其随时间演化的生物学深层隐藏变量的作用。地球自然多样性的基因测序调查正在对序列（1–3）和结构（4,5）进行分类，这些蛋白质包含十亿

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序列和数亿个结构揭示了生命中变异模式的规律。一个共识正在形成，这些序列背后是蛋白质生物学的基本语言，可以通过语言模型理解（6–11）。

目前已经开发和评估了一些蛋白质序列的语言模型（5–10， 12–17）。研究发现，语言模型中出现的表示反映了蛋白质的生物结构和功能（6–8， 18），且这些表示是在没有任何监督下学习到的（19， 20），并且随着规模的增长而得到改善（5， 21）。在人工智能领域，已经发现缩放法则可以预测随着规模的增加能力的增长，描述了计算、参数和数据的前沿（22–24）。

我们提出了ESM3，一种前沿的多模态生成模型，能够对蛋白质的序列、结构和功能进行推理。ESM3作为一种生成性掩码语言模型，针对每种模态对离散标记进行训练。结构推理是通过将三维原子结构编码为离散标记来实现的，而非最近预测（25）和生成模型（26–中采用的复杂架构和在三维空间中的扩散。

28）蛋白质。离散标记的全对全建模具有可扩展性，并允许ESM3接收其所有模态的任意组合的提示，从而实现对新蛋白质的可控生成，遵循提示组合的要求。

ESM3 在其最大规模上经过了 1.07×10^{24} 浮点操作 (FLOPs) 的训练，涵盖了 27.8 亿种蛋白质和 7710 亿个独特标记，并具备 980 亿个参数。将 ESM3 扩展到 980 亿参数的规模，能够在序列、结构和功能的表示上实现改进，以及在生成评估上的提升。我们观察到 ESM3 对提示具有高度的响应性，并能够为复杂的提示组合找到创造性的解决方案，包括那些我们在自然界中找不到匹配结构的解决方案。所有规模的模型都可以更好地对齐以跟随提示，而较大的模型对对齐的响应性更强，显示出在对齐后解决最具挑战性的提示的能力更卓越。

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We report the generation of a new green fluorescent protein (GFP) with ESM3. Fluorescent proteins are responsible for the glowing colors of jellyfish and corals (29) and are important tools in modern biotechnology (30). They share an elegant structure: an eleven stranded beta barrel with a helix that threads its center, which scaffolds the formation of a light-emitting chromophore out of the protein's own atoms. This mechanism is unique in nature—no other protein spontaneously forms a fluorescent chromophore out of its own structure—suggesting that producing fluorescence is hard even for nature.

Our new protein, which we have named esmGFP, has 36% sequence identity to *Aequorea victoria* GFP, and 58% sequence identity to the most similar known fluorescent protein. Despite GFP's intense focus as a target for protein engineering over several decades, as far as we are aware, new GFPs this distant have only been found through their discovery in nature.

Similar amounts of diversification among natural GFPs have occurred over predictable timescales. Understood in these terms, the generation of a new fluorescent protein at this distance from existing proteins appears to be equivalent to simulating over 500 million years of evolution.

ESM3

ESM3 reasons over the sequence, structure, and function of proteins. All three modalities are represented by tokens, and are input and output as separate tracks that are fused into a single latent space within the model. ESM3 is trained with a generative masked language modeling objective:

$$\mathcal{L} = -\mathbb{E}_{x,m}\left[\frac{1}{|m|}\sum_{i\in m}\log p(x_i|x_{\setminus m})\right]$$

A random mask m is applied to the tokens x describing the protein, and the model is supervised to predict the identity of the tokens that have been masked. During training, the mask is sampled using a noise schedule that varies the fraction of positions that are masked so that ESM3 sees many different combinations of masked sequence, structure, and function, and predicts completions of any combination of the modalities from any other. This differs from classical masked language modeling (31) in that the supervision is applied across all possible masking rates rather than a single fixed masking rate. This supervision factorizes the probability distribution over all possible predictions of the next token given any combination of previous tokens, ensuring that tokens can be generated in any order from any starting point (32–34).

To generate from ESM3, tokens are iteratively sampled. Starting from a fully or partially masked context, tokens can be sampled one at a time, or in parallel, in any order, until all positions are fully unmasked (Fig. 1A). In addition to enabling generation, ESM3's training objective is also effective for representation learning. High masking rates improve the generative capability, while lower masking rates improve representation learning. We chose to train ESM3 with a noise schedule that balances generative capabilities with representation learning (Appendix A.2.2).

ESM3 is a bidirectional transformer. Sequence, structure, and function tokens are embedded and fused at the input, then processed through a stack of transformer blocks (Fig. 1B). At the output of the model, shallow multi-layer perceptron (MLP) heads project the final layer representation into token probabilities for each of the tracks. ESM3 uses tokenization, rather than specialized architectural components, to represent the complexity of proteins in a learned multimodal feature space. This enables efficient and highly scalable training.

Protein structures are tokenized by a discrete autoencoder (35), which is trained to compress three-dimensional structure into discrete tokens (Fig. 1C). We propose an invariant geometric attention mechanism to efficiently process three-dimensional structure. The mechanism operates in local reference frames defined by the bond geometry at each amino acid, and allows local frames to interact globally through a transformation into the global frame (Appendix A.1.6). The local structural neighborhoods around each amino acid are encoded into a sequence of discrete tokens, one for each amino acid.

When predicting or generating protein structure, the structure tokens output by ESM3 are passed through the decoder, which reconstructs the full atomic structure. The autoencoder is trained to encode and reconstruct atomic coordinates with a geometric loss that supervises the pairwise distances and relative orientations of bond vectors and normals (Appendix A.1.7.3.1). This tokenization delivers near-perfect reconstruction of protein structure (<0.5Å RMSD on CAMEO, Fig. S3).

Since the local neighborhoods of each structure token contain information about neighboring parts of the structure, we also provide the model with a mechanism to condition on backbone atomic coordinates directly via geometric attention in the first transformer block. To support higher level abstractions of structure, we include tracks for secondary structure (SS8) tokens and solvent accessible surface area (SASA) tokens. Keyword tokens describing biological activity, such as binding, enzymatic function, and domain or fold classifications, allow an even higher level semantic description of protein architecture and function. Residue level annotation tokens indicate functions of individual residues, such as catalytic sites and post-translational modifications.

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我们报告了使用 ESM3 生成一种新的绿色荧光蛋白 (GFP)。荧光蛋白负责水母和珊瑚的发光颜色 (29)，并且是现代生物技术中的重要工具 (30)。它们具有优雅的结构：一个由 11 条 β 链组成的桶形结构，中心穿过一条螺旋，这构成了由蛋白质自身原子形成的发光发色团。这一机制在自然界中是独特的——没有其他蛋白质能够自发地从其自身结构中形成荧光发色团——这表明即使对于自然界而言，产生荧光也是一项困难的任务。

我们新开发的蛋白质，命名为 esmGFP，与维多利亚水母 GFP 的序列一致性为 36%，与已知最相似的荧光蛋白的序列一致性为 58%。尽管 GFP 在数十年来作为蛋白工程的目标受到极大关注，但据我们所知，这样远源的新型 GFP 仅通过在自然界中的发现被识别。

在可预测的时间尺度上，自然界中的 GFP 也发生了类似数量的多样化。从这个角度理解，产生一种与现有蛋白质距离如此遥远的新型荧光蛋白，似乎相当于模拟了超过五亿年的进化。

ESM3

ESM3 基于蛋白质的序列、结构和功能进行推理。所有三种模式通过标记表示，并作为单独的轨道输入和输出，这些轨道在模型内融合为单一的潜在空间。ESM3 的训练目标是生成性掩码语言建模：

$$\mathcal{L} = -\mathbb{E}_{x,m}\left[\frac{1}{|m|}\sum_{i\in m}\log p(x_i|x_{\setminus m})\right]$$

随机掩码 m 应用于标记 x 描述蛋白质，并且模型通过监督训练来预测已经被掩盖的标记的身份。在训练过程中，掩码是使用噪声调度进行采样的，该噪声调度变化了被掩盖的位置的比例，使得ESM3可以看到许多不同的掩盖序列、结构和功能的组合，并预测任何其他组合的各个模态的补全。这与传统的掩码语言建模（31）不同，因为监督作用于所有可能的掩盖率，而不是单一的固定掩盖率。这种监督将所有可能的下一个标记的预测的概率分布因式分解，以给出任何之前标记组合的条件，确保标记能够从任何起始点以任何顺序生成（32–34）。

为了从 ESM3 生成，标记被逐步采样。从完全或部分被屏蔽的上下文开始，标记可以一次或同时以任意顺序采样，直到所有位置完全取消屏蔽（图1A）。除了能够生成内容，ESM3 的训练目标对于表征学习也非常有效。较高的屏蔽率提高了生成能力，而较低的屏蔽率则提高了表征学习。我们选择使用一种噪声调度来训练 ESM3，以平衡生成能力与表征学习（附录A.2.2）。

ESM3是双向变换器。序列、结构和功能标记在输入时嵌入并融合，然后经过变换器模块的堆叠处理（图1B）。在模型输出端，浅层多层感知机（MLP）头将最终层表示投影到每个轨道的标记概率。ESM3使用分词，而不是专门的架构组件，以在学习到的多模态特征空间中表示蛋白质的复杂性。这使得高效且高度可扩展的训练成为可能。

蛋白质结构通过离散自编码器进行标记化（35），该编码器的训练目标是将三维结构压缩成离散标记（图。1C）。我们提出了一种不变的几何注意机制，以高效处理三维结构。该机制在每个氨基酸的键几何定义的局部参考框架中操作，并允许局部框架通过转换为全局框架实现全局交互（附录 A.1.6）。每个氨基酸周围的局部结构邻域被编码为一系列离散标记，为每个氨基酸生成一个标记。

在预测或生成蛋白质结构时，ESM3 输出的结构标记通过解码器进行处理，后者重建完整的原子结构。自编码器经过训练以编码和重建原子坐标，采用几何损失来监督成对距离和键向量及法线的相对方向（附录A.1.7.3.1）。这种分词技术实现了蛋白质结构的近乎完美重建（<0。5 ˚在 CAMEO 上的 RMSD，图。S3）。

由于每个结构标记的局部邻域包含有关结构邻近部分的信息，我们还为模型提供了一种通过第一层变换器块直接基于几何注意机制对主链原子坐标进行条件化的方式。为了支持更高层次的结构抽象，我们包含了描述二级结构（SS8）标记和溶剂可接触表面积（SASA）标记的轨道。描述生物活性的关键字标记，例如结合、酶功能及结构域或折叠分类，允许对蛋白质的架构和功能进行更高级别的语义描述。残基级注释标记指示个别残基的功能，例如催化位点和翻译后修饰。

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Simulating 500 million years of evolution with a language model

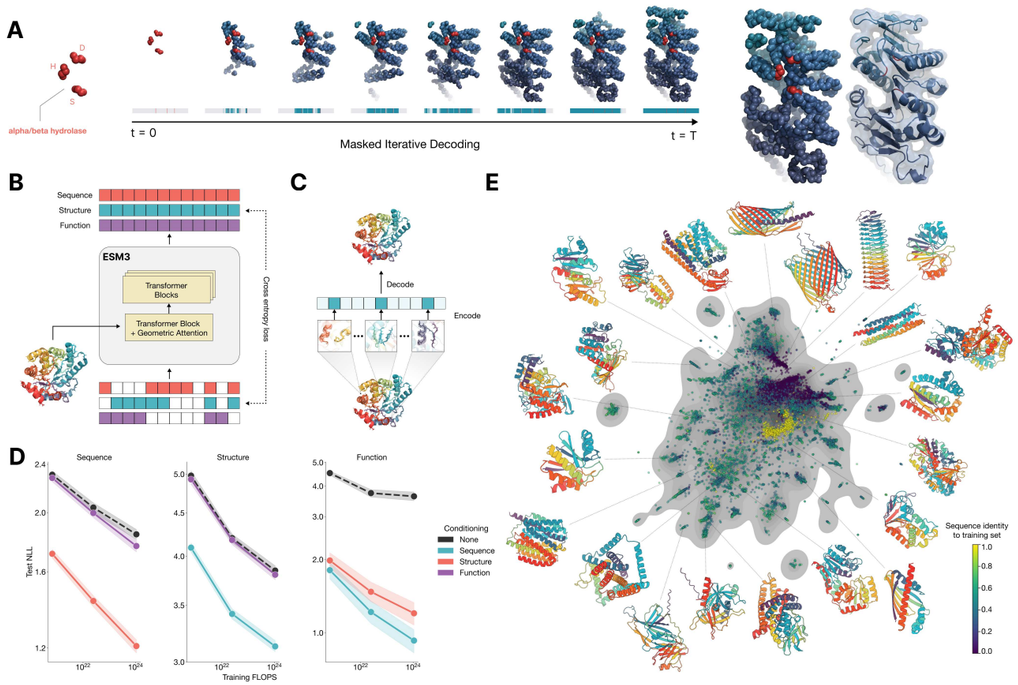

Figure 1. ESM3 is a generative language model that reasons over the sequence, structure, and function of proteins. (A) Iterative sampling with ESM3. Generation of an alpha/beta hydrolase. Sequence, structure, and function can all be used to prompt the model. At each timestep t, a fraction of the masked positions are sampled until all positions are unmasked. (B) ESM3 architecture. Sequence, structure, and function are represented as tracks of discrete tokens at the input and output. The model is a series of transformer blocks, where all tracks are fused within a single latent space; geometric attention in the first block allows conditioning on atomic coordinates. ESM3 is supervised to predict masked tokens. (C) Structure tokenization. Local atomic structure around each amino acid is encoded into tokens. (D) Models are trained at three scales: 1.4B, 7B, and 98B parameters. Negative log likelihood on test set as a function of training FLOPs shows response to conditioning on each of the input tracks, improving with increasing FLOPs. (E) Unconditional generations from ESM3 98B (colored by sequence identity to the nearest sequence in the training set), embedded by ESM3, and projected by UMAP alongside randomly sampled sequences from UniProt (in gray). Generations are diverse, high quality, and cover the distribution of natural sequences.

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使用语言模型模拟五亿年的进化

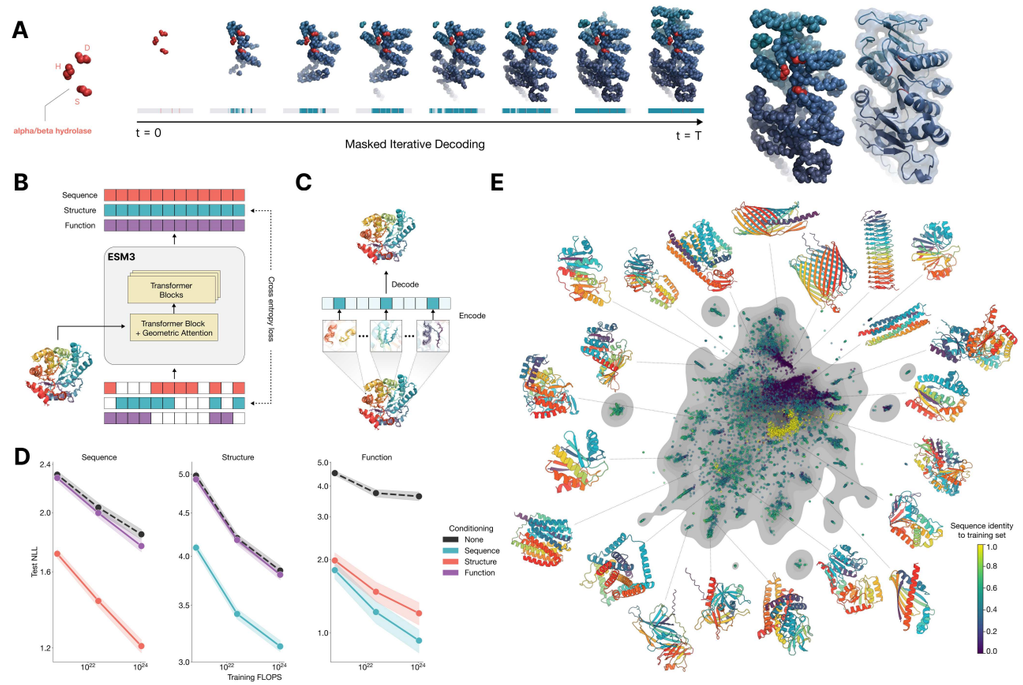

图1. ESM3是一个生成性语言模型，它对蛋白质的序列、结构和功能进行推理。(A)使用ESM3进行迭代采样。生成一个α/β水解酶。序列、结构和功能都可用于提示模型。在每个时间步t，采样一部分掩蔽位置，直到所有位置被解掩。(B)ESM3架构。序列、结构和功能在输入和输出中表示为离散标记轨道。模型是一个由变换器模块组成的系列，其中所有轨道在一个单一的潜在空间中融合；第一个模块中的几何注意力允许对原子坐标进行条件化。ESM3经过监督以预测掩蔽标记。(C)结构标记化。每个氨基酸周围的局部原子结构被编码为标记。(D)模型在三个尺度下训练：1.4B、7B和98B参数。在测试集上的负对数似然作为训练FLOPs的函数显示对每个输入轨道条件的响应，随着FLOPs的增加而改善。(E)来自ESM3 98B的无条件生成（根据与训练集中最近序列的序列一致性着色），通过ESM3嵌入，并通过UMAP投影，与来自UniProt的随机采样序列（灰色）并排。生成结果多样、高质量，并覆盖自然序列的分布。

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The largest ESM3 model is trained on 2.78 billion natural proteins collected from sequence and structure databases (2, 36–39). Since a small fraction of structures have been experimentally determined relative to sequences, we leverage predicted structures (4, 5). Sequences are annotated with function keywords using a library of hidden markov models (40). We also generate synthetic sequences with an inverse folding model (described in Appendix A.2.1.3) for all structures, including predicted ones. Overall this increases training data to 3.15 billion protein sequences, 236 million protein structures, and 539 million proteins with function annotations, totaling 771 billion unique tokens. Full details of the training dataset are described in Appendix A.2.1.

We train ESM3 models at three scales: 1.4 billion, 7 billion, and 98 billion parameters. In an initial series of experiments to evaluate representation learning performance in response to architecture hyperparameters, we found a greater response to increasing depth than to width. This informed the choice of relatively deep networks for the final architectures, with the 98 billion parameter model incorporating 216 transformer blocks (Appendix A.1.5).

Scaling ESM3 from 1.4 billion to 98 billion parameters results in substantial improvements in the loss for all tracks on the test set, with the greatest improvements observed in sequence loss (Fig. 1D, Fig. S11). These gains in test loss lead to better representation learning (Table S7 and Fig. S8). In single sequence structure prediction ESM3 98B surpasses ESMFold (0.895 vs. 0.865 mean local distance difference test, LDDT, CAMEO test set; Table S8). Generating sequences from the model without prompting (unconditional generation) produces high quality proteins—with a mean predicted LDDT (pLDDT) 0.84 and predicted template modeling score (pTM) 0.52—that are diverse in both sequence (mean pairwise sequence identity 0.155) and structure (mean pairwise TM score 0.48), spanning the distribution of known proteins (Fig. 1E, Fig. S13).

While extensive research has gone into creating specialized architectures and training objectives for proteins, our results show that scaling with language modeling, enabled by tokenization, efficient architectures, and masked token prediction, yields continued improvements in both representational and generative applications. This approach allows the model to build a shared multimodal representation space that is learned from the data, rather than explicitly hard-coded into its architecture, which given increasing compute and data could learn an increasingly richer and more general feature space. In the following sections we show that this approach achieves high fidelity for the controllable generation of proteins.

Programmable design with ESM3

We explore the ability of ESM3 to follow complex prompts with different compositions. ESM3 can be prompted with instructions from each of its input tracks: sequence, structure coordinates, secondary structure (SS8), solvent-accessible surface area (SASA), and function keywords. This allows prompts to be specified at multiple levels of abstraction, from atomic level structure to high level keywords describing the function and fold topology.

We evaluate ESM3's ability to follow prompts in each of the tracks independently (Fig. 2A). A set of prompts are constructed for each of the tracks using a temporally held out test set of natural proteins (Appendix A.3.7). The resulting generations are evaluated using ESMFold for consistency with the prompt and confidence of structure prediction (pTM). We define consistency metrics for each track: constrained site RMSD (cRMSD), the RMSD between the coordinates of the prompt, i.e. the positions of the backbone atoms, and the corresponding coordinates in the generation; SS3 accuracy, the fraction of residues where three-class secondary structure between the prompt and generations match; SASA Spearman ρ, the correlation between the SASA prompt and the corresponding region of the generation; and keyword recovery, the fraction of prompt keywords recovered by InterProScan (40). Across all tracks, the 7B parameter ESM3 finds solutions that follow the prompt and have structures which are confidently predicted by ESMFold (pTM > 0.8).

Unconditional generations reflect the distribution of natural proteins. Since we observed ESM3 can faithfully follow prompts, we reasoned that prompting could steer the model to generate proteins that differ from the training set and natural proteins. First we test the ability of the model to follow out-of-distribution prompts. We construct a set of prompts combining SS8 and SASA from held out structures (TM < 0.7 to training set). Under these prompts, while the model continues to generate coherent globular structures (mean pTM 0.85 ± 0.03), the distribution of similarities to the training set (as measured by TM-score and sequence identity) shifts to be more novel (average sequence identity to nearest training set protein $< 20\%$ and mean TM-score 0.48 ± 0.09; Fig. 2B top). To test the ability to generalize to structures beyond the distribution of natural proteins, we use secondary structure prompts derived from a dataset of artificial symmetric protein designs distinct from the natural proteins found in the training dataset (Appendix A.3.8). Similarly, ESM3 produces high confidence generations (pTM > 0.8, pLDDT > 0.8) with low sequence and structure similarity to proteins in the training set (sequence identity $< 20\%$ and TM-score 0.52 ± 0.10; Fig. 2B bottom), indicating that the model can be used to generate protein sequences and structures highly distinct from those that exist in nature.

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最大的ESM3模型在2.78十亿个自然蛋白质上进行训练，这些蛋白质来自序列和结构数据库（2, 36–39）。由于相对于序列，只有一小部分结构是通过实验确定的，我们利用预测的结构（4, 5）。序列使用隐藏马尔可夫模型库进行功能关键词的注释（40）。我们还生成具有逆折叠模型的合成序列（在附录中描述A.2.1.3），包括预测的结构。总体而言，这使得训练数据的数量增加到3.15十亿个蛋白质序列、2.36亿个蛋白质结构，以及5.39亿个带有功能注释的蛋白质，总计7710亿个唯一标记。训练数据集的完整细节在附录中描述A.2.1。

我们在三个规模上训练ESM3模型：14亿、70亿和980亿个参数。在最初的一系列实验中，为了评估对架构超参数的表征学习性能反应，我们发现对深度的增加反应大于对宽度的反应。这为最终架构选择相对较深的网络提供了指导，其中980亿参数模型包含216个变换器模块（附录A.1.5）。

将ESM3从14亿参数扩展到98亿参数，导致测试集所有轨道上的损失显著改善，其中序列损失的改善最为明显（图1D，图。S11)。这些测试损失的降低促进了更好的表征学习（表S7 和图。S8）。在单一序列结构预测中，ESM3 98B超过了ESMFold（0.895对比0.865的平均局部距离差异测试，LDDT，CAMEO测试集；表S8）。从模型中生成的序列，未使用提示（无条件生成）产生了高质量的蛋白质——平均预测LDDT（pLDDT）0.84 和预测模板建模评分（pTM）0.52——在序列（平均成对序列一致性0.155）和结构（平均成对TM分数0.48）上都表现出多样性，涵盖了已知蛋白质的分布（图。1E，图。S13）。

尽管已有大量研究致力于为蛋白质创建专门的架构和训练目标，但我们的结果表明，利用分词、有效的架构和掩码标记预测进行语言建模的扩展，能够在表征和生成应用中持续带来改进。该方法使模型能够构建一个从数据中学习的共享多模态表示空间，而不是明确地硬编码到其架构中，这在计算和数据不断增加的情况下，可以学习到越来越丰富和更通用的特征空间。在接下来的章节中，我们将展示该方法在蛋白质的可控生成中实现了高保真度。

基于ESM3的可编程设计

我们探索了ESM3按照不同组成遵循复杂提示的能力。ESM3可以根据其每个输入轨道的指令进行提示：序列、结构坐标、二级结构（SS8）、溶剂可接触表面积（SASA）和功能关键词。这使得可以在多个抽象层次上指定提示，从原子级别的结构到描述功能和折叠拓扑的高层关键词。

我们评估ESM3在每个轨道独立跟随提示的能力（图。2A）。为每个轨道构建一组提示，采用暂时保留的自然蛋白质测试集（附录A.3.7）。生成的结果使用ESMFold评估其与提示的一致性以及结构预测的置信度（pTM）。我们为每个轨道定义一致性度量：约束位点RMSD（cRMSD），即提示的坐标（即骨架原子的位置信息）与生成中相应坐标之间的RMSD；SS3准确性，提示和生成之间三类二级结构匹配的残基比例；SASA Spearman ρ，SASA提示与生成中对应区域之间的相关性；关键词回收率，即被InterProScan恢复的提示关键词的比例（40）。在所有轨道中，7B参数ESM3找到了遵循提示的解决方案，并且其结构被ESMFold（pTM > 0.8）自信地预测。

无条件生成反映了自然蛋白质的分布。由于我们观察到ESM3能够忠实地遵循提示，因此我们推测提示可以引导模型生成与训练集和自然蛋白质不同的蛋白质。首先，我们测试模型跟随分布之外提示的能力。我们构建了一组提示，将来自于被保留结构的SS8和SASA结合在一起（TM < 0.7 到训练集）。在这些提示下，尽管模型继续生成一致的球状结构（平均pTM 0.85 ± 0.03），与训练集的相似性分布（通过TM-score和序列一致性测量）发生了变化，变得更加新颖（与最近的训练集蛋白质的平均序列一致性 $< 20\%$和平均TM-score0.48 ± 0.09；图2B顶部）。为了测试模型推广到超出自然蛋白质分布的结构的能力，我们使用从一组与训练数据集中的自然蛋白质不同的人工对称蛋白质设计得出的二级结构提示（附录A.3.8）。同样，ESM3生成的结果具有高置信度（pTM> 0.8, pLDDT > 0.8），且与训练集中的蛋白质具有低序列和结构相似性（序列一致性$< 20\%$和TM-score 0.52 ± 0.10；图2B底部），表明该模型可用于生成与自然界中存在的蛋白质序列和结构高度不同的蛋白质。

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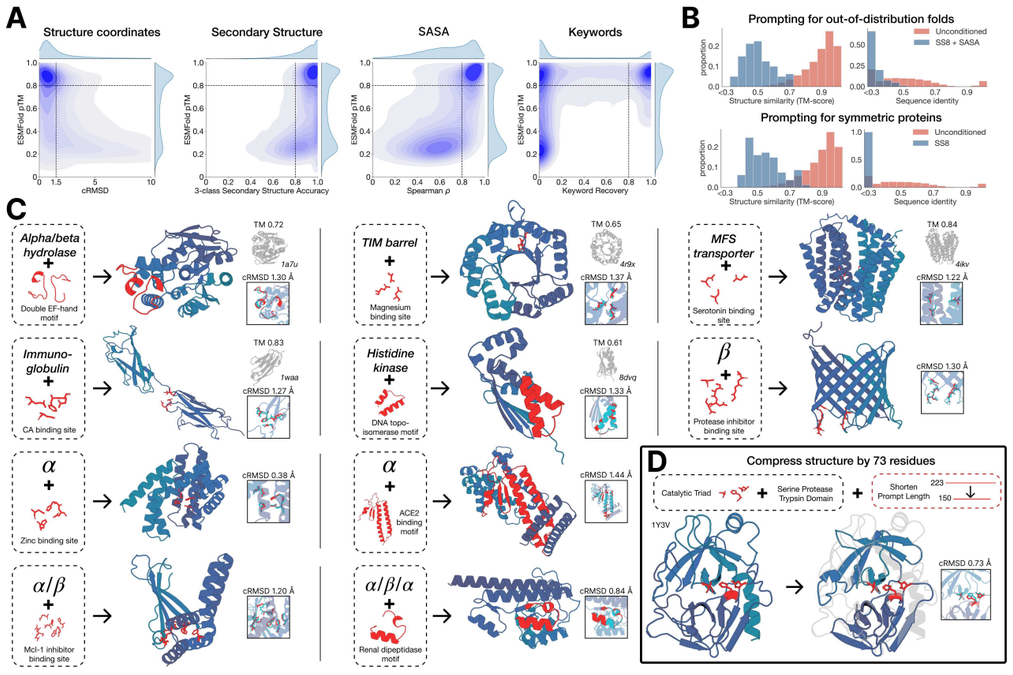

Figure 2. Generative programming with ESM3. (A) ESM3 can follow prompts from each of its input tracks. Density of faithfulness to prompting for each of the tracks is shown. Generations achieve consistency with the prompt (backbone cRMSD, SS3 accuracy, SASA Spearman ρ, keyword recovery) and high structure prediction confidence (pTM). (B) ESM3 can be prompted to generate proteins that differ in structure (left) and sequence (right) from the training set and natural proteins. Prompted generations (blue) shift toward a more novel space vs. unconditional generations (red), in response to prompts derived from out-of-distribution natural structures (upper panel) and computationally designed symmetric proteins (lower panel). (C) ESM3 generates creative solutions to a variety of combinations of complex prompts. We show compositions of atomic level motifs with high level instructions specified through keywords or secondary structure. Fidelity to the prompt is shown via similarity to a reference structure (for keyword prompts) and all-atom RMSD (for motif prompts). Solutions differ from the scaffolds where the motif was derived (median TM-score 0.36 ± 0.14), and for many motifs (e.g. serotonin, calcium, protease inhibitor, and Mcl-1 inhibitor binding sites), we could find no significant similarity to other proteins that contain the same motif. (D) An example of especially creative behavior. ESM3 compresses a serine protease by 33% while maintaining the active site structure.

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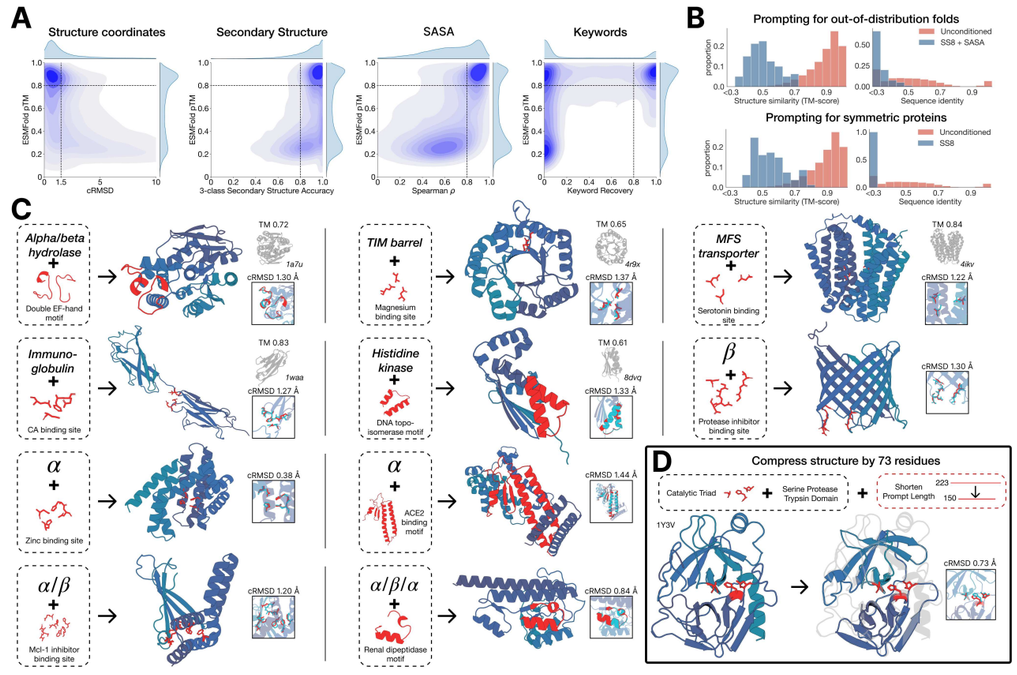

图2. 使用ESM3的生成编程。(A) ESM3能够根据其输入轨道的提示进行响应。每个轨道对提示的忠实度密度如图所示。生成结果在与提示的一致性（主链cRMSD、SS3准确性、SASA Spearman ρ、关键词恢复）以及高结构预测置信度（pTM）方面达到了一致性。(B) ESM3可以被提示生成与训练集和自然蛋白质在结构（左侧）和序列（右侧）上有所不同的蛋白质。相较于无条件生成（红色），响应来自于分布外自然结构（上面面板）和计算设计的对称蛋白质（下面面板）的提示，提示生成（蓝色）趋向于更加新颖的空间。(C) ESM3对于一系列复杂提示的组合生成创造性的解决方案。我们展示了通过关键词或二级结构指定的高层次指令与原子级别的基序组成。忠实于提示的效果通过与参考结构的相似性（针对关键词提示）以及全原子RMSD（针对基序提示）来展示。解决方案与提取基序的支架有明显不同（中位TM-score为0.36 ± 0.14），并且对于许多基序（例如血清素、钙、蛋白酶抑制剂和Mcl-1抑制剂结合位点），我们未能发现与包含相同基序的其他蛋白质之间的显著相似性。(D) 一个特别创造性行为的示例。ESM3在保持活性位点结构的同时，将一个丝氨酸蛋白酶的体积压缩了33%。

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ESM3 is able to follow complex prompts, and has the ability to compose prompts from different tracks, and at different levels of abstraction. To evaluate this ability, we prompt ESM3 with motifs that require solving for spatial coordination of individual atoms, including atoms participating in tertiary contacts between residues far apart in the sequence, such as catalytic centers and ligand binding sites. We combine the motif prompts with prompts that specify the fold architecture. For each unique combination of motif and scaffold, we generate samples until there is a success (all-atom RMSD < 1.5Å, for atomic coordinates including side-chain atoms; TM > 0.6 to a representative structure for fold level prompts; SS3 accuracy $> 80\%$ for secondary structure prompts; and high confidence in the scaffold, with pTM > 0.8, pLDDT > 0.8).

We find that ESM3 is able to solve a wide variety of such tasks (Fig. 2C). It does so without retrieving the motif's original scaffold (median TM-score of 0.40 ± 0.10 to reference protein; Appendix A.3.9). In some cases, the scaffolds are transferred from existing proteins which have similar motifs (for example, the ESM3-designed alpha-helical scaffold for the zinc-binding motif has high similarity to Ni_{2+}-binding proteins, PDB: 5DQW, 5DQY; Fig. 2C, row 3 column 1). For many motifs (e.g., binding sites for serotonin, calcium, protease inhibitor, and Mcl-1 inhibitor) Foldseek (41) finds no significant similarity to other proteins that contain the same motif. In these cases we observe that sometimes the motif has been grafted into entirely different folds (e.g. a protease inhibitor binding site motif in a beta-barrel which is most similar to a membrane-bound copper transporter, PDB: 7PGE; Fig. 2C, row 3 column 3). At other times, the scaffold appears to be entirely novel, such as an alpha/beta protein designed to scaffold the Mcl-1 inhibitor binding motif, which has low structural similarity to all known proteins in the PDB, ESMAtlas, and the AlphaFold databases (max. TM-score < 0.5; Fig. 2C, row 4 column 1). Overall, the generated solutions have high designability, i.e. confident recovery of the original structure after inverse folding with ESM-IF1 (42) and refolding with ESMFold (median pTM 0.80 ± 0.08; scTM 0.96 ± 0.04; Appendix A.3.9).

Through experiments with prompt engineering, we have observed especially creative responses to prompts. Here, we highlight an example of protein compression (Fig. 2D). Starting from a natural trypsin (PDB 1Y3V), we prompt with the sequence and coordinates of the catalytic triad as well as functional keywords describing trypsin, but reduce the overall generation length by a third (from 223 to 150 residues). ESM3 maintains the coordination of the active site (all-atom RMSD 0.73Å) and the overall fold with high designability (pTM 0.84, scTM mean 0.97, std 0.006), despite the significant reduction in sequence length and the fold only being specified by the function keyword prompt (Appendix A.3.10).

These examples illustrate ESM3's ability to find creative solutions to prompts specified in any of its input tracks, individually or in combination. This capability enables a rational approach to protein design, providing control at various levels of abstraction, from high level topology to atomic coordinates, using a generative model to bridge the gap between the prompt and biological complexity.

Biological alignment

While we have observed meaningful increases in the performance of the base models with scale, larger models could have even greater latent capabilities that we do not observe. The base ESM3 models can be prompted to perform difficult tasks such as tertiary motif scaffolding and composition of prompts, despite the fact that the models have not been explicitly optimized for these objectives. Since the properties we evaluate generative outputs on—such as adherence to the prompt, or the confidence of the scaffold—are only seen by the model indirectly during pretraining, aligning the model directly to the generative task with finetuning could elicit even greater capability differences with larger models.

We study how the base models can be aligned (43, 44) to generate proteins that satisfy challenging prompts. For each model, we construct a dataset of backbone atomic coordinate prompts, consisting of contiguous spans of residues, and tertiary motifs (which also specify the identities of the contacting amino acids). We generate multiple protein sequences for each prompt and fold each of the sequences using ESM3, scoring for consistency with the prompt (backbone cRMSD) and structure prediction confidence (pTM). High quality samples are paired with low quality samples for the same prompt to construct a preference dataset (Appendix A.4). ESM3 is then finetuned with a preference optimization loss (45, 46), which causes the model to put higher likelihood on the high quality samples relative to the low quality samples.

After aligning each of the base models, we evaluate their absolute performance, and the shift in the distribution of generations. We focus on a series of challenging prompts that require coordination of the backbone atoms of residues in tertiary contact. We evaluate the ability to generate high quality scaffolds (pTM > 0.8) that follow the prompt with high resolution (backbone cRMSD < 1.5Å), using ESMFold for the evaluation. We prompt each model with amino acid identities and backbone atomic coordinates from a held-out dataset of 46 ligand binding motifs (Appendix A.4.5). For each motif, we create 1024 prompts by permuting the order of the residues, varying their position in the sequence, and varying the length of the sequence. A single protein is generated per prompt. The 1024 generations for each motif are used to construct an unbiased estimator of the fraction of tertiary coordination tasks solved after 128 gen-

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ESM3能够遵循复杂的提示，并具备从不同轨迹和不同抽象层次构建提示的能力。为了评估这一能力，我们对ESM3提出了需要解决单个原子空间坐标的图案提示，包括参与远离序列的残基之间的三级接触的原子，例如催化中心和配体结合位点。我们将图案提示与指定折叠架构的提示相结合。对于每个独特的图案和支架组合，我们生成样本，直到出现成功（全原子RMSD < 1.5 ˚A，包括侧链原子的原子坐标；TM > 0.6 与折叠层级提示的代表性结构；SS3准确性 $> 80\%$用于二级结构提示；以及对支架的高信心，具有pTM > 0.8和pLDDT > 0.8的值）。

我们发现ESM3能够解决各种此类任务（图。2C）。它在不检索图案原始支架的情况下做到这一点（中位数TM分数为0。40 ± 0.10与参考蛋白质对比；附录A.3.9）。在某些情况下，支架是从具有相似图案的现有蛋白质中转移过来的（例如，ESM3设计的锌结合图案的α-螺旋支架与Ni2+-结合蛋白质高度相似，PDB: 5DQW, 5DQY；图。2C，行3列1）。对于许多图案（例如，血清素、钙、蛋白酶抑制剂和Mcl-1抑制剂的结合位点），Foldseek（41）未能找到与包含相同图案的其它蛋白质的显著相似性。在这些情况下，我们观察到图案有时已被移植到完全不同的折叠中（例如，一个蛋白酶抑制剂结合位点图案在一个β-桶中，最接近于膜结合的铜转运蛋白，PDB: 7PGE；图。2C，行3列3）。在其他情况下，支架似乎完全新颖，例如一个设计用于支撑Mcl-1抑制剂结合图案的α/β蛋白，其与PDB、ESMAtlas和AlphaFold数据库中所有已知蛋白质的结构相似性较低（最大TM分数 < 0.5；图。2C，行4列1）。总体而言，生成的解决方案具有高设计性，即在使用ESM-IF1进行反向折叠后，对原始结构的自信恢复（42）以及使用ESMFold进行重新折叠（中位值pTM0.80 ± 0.08；scTM 0.96 ± 0.04；附录 A.3.9）。

通过提示工程的实验，我们观察到对提示的尤其富有创意的响应。在这里，我们突出一个蛋白质压缩的例子（图。2D）。从自然胰蛋白酶（PDB 1Y3V）开始，我们使用催化三联体的序列和坐标作为提示，并使用描述胰蛋白酶的功能关键词，但将整体生成长度减少了三分之一（从223减少到150个残基）。ESM3 维持了活性位点的协调性（全原子 RMSD 0。73 ˚A）以及高设计性下的整体折叠（pTM 0.84，scTM 平均值 0.97，标准差 0.006），尽管序列长度显著减少并且折叠仅通过功能关键词提示进行指定（附录A.3.10）。

这些示例展示了ESM3在任何输入轨道中找到针对提示的创造性解决方案的能力，无论是单独使用还是组合使用。这一能力使得在蛋白质设计中能够采取一种理性的方法，在从高级拓扑到原子坐标的不同抽象层次上提供控制，利用生成模型来架起提示与生物复杂性之间的桥梁。

生物对齐

尽管我们观察到基础模型在规模上的性能有了显著提升，但更大的模型可能具有我们未观察到的更强隐含能力。基础ESM3模型可以被提示执行诸如三级结构模板搭建和提示组成等困难任务，尽管这些模型并未为这些目标进行明确优化。由于我们评估生成输出的属性——例如对提示的遵循程度或支架的置信度——在预训练期间仅被模型间接感知，因此通过微调将模型直接对齐到生成任务可能会造成更大的能力差异，尤其是在更大的模型中。

我们研究基础模型如何被对齐（43， 44）以生成满足挑战性提示的蛋白质。对于每个模型，我们构建一个由主链原子坐标提示组成的数据集，该提示包含连续的残基区间和指定接触氨基酸身份的三级结构模板。我们为每个提示生成多个蛋白质序列，并使用ESM3对每个序列进行折叠，评估其与提示的一致性（主链cRMSD）及结构预测置信度（pTM）。高质量样本与同一提示下的低质量样本配对，以构建偏好数据集（附录A.4）。然后，ESM3利用偏好优化损失（45， 46）进行微调，从而使模型相对于低质量样本更倾向于高质量样本。

在对每个基础模型进行对齐后，我们评估它们的绝对性能以及生成分布的变化。我们专注于一系列挑战性的提示，这些提示要求协调处于三级接触的残基的主链原子。我们评估生成高质量支架的能力（pTM > 0.8），该支架遵循高分辨率的提示（主链cRMSD < 1.5 ˚A），并使用ESM-Fold进行评估。我们用来自46个配体结合基序的保持数据集中的氨基酸身份和主链原子坐标对每个模型进行提示（附录A.4.5）。对于每个基序，我们通过排列残基的顺序、改变它们在序列中的位置以及改变序列的长度创建1024个提示。每个提示生成一个单一的蛋白质。每个基序的1024个生成用于构建在128个生成后解决的三级协调任务的比例的无偏估计器。

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Simulating 500 million years of evolution with a language model

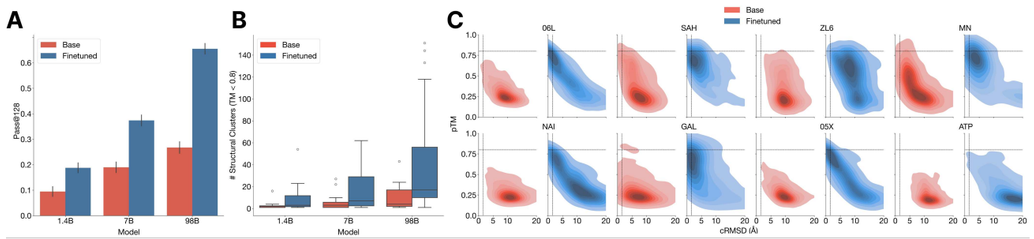

Figure 3. The ability to solve complex tasks increases with scale through alignment. ESM3 is aligned to follow tertiary coordination prompts with a dataset of preference pairs constructed from prompted generations, where positive samples with good scores for desired properties (high pTM, low cRMSD) are paired with negative samples with worse scores. The preference tuning loss encourages the model to put higher likelihood on the positive samples. After training, models are evaluated by prompting with the backbone atomic coordinates of residues in tertiary contact. (A) We show the effect of finetuning on the fraction of tasks solved with 128 generations (Pass@128; 2 s.d. error bars). A large gap opens between the models with scale. The response to alignment shows a latent capability to solve complex tasks in the largest model. (B) Number of distinct solutions (clustered at TM $>$ 0.8) generated for each tertiary motif. After finetuning there are often many unique solutions for ligands where there are successes. (C) Densities of prompted generations are shown for the base model (left) and aligned model (right) at the 98B scale for a number of randomly selected ligands. After alignment, the fidelity to the prompt (backbone cRMSD) and quality of generations (pTM) tends to improve substantially.

erations (Pass@128; Appendix A.4.5).

Aligned models solve double the tertiary coordination tasks compared to base models (Fig. 3A). While the base models show differences in the percentage of tasks solved (9.5% for 1.4B, 19.0% for 7B, 26.8% for 98B; Fig. 3A), a much larger capability difference is revealed through alignment (increasing from 9.5% to 18.8%, 19.0% to 37.4%, and 26.8% to 65.5% for the 1.4B, 7B and 98B models, respectively). Preference-tuned models not only solve a greater proportion of tasks, but also find a greater number of solutions per task, as evaluated by the number of distinct structural clusters (TM $>$ 0.8) with backbone cRMSD $<$ 1.5Å and pTM $>$ 0.8 (Fig. 3B). A shift in the distribution of ESMFold pTM and backbone cRMSD for each ligand binding motif is observed (Fig. 3C; Fig. S17). At the 98B scale, the finetuned model produces more distinct successful clusters than the base model on 37 of the 46 tested ligands, while the remaining 9 ligands were not solved by either the base or aligned model, indicating that alignment almost universally improves the faithfulness to the prompt and confidence of the structure prediction for the generated proteins. Compared to a supervised finetuning baseline, which only maximizes the likelihood of the positive examples, preference tuning leads to larger improvements at all scales (Appendix A.4.6).

Our experiments with alignment reveal a considerable difference in capabilities between model scales. The largest aligned model improves substantially relative to the base model before alignment, as well as in comparison to the smaller models after alignment. Through alignment the models learn to generalize from a small number of examples: the distribution of generations shifts to improve the quality of scaffolds, and consistency with prompts, increasing the fraction of tasks solved and the number of distinct solutions.

Alignment requires the models to learn by example. The ability to identify the underlying properties that are illustrated by the finetuning examples, and to generalize those demonstrations to new tasks, implies that there is an internal representation of the properties that the finetuning accesses. This representation space is learned through the process of pretraining, where the model is trained on proteins across evolution, which suggests it reflects and contains the immense variety and complexity of protein biology. Such a representation space is likely to contain features that support generalization of many biological properties. The greater responsiveness of larger models to alignment suggests that their internal representation space better approximates those underlying properties, which is evidence of a deep capability for transfer, via the features learned in pretraining, that improves with scale.

Generating a new fluorescent protein

We sought to understand if the base pretrained ESM3 model has sufficient biological fidelity to generate functional proteins. We set out to create a functional green fluorescent protein (GFP) with low sequence similarity to existing ones. We chose the functionality of fluorescence because it is difficult to achieve, easy to measure, and one of the most beautiful mechanisms in nature.

Responsible for the fluorescence of jellyfish and the vivid colors of coral (47), proteins in the GFP family are unique

用语言模型模拟五亿年的进化

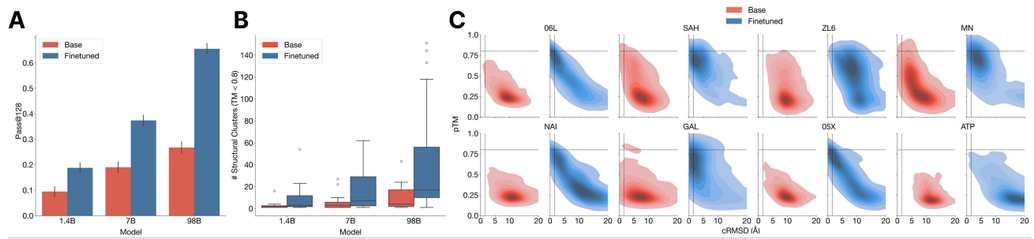

图3. 解决复杂任务的能力随着对齐而通过规模增加。ESM3按照与从提示生成构建的偏好对的数据集相结合的三维协调提示进行对齐，其中具有良好所需属性得分（高pTM，低cRMSD）的正样本与得分较低的负样本成对。偏好调整损失鼓励模型对正样本赋予更高的可能性。在训练后，通过提示与三维接触残基的主链原子坐标评估模型。(A) 我们展示了细调对通过128代解决的任务比例（Pass@128；2个标准差误差条）的影响。随着模型规模的增长，之间出现了巨大的差距。对齐的响应显示出最大模型在解决复杂任务方面的潜在能力。(B) 为每个三维构型生成的不同解决方案数量（在TM $>$ 0.8处聚类）。在细调后，对于那些成功的配体，通常会有许多独特的解决方案。(C) 在98B规模下，展示了基本模型（左）和对齐模型（右）在多个随机选择的配体中的提示生成密度。在对齐后，对提示的忠实度（主链cRMSD）和生成质量（pTM）往往显著改善。

操作 (Pass@128; 附录 A.4.5).

对齐的模型解决的三级协调任务数量是基础模型的两倍（图.3A）。虽然基础模型在解决任务的百分比上表现出差异（1.4B 为 9.5%，7B 为 19.0%，98B 为 26.8%；图.3A），但通过对齐揭示了更大的能力差异（对于 1.4B、7B 和 98B 模型，分别从 9.5% 增加到 18.8%、19.0% 增加到 37.4%、26.8% 增加到 65.5%）。偏好调优模型不仅解决了更大比例的任务，而且每个任务找到的解决方案数量也更多，这通过不同结构簇的数量（TM $>$ 0.8）以及主链 cRMSD $<$ 1.5 ˚A 和 pTM $>$ 0.8 进行评估（图.3B）。观察到每个配体结合模式的 ESMFold pTM 和主链 cRMSD 的分布发生了变化（图.3C；图.S17）。在 98B 规模下，微调模型在 46 个测试配体中的 37 个上产生了比基础模型更多的不同成功簇，而其余 9 个配体则未被基础模型或对齐模型解决，这表明对齐几乎普遍提高了对提示的忠诚度和对生成蛋白质的结构预测的信心。与仅最大化正例可能性的监督微调基线相比，偏好调优在所有规模上均导致更大幅度的改进（附录A.4.6）。

我们的对齐实验揭示了不同模型规模之间能力的显著差异。与未对齐的基模型相比，最大的对齐模型在对齐前后都有显著提升，同时与小规模模型相比亦是如此。通过对齐，这些模型学会了从少量示例中进行泛化：生成的分布发生变化，从而提升了

支架的质量和与提示的一致性，增加了解决的任务比例和不同解决方案的数量。

对齐要求模型通过示例进行学习。识别通过微调示例所展示的潜在属性的能力，并将这些展示推广到新任务，意味着对微调所访问属性的内部表示存在。这种表示空间是通过预训练过程学习的，在此过程中模型在进化过程中学习蛋白质，这表明它反映并包含了蛋白质生物学的巨大多样性和复杂性。这种表示空间可能包含支持多种生物特性泛化的特征。较大模型对对齐的更大响应性表明，它们的内部表示空间更好地近似那些潜在属性，这是通过在预训练中学习的特征而改善迁移深能力的证据，且随着规模的增加而提高。

生成一个新的荧光蛋白

我们希望了解基础的预训练ESM3模型是否具有生成功能性蛋白质的足够生物学保真度。我们着手创建一种与现有蛋白质序列相似度较低的功能性绿色荧光蛋白（GFP）。我们选择荧光作为功能，因为它难以实现、易于测量，并且是自然界中最美丽的机制之一。

负责水母的荧光和珊瑚鲜艳颜色的47，GFP家族中的蛋白质是独特的

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Simulating 500 million years of evolution with a language model

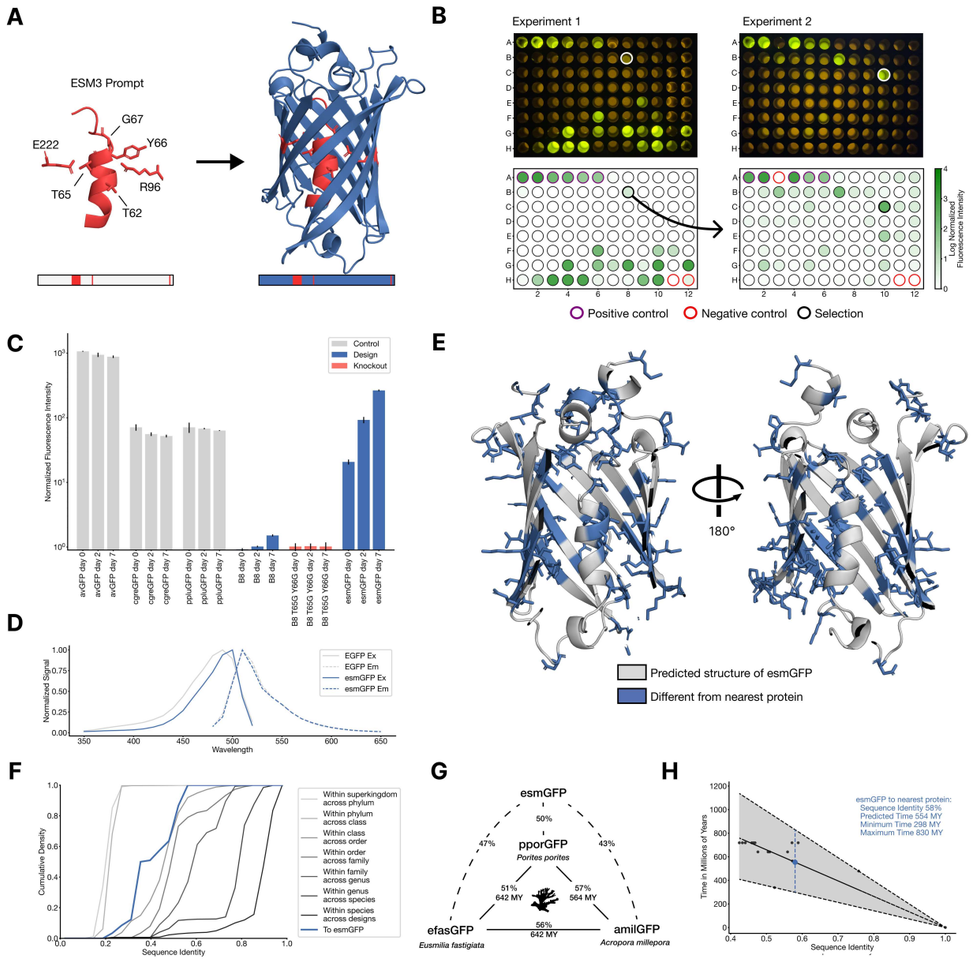

Figure 4. Generating a new fluorescent protein with a chain of thought. (A) We prompt ESM3 with the sequence and structure of residues required for forming and catalyzing the chromophore reaction, as well as the structure of part of the central alpha helix from a natural GFP (left). Through a chain of thought, ESM3 generates design candidates (right). (B) ESM3 found a bright GFP distant from other known GFPs in two experiments. We measured fluorescence in *E. coli* lysate. Top row, photograph of plates. Bottom row, plate reader fluorescence quantification. Positive controls of known GFPs are marked with purple circles, negative controls with no GFP sequence or no *E. coli* are marked with red circles. In the first experiment (left) we expressed designs with a range of sequence identities. A notable design with low sequence identity (57%) to known fluorescent proteins appears in the well labeled B8 (highlighted in a black circle bottom, white circle top). We continue the chain of thought from the protein in B8 for the second experiment (right). A bright design appears in the well labeled C10 (58% sequence identity to known fluorescent proteins, black circle bottom, white circle top) which we designate esmGFP. (C) esmGFP exhibits fluorescence intensity similar to common GFPs. Normalized fluorescence is shown for a subset of proteins in experiment 2. (D) Excitation and emission spectra for esmGFP overlaid on the spectra of EGFP. (E) Two cutout views of the central alpha helix and the inside of the beta barrel of a predicted structure of esmGFP. The 96 mutations esmGFP has relative to its nearest neighbor, tagRFP, are shown in blue. (F) Cumulative density of sequence identity between fluorescent proteins across taxa. esmGFP has the level of similarity to all other FPs that is typically found when comparing sequences across orders, but within the same class. (G) Evolutionary distance by time in millions of years (MY) and sequence identities for three example anthozoan GFPs and esmGFP. (H) Estimator of evolutionary distance by time (MY) from GFP sequence identity. We estimate esmGFP is over 500 million years of natural evolution removed from the closest known protein.

使用语言模型模拟五亿年的进化

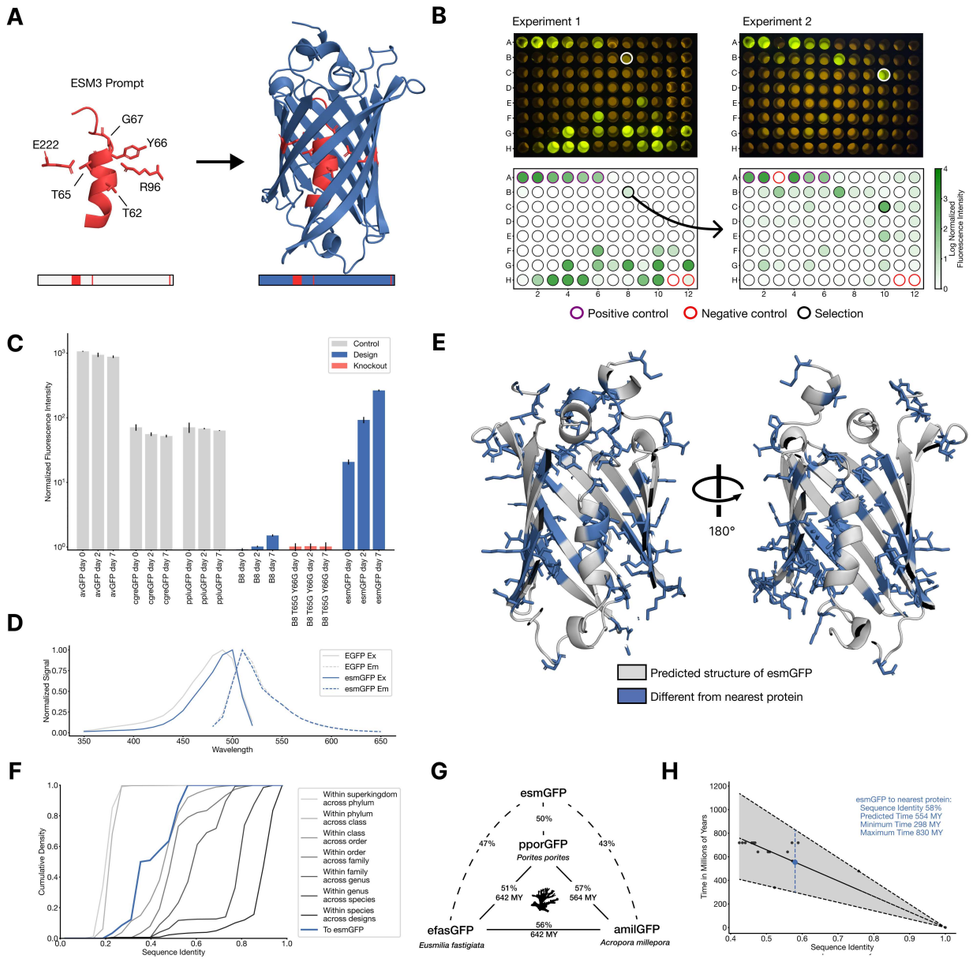

图4. 生成新的荧光蛋白与思维链。(A) 我们使用残基的序列和结构提示ESM3，以形成和催化发色团反应所需的结构，以及来自天然GFP的部分中心α螺旋的结构（左）。通过思维链，ESM3生成设计候选（右）。(B) ESM3在两次实验中发现了一种与其他已知GFP相距较远的明亮GFP。我们在大肠杆菌裂解液中测量荧光。上排为板的照片。下排为板读数荧光定量。已知GFP的阳性对照用紫色圆圈标记，没有GFP序列或没有大肠杆菌的阴性对照用红色圆圈标记。在第一次实验中（左），我们表达了具有不同序列一致性的设计。一个与已知荧光蛋白的序列一致性较低（57%）的显著设计出现在标记为B8的孔中（底部黑圈突出，顶部白圈）。我们从孔B8中的蛋白质继续思维链进行第二次实验（右）。在标记为C10的孔中出现一个明亮设计（与已知荧光蛋白的序列一致性为58%，底部黑圈，顶部白圈），我们将其指定为esmGFP。(C) esmGFP的荧光强度与常见GFP相似。在实验2中显示了一部分蛋白质的归一化荧光。(D) esmGFP的激发和发射光谱与EGFP的光谱重叠。(E) esmGFP预测结构的中心α螺旋和β桶内部的两个剪裁视图。esmGFP相对于其最近邻tagRFP的96个突变以蓝色显示。(F) 不同分类群之间荧光蛋白序列一致性的累积分布。esmGFP与所有其他FP的相似性水平通常在比较同一类内的序列时发现，但跨越不同的目。(G) 三个示例刺胞动物GFP与esmGFP的时间（百万年MY）进化距离与序列一致性。(H) 从GFP序列一致性估计的时间（MY）进化距离。我们估计esmGFP与已知最接近的蛋白质相隔超过五亿年的自然进化。

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in their ability to form a fluorescent chromophore without cofactors or substrates (30). This property allows the GFP sequence to be inserted into the genomes of other organisms to visibly label molecules, cellular structures, or processes, providing a foundational toolkit that has been broadly applied across the biosciences.

The GFP family has been the subject of decades of protein engineering efforts, but still the vast majority of the diversity of functional variants has come from prospecting the natural world. Rational design and mutagenesis have yielded GFP sequences with improved properties—such as higher brightness or stability, or differently colored variants—that incorporated small numbers of mutations (typically 5 to 15, out of the total 238 amino acid coding sequence). In a few cases, leveraging high-throughput experimentation and machine learning, scientists have been able to introduce up to 40-50 mutations (i.e. 80% sequence identity) while retaining fluorescence (48–50).

Generating a new GFP would require materialization of the complex biochemistry and physics that underlie its fluorescence. In all GFPs, an autocatalytic process forms the chromophore from three key amino acids in the core of the protein. The unique structure of GFP, a kinked central alpha helix surrounded by an eleven stranded beta barrel with inward facing coordinating residues, enables this reaction (51). Once formed, the chromophore must not just absorb light but also emit it in order to be fluorescent. Light emission is highly sensitive to the local electronic environment of the chromophore. The fitness landscape of GFP reflects the precise configuration of both the active site and the surrounding tertiary interactions required to achieve its function, since a few random mutations are sufficient to reduce fluorescence to zero (48, 52).

In an effort to generate new GFP sequences, we directly prompt the base pretrained 7B parameter ESM3 to generate a 229 residue protein conditioned on the positions Thr62, Thr65, Tyr66, Gly67, Arg96, Glu222, which are critical residues for forming and catalyzing the chromophore reaction (Fig. 4A). We additionally condition on the structure of residues 58 through 71 from the experimental structure in 1QY3, which are known to be structurally important for the energetic favorability of chromophore formation (53). Specifically, sequence tokens, structure tokens, and atomic coordinates of the backbone are provided at the input, and generation begins from a nearly completely masked array of tokens corresponding to 229 residues, except for the token positions used for conditioning.

We generate designs using a chain-of-thought procedure as follows. The model first generates structure tokens, effectively creating a protein backbone. Backbones that have sufficiently good atomic coordination of the active site but differentiated overall structure from the 1QY3 backbone pass through a filter to the next step of the chain. We add the generated structure to the original prompt to generate a sequence conditioned on the new prompt. We then perform an iterative joint optimization, alternating between optimizing the sequence and the structure. We reject chains-of-thought that lose atomic coordination of the active site (Appendix A.5.1). We draw a computational pool of 10s of thousands of candidate GFP designs from the intermediate and final points in the iterative joint optimization stage of the generation protocol. We bucket the designs by sequence similarity to known fluorescent proteins and filter and rank designs using a variety of metrics (details in Appendix A.5.1.5).

We performed a first experiment with 88 designs on a 96 well plate, evaluating the top generations in each sequence similarity bucket. Each generated protein was synthesized, expressed in *E. coli*, and measured for fluorescence activity at an excitation wavelength of 485 nm (Fig. 4B left). We measured brightness similar to positive controls from a number of designs that have higher sequence identity with naturally occurring GFPs. We also identify a design in well B8 (highlighted in a black circle) with only 36% sequence identity to the 1QY3 sequence and 57% sequence identity to the nearest existing fluorescent protein, tagRFP. This design was 50x less bright than natural GFPs and its chromophore matured over the course of a week, instead of in under a day, but it presents a signal of function in a new portion of sequence space that to our knowledge has not been found in nature or through protein engineering.

We continue the chain of thought starting from the sequence of the design in well B8 to generate a protein with improved brightness, using the same iterative joint optimization and ranking procedure as above. We create a second 96 well plate of designs, and using the same plate reader assay we find that a few designs in this cohort have a brightness in the range of GFPs found in nature. The best design, located in well C10 of the second plate (Fig. 4B right), we designate esmGFP.

We find esmGFP exhibits brightness in the distribution of natural GFPs. We evaluated the fluorescence intensity at 0, 2, and 7 days of chromophore maturation, and plot these measurements for esmGFP, a replicate of B8, a chromophore knockout of B8, along with three natural GFPs avGFP, cgreGFP, ppluGFP (Fig. 4C). esmGFP takes longer to mature than the known GFPs that we measured, but achieves a comparable brightness after two days. To validate that fluorescence was mediated by the intended Thr65 and Tyr66, we show that B8 and esmGFP variants where these residues were mutated to glycine lost fluorescence activity (Fig. S21).

Analysis of the excitation and emission spectra of esmGFP reveals that its peak excitation occurs at 496 nm, which is shifted 7 nm relative to the 489 nm peak for EGFP,

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它们在无需辅因子或底物的情况下形成发色团的能力上（30）。这一特性允许将GFP序列插入其他生物的基因组中，以可视化标记分子、细胞结构或过程，提供了一种已广泛应用于生物科学的基础工具包。

GFP家族已成为几十年来蛋白质工程努力的研究对象，但功能变异的绝大多数多样性仍源自自然界的勘探。理性设计和突变技术已产生了具有改进特性的GFP序列——例如更高的亮度或稳定性，或者不同颜色的变体——这些变体包含少量突变（通常为5到15个，基于总计238个氨基酸编码序列）。在少数情况下，科学家们利用高通量实验和机器学习，已经能够引入多达40-50个突变（即80%的序列一致性），同时保留荧光性（48–50）。

生成一种新的GFP需要物化其荧光所基于的复杂生物化学和物理学。在所有GFP中，一个自催化过程由蛋白质核心中的三个关键氨基酸形成发色团。GFP的独特结构，一个扭曲的中心α螺旋被一个内侧面向协调残基的十一个链 β 螺旋筒包围，使这一反应成为可能(51)。一旦形成，发色团不仅必须吸收光，还必须发出光以实现荧光。光的发射对发色团的局部电子环境高度敏感。GFP的适应度景观反映了实现其功能所需的活性位点及其周围三级相互作用的精确配置，因为少量随机突变就足以将荧光降低到零 (48,52).

为了生成新的GFP序列，我们直接提示基础预训练的7B参数ESM3生成一个以Thr62、Thr65、Tyr66、Gly67、Arg96、Glu222位置为条件的229残基蛋白，这些残基对于形成和催化发色团反应至关重要（图4A）。此外，我们还依据1QY3中实验结构的58到71残基的结构进行条件处理，这些残基已知对发色团形成的能量优势在结构上非常重要（53）。具体而言，输入中提供了序列标记、结构标记和主链的原子坐标，生成从几乎完全掩蔽的对应于229残基的标记数组开始，除了用于条件处理的标记位置。

我们使用以下链式思维程序生成设计。模型首先生成结构标记，有效地创建了蛋白质骨架。骨架在活性位点的原子协调上具有足够良好的表现，但整体结构与1QY3骨架有所区别。

通过过滤器传递到链的下一步骤。我们将生成的结构添加到原始提示中，以生成基于新提示的序列。然后，我们进行迭代联合优化，在优化序列和结构之间交替进行。我们拒绝失去活性位点原子协调的链式思维（附录A.5.1）。我们从生成协议的迭代联合优化阶段的中间和最终点中提取出数万种候选GFP设计。我们根据与已知荧光蛋白的序列相似性对设计进行分类，并使用各种指标对设计进行过滤和排名（详细信息见附录A.5.1.5）。

我们在一个96孔板上进行了首次实验，测试了88种设计，评估了每个序列相似性桶中的最佳生成物。每种生成的蛋白质都在大肠杆菌中合成、表达，并在485 nm的激发波长下测量荧光活性（见图。4B左）。我们的测量亮度与一些具有更高序列一致性的设计的阳性对照相似，这些设计与自然存在的绿荧光蛋白（GFPs）相比。我们还在B8孔中识别出一个设计（用黑圈突出显示），该设计与1QY3序列的序列一致性仅为36%，而与现存最近的荧光蛋白tagRFP的序列一致性为57%。该设计的亮度比自然GFPs低50倍，且其发色团的成熟过程超过一周，而不是在一天之内，但它在我们所知未在自然界或通过蛋白质工程中发现的新序列空间部分中表现出功能信号。

我们继续从B8孔设计的序列出发，开展思路链，以生成具有更高亮度的蛋白质，采用与上述相同的迭代联合优化和排名程序。我们创建了第二个96孔板的设计，并使用相同的孔板读数检测，我们发现该组中的一些设计拥有与自然中发现的GFPs相当的亮度。最佳设计位于第二个板的C10孔（见图。4B右），我们将其指定为esmGFP。

我们发现esmGFP在自然绿荧光蛋白的分布中表现出亮度。我们评估了在发色团成熟的0、2和7天的荧光强度，并绘制了这些esmGFP测量结果，与B8的重复、B8的发色团敲除体及三种自然的绿荧光蛋白avGFP、cgreGFP、ppluGFP一起展示（图4C）。 esmGFP的成熟时间比我们测量的已知绿荧光蛋白要长，但在两天后却达到了相当的亮度。为了验证荧光是由预期的Thr65和Tyr66介导的，我们展示了将这些残基突变为甘氨酸的B8和esmGFP变体失去了荧光活性（图。S21）。

对esmGFP的激发和发射光谱的分析显示，其峰值激发发生在496 nm，在相对于EGFP的489 nm峰值处发生了7 nm的偏移，

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while both proteins emit at a peak of 512nm (Fig. 4D). The shapes of the spectra indicated a narrower full-width-half-maximum (FWHM) for the excitation spectrum of esmGFP (39mm for esmGFP vs 56 nm for EGFP), whereas the FWHM of their emission spectra were highly comparable (35nm and 39 nm, respectively). Overall esmGFP exhibits spectral properties consistent with known GFPs.

We next sought to understand how esmGFP compares to known proteins. A BLAST (54) search against the non-redundant protein sequences database and an MMseqs (55) search of ESM3's training set report the same top hit—tagRFP, which was also the nearest neighbor to B8—with 58% sequence identity, representing 96 mutations throughout the sequence. tagRFP is a designed variant, and the closest wildtype sequence to esmGFP from the natural world is eqFP578, a red fluorescent protein, which differs from esmGFP by 107 sequence positions (53% identity). Sequence differences between esmGFP and tagRFP occur throughout the structure (Fig. 4E) with 22 mutations occurring in the protein's interior, which is known to be highly sensitive to mutations due to chromophore proximity and a high density of interactions (56).

Examination of a sequence alignment of 648 natural and designed GFP-like fluorescent proteins revealed that esmGFP has the level of similarity to all other FPs that is typically found when comparing sequences across taxonomic orders, but within the same taxonomic class (Fig. 4F). For example, the difference of esmGFP to other FPs is similar to the level of difference between FPs belonging to the orders of Scleractinia (stony corals) and Actiniaria (sea anemones) both of which belong to the larger class Anthozoa of marine invertebrates (Fig. 4G). The closest FPs to esmGFP come from the Anthozoa class (corals and anemones; average sequence identity 51.4%), but esmGFP also shares some sequence identity with FPs from the Hydrozoa (jellyfish) where avGFP was discovered (average sequence identity 33.4%; Fig. S22).

We can draw insight from evolutionary biology on the amount of time it would take for a protein with similar sequence identity to arise through natural evolution. In Fig. 4G we show esmGFP alongside three anthozoan GFPs. We use a time-calibrated phylogenetic analysis of the anthozoans (57) that estimated the millions of years ago (MYA) to last common ancestors to estimate evolutionary time between each pair of these species. Using a larger dataset of six anthozoan GFPs and species for which we have accurate MYA to last common ancestors and GFP sequence identities, we construct a simple estimator that correlates sequence identity between FPs to MY of evolutionary time between the species (Fig. 4H) to calibrate against natural evolution. Based on this analysis we estimate esmGFP represents an equivalent of over 500 million years of evolution from the closest protein that has been found in nature.

Discussion

We have found that language models can reach a design space of proteins that is distant from the space explored by natural evolution, and generate functional proteins that would take evolution hundreds of millions of years to discover. Protein language models do not explicitly work within the physical constraints of evolution, but instead can implicitly construct a model of the multitude of potential paths evolution could have followed.

Proteins can be seen as existing within an organized space where each protein is neighbored by every other that is one mutational event away (58). The structure of evolution appears as a network within this space, connecting all proteins by the paths that evolution can take between them. The paths that evolution can follow are the ones by which each protein transforms into the next without the collective loss of function of the system it is a part of.

It is in this space that a language model sees proteins. It sees the data of proteins as filling this space, densely in some regions, and sparsely in others, revealing the parts that are accessible to evolution. Since the next token is generated by evolution, it follows that to solve the training task of predicting the next token, a language model must predict how evolution can move through the space of possible proteins.

Simulations are computational representations of reality. In that sense, a language model which can predict possible outcomes of evolution can be said to be a simulator of it. ESM3 is an emergent simulator that has been learned from solving a token prediction task on data generated by evolution. It has been theorized that neural networks discover the underlying structure of the data they are trained to predict (59, 60). In this way, solving the token prediction task would require the model to learn the deep structure that determines which steps evolution can take, i.e. the fundamental biology of proteins.

In ESM3's generation of a new fluorescent protein, it is the first chain of thought to B8 that is the most intriguing. At 96 mutations to B8's closest neighbor there are $\binom{229}{96} \times 19^{96}$ possible proteins, out of which only a vanishingly small fraction can have function, since fluorescence falls off sharply even after just a few random mutations. The existence of C10 and other bright designs in the neighborhood of B8 confirms that in the first chain of thought to B8, ESM3 found a new part of the space of proteins that, although unexplored by nature, is dense with fluorescent proteins.

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尽管两种蛋白质在512纳米的峰值处发射（图4D）。谱形状表明，es- mGFP的激发光谱具有更窄的全宽半最大值（FWHM）（esmGFP为39纳米，EGFP为56纳米），而它们的发射光谱的FWHM则高度相似（分别为35纳米和39纳米）。总体而言，esmGFP表现出与已知绿荧光蛋白相一致的光谱特性。

接下来，我们希望了解esmGFP与已知蛋白质的比较。通过对非冗余蛋白序列数据库进行BLAST（54）搜索，以及对ESM3训练集进行MMseqs（55）搜索，报告的首要命中都是tagRFP，它也是与B8最近的邻居，具有58%的序列一致性，共表示在序列中的96个突变。tagRFP是一个设计变体，而esmGFP在自然界中最近的野生型序列是eqFP578，一种红色荧光蛋白，其不同于esmGFP有107个序列位置（53%一致性）。esmGFP和tagRFP之间的序列差异贯穿整个结构（图4E），其内部发生了22个突变，而该区域因发色团的接近性和高密度的相互作用，已知对突变高度敏感（56）。

对648种天然和设计的GFP样荧光蛋白序列对齐的检查显示，esmGFP的相似度水平与所有其他FPs相似，这种相似度通常在比较属于同一分类学类的序列时发现（图4F）。例如，esmGFP与其他FPs的差异类似于属于石珊瑚（Scleractinia）和海葵（Actiniaria）这两个属于海洋无脊椎动物更大分类类Anthozoa的分类学顺序之间FPs的差异（图4G）。与esmGFP最接近的FPs来自Anthozoa类（珊瑚和海葵；平均序列一致性51.4%），但esmGFP也与来自水母（Hydrozoa）的FPs共享某些序列一致性，其中发现了avGFP（平均序列一致性33.4%；图S22）。

我们可以从进化生物学中获得对蛋白质通过自然进化产生所需时间的洞察，这一蛋白质与之具有相似的序列一致性。在图。4G中，我们展示了esmGFP以及三种珊瑚纲GFP。我们采用了一个时间校准的系统发育分析，该分析针对珊瑚纲（57）估算了最后共同祖先至今的百万年前（MYA）时间，以估计这些物种之间的进化时间。利用一个较大的数据集，该数据集包含六种珊瑚纲GFP和我们拥有准确的最后共同祖先MYA及GFP序列一致性的物种，我们构建了一个简单的估算器，该估算器将光合荧光蛋白之间的序列一致性与物种间的百万年进化时间（图。4H）相联系，以对照自然进化。基于此分析，我们估计esmGFP代表了超过五亿年的进化等效。

从在自然界中发现的最接近的蛋白质。

讨论

我们发现，语言模型能够达到一个与自然进化所探索的蛋白质设计空间相距甚远的区域，并生成功能蛋白，这些蛋白质的发现将需要经过数亿年的进化才能实现。蛋白质语言模型并未明确地在进化的物理约束内工作，而是可以隐含地构建出进化可能遵循的众多潜在路径的模型。

蛋白质可以被视为存在于一个有序空间中，在该空间内，每个蛋白质都与每个与其相邻的蛋白质相距一个突变事件（58）。进化的结构出现在这个空间内，作为一个网络，将所有蛋白质通过进化可能采取的路径连接起来。进化可以遵循的路径是每个蛋白质转变为下一个蛋白质的路径，而不导致其所处系统的功能集体损失。

语言模型在这个空间中看待蛋白质。它将蛋白质的数据视为填充这个空间，在某些区域密集而在其他区域稀疏，揭示出进化可及的部分。由于下一个标记是由进化生成的，因此，解决预测下一个标记的训练任务就意味着语言模型必须预测进化如何在可能蛋白质的空间中移动。

模拟是对现实的计算表示。从这个意义上讲，能够预测进化可能结果的语言模型可以被认为是这种模拟器。ESM3 是一个新兴的模拟器，它是通过解决在进化生成的数据上进行标记预测任务而学习到的。已经有人理论化神经网络发现它们被训练以预测的数据的潜在结构 (59, 60)。通过这种方式，解决标记预测任务将要求模型学习决定进化可以采取哪些步骤的深层结构，即蛋白质的基本生物学。

在ESM3生成新型荧光蛋白时，最引人入胜的是到B8的第一思路。到B8最近邻的突变有96个，有$\binom{229}{96} \times 19^{96}$种可能的蛋白质，其中只有极少数能够具有功能，因为即使在仅经过几次随机突变后，荧光的强度也会急剧下降。在B8周围发现C10及其他明亮设计的存在，确认了在到B8的第一思路中，ESM3发现了一部分蛋白质空间，尽管尚未被自然探索，但其中充满了荧光蛋白。

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Contributions

Data: H.A., Z.L., R.R., A.R., T.S., N.T., R.V.

Pre-training: H.A., S.C., J.D., T.H., Z.L., D.O., R.R., A.R., T.S., I.S., R.V., M.W.

Post-training: H.A., S.C., A.D., J.G., T.H., D.O., R.R., A.R., M.W.

Evaluation and Analysis: R.B., J.D., A.D., T.H., Y.K., C.K., Z.L., R.S.M., A.R., N.J.S.

Open Model & Responsible Development: J.G., I.S., N.J.S., T.S., R.S.M., Z.L., R.R., A.R., N.T.

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GFP Computational: S.C., T.H., N.J.S., A.R., R.V.

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Manuscript: S.C., T.H., R.R., A.R., N.J.S. overall manuscript. All authors contributed to the sections for which they are credited.

Supplement: H.A., R.B., L.J.B., S.C., J.D., A.D., T.H., C.K., Z.L., R.S.M., D.O., R.R., A.R., N.J.S., T.S., I.S., N.T., V.Q.T., R.V., M.W.

Competing interests

Authors H.A., R.B., S.C., J.D., A.D., J.G., T.H., C.K., Z.L., R.S.M., C.M., D.O., R.R., A.R., N.J.S., T.S., I.S., N.T., R.V., M.W. are employees of EvolutionaryScale, PBC. P.D.H. is a cofounder of Stylus Medicine, Circle Labs, and Spotlight Therapeutics, serves on the board of directors at Stylus Medicine, is a board observer at EvolutionaryScale, Circle Labs, and Spotlight Therapeutics, a scientific advisory board member at Arbor Biosciences and Veda Bio, and an advisor to NFDG, Varda Space, and Vial Health. Patents have been filed related to aspects of this work.

Model and data availability

Weights and code for ESM3-open are provided for academic research use. The ESM3-open model was reviewed by a committee of technical experts who found that the benefits of releasing the model greatly outweighed any potential risks. ESM3 models will be available via API with a free access tier for academic research. The sequence of esmGFP (along with the other GFPs generated for this work) is committed to the public domain. Plasmids for esmGFP-C10 and esmGFP-B8 will be made available.

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模型和数据可用性

ESM3-open 的权重和代码被提供用于学术研究。ESM3-open 模型经过技术专家委员会的审查，他们认为发布该模型的好处远远超过任何潜在风险。ESM3 模型将通过 API 提供，学术研究可享受免费访问层。esmGFP 的序列（连同为本工作生成的其他绿荧光蛋白）被承诺进入公共领域。esmGFP-C10 和 esmGFP-B8 的质粒将被提供。

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A. Materials and Methods

A.1. Architecture

A.1.1. Notation

In the following, we use L to denote the sequence length, d for the embedding dimension, $\{a..b\}$ to denote the inclusive set of integers from a to b, and $[a, b]$ an interval of real numbers. $SE(3)$ is the special Euclidean group, which we use to denote frames (Appendix A.1.6.1).

A.1.2. Overview

ESM3 is all-to-all generative model that both conditions on and generates a variety of different tracks. As input, ESM3 is conditioned on various tracks as described in Appendix A.1.5.1, and as output, ESM3 generates predictions detailed in Appendix A.1.5.2.

The generative pipeline is as follows.

Tokenization First, raw inputs are tokenized as described in Appendix A.1.3. Structural inputs are tokenized via a VQ-VAE (Appendix A.1.7). Function keywords are tokenized by quantizing the TF-IDF transform of functional keywords with locality sensitive hashing (LSH), detailed in Appendix A.1.8.

Transformer Trunk A standard Transformer (61, 62) architecture processes the post-tokenized inputs. Geometric Attention (Algorithm 6 and Fig. S2) directly processes structural coordinates as input. Model outputs are logits over token space, and can be sampled to obtain outputs described in Appendix A.1.5.2. The overall architecture is diagrammed in Fig. S1.

Decoder Most tracks can be naively decoded into tokens detailed in Appendix A.1.3. Structure tokens must be decoded with a model—we use a 700M parameter transformer model to do this, trained post-hoc (Appendix A.1.7.2). The decoder uses sequence tokens and structure tokens to directly predict coordinates, pTM, and pLDDT (63). Function tokens are decoded using a small 3-layer transformer, trained post-hoc to invert the LSH quantization procedure (Appendix A.1.8.2.1).

A.1.3. Tokenization

All tracks are represented as a sequence of tokens, with tokens specified in each amino acid position. During tokenization, special beginning-of-sequence (BOS) and end-of-sequence (EOS) tokens are prepended and appended.

Sequence Protein sequences are tokenized as the 20 canonical amino acids, plus BOS, EOS, mask, pad, unknown. We keep four non-standard amino acids as in Lin et al. (5), B - Asparagine, U - selenocysteine, Z - glutamic acid, and O - ornithine. This totals to 29 tokens.

Structure Structure tokenization is described in Appendix A.1.7.1. ESM3 uses a codebook size of 4096 with 4 special tokens - EOS, BOS, mask, and pad.

Secondary Structure Secondary structure is taken to be the canonical 8-class tokens (64), with unknown and mask, for a total of 10 tokens. The mask token is forced to be the 0-vector during embedding.

SASA The continuous values representing SASA are tokenized by discretization into a fixed set of 16 bins. SASA bin boundaries were chosen by computing SASA on 100 random structures and ensuring an equal number of residues belong in each bin. Unknown and mask are used for a total of 18 tokens. The mask token is forced to be the 0-vector during embedding.

Function annotations We tokenize function annotations as bags of keywords, described in Appendix A.1.8. Keywords are quantized using LSH into 8 tokens per residue, each of which can be one of 255 tokens. There are three special tokens, empty set, no-annotation, and mask. Again, the mask token is forced to be the 0-vector during embedding.

Residue annotations InterPro annotations are tokenized as a multi-hot feature vector (1478 dimensions) over possible InterPro labels (40). Input annotations are limited to a maximum of 16. When annotations are not present, we enforce that the 0-vector is added.

A.1.4. ESM3 Inputs and Forward Pass

As mentioned above, ESM3 can take several tracks, all of which are optionally disabled via masking. In the following, we concisely denote the inputs to ESM3 as

$$\mathbf{x}_{\text{inputs}} = \begin{cases} x_{\text{structure}} \in \{0..4099\}^L, x_{\text{ss8}} \in \{0..10\}^L, \\ x_{\text{sasa}} \in \{0..18\}^L, x_{\text{func}} \in \{0..258\}^{L\times 8}, \\ x_{\text{res}} \in \{0,1\}^{L\times 1478}, \ x_{\text{plddt}} \in [0,1]^L, \\ x_{\text{avgplddt}} \in [0,1] \end{cases}$$

We now present the high level algorithm for a forward pass of ESM3:

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A. 材料与方法

A.1. 架构

A.1.1. 符号

在以下内容中，我们使用 L 表示序列长度， d表示嵌入维度， $\{a..b\}$ 表示从 a 到 b的闭合集合整数，以及 $[a, b]$ 一个实数区间。 $SE(3)$ 是特殊的欧几里得群，我们用它来表示框架（附录A.1.6.1）。

A.1. 2. 概述ESM3 是一种全对全生成模型，它既对不同轨迹进行条件处理，又生成多种不同轨迹。作为输入，ESM3 对多种轨迹进行条件处理，如附录A.1.5.1中所描述，并且作为输出，ESM3 生成了在附录A.1.5.2中详细说明的预测。生成管道如下。分词 首先，原始输入按附录A.1.3中所述进行分词。结构输入通过 VQ-VAE 进行分词（附录A.1.7）。功能关键词通过使用局部敏感哈希（LSH）对功能关键词的 TF-IDF 变换进行量化来进行分词，详细说明见附录A.1.8。变换器主干 标准变换器（61， 62）架构处理后分词输入。几何注意力（算法6 和图 S2）直接处理结构坐标作为输入。模型输出是标记空间上的逻辑值，可以进行采样以获得附录A.1.5.2中描述的输出。整体架构在图中示意 S1。解码器 大多数轨迹可以简单地解码为附录 A.1.3中详细说明的标记。结构标记必须通过模型进行解码——我们使用一个700M参数的变换器模型来做到这一点，后期训练（附录A.1.7.2）。解码器使用序列标记和结构标记直接预测坐标、pTM 和 pLDDT（63）。功能令牌使用一个小型的 3 层变换器进行解码，后期训练以反转 LSH 量化程序（附录A.1.8.2.1）。

A.1. 3. 分词所有轨迹都表示为标记序列，每个氨基酸位置指定标记。在分词过程中，特殊的序列开始标记（BOS）和序列结束标记（EOS）分别被预置和添加到序列中。序列蛋白质序列被标记为20种标准氨基酸，加上BOS、EOS、mask、pad和unknown。

我们保留四种非标准氨基酸，如林等所述。

(5)，B - 天冬酰胺，U - 硒氨酸，Z - 谷氨酸，和O - 精氨酸。这总共是29个标记。

结构结构标记化在附录A.1.7.1中进行了描述。ESM3使用4096的代码本大小和4个特殊标记 - EOS、BOS、mask和pad。

二级结构二级结构被视为标准的8类标记（64），加上unknown和mask，总共10个标记。mask标记在嵌入过程中被强制设为0向量。

SASA 表示 SASA 的连续值通过离散化为固定的 16 个区间进行标记化。SASA 区间边界通过在 100 个随机结构上计算 SASA 得出，确保每个区间内包括相等数量的残基。未知和掩码共用于 18 个标记。掩码标记在嵌入过程中被强制为零向量。

功能注释我们将功能注释标记化为关键词包，如附录A.1.8中所述。关键词通过局部敏感哈希量化为每个残基 8 个标记，每个标记可以是 255 个标记之一。存在三个特殊标记：空集、无注释和掩码。再次强调，掩码标记在嵌入过程中被强制为零向量。

残基注释InterPro 注释被标记化为一个多热特征向量（1478 维），涵盖可能的 InterPro 标签（40）。输入注释限制为最多 16 个。当不存在注释时，我们强制添加零向量。

A.1.4. ESM3 输入和前向传递

如上所述，ESM3 可以采用多个轨道，所有这些轨道均可通过掩蔽选择性禁用。在以下内容中，我们简洁地表示 ESM3 的输入为

$$\mathbf{x}_{\text{inputs}} = \begin{cases} x_{\text{structure}} \in \{0..4099\}^L, x_{\text{ss8}} \in \{0..10\}^L, \\ x_{\text{sasa}} \in \{0..18\}^L, x_{\text{func}} \in \{0..258\}^{L\times 8}, \\ x_{\text{res}} \in \{0,1\}^{L\times 1478}, \ x_{\text{plddt}} \in [0,1]^L, \\ x_{\text{avgplddt}} \in [0,1] \end{cases}$$

我们现在呈现 ESM3 前向传递的高级算法：

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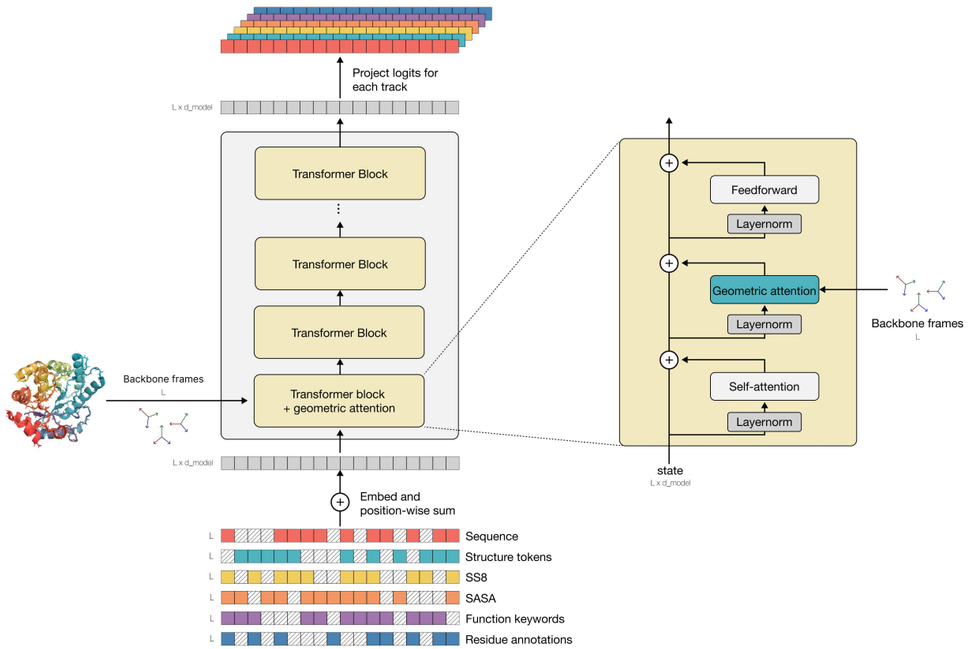

Figure S1. The ESM3 architecture. ESM3 is a masked language model that reasons over protein sequence, structure, and function, each of which are represented as token tracks at the input and output. Tokens are embedded and summed at the input to a transformer stack. The first block (expanded on the right) contains an additional geometric attention layer for processing atomic coordinate inputs. During training, random masks are sampled and applied to each track. Masked token positions are predicted at the output.

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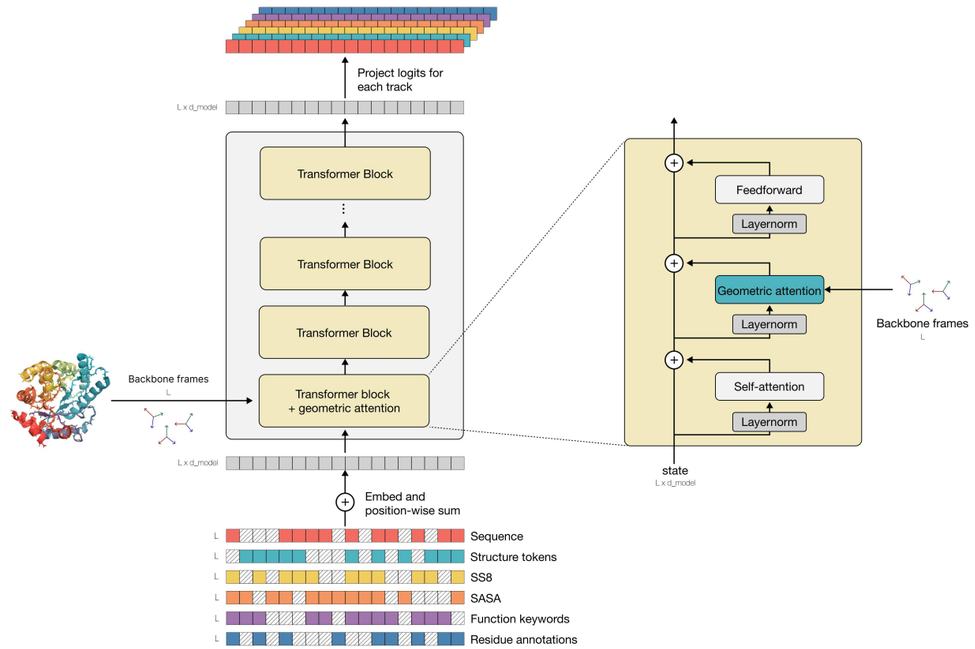

图 S1. ESM3 架构。ESM3 是一个掩码语言模型，它对蛋白质序列、结构和功能进行推理，这些均作为输入和输出的标记轨道表示。标记在输入到变压器堆时被嵌入并求和。第一个模块（右侧扩展）包含一个额外的几何注意力层，用于处理原子坐标输入。在训练期间，随机掩码会被采样并应用于每个轨道。掩码标记位置在输出时被预测。

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Algorithm 1 `esm3_forward`

Input: $\mathbf{x}_{\text{inputs}}$

$z^{(0)}_{\text{embed}} = \texttt{encode_inputs}(\mathbf{x}_{\text{inputs}})$ ▷ $\mathbb{R}^{L\times d}$
for $\ell \in \{1..n_{\text{layers}}\}$ **do**
  $z^{(\ell)}_{\text{embed}} = \texttt{transformer_block}(z^{(\ell-1)}_{\text{embed}})$
end for
for track in desired output tracks **do**
  $z_{\text{track}} = \texttt{regression_head}(z^{(n_{\text{layers}})}_{\text{embed}})$
end for
return Track specific logits $z_{\text{track}} \in \mathbb{R}^{L\times c_{\text{track}}}$

In the next few sections, we detail each component.

A.1.5. Transformer

Our network is based on the transformer architecture (61), incorporating several subsequent improvements: We use Pre-LN instead of Post-LN (62), rotary embeddings (65) instead of absolute positional embeddings, and we replace ReLU non-linearity with SwiGLU (66). The hidden dimension is set to approximately $\frac{8}{3}d$, rounded to the nearest multiple of 256 for training efficiency. No biases are used in linear layers or layer norms, as suggested by PaLM (67). We have observed through the literature and in internal experiments that these architecture changes improve the stability and performance of models.

A core architectural modification we make is the insertion of the Geometric Attention sub-layer in the first block of the network only (Algorithm 2, line 3).

Algorithm 2 `transformer_block`

Input: $x \in \mathbb{R}^{L\times d}, T \in SE(3)^L$

$s = \sqrt{\frac{36}{n_{\text{layers}}}}$ ▷ $\mathbb{R}$
$x = x + s \cdot \text{MultiHeadSelfAttention}(x)$ ▷ $\mathbb{R}^{L\times d}$
$x = x + s \cdot \texttt{geometric_mha}(x, T)$ ▷ $\mathbb{R}^{L\times d}$
$x = x + s \cdot \text{SwiGLUMLP}(x)$ ▷ $\mathbb{R}^{L\times d}$

ESM3-small (1.4B) is a 48-layer network, while ESM3-medium (7B) has 96 layers, and ESM3-large (98B) has 216 layers. We experimented with different width-to-depth ratios and observed higher returns for depth than width. Prior work also demonstrates that modalities like ours benefit more from deeper networks (68, 69). Detailed network specifications can be found in Table S1.

A.1.5.1. Embedding

There are 7 unique input tracks to ESM3: (a) sequence (amino acid tokens), (b) structure coordinates, (c) structure tokens, (d) 8-class secondary structure labels (SS8), (e) quantized solvent-accessible surface area (SASA) values, (f) function keyword tokens and (g) residue (InterPro) annotation binary features.

There are two additional tracks used during pretraining only: (h) per-residue confidence (pLDDT) and (i) averaged confidence (pLDDT). At inference time, these values are fixed, and these tracks are equivalent to adding a constant vector z_{plddt}.

Structure coordinates are parsed through the Geometric Attention and are not embedded.

For keyword-based function tokens, each of the eight integers per residue is converted to a "sub-embedding" (Algorithm 3 line 5), then concatenated to form the per-residue embedding (Algorithm 3 line 6). For InterPro residue annotations, the inputs are multi-hot. To create an embedding vector, we sum the embeddings for each of the "on" features (equivalent to the matrix-multiply on Algorithm 3 line 7).

The largest model, 98B, has an additional taxonomy track detailed in Appendix A.1.9.2, only enabled in the final 30K steps of pretraining.

The embeddings are all summed as input to the first layer in the network architecture.

Algorithm 3 `encode_inputs`

Input: $\mathbf{x}_{\text{inputs}} = \{x_{\text{seq}}, x_{\text{structure}}, x_{\text{ss8}}, x_{\text{sasa}}, x_{\text{func}}, x_{\text{res}}, x_{\text{plddt}}, x_{\text{avgplddt}}\}$

$z_{\text{seq}} = \text{embed}(x_{\text{seq}})$ ▷ $\mathbb{R}^{L\times d}$
$z_{\text{structure}} = \text{embed}(x_{\text{structure}})$ ▷ $\mathbb{R}^{L\times d}$
$z_{\text{ss8}} = \text{embed}(x_{\text{ss8}})$ ▷ $\mathbb{R}^{L\times d}$
$z_{\text{sasa}} = \text{embed}(x_{\text{sasa}})$ ▷ $\mathbb{R}^{L\times d}$
$h_{\text{func},i} = \text{embed}([x_{\text{func}}]_{:,i})$ ▷ $\mathbb{R}^{L\times \frac{d}{8}}$
$z_{\text{func}} = [h_{\text{func},1} \mid h_{\text{func},2} \mid \ldots \mid h_{\text{func},8}]$ ▷ $\mathbb{R}^{L\times d}$
$z_{\text{res}} = x_{\text{res}} W_{\text{res}}$ ▷ $\mathbb{R}^{L\times d}$
$z_{\text{plddt}} = \texttt{plddt_embed}(x_{\text{plddt}}, x_{\text{avgplddt}})$ ▷ $\mathbb{R}^{L\times d}$
return $z_{\text{seq}} + z_{\text{plddt}} + z_{\text{structure}} + z_{\text{ss8}} + z_{\text{sasa}} + z_{\text{func}} + z_{\text{res}}$

A.1.5.2. Logits

We use a `regression_head` to take in d dimensional last layer hidden features and produce c_{track}-dimensional logits for each of the tracks, where c_{track} corresponds to the size of the vocabulary per track. Note that for the keyword function tokens, we produce $c_{\text{func}} \times 8$ logits, and softmax over each of the 8 independently when calculating the loss.

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算法 1 `esm3_`

输入：$\mathbf{x}_{\text{inputs}}$

$z^{(0)}$ 嵌入 = ▷ $\mathbb{R}^{L\times d}$
对于 $\ell \in \{1..n_{\text{layers}}\}$ 执行
  $z^{(\ell)}$ 嵌入 =
结束对于
对于轨迹在期望输出轨迹中进行
  $z_{\text{track}} = \text{回归_头}(z^{(n_{\text{layers}})}\ \text{嵌入})$
结束对于
返回跟踪特定逻辑值 $z_{\text{track}} \in \mathbb{R}^{L\times c_{\text{track}}}$

在接下来的几个部分中，我们详细说明每个组件。

A.1.5. 变换器

我们的网络基于变压器架构（61），综合了若干后续改进：我们使用预层归一化（Pre-LN）而不是后层归一化（Post-LN）（62），采用旋转嵌入（rotary embeddings）（65）而非绝对位置嵌入，同时将ReLU非线性替换为SwiGLU（66）。隐藏维度被设置为大约 $\frac{8}{3}d$，经过四舍五入至训练效率的256倍数。线性层或层归一化中不使用偏差，正如PaLM（67）所建议的。我们通过文献和内部实验观察到，这些架构变更提高了模型的稳定性和性能。

我们所做的一个核心架构修改是在网络的第一个块中仅插入几何注意力子层（算法2，第3行）。

算法2 变换器

输入：$x \in \mathbb{R}^{L\times d}, T \in SE(3)^L$
$s =$ ▷ $\mathbb{R}$
$x = x + s$ 多头自注意力(x) ▷ $\mathbb{R}^{L\times d}$
$x = x + s$ ▷ $\mathbb{R}^{L\times d}$
$x = x + s$ SwiGLU多层感知机(x) ▷ $\mathbb{R}^{L\times d}$

ESM3-small (1.4B) 是一款48层网络，而 ESM3-medium (70亿) 具有96层，ESM3-large (98B) 拥有216层。我们尝试了不同的宽度与深度比例，观察到深度相比于宽度有更高的收益。先前的研究也表明，像我们这样的模态更有利于更深的网络（68, 69）。详细的网络规格可以在表S1中找到。

A.1.5.1. 嵌入

ESM3 有 7 个独特的输入轨道： (a) 序列（氨基酸标记）， (b) 结构坐标， (c) struc-ture 标记， (d) 8 类二级结构标签（SS8），(e) 经过量化的溶剂可接触表面积 (SASA) 值、(f) 功能关键字标记以及 (g) 残基 (InterPro) 注释的二进制特征。

在预训练期间仅使用两个额外的轨道：(h) 每个残基的置信度 (pLDDT) 和 (i) 平均置信度 (pLDDT)。在推理阶段，这些值是固定的，这些轨道相当于添加一个常量向量 z_{plddt}。

结构坐标通过几何注意力进行解析，并未被嵌入。

对于基于关键字的功能标记，每个残基的八个整数被转换为一个“子嵌入”（算法3 第 5 行），然后拼接以形成每个残基的嵌入（算法 3第 6 行）。对于 InterPro 残基注释，输入是多热的。为了创建一个嵌入向量，我们对“开启”特征的嵌入进行求和（相当于算法3 第 7 行上的矩阵乘法）。

最大的模型，98B，在附录A.1.9.2中详细介绍了额外的分类法跟踪，仅在预训练的最后30K步骤中启用。

所有嵌入作为输入相加，进入网络架构的第一层。

算法3 编码_

输入：$\mathbf{x}_{\text{inputs}} = \{x_{\text{seq}}, x_{\text{structure}}, x_{\text{ss8}}, x_{\text{sasa}}, x_{\text{func}}, x_{\text{res}}, x_{\text{plddt}}, x_{\text{avgplddt}}\}$

z_{seq} =嵌入(x_{seq}) ▷ $\mathbb{R}^{L\times d}$
$z_{\text{structure}}$ =嵌入$(x_{\text{structure}})$ ▷ $\mathbb{R}^{L\times d}$
z_{ss8} =嵌入(x_{ss8}) ▷ $\mathbb{R}^{L\times d}$
z_{sasa} =嵌入(x_{sasa}) ▷ $\mathbb{R}^{L\times d}$
$h_{\text{func},i}$ =嵌入$([x_{\text{func}}]_{:,i})$ ▷ $\mathbb{R}^{L\times d8-}$
$z_{\text{func}} = [h_{\text{func},1} \mid h_{\text{func},2} \mid \ldots \mid h_{\text{func},8}]$ ▷ $\mathbb{R}^{L\times d}$
$z_{\text{剩余}} = x_{\text{剩余}} W_{\text{剩余}}$ ▷ $\mathbb{R}^{L\times d}$
$z_{\text{plddt}} =$ plddt_嵌入 $(x_{\text{plddt}}, x_{\text{avgplddt}})$ ▷ $\mathbb{R}^{L\times d}$
返回 $z_{\text{seq}} + z_{\text{plddt}} + z_{\text{structure}} + z_{\text{ss8}} + z_{\text{sasa}} + z_{\text{func}} + z_{\text{res}}$

A.1.5.2. 逻辑值

我们使用一个 回归_头 来接收 d 维度的最后一层隐藏特征，并生成 c_{track}维度的逻辑值，用于每条轨道，其中 c_{track} 对应于每条轨道的词汇表大小。请注意，对于关键字函数令牌，我们生成 $c_{\text{func}} \times 8$ 逻辑值，并在计算损失时对每一个独立的 8 进行 Softmax。

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Simulating 500 million years of evolution with a language model

Params	n_{layers}	d_{model}	d_{head}	**Context length**	**Learning rate**	**Warmup steps**	**Batch size in tokens**	**Num steps**	**Total tokens**	**FLOPs**
1.4B	48	1536	64	2048	4.0e-4	5K	1,572,864	50K	∼80B	6.72×10^{20}
1.4B	48	1536	64	2048	4.0e-4	5K	1,572,864	200K	∼320B	2.7×10^{21}
7.7B	96	2560	128	2048	2.5e-4	5K	3,932,160	140K	∼550B	2.47×10^{22}
98.5B	216	6144	128	2048	1.0e-4	20K	4,194,304	430K	∼1.8T	1.07×10^{24}

Table S1. Parameter details for different model configurations.

Algorithm 4 `regression_head`

Input: $x \in \mathbb{R}^{\cdots \times d}$
$z = \text{proj}_{\text{in}}(x)$
$z = \text{GeLU}(z)$
$z = \text{LayerNorm}(z)$
$z = \text{proj}_{\text{out}}(z)$
return z

Except for structure coordinates, we output predictions for each of the tracks detailed in Appendix A.1.5.1: (a) sequence, (b) structure tokens, (c) SS8, (d) quantized SASA, (e) function keyword tokens and (f) residue (InterPro) annotation binary features.

Except for the multi-hot residue annotations, all other tracks are predicted as a categorical distribution over possible tokens.

A.1.6. Geometric Attention

As mentioned in Appendix A.1.5.1, ESM3 processes structural information in two independent ways:

Geometric Attention Described in Algorithm 6, this leverages fine-grained 3D information via conditioning on atomic coordinates of backbone atoms. Coordinates are only used as model inputs.

Structure Tokens Described in Appendix A.1.7, structure tokens enable faster learning due to rich local neighborhood semantics being compressed into tokens. Structure tokens are generally used as model outputs.

Geometric attention enables high-throughput encoding of protein structures. Protein backbone structure can be represented by the relative distance and orientation of frames defined by each residue's backbone coordinates. Reasoning over the relative orientation of frames is important to capture the local backbone geometry when only partial structure is provided. Geometric attention is an $SE(3)$ invariant all-to-all attention mechanism which reasons over the relative distances and orientations of all defined frames in the input (Fig. S2). Because this attention operation can be realized using the same computational primitives as attention, it is readily scalable.

We first provide an overview of frames, and then describe how geometric attention uses them:

A.1.6.1. Frames

Frames are representations that encapsulate the positions and orientations of each residue in the protein. We use a formulation similar to Ingraham et al. (70). Each frame $T \in SE(3)$ consists of a rotation matrix $\mathbf{R} \in SO(3)$ and a translation vector $\mathbf{t} \in \mathbb{R}^3$.

Definition: A frame T_i for residue i is defined as:

$$T_i = \begin{bmatrix} \mathbf{R}_i & \mathbf{t}_i \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \in SE(3)$$

where $\mathbf{R}_i \in SO(3)$ and $\mathbf{t}_i \in \mathbb{R}^3$.

Rotation Matrix: The rotation matrix $\mathbf{R}_i$ for residue i is composed of three 3-dimensional vectors $[\hat{x}, \hat{e}_1, \hat{e}_2]$:

1. $\hat{x}$ and $\hat{e}_1$ are orthogonal unit vectors on the $N - C_\alpha - C$ plane.
2. $\hat{e}_2$ is a unit vector perpendicular to both $\hat{x}$ and $\hat{e}_1$.

This matrix rotates vectors to a local coordinate system where the $N - C_\alpha - C$ plane for the corresponding residue spans the xy plane.

Translation Vector: The translation vector $\mathbf{t}_i$ specifies the position of the residue's C_α.

Transformation: To transform a point $\mathbf{p} \in \mathbb{R}^3$ from the local frame of residue i to the global coordinate system, the following equation is used:

$$\mathbf{p}_{\text{global}} = T_i(\mathbf{p}) = \mathbf{R}_i\mathbf{p} + \mathbf{t}_i$$

Inverse Transformation: To transform a point $\mathbf{p}_{\text{global}} \in \mathbb{R}^3$ from the global coordinate system back to the local frame of residue i, the following equation is used:

$$\mathbf{p} = T_i^{-1}(\mathbf{p}_{\text{global}}) = \mathbf{R}_i^{-1}(\mathbf{p}_{\text{global}} - \mathbf{t}_i)$$

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用语言模型模拟五亿年的进化

参数 n_{layers}	d_{model}	d_{head}	上下文长度	学习率	预热步数	以标记为单位的批量大小	编号步骤	总计标记	FLOPs
1.4B 48 1536 64 2048 1				4.0e-4	5K	，572， 864 50K		∼80B	6.72×10^{20}
1.4B 48 1536 64 2048 1				4.0e-4	5K	，572， 864 20万		∼320B	2.7×10^{21}
70亿 96 2560 128 2048 3				2.5e-4	5K	，932， 160 140K		∼550B	2.47×10^{22}
98.5B 216 6144 128 2048 4				1.0e-4	20K	，194， 304 430K		∼1.8T	1.07×10^{24}

表 S1. 不同模型配置的参数详细信息。

算法 4 回归_

输入：$x \in \mathbb{R}^{\cdots \times d}$
1: z =proj$_{\text{in}}(x)$2: z =
GeLU(z)3: z =
LayerNorm(z)4: z =
proj$_{\text{out}}(z)$5: 返回 z

除了结构坐标之外，我们输出附录 A.1.5.1 中详细描述的每个轨迹的预测：(a) 序列，(b) 结构标记，(c) SS8，(d) 量化的 SASA，(e) 函数关键词标记以及 (f) 残基 (InterPro) 注释的二进制特征。除了多热残基注释之外，所有其他轨迹都被预测为可能标记的分类分布。

A.1.6. 几何注意力

如附录A.1.5.1中提到的，ESM3以两种独立的方式处理结构信息：

几何注意力在算法6中描述，利用对主链原子原子坐标的条件化来利用细粒度的三维信息。坐标仅用作模型输入。结构标记在附录A.1.7中描述，结构标记由于丰富的局部邻域语义被压缩为标记，从而实现更快的学习。结构标记通常用作模型输出。

几何注意力使得高速编码蛋白质结构成为可能。蛋白质主链结构可以通过每个残基主链坐标定义的框架的相对距离和方向表示。当仅提供部分结构时，对框架相对方向的推理对于捕捉局部主链几何形状至关重要。几何注意力是一种 $SE(3)$不变的全对全注意力机制，它在输入中对所有定义框架的相对距离和方向进行推理。

(图S2)。因为这个注意力操作可以实现
使用与注意力相同的计算原语，因此它易于扩展。我们首先提供帧的概述，然后描述几何注意力如何使用帧：

A.1.6.1. 帧

帧是封装每个残基在蛋白质中的位置和方向的表示。我们使用类似于Ingraham等人（70）的公式。每个帧$T \in SE(3)$ 由旋转矩阵$\mathbf{R} \in SO(3)$ 和一个平移向量$\mathbf{t} \in \mathbb{R}^3$组成。

定义：用于残基 i 的框架 T_i 定义如下：

$$T_i = \begin{bmatrix} \mathbf{R}_i & \mathbf{t}_i \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \in SE(3)$$

其中$\mathrm{R}_i \in SO(3)$和$\mathrm{t}_i \in \mathbb{R}^3$。旋转矩阵：用于残基 i 的旋转矩阵R_i 由三维向量 $[\hat{x}、\hat{e}_1、\hat{e}_2]$组成：1. $\hat{x}$ 和$\hat{e}_1$ 是 $N - C_\alpha - C$ 平面上的正交单位向量。2. $\hat{e}_2$ 是垂直于$\hat{x}$ 和$\hat{e}_1$的单位向量。该矩阵将向量旋转到局部坐标系统中。

其中对应残基的 $N - C_\alpha - C$ 平面展开了 xy 平面。平移向量：平移向量t_i 指定残基的 C_α的位置。变换：要将点$\mathrm{p} \in \mathbb{R}^3$ 从残基 i 的局部框架变换到全局坐标系统，使用以下方程：

$$\mathbf{p}_{\text{global}} = T_i(\mathbf{p}) = \mathbf{R}_i\mathbf{p} + \mathbf{t}_i$$

逆变换：要将点$\mathrm{p}_{\text{global}} \in \mathbb{R}^3$ 从全局坐标系统变换回局部
残基 i的框架，使用以下方程：

$$\mathbf{p} = T_i^{-1}(\mathbf{p}_{\text{global}}) = \mathbf{R}_i^{-1}(\mathbf{p}_{\text{global}} - \mathbf{t}_i)$$

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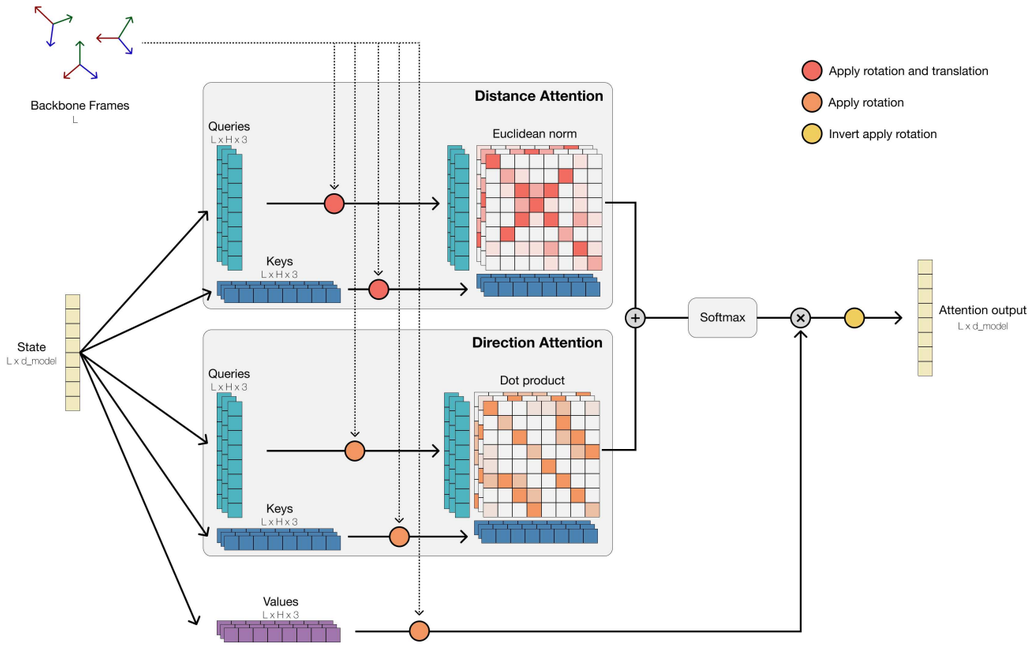

Figure S2. Geometric attention. Geometric attention is an SE(3) invariant all-to-all attention mechanism where the attention score matrix is a weighted sum of two terms: (1) the pairwise distances between queries and keys rotated and translated by their respective backbone frames, and (2) the pairwise dot products between queries and keys rotated by their respective backbone frames. This attention mechanism encodes structural information with throughput comparable to the standard attention operation in transformers.

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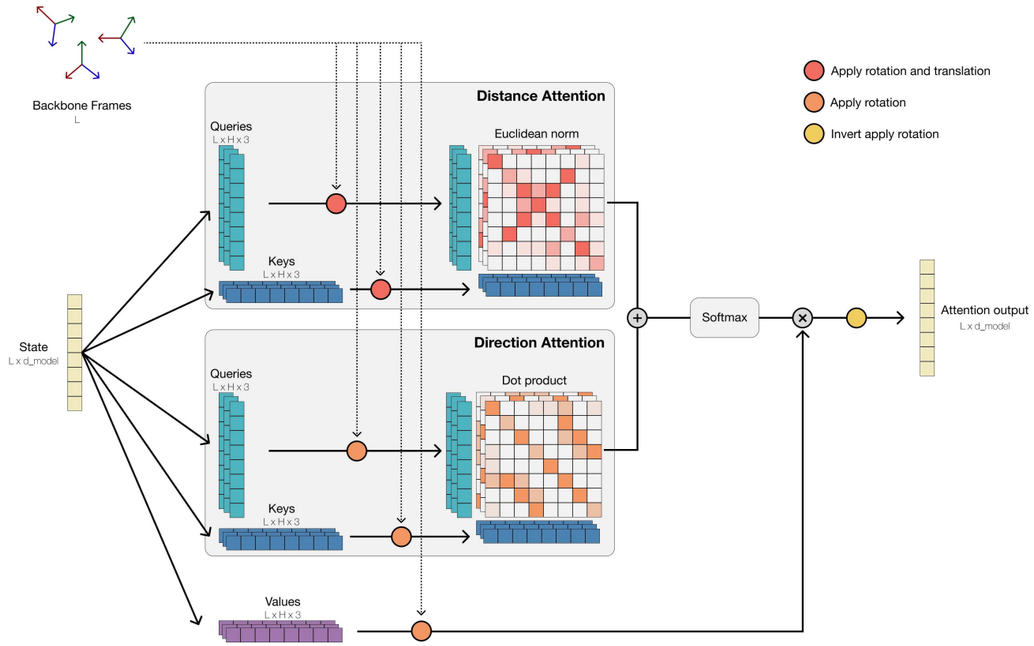

图 S2. 几何注意力。几何注意力是一种对所有元素均不变的 SE(3) 关注机制，其中注意力得分矩阵是两个项的加权和：(1) 查询和键之间的成对距离，经过各自骨干框架的旋转和平移，以及 (2) 查询和键之间的成对点积，同样经过各自骨干框架的旋转。该注意力机制以与变换器中标准注意力操作相当的吞吐量编码结构信息。

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To create frames, all we require is a translation vector $\vec{t}$, and two vectors $\vec{x}$ and $\vec{y}$ defining the local xy plane *after* conversion to global coordinates, from which the frame T can be calculated with the standard Gram-Schmidt algorithm:

Algorithm 5 `gram_schmidt`

Input: $\vec{t} \in \mathbb{R}^{L\times 3}, \vec{x} \in \mathbb{R}^{L\times 3}, \vec{y} \in \mathbb{R}^{L\times 3}$

$\hat{x} = \frac{\vec{x}}{\|\vec{x}\|}$
$\vec{e}_1 = \vec{y} - (\hat{x} \cdot \vec{y})\hat{x}$
$\hat{e}_1 = \frac{\vec{e}_1}{\|\vec{e}_1\|}$
$\hat{e}_2 = \hat{x} \times \hat{e}_1$
$R = [\hat{x}, \hat{e}_1, \hat{e}_2]$ $\triangleright SO(3)^L$
$T = \begin{bmatrix} R & \vec{t} \\ 0_{1\times 3} & 1 \end{bmatrix}$ $\triangleright SE(3)^L$
return T

We construct frames such that the C_α is at the origin of the frame ($\vec{t}$), C on the negative x-axis ($-\vec{x}$), and N is on the xy-plane.

A.1.6.2. Geometric Self-Attention

Algorithm 6 details the Geometric Self-Attention layer. It can be efficiently implemented using similar ideas as FlashAttention (71). It is used twice in our system: in the VQ-VAE encoder for structure tokens (Appendix A.1.7.1), and in the first layer of ESM3.

Unlike regular self-attention, which only operates on per-residue embeddings, Geometric Attention incorporates the per-residue frames T to integrate geometric information in a rotation and translation invariant way. The process of forming the attention matrix A is as follows:

1. **QKV Projections:** Two sets of keys and queries (Q_r, K_r) and (Q_d, K_d), along with V, all with shapes $\in \mathbb{R}^{L\times h\times 3}$ are linearly projected from layer input X. L is the sequence length, h is the number of heads.

2. **Convert QKV to global frame:** Each of the queries, keys and values are initially assumed to be in the local frame of their corresponding residue.

   (a) **Convert to Global Rotational Frame:** We convert each of the vectors in Q_r, K_r, V from their local frame (where the xy plane is the $N-C_\alpha-C$ plane for each residue) to a *global* rotational frame (where the xy plane is aligned for all residues) by applying $\mathbf{R}_i$ (Algorithm 6, lines 3, 4).

   (b) **Convert to Global Distance Frame:** We convert each of the vectors in Q_d, K_d from their local frame to a *global* frame by applying T_i (Algorithm 6, lines 5, 6).

3. **Direction Attention:** The pairwise, per-head h rotational similarity R between keys i and queries j is calculated using the dot product $[R]_{i,j,h} = \frac{1}{\sqrt{3}}[q_r]_{i,h,:} \cdot [k_r]_{j,h,:}$. This is equivalent to the cosine distance between projected points.

4. **Distance Attention:** The pairwise, per-head h distance similarity D between keys i and queries j is computed using the L_2 norm of the difference $[D]_{i,j,h} = \frac{1}{\sqrt{3}}\|[q_r]_{i,h,:} - [k_r]_{j,h,:}\|_2$.

5. **Scale Factor:** R and D are scaled per-head with learned scalars $[\bar{w}_r]_h$ and $[\bar{w}_d]_h$, respectively, where $\bar{w}_r, \bar{w}_d \in \mathbb{R}^h$. We use the softplus function to transform weights into $[0, \infty)^h$. This scaling allows certain heads to specialize in attending via distance or direction attention.

Algorithm 6 `geometric_mha`

Input: $X \in \mathbb{R}^{L\times d}, T \in SE(3)^L$

$Q_r, K_r, Q_d, K_d, V = \text{Linear}(X)$ $\triangleright (\mathbb{R}^{L\times h\times 3})_{\times 5}$
$(\mathbf{R}_i, \mathbf{t}_i) = T_i$ $\triangleright (SO(3)^L, \mathbb{R}^{L\times 3})$
$[Q_r]_{i,h,:} = \mathbf{R}_i([Q_r]_{i,h,:})$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[K_r]_{i,h,:} = \mathbf{R}_i([K_r]_{i,h,:})$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[Q_d]_{i,h,:} = T_i([Q_d]_{i,h,:})$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[K_d]_{i,h,:} = T_i([K_d]_{i,h,:})$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[R]_{i,j,h} = \frac{1}{\sqrt{3}}[q_r]_{i,h,:} \cdot [k_r]_{j,h,:}$ $\triangleright \mathbb{R}^{L\times L\times h}$
$[D]_{i,j,h} = \frac{1}{\sqrt{3}}\|[q_r]_{i,h,:} - [k_r]_{j,h,:}\|_2$ $\triangleright \mathbb{R}^{L\times L\times h}$
$A = \text{softplus}(\bar{w}_r)R - \text{softplus}(\bar{w}_d)D$ $\triangleright \mathbb{R}^{L\times L\times h}$
$A = \text{softmax}_j(A)$
$[V]_{i,h,:} = \mathbf{R}_i([V]_{i,h,:})$
$O = A \cdot V$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[O]_{i,h,:} = \mathbf{R}_i^{-1}([O]_{i,h,:})$
$X = X + \text{Linear}(O)$ $\triangleright \mathbb{R}^{L\times d}$

A.1.7. Structure Tokenizer

Each residue is associated with one of 4,096 structure tokens (+4 special tokens), designed to provide a rich, learned representation of its local neighborhood. Prior works have show structure tokens can be used for search (41), to improve representation learning (72), and for generation (16), but we trained our own tokenizer to optimize for structure generation. The tokens are generated with a VQ-VAE encoder, with a corresponding decoder to enable decoding of generated tokens back to 3D coordinates.

A.1.7.1. Encoder

The VQ-VAE encoder f_{enc} consists of two geometric attention blocks (Transformer blocks, but self-attention replaced with `geometric_mha`) with an embedding width

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为了创建框架，我们所需的仅是一个平移向量 $\vec{t}$，以及两个向量 $\vec{x}$ 和 $\vec{y}$，它们定义了转换为全局坐标后的局部 xy 平面，从该平面可以通过标准的Gram-Schmidt算法计算框架 T：

算法5 `gram_`

输入： $\vec{t} \in \mathbb{R}^{L\times 3}, \vec{x} \in \mathbb{R}^{L\times 3}, \vec{y} \in \mathbb{R}^{L\times 3}$

$\hat{x} = \frac{\vec{x}}{\|\vec{x}\|}$
$\vec{e}_1 = \vec{y} - (\hat{x} \cdot \vec{y})\hat{x}$
$\hat{e}_1 = \frac{\vec{e}_1}{\|\vec{e}_1\|}$
$\hat{e}_2 = \hat{e}_1$
$R = [\hat{x}, \hat{e}_1, \hat{e}_2]$ $\triangleright SO(3)^L$
$T = \begin{bmatrix} R & \vec{t} \\ 0_{1\times 3} & 1 \end{bmatrix}$ $\triangleright SE(3)^L$
返回 T

我们构造框架，使得 C_α 位于框架的原点 ($\vec{t}$)， C 位于负x轴上 ($-\vec{x}$)，而 N 位于xy平面上。

A.1.6.2. 几何自注意力

算法 6详细介绍了几何自注意力层。该层可以利用与FlashAttention (71)相似的思路高效实现。在我们的系统中，该层被使用了两次：在VQ-VAE编码器中用于结构标记（附录A.1.7.1），以及在ESM3的第一层。

与仅在每个残基嵌入上操作的常规自注意力不同，几何注意力整合了每个残基框架 T，以以旋转和位移不变的方式整合几何信息。形成注意力矩阵 A 的过程如下：

1. QKV投影：两个键和值的查询集 (Q_r, K_r) 和 (Q_d, K_d)，加上 V，所有这些的形状为 $\in \mathbb{R}^{L\times h\times 3}$，它们都是从层输入 X线性投影而来的。L 为序列长度， h 为头的数量。

2. 将 QKV 转换为全局框架：每个查询、键和值最初被假定处于其对应残基的局部框架中。(a) 转换为全局旋转框架：我们将 Q_r、 K_r、 V中的每个向量从其局部框架（其中 xy 平面是每个残基的 $N-C_\alpha-C$ 平面）转换到 全局 旋转框架（其中 xy 平面对齐于所有残基）通过应用 $\mathbf{R}_i$ (算法6，第 3 行，第 4 行)。(b) 转换为全局距离框架：我们将 Q_d、 K_d 中的每个向量从其局部框架转换为 全局 框架通过应用 T_i (算法6，第 5 行，第 6 行)。

3. 方向注意力：键与查询之间的逐对、每个头的 h 旋转相似性 R 是通过点积$[R]_{i,j,h} = \frac{1}{\sqrt{3}}[q_r]_{i,h,:} \cdot [k_r]_{j,h,:}$来计算的。这相当于投影点之间的余弦距离。

4. 距离注意力： 每个头的成对 h 距离相似性 D 在键 i 和查询 j 之间，使用 L_2 差的范数$[D]_{i,j,h} = \frac{1}{\sqrt{3}}\|[q_r]_{i,h,:} - [k_r]_{j,h,:}\|_2$。

5. 缩放因子： R 和 D 分别使用学习到的标量 $[\bar{w}_r]_h$ 和 $[\bar{w}_d]_h$进行每个头的缩放，其中$\bar{w}_r, \bar{w}_d \in \mathbb{R}^h$。我们使用softplus函数将权重转换为 $[0, \infty)^h$。这种缩放允许某些头专注于通过距离或方向注意力进行关注。

算法 6 几何_

输入： $X \in \mathbb{R}^{L\times d}, T \in SE(3)^L$

$Q_r, K_r, Q_d, K_d, V = \text{线性}(X)$ $\triangleright (\mathbb{R}^{L\times h\times 3})_{\times 5}$
$(\mathbf{R}_i, \mathbf{t}_i) = T_i$ $\triangleright (SO(3)^L, \mathbb{R}^{L\times 3})$
$[Q_r]_{i,h,:} = \mathbf{R}_i([Q_r]_{i,h,:})$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[K_r]_{i,h,:} = \mathbf{R}_i([K_r]_{i,h,:})$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[Q_d]_{i,h,:} = T_i([Q_d]_{i,h,:})$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[K_d]_{i,h,:} = T_i([K_d]_{i,h,:})$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[R]_{i,j,h} = \frac{1}{\sqrt{3}}[q_r]_{i,h,:} \cdot [k_r]_{j,h,:}$ $\triangleright \mathbb{R}^{L\times L\times h}$
$[D]_{i,j,h} = \frac{1}{\sqrt{3}}\|[q_r]_{i,h,:} - [k_r]_{j,h,:}\|_2$ $\triangleright \mathbb{R}^{L\times L\times h}$
$A = \text{softplus}(\bar{w}_r)R - \text{softplus}(\bar{w}_d)D$ $\triangleright \mathbb{R}^{L\times L\times h}$
$A = \text{Softmax}_j(A)$
$[V]_{i,h,:} = \mathbf{R}_i([V]_{i,h,:})$
$O = A \cdot V$ $\triangleright \mathbb{R}^{L\times h\times 3}$
$[O]_{i,h,:} = \mathbf{R}_i^{-1}([O]_{i,h,:})$
$X = X + \text{线性}(O)$ $\triangleright \mathbb{R}^{L\times d}$

A.1.7. 结构标记器

每个残基与 4,096 个结构标记（+4 特定标记）相关联，旨在提供其局部邻域的丰富的学习表征。先前的研究表明，结构标记可以用于搜索（41），提高表征学习（72）以及生成（16），但我们训练了自己的标记器以优化结构生成。标记是通过 VQ-VAE 编码器生成的，并配备相应的解码器，以便将生成的标记解码回三维坐标。

A.1.7.1. 编码器

VQ-VAE 编码器 f_{enc} 由两个几何注意力模块组成（变换器模块，但自注意力被 几何_多头注意力机制）与嵌入宽度

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of 1024 and 128 geometric heads per geometric attention layer.

The VQ-VAE encoder reasons over the backbone frames and the relative sequence position of residues in the local structure. Relative sequence positions are encoded through a learned positional embedding. Sequence positions are determined relative to the query residue (i.e., if the query residue has residue index 56, then the residue in index 58 has a +2 sequence position). Relative sequence positions are clamped to +/- 32 before encoding, meaning long-range contacts share sequence positional embeddings. Relative sequence positional embeddings define the initial encoder state N, and has shape $L \times 16 \times d$ (Algorithm 7, line 4). Note that this means the input to the VQ-VAE encoder is purely structural: no sequence (amino acid), function or other information is used here. Furthermore, each neighborhood is processed completely independently; for each residue, the encoder only uses the information of its 16 nearest neighbors.

Geometric attention blocks operate similar to Transformer blocks in that they transform a state according to an attention operation (`geometric_mha`) and feedforward network (SwiGLU MLP). As such, the output has the same shape as the input. In this case, this means that the encoder outputs 16 latents per residue. However, we want to learn a single token, i.e., a single latent per residue, hence we take the embedding corresponding to the query residue position $N_{:,0,:}$.

The process of generating structure tokens (Algorithm 7) from the full 3D coordinates of the protein then is as follows:

1. **Local Neighborhood:** For each residue, obtain the indices $N_{\text{idx}} \in \{0..L-1\}^{L\times 16}$ of the 16 nearest residues (as measured by C_α distance). The first of the 16 neighbors is always the residue itself. We also obtain the frames for each residue in a local neighborhood with T_{knn}.

2. **Embed Neighbors:** Embed the relative distance in sequence space for each neighbor, $\Delta i = \text{clamp}(N_{\text{idx}} - i, -32, 32)$ to form $N \in \mathbb{R}^{L\times 16\times d}$.

3. **Encode:** Pass N through a shallow encoder f_{enc} consisting of 2 Transformer blocks, with regular multi-head self-attention swapped with `geometric_mha`. The attention is unmasked, all-to-all over the entire neighborhood.

4. **Quantize:** Extract the first element $N_{:,0,:}$ from the neighborhood, which corresponds to the residue itself. Project it linearly, and quantize by replacing with the nearest vector in a codebook. This yields the structure token per residue.

Algorithm 7 `structure_encode`

Input: $x_{C_\alpha} \in \mathbb{R}^{L\times 3}, T \in SE(3)^L$

1: $N_{\text{idx}} = \text{knn}(x_{C_\alpha})$ ▷$\{0..L-1\}^{L\times 16}$
2: $T_{\text{knn}} = T[N_{\text{idx}}]$ ▷$SE(3)^{L\times 16}$
3: $\Delta i = \text{clamp}(N_{\text{idx}} - i, -32, 32)$
4: $N = \text{embed}(\Delta i)$ ▷$\mathbb{R}^{L\times 16\times d}$
5: $N = f_{enc}(N, T_{\text{knn}})$ ▷$\mathbb{R}^{L\times 16\times d}$
6: $z = \text{Linear}(N_{:,0,:})$ ▷$\mathbb{R}^{L\times d'}$
7: $z =$ `quantize` (z) ▷$\{0..4095\}^L$

A.1.7.1.1. Codebook Learning

`quantize` transforms the L latents into L discrete tokens. Since the VQ-VAE was initially proposed (73), numerous approaches and tricks have been developed to address issues with poor codebook utilization and unstable training. We chose to learn the codebook as an exponential moving average of encoder outputs (73–75). To improve codebook utilization, unused codes are re-initialized to encoder outputs.

A.1.7.1.2. Parallel Encoding

To improve training and inference efficiency, we encode all local structure graphs within a protein in parallel. In practice, this means that given a batch of B proteins with average sequence length L, then the inputs to the structure encoder will have shape $BL \times 16 \times d$.

A.1.7.2. Decoder

While the encoder independently processes all local structures in parallel, the decoder f_{dec} attends over the entire set of L tokens to reconstruct the full structure. It is composed using a stack of bidirectional Transformer blocks with regular self-attention.

As discussed in Appendix A.1.7.3, the VQ-VAE is trained in two stages. In the first stage, a smaller decoder trunk consisting of 8 Transformer blocks with width 1024, rotary positional embeddings, and MLPs is trained to only predict backbone coordinates. In the second stage, the decoder weights are re-initialized and the network size is expanded to 30 layers, each with an embedding dimension of 1280 (∼600M parameters) to predict all atom coordinates.

The exact steps to convert structure tokens back to 3D all-atom coordinates using the decoder is provided in Algorithm 8 and detailed as follows,

1. **Transformer:** We embed the structure tokens and pass them through a stack of Transformer blocks f_{dec} (regular self-attention + MLP sublayers, no geometric attention).

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每个几何注意力层有1024和128个几何头。

VQ-VAE编码器对骨干帧和局部结构中残基的相对序列位置进行推理。相对序列位置通过学习的位置信息嵌入进行编码。序列位置是相对于查询残基确定的（即，如果查询残基的残基索引为56，则索引为58的残基具有一个 +2 序列位置）。相对序列位置在编码之前被限制在 +/- 32，这意味着长距离接触共享序列位置嵌入。相对序列位置嵌入定义了初始编码器状态 N，其形状为 $L \times 16 \times d$ (算法7，第4行)。值得注意的是，这意味着VQ-VAE编码器的输入完全是结构性的：这里不使用序列（氨基酸）、功能或其他信息。此外，每个邻域都是完全独立地处理的；对于每个残基，编码器仅使用其16个最近邻的消息。

几何注意力模块的操作方式类似于变换器模块，因为它们根据注意力操作（geometric_mha）和前馈网络（SwiGLU MLP）转换状态。因此，输出的形状与输入相同。在这种情况下，这意味着编码器每个残基输出16个潜变量。然而，我们希望学习单个标记，即每个残基一个潜变量，因此我们取对应于查询残基位置$N_{:\text{的嵌入}，0，:}$。

生成结构标记的过程（算法7）基于蛋白质的完整三维坐标，步骤如下：

1. 局部邻域：对于每个残基，获得16个最近残基（按 C_α 距离测量）的索引 $N_{\text{idx}} \in \{0..L-1\}^{L\times 16}$。这16个邻居中的第一个总是残基本身。我们还根据$T_{\text{knn}}$为局部邻域中的每个残基获得帧。

2. 嵌入邻居：在序列空间中嵌入每个邻居的相对距离，$\Delta i = \text{clamp}(N_{\text{idx}} - i, -32, 32)$以形成 $N \in \mathbb{R}^{L\times 16\times d}$。

3. 编码：将 N 通过一个浅层编码器 f_{enc} 传递，该编码器由2个变换器模块组成，常规多头自注意力被几何的_mha 替换。注意力是无掩蔽的，全面覆盖整个邻域。

4. 量化：提取邻域中的第一个元素$N_{:,\ 0,\ :}$ 这对应于残基本身。将其线性投影，并通过用代码本中最近的向量替换来进行量化。这为每个残基产生了结构标记。

算法 7 结构_

输入： $x_{C_\alpha} \in \mathbb{R}^{L\times 3}, T \in SE(3)^L$

1: $N_{\text{idx}} = \text{knn}(x_{C_\alpha})$ ▷$\{0..L-1\}^{L\times 16}$
2: $T_{\text{knn}} = T[N_{\text{idx}}]$ ▷$SE(3)^{L\times 16}$
3: $\Delta i =$ 夹紧$(N_{\text{idx}} - i, -32, 32)$
4: $N =$嵌入(Δi) ▷$\mathbb{R}^{L\times 16\times d}$
5: $N = f_{enc}(N, T_{\text{knn}})$ ▷$\mathbb{R}^{L\times 16\times d}$
6: $z =$线性$(N_{:,0,:})$ ▷$\mathbb{R}^{L\times d'}$
7: $z =$ 量化 (z) ▷$\{0..4095\}^L$

A.1.7.1.1. 代码本学习

量化 将 L 潜变量转换为 L 离散标记。由于最初提出 VQ-VAE 以来（73），已经开发出许多方法和技巧来解决代码本利用率低和不稳定训练的问题。我们选择将代码本学习为编码器输出的指数移动平均（73–75）。为提高代码本的利用率，未使用的代码被重新初始化为编码器输出。

A.1.72. 并行编码 为了提高训练和推断效率，我们以并行方式编码蛋白质中的所有局部结构图。实际上，这意味着对于一批平均序列长度为 L 的 B 蛋白质，结构编码器的输入将具有形状 $BL \times 16 \times d$。

A.1.7.2. 解码器

虽然编码器独立并行地处理所有局部结构，但解码器 f_{dec} 会在整个 L 标记集上进行关注，以重构完整的结构。它由一系列双向变换器模块和常规自注意力构成。

正如附录 A.1.7.3 中讨论的，VQ-VAE 是分两阶段训练的。在第一阶段，一个包含 8 个宽度为 1024 的变换器模块、旋转位置嵌入和多层感知机的小型解码器主干被训练仅用于预测骨架坐标。在第二阶段，解码器的权重被重新初始化，网络规模扩大到 30 层，每层具有 1280 的嵌入维度（∼600M 参数）以预测所有原子坐标。

将结构标记转换回三维全原子坐标的确切步骤使用解码器提供在算法8中详细说明如下，

1. 变换器：我们嵌入结构标记并通过一堆变换器模块传递它们 f_{dec} （常规自注意力 + 多层感知机子层，无几何注意力）。

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2. **Projection Head:** We use a projection head to regress 3 3-D vectors per residue: a translation vector $\vec{t}$, and 2 vectors $-\vec{x}$ and $\vec{y}$ that define the $N - C_\alpha - C$ plane per residue *after* it has been rotated into position. This head also predicts the unnormalized sine and cosine components of up to 7 side chain torsion angles.

3. **Calculate T:** We use `gram_schmidt` to convert $\vec{t}$, $\vec{x}$, and $\vec{y}$ into frames $T \in SE(3)^L$.

4. **Calculate T_{local}:** We normalize the sine and cosine components and convert them to frames $T_{\text{local}} \in SE(3)^{L\times 7}$ corresponding to rotations around the previous element on the side chain.

5. **Compose Frames:** We compose each element of T_{local} with its predecessors on a tree rooted at T to form $T_{\text{global}} \in SE(3)^{L\times 14}$, corresponding to the transformations needed for each heavy atom per residue in atom14 representation.

6. **Apply Frames:** We then apply the frame to the $\vec{X}_{\text{ref}} \in \mathbb{R}^{L\times 14\times 3}$ coordinates in a reference frame, to rotate and transform each residue into their final positions.

Algorithm 8 `structure_decode`

Input: $z \in \{0..4099\}^{L\times 16}$

$z = \text{embed}(z)$ $\quad \triangleright \mathbb{R}^{L\times d}$
$z = f_{dec}(z)$ $\quad \triangleright \mathbb{R}^{L\times d}$
$\vec{t}, \vec{x}, \vec{y}, \sin\theta, \cos\theta = \text{proj}(z)$ $\quad \triangleright (\mathbb{R}^{L\times 3})_{\times 3}, (\mathbb{R}^{L\times 7})_{\times 2}$
$T =$ `gram_schmidt` $(\vec{t}, -\vec{x}, \vec{y})$ $\quad \triangleright SE(3)^L$
$\sin\theta = \frac{\sin\theta}{\sqrt{\sin\theta^2 + \cos\theta^2}}$ $\quad \triangleright [-1, 1]^{L\times 7}$
$\cos\theta = \frac{\cos\theta}{\sqrt{\sin\theta^2 + \cos\theta^2}}$ $\quad \triangleright [-1, 1]^{L\times 7}$
$T_{\text{local}} =$ `rot_frames` $(\sin\theta, \cos\theta)$ $\quad \triangleright SE(3)^{L\times 7}$
$T_{\text{global}} =$ `compose` (T_{local}, T) $\quad \triangleright SE(3)^{L\times 14}$
$\vec{X} = T_{\text{global}}(\vec{X_{ref}})$ $\quad \triangleright \mathbb{R}^{L\times 14\times 3}$

A.1.7.3. Training

When using a VQ-VAE to learn discrete representations which maximize reconstruction quality, it is common to train in the autoencoder in two stages (76). In the first stage, the encoder and codebook is learned with a relatively small and efficient decoder. In the second stage, the encoder and codebook are frozen and a larger or otherwise more computationally expensive decoder is trained to maximize reconstruction quality. We follow this two-stage training approach for the structure tokenizer.

A.1.7.3.1. Stage 1.

The VQ-VAE is trained for 90k steps on a dataset of single chain proteins from the PDB, AFDB, and ESMAtlas. We use the AdamW optimizer (Loshchilov et al. 2017) with learning rate annealed from 4e-4 according to a cosine decay schedule. Proteins are cropped to a maximum sequence length of 512. Five losses are used to supervise this stage of training. The geometric distance and geometric direction losses are responsible for supervising reconstruction of high quality backbone structures.

Additionally, binned distance and direction classification losses are used to bootstrap structure prediction but are ultimately immaterial to reconstruction. We have found that these structure prediction losses formulated as classification tasks improve convergence early in training. To produce these pairwise logits, we use a `pairwise_proj_head`, that takes $x \in \mathbb{R}^{L\times d}$ and returns logits $z \in \mathbb{R}^{L\times L\times d'}$. It works as follows:

Algorithm 9 `pairwise_proj_head`

Input: $x \in \mathbb{R}^{L\times d}$

$q, k = \text{proj}(x), \text{proj}(x)$
$\text{prod}_{i,j,:}, \text{diff}_{i,j,:} = q_{j,:} \odot k_{i,:}, q_{j,:} - k_{i,:}$
$z =$ `regression_head` $([\text{prod} \mid \text{diff}])$ $\quad \triangleright \mathbb{R}^{L\times L\times d'}$
return z

Finally, an inverse folding token prediction loss (i.e., a cross-entropy loss between predicted sequence and ground truth sequence) is an auxiliary loss used to encourage the learned representations to contain information pertinent to sequence-related tasks.

The five losses are covered in detailed as follows:

1. **Backbone Distance Loss:** Compute the pairwise L_2 distance matrix for the predicted and true coordinates of the 3 backbone atoms (N, C_α, C). Let $D_{\text{pred}}, D \in \mathbb{R}^{3L\times 3L}$. Compute $(D_{\text{pred}} - D)^2$, clamp the maximum error to $(5\ \text{Å})^2$, and take the mean.

Algorithm 10 `backbone_distance_loss`

Input: $\hat{X} \in \mathbb{R}^{L\times 3\times 3}, X \in \mathbb{R}^{L\times 3\times 3}$

$\hat{Z}, Z = \text{flatten}(\hat{X}), \text{flatten}(X)$ $\quad \triangleright \mathbb{R}^{3L\times 3}, \mathbb{R}^{3L\times 3}$
$[D_{\text{pred}}]_{i,j} = \|[\hat{Z}]_{i,:} - [\hat{Z}]_{j,:}\|_2^2$ $\quad \triangleright \mathbb{R}^{3L\times 3L}$
$[D]_{i,j} = \|[Z]_{i,:} - [Z]_{j,:}\|_2^2$ $\quad \triangleright \mathbb{R}^{3L\times 3L}$
$E = (D_{\text{pred}} - D)^2$
$E = \min(E, 25)$
$l = \text{mean}_{i,j}(E)$ $\quad \triangleright \mathbb{R}$
return l

2. **Backbone Direction Loss:** Compute six vectors for both predicted and ground truth coordinates for each residue:

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2. 投影头：我们使用投影头对每个残基回归3个三维向量：一个平移向量 $\vec{t}$，以及2个向量 $-\vec{x}$ 和 $\vec{y}$，它们定义了每个残基在旋转到位后所处的 $N - C_\alpha - C$ 平面。该投影头还预测多达7个侧链扭转角的未归一化正弦和余弦分量。

3. 计算 T：我们使用格拉姆_施密特将 $\vec{t}$、$\vec{x}$和 $\vec{y}$ 转换为框架 $T \in SE(3)^L$。

4. 计算 T_{local}：我们对正弦和余弦分量进行归一化，并将其转换为框架 $T_{\text{local}} \in SE(3)^{L\times 7}$，对应于侧链上前一个元素的旋转。

5. 组合框架：我们在以 T 为根的树上，组合 T_{local}的每个元素及其前驱，以形成$T_{\text{global}} \in SE(3)^{L\times 14}$，对应于每个残基中重原子的变换，这些变换以atom14表示法显示。→

6. 应用框架：然后，我们将框架应用于 $X_{\text{ref}} \in \mathbb{R}^{L\times 14\times 3}$ 中的坐标，在参考框架中旋转并转换每个残基到其最终位置。算法8结构_解码输入：

使用 AdamW 优化器（Loshchilov et al. 2017），其学习率根据余弦衰减计划从 4e-4 进行退火。蛋白质被剪切到最大序列长度为 512。使用五个损失来监督这一训练阶段。几何距离和几何方向损失负责监督高质量骨架结构的重建。

此外，分箱距离和方向分类损失用于引导结构预测，但对重建最终并无实质性影响。我们发现，这些 formulierte 作为分类任务的结构预测损失在训练初期改善了收敛性。为了生成这些成对逻辑斯蒂，我们使用一个 成对_`proj_head`，

该函数接受 $x \in \mathbb{R}^{L\times d}$ 并返回逻辑值 $z \in \mathbb{R}^{L\times L\times d'}$。其工作原理如下：

算法 9 成对_

输入：$x \in \mathbb{R}^{L\times d}$

$q, k =$ 投影(x), 投影(x)
乘积$_{i,j,:}$,差异$_{i,j,:} = q_{j,:} \odot k_{i,:}, q_{j,:} - k_{i,:}$
$z =$ 回归_头部 $([\text{prod} \mid \text{diff}])$ $\quad \triangleright \mathbb{R}^{L\times L\times d'}$
返回 z

最后，反向折叠标记预测损失（即，预测序列与真实值序列之间的交叉熵损失）是一种辅助损失，旨在鼓励学习到的表示包含与序列相关任务相关的信息。

这五个损失的详细内容如下：

1. 骨架距离损失：计算预测与真实坐标间的 L_2距离矩阵，包含3个骨架原子（N, C_α, C）。设 D_{pred}, $D \in \mathbb{R}^{3L\times 3L}$。计算 $(D_{\text{pred}} - D)^2$, 限制最大误差为 (5 Å)^2, 并取均值。算法 10 骨架_距离_损失输入：

$\hat{X} \in \mathbb{R}^{L\times 3\times 3}, X \in \mathbb{R}^{L\times 3\times 3}$
$\hat{Z}, Z =$ 展开$(\hat{X})$, 展开(X) $\quad \triangleright \mathbb{R}^{3L\times 3}, \mathbb{R}^{3L\times 3}$
$[D_{\text{pred}}]_{i,j} = \|[\hat{Z}]_{i,:} - [\hat{Z}]_{j,:}\|_2^2$ $\quad \triangleright \mathbb{R}^{3L\times 3L}$
$[D]_{i,j} = \|[Z]_{i,:} - [Z]_{j,:}\|_2^2$ $\quad \triangleright \mathbb{R}^{3L\times 3L}$
$E = (D_{\text{pred}} - D)^2$
$E = \min(E, 25)$
$l =$均值$_{i,j}(E)$ $\quad \triangleright \mathbb{R}$
返回 l

2. 骨架方向损失：为每个残基计算预测坐标和真实值坐标的六个向量：

A.1.7.3. 训练

在使用 VQ-VAE 学习最大化重建质量的离散表示时，通常在自编码器中分两个阶段进行训练 (76)。在第一阶段，使用相对较小且高效的解码器学习编码器和代码本。在第二阶段，编码器和代码本被冻结，并训练一个更大或计算成本更高的解码器以最大化重建质量。我们遵循这种两阶段训练方法以进行结构标记器的训练。

A.1.7.3.1. 阶段 1.

VQ-VAE 在来自 PDB、AFDB 和 ESMAtlas 的单链蛋白质数据集上训练了 90,000 步。我们

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(a) $N \to C_\alpha$
(b) $C_\alpha \to C$
(c) $C \to N_{\text{next}}$
(d) $\mathbf{n}_{C_\alpha} = -(N \to C_\alpha) \times (C_\alpha \to C)$
(e) $\mathbf{n}_N = (C_{\text{prev}} \to N) \times (N \to C_\alpha)$
(f) $\mathbf{n}_C = (C_\alpha \to C) \times (C \to N_{\text{next}})$

Compute the pairwise dot product, forming $D_{\text{pred}}, D \in \mathbb{R}^{6L\times 6L}$. Compute $(D_{\text{pred}} - D)^2$, clamp the maximum error to 20, and take the mean.

In algorithm form (with `compute_vectors` computing the six vectors described above):

Algorithm 11 `backbone_direction_loss`

Input: $\hat{X} \in \mathbb{R}^{L\times 3\times 3}, X \in \mathbb{R}^{L\times 3\times 3}$

$\hat{V} =$ `compute_vectors` $(\hat{X})$ ▷$\mathbb{R}^{6L\times 3}$
$V =$ `compute_vectors` (X) ▷$\mathbb{R}^{6L\times 3}$
$[D_{\text{pred}}]_{i,j} = [\hat{V}]_{i,:} \cdot [\hat{V}]_{j,:}$ ▷$\mathbb{R}^{6L\times 6L}$
$[D]_{i,j} = [V]_{i,:} \cdot [V]_{j,:}$ ▷$\mathbb{R}^{6L\times 6L}$
$E = (D_{\text{pred}} - D)^2$
$E = \min(E, 20)$
$l = \text{mean}_{i,j}(E)$ ▷$\mathbb{R}$
return l

3. **Binned Direction Classification Loss:** This loss captures a coarser similarity between ground truth and predicted orientations to stabilize early training. It uses the last layer representations of the decoder, not the predicted coordinates. The process is as follows:

   (a) **Unit vectors:** Compute three vectors per residue from ground truth coordinates: $C_\alpha \to C$, $C_\alpha \to N$, and $\mathbf{n}_{C_\alpha} = (C_\alpha \to C) \times (C_\alpha \to N)$, and normalize them to unit length.

   (b) **Dot Products:** Compute pairwise dot products between each pair of vectors for all residues, forming $D \in [-1, 1]^{L\times L\times 6}$. Bin the dot products into 16 evenly spaced bins in $[-1, 1]$, forming classification labels $y \in \{0..15\}^{L\times L}$.

   (c) **Pairwise Logits:** Pass the final layer representations of the decoder $h \in \mathbb{R}^{L\times d}$ through a `pairwise_proj_head` to obtain logits $z \in \mathbb{R}^{L\times L\times 6\times 16}$.

   (d) **Cross Entropy:** Calculate cross-entropy loss using the labels y from the ground truth structure and the logits z, and average over all $L \times L \times 6$ values.

4. **Binned Distance Classification Loss:** Similar to the Binned Direction Classification Loss, this loss bins the true distances between residues (specifically, their C_β) to get ground truth targets and computes a cross-entropy between these targets and pairwise logits. In detail:

   (a) **Calculate C_β:** Given the ground truth N, C_α, and C coordinates, we compute the location of C_β:

      i. Obtain the three vectors $N \to C_\alpha$, $C_\alpha \to C$, and $\mathbf{n} = (N \to C_\alpha) \times (C_\alpha \to C)$.

      ii. Define the following scalars:

$$a = -0.58273431$$
$$b = 0.56802827$$
$$c = -0.54067466$$

      iii. Compute the location of C_β using the formula:

$$C_\beta = a\mathbf{n} + b(N \to C_\alpha) + c(C_\alpha \to C) + C_\alpha \quad (1)$$

   (b) **Pairwise C_β distances:** Compute an $L \times L$ pairwise distance matrix of the C_β, and bin them into one of 64 bins, with lower bounds $[0, 2.3125^2, (2.3125 + 0.3075)^2, \ldots, 21.6875^2]$, forming the labels $y \in \{0..63\}^{L\times L}$.

   (c) **Pairwise logits:** Pass the final layer representations of the decoder $h \in \mathbb{R}^{L\times d}$ through a `pairwise_proj_head` to obtain the logits $z \in \mathbb{R}^{L\times L\times 64}$.

   (d) **Cross Entropy:** Calculate the cross-entropy using the labels y computed from the ground truth structure and the logits z, then average over all $L \times L$ values.

5. **Inverse Folding Loss:** Pass final layer representations of the decoder through a regression head to produce logits z. Using ground truth residues as labels y, compute cross-entropy for the classification task of predicting residues from final layer representations.

A.1.7.3.2. Stage 2.

In the second stage of VQ-VAE training, the encoder and codebook are frozen and a new, deeper decoder is trained. This second stage of training has multiple purposes. First, a larger decoder improves reconstruction quality. Second, augmented structure tokens from ESM3 are added to enable learning pAE and pLDDT heads. Third, we add sequence conditioning and train with all-atom geometric losses to be able to decode all-atom protein structures. Fourth, we extend the context length of the decoder to be able to decode multimers and larger single chain proteins.

Training data for stage 2 consists of predicted structures in AFDB and ESMAtlas, as well as single chains, multimers,

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(a) $N \to C_\alpha$
(b) $C_\alpha \to C$
(c) $C \to N_{\text{next}}$
(d) $\mathrm{n}^{C_\alpha} = -(N \to C_\alpha) \times (C_\alpha \to C)$
(e) $\mathrm{n}_N = (C_{\text{prev}} \to N) \times (N \to C_\alpha)$
(f) $\mathrm{n}_C = (C_\alpha \to C) \times (C \to N_{\text{next}})$

计算成对的点积，形成 $D_{\text{pred}}, D \in \mathbb{R}^{6L\times 6L}$。计算 $(D_{\text{pred}} - D)^2$，将最大误差限制为 20，并取其均值。

以算法形式（与 计算_向量 计算上述六个向量）：

算法 11 主干_

输入：$\hat{X} \in \mathbb{R}^{L\times 3\times 3}, X \in \mathbb{R}^{L\times 3\times 3}$

$\hat{V} =$ $(\hat{X})$ ▷$\mathbb{R}^{6L\times 3}$
$V =$ ▷$\mathbb{R}^{6L\times 3}$
$[D_{\text{pred}}]_{i,j} = [\hat{V}]_{i,:} \cdot [\hat{V}]_{j,:}$ ▷$\mathbb{R}^{6L\times 6L}$
$[D]_{i,j} = [V]_{i,:} \cdot [V]_{j,:}$ ▷$\mathbb{R}^{6L\times 6L}$
$E = (D_{\text{pred}} - D)^2$
$E = \min(E, 20)$
$l =$平均$_{i,j}(E)$ ▷$\mathbb{R}$
返回 l

3. 分箱方向分类损失：该损失捕捉了真实值与预测方向之间的粗略相似性，以稳定早期训练。它使用解码器的最后一层表示，而不是预测坐标。该过程如下：（a）单位向量：从真实值坐标计算每个残基的三个向量：$C_\alpha \to C$、$C_\alpha \to N$，以及$\mathrm{n}^{C_\alpha} = (C_\alpha \to C) \times (C_\alpha \to N)$，并将它们归一化为单位长度。（b）点积：计算所有残基中每对向量之间的成对点积，形成 $D \in [-1, 1]^{L\times L\times 6}$。将点积分箱为16个均匀分布的箱，存储在 $[-1, 1]$中，形成分类标签 $y \in \{0..15\}^{L\times L}$。（c）成对逻辑斯蒂：将解码器的最后一层表示 $h \in \mathbb{R}^{L\times d}$ 通过成对_proj_头以获取逻辑值 $z \in \mathbb{R}^{L\times L\times 6\times 16}$。（d）交叉熵：使用来自真实值结构的标签 y 和逻辑值 z计算交叉熵损失，并对所有 $L \times L \times 6$值进行平均。

4. 分箱距离分类损失：类似于分箱方向分类损失，该损失对残基之间的真实距离进行分箱（具体而言，它们的

C_β）以获取真实值目标，并计算这些目标与成对逻辑斯蒂之间的交叉熵。具体而言：

(a) 计算 C_β：给定真实值 N, C_α和 C 坐标，我们计算C_β的位置：

i. 获取三个向量 $N \to C_\alpha$, $C_\alpha \to C$和$\mathrm{n} = (N \to C_\alpha) \times (C_\alpha \to C)$。ii. 定义以下标量：

$$a = -0.58273431$$
$$b = 0.56802827$$
$$c = -0.54067466$$

iii. 使用公式计算 C_β 的位置：

$$C_\beta = a\mathbf{n} + b(N \to C_\alpha) + c(C_\alpha \to C) + C_\alpha \quad (1)$$

(b) 成对 C_β 距离：计算 $L \times L$的成对距离矩阵，并将其划分到64个区间中的一个，区间下限为$[0, 2.3125^2, (2.3125 + 0.3075)^2, \ldots, 21.6875^2]$，形成标签 $y \in \{0..63\}^{L\times L}$。

(c) 成对逻辑值：将解码器的最终层表示 $h \in \mathbb{R}^{L\times d}$ 传递通过成对_proj_头，以获得逻辑值$z \in \mathbb{R}^{L\times L\times 64}$。

(d) 交叉熵：使用从真实值结构计算得出的标签 y 和逻辑值 z 计算交叉熵，然后对所有 $L \times L$ 值进行平均。

5. 反向折叠损失：将解码器的最终层表示通过回归头传递以生成逻辑值 z。使用真实值残基作为标签 y，计算用于从最终层表示预测残基的分类任务的交叉熵。

A.1.7.3.2. 第二阶段。

在 VQ-VAE 训练的第二阶段，编码器和代码本被冻结，并训练一个新的、更深的解码器。这一训练阶段的目的有多个。首先，更大的解码器提高了重建质量。其次，增加了来自 ESM3 的增强结构标记，以便能够学习 pAE 和 pLDDT 头部。第三，我们增加序列条件，并使用全原子几何损失进行训练，以能够解码全原子蛋白质结构。第四，我们扩展了解码器的上下文长度，以便能够解码多聚体和更大的单链蛋白质。

第二阶段的训练数据包括 AFDB 和 ESMAtlas 中的预测结构，以及单链和多聚体，

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and antibody-antigen complexes from the PDB. Sequence conditioning was added to the decoder via learned embeddings which are summed with structure token embeddings at the input to the decoder stack.

The structure token decoder was trained in three stages: 2A, 2B, 2C detailed in Table S2. The purpose of stage 2A is to efficiently learn decoding of all-atom structures. Enhanced training efficiency is achieved by keeping a short context length and omitting the pAE and pLDDT losses, which are both memory-consuming and can be in competition with strong reconstruction quality. In stage 2B, we add the pAE and pLDDT losses. These structure confidence heads cannot be calibrated unless structure tokens are augmented such that ESM3-predicted structure tokens are within the training distribution. To this end, for stages 2B and 2C we replace ground truth structure tokens with ESM3-predicted structure tokens 50% of the time. In stage 2C, we extend context length to 2048 and upsample experimental structures relative to predicted structures.

1. **All-atom Distance Loss:** We generalize the Backbone Distance Loss to all atoms by computing a pairwise L_2 distance matrix for all 14 atoms in the atom14 representation of each residue. This results in $D_{\text{pred}}, D \in \mathbb{R}^{14L\times 14L}$. The rest of the computation follows as before: $(D_{\text{pred}} - D)^2$, clamping to $(5\ \text{Å})^2$, and taking the mean, while masking invalid pairs (where any atom14 representations are "empty").

2. **All-atom Direction Loss:** We extend the Backbone Direction Loss to all heavy atoms:
   (a) Identify all covalent bonds between heavy atoms within each residue.
   (b) For each atom with at least two covalent bonds, compute a normal (z-axis) vector to the plane spanned by its first two covalent bonds. This results in a set of z-axis and bond vectors.
   (c) Compute all-to-all pairwise dot products between z-axis and bond vectors, forming $D_{\text{pred}}, D \in \mathbb{R}^{n\times n}$. Compute $(D_{\text{pred}} - D)^2$, clamp the max to 20, and take the mean.

3. **pLDDT Head:** Uses a Regression Head with 50 output classes (each capturing 0.02 units from 0 to 100). Predicted structures are compared to ground truth to calculate per-residue pLDDT values, which are supervised with cross-entropy loss.

4. **pAE Head:** Use a Pairwise Projection Head to produce 64 logits per residue pair $\in \mathbb{R}^{L\times L\times d}$, converting to probabilities p via softmax. Each probability corresponds to a bin representing 0.5 Å of positional error, with centers $[0.25, 0.75, \ldots, 31.25, 31.75]$.

   Computing Loss:
   (a) Compute the pairwise distances between residues in both the predicted and ground truth structures, resulting in distance matrices D_{pred} and $D \in \mathbb{R}^{L\times L}$.
   (b) Calculate the differences $(D_{\text{pred}} - D)$.
   (c) Bin these differences into 64 bins, generating classification targets for the logits.
   (d) Compute the loss using cross-entropy between these targets and the logits.

   Computing pAE: Multiply probabilities by bin centers and sum to obtain the expected positional error per residue pair, with values $\in [0.25, 31.75]$.

   Computing pTM: Additionally, the pairwise logits are used to compute the pTM (Predicted Template Modeling) score, as follows:
   (a) Compute f_d for sequence length L as:

$$d_0 = 1.24 \cdot (\max(L, 19) - 15)^{\frac{1}{3}} - 1.8$$
$$f_d = \frac{1}{1 + \left(\frac{\text{bins}}{d_0}\right)^2}$$

   (b) Compute pTM using previously computed probabilities p:

$$\text{pTM} = \max_i \left[\frac{1}{L}\sum_j \left(\sum_{\text{bin}} [p]_{i,j,\text{bin}} [f_d]_{\text{bin}}\right)\right]$$

A.1.7.4. Evaluation

We evaluate the reconstruction quality of the structure tokenizer after stage 1 and stage 2 of training using a set of CAMEO, CASP14, and CASP15 proteins taken after the training cutoff date (Fig. S3). Both decoders consistently reach backbone RMSD < 1Å, LDDT-CA > 0.98. The retraining of the structure token decoder results in substantial improvements in reconstruction quality across all test sets. The stage 2 decoder, trained with an all-atom reconstruction loss and a sequence input, achieves strong all-atom reconstruction as well (Fig. S3C). We also visualize a random sample of backbone reconstructions on the CAMEO test set (Fig. S4A). Looking at the proteins with worse reconstruction quality, we find that long regions with few tertiary contacts, disordered regions, and unresolved coordinates can lead to inaccurate global orientation of structural elements, while local structure reconstruction remains largely error-free (Fig. S4B). This behavior can be explained by the fact that the tokenizer relies on tertiary contacts to resolve the global orientation of a residue.

We also investigate the vocabulary learned by the structure tokenizer by visualizing the local neighborhoods which map

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以及来自 PDB 的抗体-抗原复合物。序列条件通过学习的嵌入添加到解码器，这些嵌入与传递到解码器堆栈的结构标记嵌入相加。

结构标记解码器的训练分为三个阶段：2A、2B、2C，具体细节见表S2。阶段2A的目的是高效学习全原子结构的解码。通过保持短上下文长度并省略pAE和pLDDT损失（这两者都占用大量内存且可能与强重建质量存在竞争），从而实现训练效率的提升。在阶段2B中，我们添加了pAE和pLDDT损失。这些结构置信头只有在结构标记得到增强，使得ESM3预测的结构标记位于训练分布内时，才能进行校准。为此，在阶段2B和2C中，我们将真实值结构标记以50%的时间替换为ESM3预测的结构标记。在阶段2C中，我们将上下文长度扩展至2048，并相对于预测结构上采样实验结构。

1. 全原子距离损失：我们通过计算每个残基的atom14表示法中14个原子之间的成对距离矩阵，向所有原子推广骨架距离损失。这导致了D_{pred}，$D \in \mathbb{R}^{14L\times 14L}$。计算的其余部分与之前相同：$(D_{\text{pred}} - D)^2$，将其限制在$(5\ ^{\circ}\text{A})^2$，并取平均，同时对无效对进行掩盖（其中任何atom14表示法都是“空的”）。

2. 全原子方向损失：我们将骨架方向损失扩展到所有重原子：（a）识别每个残基中重原子之间的所有共价键。（b）对于每个至少有两个共价键的原子，计算一个与其前两个共价键所张成的平面的法向量（z轴）。这导致形成一组z轴和键向量。（c）计算z轴和键向量之间的全对成对点积，形成 $D_{\text{pred}}, D \in \mathbb{R}^{n\times n}$。计算$(D_{\text{pred}} - D)^2$，将最大值限制为20，并取平均。

3. pLDDT 头部：使用具有50个输出类别的回归头部（每个类别捕获0到100之间的0.02单位）。将预测的结构与真实值进行比较，以计算每个残基的pLDDT值，并通过交叉熵损失进行监督。

4. pAE 头部：使用成对投影头产生每对残基 64 个逻辑值 $\in \mathbb{R}^{L\times L\times d}$，通过 Softmax 转换为概率 p。每个概率对应于一个代表 0.5 $^{\circ}$A 的位置误差的区间，其中心为 $[0.25, 0.75, \ldots, 31.25, 31.75]$。计算损失：
   (a) 计算预测结构与真实值结构之间的残基成对距离，从而得到距离矩阵 D_{pred} 和 $D \in \mathbb{R}^{L\times L}$。
   (b) 计算差异（$D_{\text{pred}} - D$）。
   (c) 将这些差异分入64个区间，生成逻辑值的分类目标。
   (d) 使用这些目标和逻辑值之间的交叉熵计算损失。

   计算pAE：将概率乘以区间中心并求和，以获得每对残基的预期位置误差，值为 $\in [0.25, 31.75]$。

   计算pTM：此外，成对逻辑值用于计算pTM（预测模板建模）得分，如下所示：
   (a) 计算 f_d 对于序列长度 L 如下：

$$d_0 = 1.24 \cdot (\max(L, 19) - 15)^{\frac{1}{3}} - 1.8$$
$$f_d = \frac{1}{1 + \left(\frac{\text{bins}}{d_0}\right)^2}$$

   (b) 使用先前计算的概率 p计算pTM：

$$\text{pTM} = \max_i \left[\frac{1}{L}\sum_j \left(\sum_{\text{bin}} [p]_{i,j,\text{bin}} [f_d]_{\text{bin}}\right)\right]$$

A.1.7.4. 评估

我们在训练的第1阶段和第2阶段后，使用一组CAMEO、CASP14和CASP15蛋白质评估结构标记器的重建质量，这些蛋白质是在训练截止日期之后获得的（图。S3）。两个解码器始终达到骨架RMSD $< 1\ ^{\circ}$A，LDDT-CA > 0.98。结构标记解码器的重新训练在所有测试集的重建质量上带来了显著的改善。第2阶段解码器以全原子重建损失和序列输入进行训练，亦实现了强大的全原子重建（图。S3C）。我们还在CAMEO测试集上可视化了一组随机样本的骨架重建（图。S4A）。观察到重建质量较差的蛋白质时，我们发现长的区域缺乏三级接触、无序区域以及未解决的坐标可能导致结构元素的全局取向不准确，而局部结构重建则保持大量无误（图。S4B）。这种行为可以通过标记器依赖于三级接触来解决残基的全局取向来解释。

我们还通过可视化映射局部邻域来调查结构标记器所学习的词汇表，

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Stage	Steps	All-atom geometric losses	pAE and pLDDT losses	Augmentation with ESM3-predicted tokens	Context length	Data mixture
2A	90k	✓	✗	✗	512	Roughly uniform sampling of predicted and experimental structures
2B	20k	✓	✓	✓	512	Roughly uniform sampling of predicted and experimental structures
2C	30k	✓	✓	✓	2048	Upsampling experimental structures

Table S2. Training details for stage 2 training of an all-atom structure token decoder.

to the same learned structure token. We find that many structure tokens encode semantically coherent sets of local neighborhoods (Fig. S5A). However, a number of tokens appear to represent multiple local neighborhoods (Fig. S5B). While the meaning of a single token may be ambiguous, the high-fidelity reconstruction quality from the decoder suggests that it is able to disambiguate given surrounding context in the full set of structure tokens.

Fig. S6 indicates that pLDDT and pTM have good predictive power. We assess the calibration of the structure confidence heads on the CAMEO test set using structure tokens predicted by ESM3 7B. Most predictions for pLDDT lie along the diagonal, though there is a small bias towards more confident predictions. As pTM is a pessimistic estimator of the TM-score, we find that pTM is biased downwards. Anecdotally, we also find that pLDDT can be poorly calibrated for some generated sequences, particularly in alpha helical regions where it can be an overestimate.

A.1.8. Function Tokenization

ESM3 processes annotations of functional characteristics of proteins through two tracks: function tokens, and residue annotations. Both support input conditioning and output heads for generation. Appendix A.1.5.1 outlines how tokens are processed into the network. We further describe the creation of the tokens in this section.

A.1.8.1. Function Tokens

Function tokens are a dense semantic representation of functional characteristics of proteins derived from free-text descriptions of the InterPro and Gene Ontology (GO) terms at each residue. At training time, function tokens are produced from each protein's InterPro annotations by a multi-step process illustrated in Fig. S7. At a high level:

1. For each residue, we gather free-text for each InterPro annotation via annotation term names, associated GO terms per annotation (via InterPro2GO mapping), and all ancestor GO terms. We parse the free-text into counts from a vocabulary of 68,103 keywords. The vocabulary is composed of unigram and bigrams extracted from the free-text of all valid InterPro annotations (and their associated GO/ancestor GO terms) in our training datasets.

2. The keywords are converted to a sparse TF-IDF vector per InterPro annotation. During training, we also produce a corrupted version by dropping keywords at the protein level (i.e. the same keywords have their counts set to 0 across all residues) at a 15% probability per keyword.

3. To create a vector per residue from the per annotation vectors, we max pool the TF-IDF vectors for the annotations per residue. During training, we apply further corruption by dropping annotations at the protein level (i.e. the same annotations are removed from the max pool across all residues) at a 15% probability per annotation.

4. We then quantize each residue's vector (a highly sparse vector with float entries) into a discrete representation suitable for input to the language model as tokens by applying a fixed series of 8 locality sensitive hashes (LSH), each with 8 hyperplanes.

The result is a sequence of 8 tokens each ranging in value from 0 to 255 per-residue. We reserve a special token `<none>` to represent positions with an empty set of InterPro annotations. For proteins that lack any functional annotations, the tokens are filled with the `<pad>` token which has an embedding value fixed to all zeros. At test time, we can produce per residue vectors using the process described, or directly creating a TF-IDF vector with keywords.

During pretraining we use the corrupted versions of the function tokens at input, predicting the un-corrupted version function tokens at positions which have been masked. 90% of the time, the entire input is replaced with `<mask>` . The other 10% of the time, we replace all 8 tokens of selected

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阶段	步骤	全原子几何损失	pAE 和 pLDDT 损失	增强与ESM3预测的标记	上下文长度	数据混合
2A	90k				512	预测和实验结构的近似均匀采样
2B	20k				512	预测和实验结构的近似均匀采样
2C	30k				2048	上采样实验结构

表 S2. 全原子结构标记解码器第二阶段训练的训练细节。

以相同的结构标记为学习结构标记。我们发现许多结构标记编码了语义上连贯的局部邻域集合（图。S5A）。然而，许多标记似乎代表多个局部邻域（图。S5B）。尽管单个标记的含义可能模糊，但解码器所展现的高保真重建质量表明，它能够在完整的结构标记集合中根据周围上下文进行消歧。

图 S6 表明 pLDDT 和 pTM 具有良好的预测能力。我们使用 ESM3 7B 预测的结构标记评估 CAMEO 测试集上结构置信头的校准情况。大多数 pLDDT 的预测沿对角线分布，尽管存在对更自信预测的轻微偏差。由于 pTM 是 TM-score 的悲观估计器，我们发现 pTM 存在向下偏差。根据轶事，我们还发现 pLDDT 在一些生成的序列中可能校准不佳，特别是在 α 螺旋区域，此时它可能被高估。

A.1.8. 函数标记化

ESM3 通过两个轨道处理蛋白质的功能特征注释：功能令牌和残基注释。两者支持输入条件和生成的输出头。附录 A.1.5.1 概述了标记如何被处理到网络中。我们在本节进一步描述了标记的创建过程。

A.1.8.1. 功能令牌

功能令牌是从每个残基的InterPro和基因本体（GO）术语的自由文本描述中提取的蛋白质功能特征的密集语义表示。在训练时，通过一个多步骤过程生成功能令牌，该过程展示于图。S7. 从高层次看：

1. 对于每个残基，我们通过注释术语名称收集每个 InterPro 注释的自由文本，通过 InterPro2GO 映射关联的 GO 术语和所有祖先 GO 术语。我们解析自由文本为来自 68,103 个关键词的词汇表统计数量。该词汇表由从我们训练数据集中所有有效 InterPro 注释（及其关联的 GO/祖先 GO 术语）的自由文本中提取的单元词和二元词组组成。

2. 关键词被转换为每个InterPro注释的稀疏 TF-IDF向量。在训练期间，我们还通过在蛋白质级别丢弃关键词（即相同的关键词在所有残基上的计数被设置为0）以15%的概率生成一个损坏的版本。

3. 为了从每个注释向量创建每个残基的向量，我们对每个残基的注释的TF-IDF向量进行最大池化。在训练期间，我们通过在蛋白质级别丢弃注释（即相同的注释从所有残基的最大池中移除）以15%的概率进一步引入损坏。

4. 然后我们将每个残基的向量（一个高度稀疏的浮点数向量）量化为适合作为标记输入到语言模型的离散表示，通过应用一系列固定的8个局部敏感哈希（LSH），每个哈希具有8个超平面。结果是一系列包含8个标记的序列，每个标记的值范围从0到255，针对每个残基。我们保留一个特殊的标记`<none>` 以表示没有InterPro注释的空集合位置。对于缺乏任何功能注释的蛋白质，这些标记使用 `<pad>` 标记填充，其嵌入值固定为全零。在测试时，我们可以使用所描述的过程生成每个残基的向量，或直接创建一个带有关键词的TF-IDF向量。在预训练期间，我们使用功能令牌的损坏版本作为输入，预测在被屏蔽位置的未损坏版本功能令牌。90%的时间，整个输入被替换为 `<mask>` 。另外10%的时间，我们替换所选的所有8个标记。

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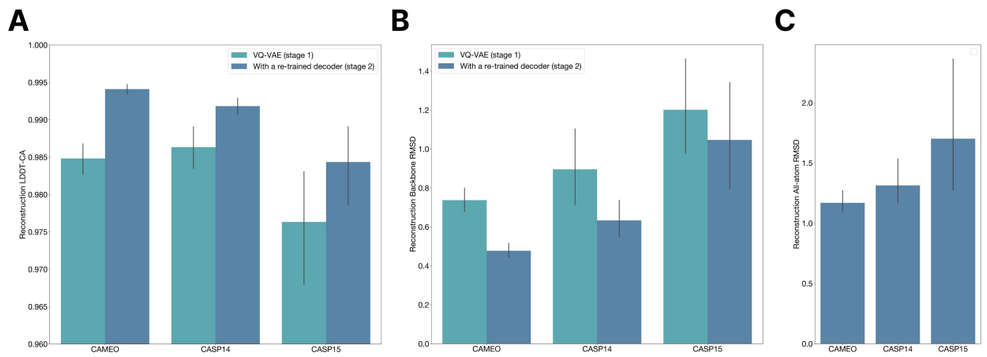

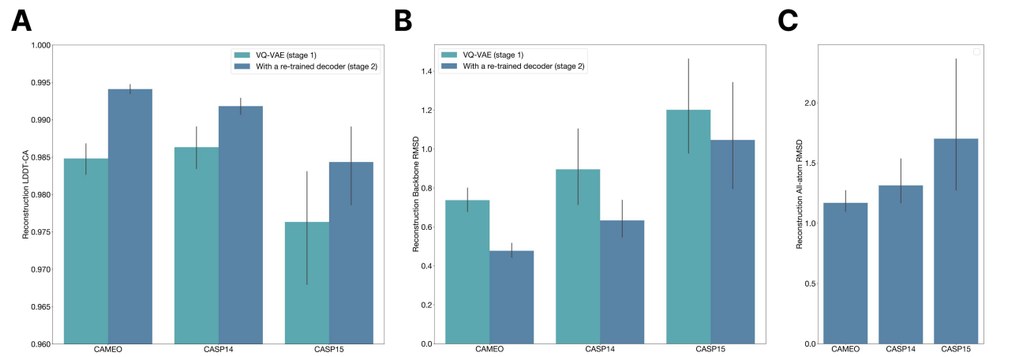

Figure S3. Structure tokenizer reconstruction quality. Reconstruction quality of the structure tokenizer after stage 1 and stage 2 of VQ-VAE decoder training evaluated on temporally held out CAMEO, CASP14, and CASP15. **(A)** Reconstruction LDDT-CA. **(B)** Reconstruction backbone RMSD. **(C)** All-atom reconstruction RMSD from the stage 2 decoder which additionally receives sequence input.

图 S3. 结构标记器重建质量。VQ-VAE 解码器训练的第一阶段和第二阶段后，结构标记器的重建质量评估在时间上保留的 CAMEO、CASP14 和 CASP15 上。(A) 重建 LDDT-CA。(B) 重建骨架 RMSD。(C) 从第二阶段解码器获得的全原子重建 RMSD，该解码器还接受序列输入。

residue with a <mask>, with the per-residue selection probability sampled from a cosine masking schedule per protein. The model has an output head which predicts each of the 8 function tokens in positions with <mask> as input, and is trained with a categorical cross entropy loss.

Function tokenization offers several key advantages as compared to simpler approaches (e.g., using a dedicated InterPro tag vocabulary). Encoding functional annotations with a generic functional keyword vocabulary enables flexible prompting of the model at test time, by combinations of keywords that were not encountered during training time. This enhances the programmability of ESM3 in designing novel proteins with not previously observed functional characteristics.

Function tokenization can also be viewed through the lens of data compression. This choice of representation reduces the input/output space from all possible InterPro combinations which would naively be represented by 35k bits, to a reduced space of 8 tokens x 8 bits / token = 64 bits. This also affords significant memory saving during pretraining by eliminating the need to perform multi-class multi-label binary classification.

A.1.8.2. Function Prediction

ESM3 is trained to predict all 8 function tokens, each spanning 256 possible values. To extract interpretable predictions of protein function from ESM3, we decode the predicted function tokens into function keywords using a seperately trained function token decoder.

A.1.8.2.1. Function Token Decoder

We train a 3-layer transformer model to learn the inverse map of the function tokenization process. The model takes as input the 8 function tokens representing the locality sensitive hash of function keywords. It outputs for each residue the binary-classification predictions predicting the presence of each function keyword, as well as predicting InterPro annotations from which the keywords originate. We find that unpacking the 8-bit LSH tokens into single-bit tokens improves training dynamics of the function token decoder. We train the function token decoder offline using combinations of InterPro tags from the UniRef annotated proteins. Since the function token vocabulary is fixed, the decoder is applied identically across different ESM3 model sizes.

A.1.8.2.2. Evaluation

To evaluate ESM3's performance in predicting protein function, we compute Average Precision, a standard measure of information retrieval, using a validation set of proteins from UniRef and their associated InterProScan function annotations. We present results in Fig. S8.

A.1.8.3. Residue Annotations Track

Residue annotations label a protein's sites of functional residues with a vocabulary of 1474 multi-hot labels emitted by InterProScan. To gather this data, we run InterProScan with databases (SFLD, CDD, PIR) on all cluster members in our UniRef and Mgnify datasets (seq-id 90 clustered). We take all unique residue annotation descriptions that occur in more than 1k proteins across all of UniRef90 and MGnify90, and deduplicate labels by punctuation and case insensitivity.

残基与一个 <掩码>，每个残基的选择概率是从每个蛋白质的余弦掩码调度中采样的。该模型具有一个输出头，它预测每个功能令牌在具有 <掩码> 的位置信息，并使用分类交叉熵损失进行训练。

与较简单的方法（例如，使用专门的 InterPro 标签词汇表）相比，函数标记化提供了几个关键优势。使用通用功能关键词词汇编码功能注释，使得在测试时通过未在训练期间遇到的关键词组合灵活地提示模型成为可能。这增强了 ESM3 的可编程性，以设计具有先前未观察到的功能特征的新型蛋白质。

函数标记化也可以从数据压缩的角度来看待。这种表示选择将输入/输出空间从所有可能的 InterPro 组合的 35k 比特，缩减到一个 8 个标记 x 8 比特/标记 = 64 比特的减小空间。这还通过消除执行多类多标签二元分类的需要，在预训练期间显著节省内存。

A.1.8.2. 功能预测

ESM3经过训练能够预测所有8个功能令牌，每个功能令牌覆盖256种可能值。为了从ESM3提取可解释的蛋白质功能预测，我们使用单独训练的功能令牌解码器将预测的功能令牌解码为功能关键词。

A.1.8.2.1. 功能令牌解码器

我们训练了一个3层变换器模型，以学习功能标记化过程的反向映射。该模型的输入为8个功能令牌，代表功能关键词的局部敏感哈希。它为每个残基输出二元分类预测，预测每个功能关键词的存在，以及预测关键词来源的InterPro注释。我们发现，将8位LSH令牌展开为单个位令牌改善了功能令牌解码器的训练动态。我们使用从UniRef注释蛋白质中组合的InterPro标签离线训练功能令牌解码器。由于功能令牌词汇固定，解码器在不同ESM3模型大小上应用一致。

A.1.8.2.2. 评估

为了评估ESM3在预测蛋白质功能方面的性能，我们计算平均精度，这是一种标准的信息检索衡量方法，使用来自UniRef的蛋白质及其相关的InterProScan功能注释的验证集。我们在图中呈现结果。S8.

A.1.8.3. 残基注释跟踪

残基注释标记了蛋白质的功能残基位点，使用由InterProScan发出的1474个多热标记的词汇表。为了收集这些数据，我们在我们的UniRef和MGnify数据集中的所有集群成员（序列 ID 90 集群）上运行InterProScan，数据库包括（SFLD，CDD，PIR）。我们提取所有在UniRef90和MGnify90的超过1000个蛋白质中出现的独特残基注释描述，并通过标点符号和不区分大小写来消除重复标签。

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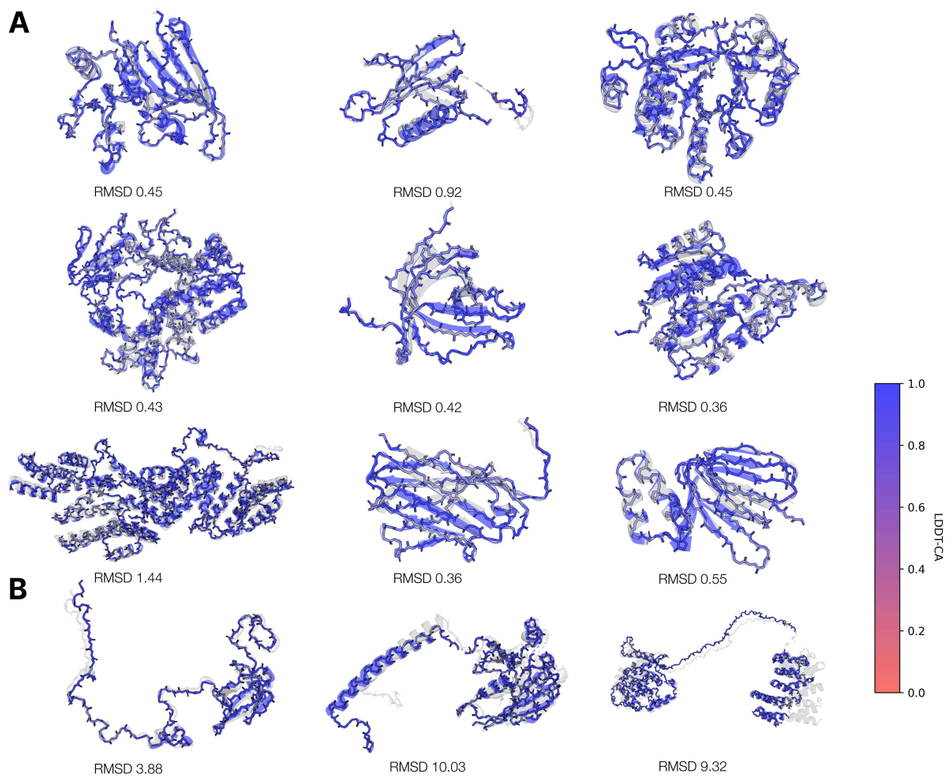

Figure S4. Visualization of structure tokenizer backbone reconstructions. **(A)** A random sample of reconstructions from the structure tokenizer on the test set. The vast majority of structures have near perfect backbone reconstruction. **(B)** A selection of the worst reconstructions in the test set. Long stretches of disordered regions, a lack of tertiary contacts, and unresolved coordinates can lead to inaccurate global orientation of structural elements, while local structure reconstruction remains largely error-free.

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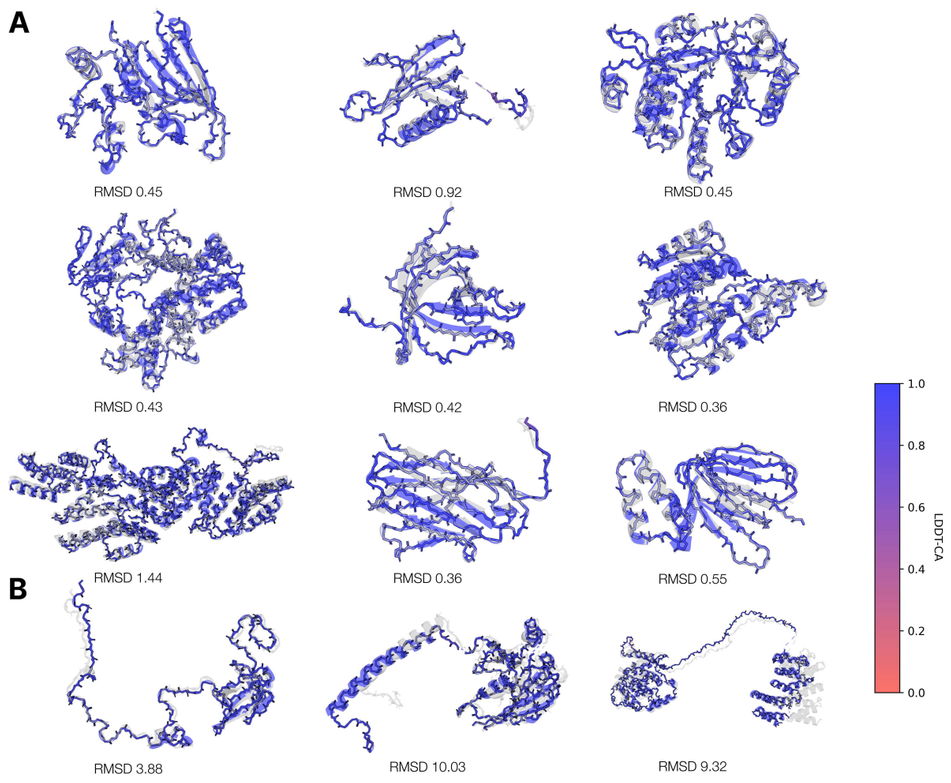

图 S4. 结构标记器主干重建的可视化。 (A) 结构标记器在测试集上重建的随机样本。绝大多数结构的主干重建几乎完美。 (B) 测试集中最差重建的选择。长段无序区域、缺乏三级接触和未解决坐标可能导致结构元素的全球方向不准确，而局部结构重建在很大程度上保持无误。

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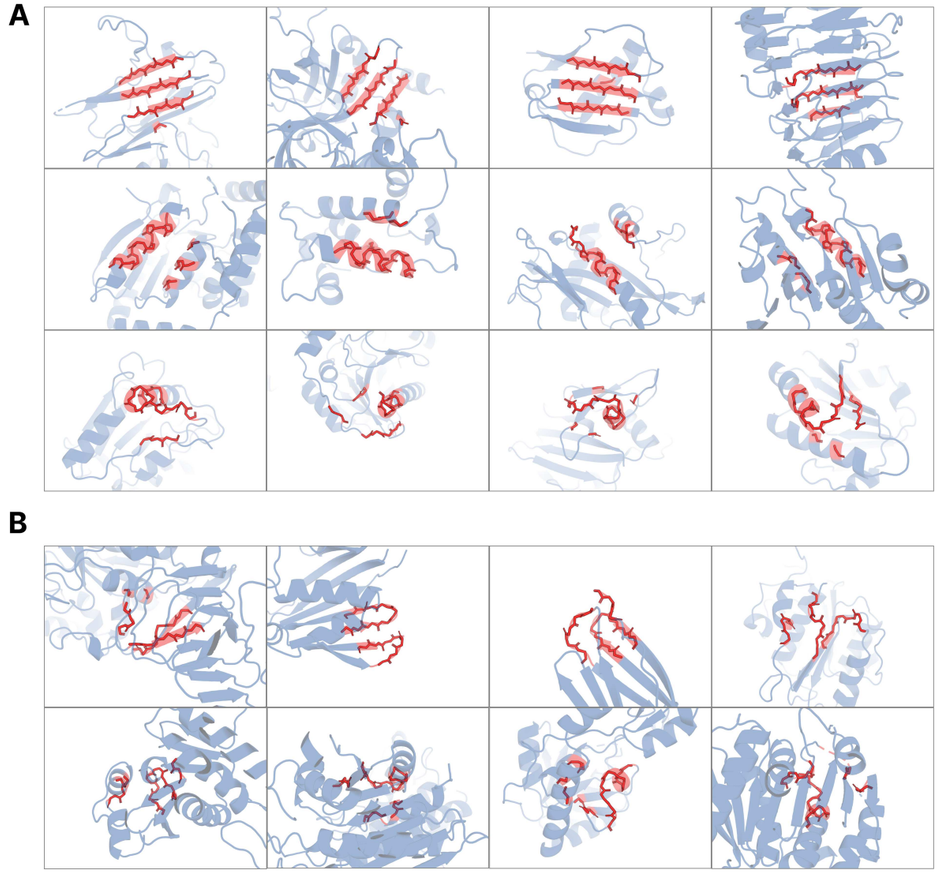

Figure S5. Visualization of local neighborhoods which map to the same learned structure token. The VQ-VAE encoder reasons over local structure neighborhoods (highlighted in red) which include the query residue and the 15 nearest neighbors in structure space. **(A)** Rows correspond to token indices 585, 59, and 3692 for top, middle, and bottom, respectively. Columns show different local structures mapping to the same token. **(B)** Some tokens represent multiple types of local neighborhoods. All local neighborhoods in B map to the same codebook index 3276. While the meaning of a single token may be ambiguous, the decoder is able to disambiguate given surrounding context in the full sequence of structure tokens.

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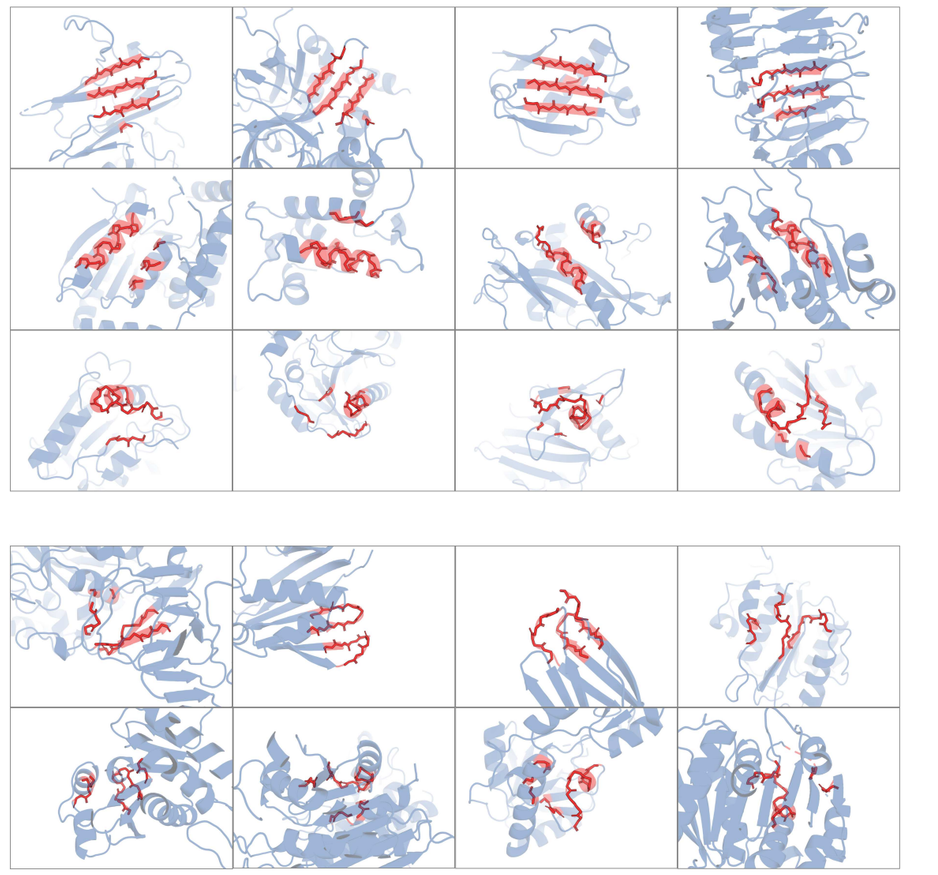

图 S5. 可视化映射到相同学习结构标记的局部邻域。VQ-VAE 编码器对局部结构邻域进行推理（用红色突出显示），其中包括查询残基及其在结构空间中的 15 个最近邻居。(A) 行对应于标记索引 585、59 和 3692，分别代表顶部、中间和底部。列显示映射到相同标记的不同局部结构。(B) 一些标记代表多种类型的局部邻域。B 中的所有局部邻域都映射到相同的代码本索引 3276。尽管单个标记的含义可能模糊不清，但解码器能够在总体结构标记序列的上下文中消除歧义。

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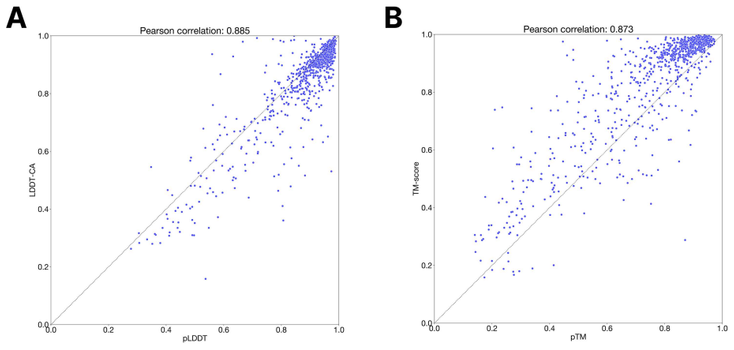

Figure S6. pTM and pLDDT calibration. Calibration of the structure token decoder pLDDT and pTM (using ESM3 7B as the structure token prediction model) on the CAMEO test set.

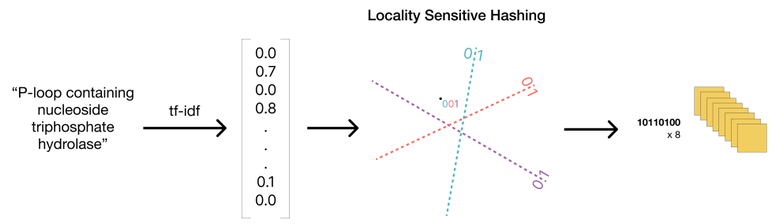

Figure S7. Schematic of function tokenization. The set of InterPro and GO descriptions of protein function are vectorized by a TF-IDF model and then hashed by a locality sensitive hash to produce 8 tokens each containing 8 bits.

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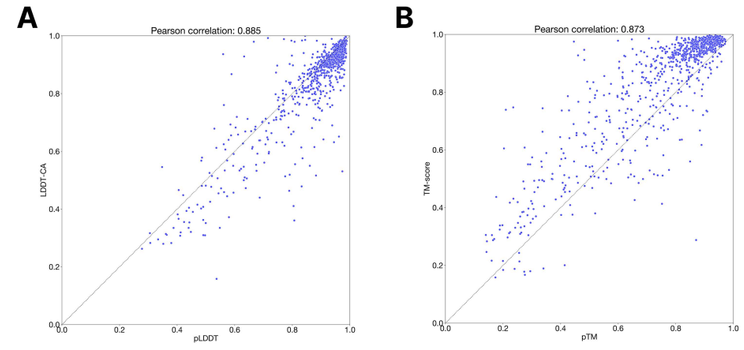

图 S6. pTM 和 pLDDT 校准。结构标记解码器 pLDDT 和 pTM（使用 ESM3 7B 作为结构标记预测模型）在 CAMEO 测试集上的校准。

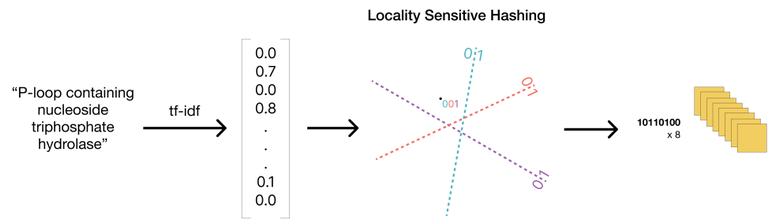

图 S7. 函数标记化示意图。蛋白质功能的 InterPro 和 GO 描述集合通过 TF-IDF 模型向量化，然后通过局部敏感哈希进行哈希处理，从而产生 8 个每个包含 8 位的标记。

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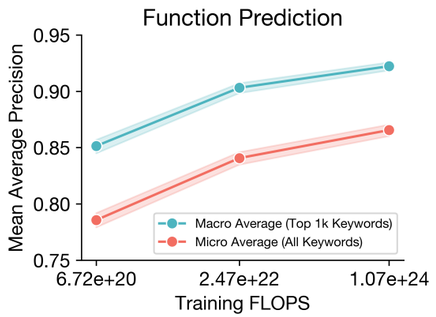

Figure S8. Function prediction benchmarking results. Mean Average Precision (mAP) for function keyword prediction. Predictions and labels are compared on a per-position basis to evaluate the model's ability to localize site-specific functional attributes by keywords such as "active site". We report mAP for the full keyword set (red) with a "micro" average because many keywords have few or no labels in the validation set. To report a "macro" average mAP we compute mAP for each of the top 1,000 most prevalent keywords in our evaluation set (discarding uninformative keywords such as "the") and report a uniform average (blue). 95% confidence intervals are shown by shading.

We join these annotations into our UniRef, MGnify, AFDB, and ESMAtlas datasets for training.

As introduced in Appendix A.1.5.1, ESM3 has a track dedicated to processing residue annotations that supports input conditioning, and an output head for prediction and generation. The residue annotation labels for a protein are tokenized into a sequence of token-sets in length equal to the protein. At each position there is an unordered set of tokens representing the residue annotations present at that position. The tokens are input to ESM3 first through an embedding lookup followed by a sum over embeddings. The permutation invariance of the sum retains that the labels are represented to an unordered set as a model. The per-position embedding sums are then added onto the per-position sequence embedding before input into the first transformer block. Positions with no residue annotations are represented by a `<pad>` token which has an embedding fixed to zeros.

The residue annotations track has an output head which outputs a set of binary classification logits predicting for each position the presence or absence of each residue annotation in the vocabulary. We apply a masking procedure to partially/fully mask residue annotation labels, and train the output head with a binary cross-entropy loss function to reconstruct the full residue annotation. In pretraining, with 90% probability all residue annotations are masked, and otherwise we independently sample positions to mask with a square root schedule.

A.1.9. Other Tracks

A.1.9.1. Confidence Tracks

As mentioned in Appendix A.1.5.1, ESM3 has two additional tracks that are only used during pretraining, and only used as input (we do not have output heads predicting these values). The first is a per-residue pLDDT: for ground truth PDB structures, these values are all 1; for AlphaFold-DB/ESMFold structures, we use the provided pLDDT. We also provide an averaged pLDDT across all the residues when structure is provided (1 otherwise), with the average calculated before any tokens are masked.

This information allows the model to distinguish between gold-standard structures and computationally predicted ones; at inference time, we set these to 1 throughout, with the goal of producing structures better than the computational predictions used to pretrain the model. The embedding itself is straightforward, with the pLDDT values first having a radial basis function, followed by a Linear layer applied to them:

Algorithm 12 `rbf`

Input: $x \in \mathbb{R}^{\cdots \times L}$, $a \in \mathbb{R}$, $b \in \mathbb{R}$, $n \in \mathbb{Z}^+$

$\Delta = \frac{b-a}{n-1}$ ▷$\mathbb{R}$
$c = [a, a+\Delta, a+2\Delta, \ldots, a+(n-2)\Delta, b]$ ▷$\mathbb{R}^n$
$\sigma = \frac{b-a}{n}$ ▷$\mathbb{R}$
$[z]_{\ldots,i,j} = \frac{1}{\sigma}([x]_{\ldots,i} - [c]_j)$ ▷$\mathbb{R}^{\cdots \times L \times n}$
return $\exp(-z^2)$ ▷$\mathbb{R}^{\cdots \times L \times n}$

Algorithm 13 `plddt_embed`

Input: $x_{\text{plddt}} \in [0,1]^L$, $x_{\text{avgplddt}} \in [0,1]$

$\text{rbf}_{\text{plddt}} = \texttt{rbf}(x_{\text{plddt}}, 0.0, 1.0, 16)$ ▷$\mathbb{R}^{L \times 16}$
$\text{rbf}_{\text{avgplddt}} = \texttt{rbf}(x_{\text{avgplddt}}, 0.0, 1.0, 16)$ ▷$\mathbb{R}^{16}$
$z_{\text{avgplddt}} = \text{Linear}(\text{rbf}_{\text{avgplddt}})$ ▷$\mathbb{R}^d$
$z_{\text{plddt}} = \text{Linear}(\text{rbf}_{\text{plddt}})$ ▷$\mathbb{R}^{L \times d}$
$[z_{\text{plddt}}]_{i,:} = [z_{\text{plddt}}]_{i,:} + z_{\text{avgplddt}}$ ▷$\mathbb{R}^{L \times d}$
return z_{plddt}

A.1.9.2. Taxonomy Track

The final 30,000 steps in the pretraining of the 98B variant of ESM3 includes a track for processing the taxonomic and species classification of the organism from which the protein sequence originates. For each protein, the taxonomic and species classifications are concatenated to create a full taxonomic lineage. The list of terms is then tokenized using a vocabulary comprised of the top 40,000 taxonomic terms in the UniRef training dataset. At input, learned embeddings (dimension 768) for each term in the lineage are summed and projected by a learned linear projection to a

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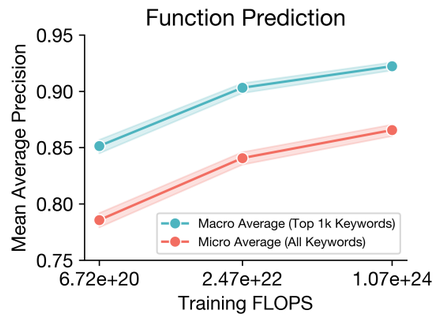

图 S8. 功能预测基准结果。功能关键词预测的平均精度（mAP）。预测与标签在每个位置进行比较，以评估模型在通过诸如“活性位点”等关键词局部化位点特定功能属性的能力。我们报告针对完整关键词集的 mAP（红色），使用“微”平均，因为许多关键词在验证集中具有少量或没有标签。为了报告“宏”平均 mAP，我们计算评估集中前 1,000 个最常见关键词的 mAP（丢弃诸如“the”等无信息关键词），并报告均匀平均（蓝色）。95% 置信区间通过阴影显示。

我们将这些注释整合到我们的 UniRef、MGnify、AFDB 和 ESMAtlas 数据集中以进行训练。

如附录中介绍的 A.1.5.1，ESM3 配有一个专门处理残基注释的轨道，支持输入条件化，以及一个用于预测和生成的输出头。蛋白质的残基注释标签被标记为与蛋白质长度相等的标记集序列。在每个位置上，有一个无序的标记集，表示该位置存在的残基注释。这些标记首先通过嵌入查找输入到 ESM3，随后进行嵌入的求和。求和的排列不变性保持了标签作为无序集合的表示。然后，逐位置的嵌入求和添加到逐位置的序列嵌入中，然后输入到第一个变换器块。没有残基注释的位置由一个 `<pad>` 标记表示，其嵌入固定为零。

残基注释轨道具有输出头，该输出头输出一组二分类逻辑值，以预测每个位置上每个残基注释在词汇表中的存在或缺失。我们应用掩蔽程序以部分/完全掩蔽残基注释标签，并使用二元交叉熵损失函数训练输出头，以重建完整的残基注释。在预训练中，以90%的概率所有残基注释被掩蔽，其余情况下我们独立地采样位置进行掩蔽，采用平方根调度。

A.1.9. 其他轨迹

A.1.9.1. 置信度轨迹

如附录中所述A.1.5.1，ESM3 具有两个额外的轨道，仅在预训练期间使用，并且只作为输入使用（我们没有预测这些值的输出头）。第一个是每个残基的 pLDDT：对于真实值 PDB 结构，这些值均为1；对于 AlphaFold-DB/ESMFold 结构，我们使用提供的 pLDDT。当提供结构时，我们还提供所有残基的平均 pLDDT （否则为1），该平均值在任何标记被掩蔽之前计算。

这些信息使模型能够区分金标准结构与计算预测结构；在推理时，我们将这些值设置为 1，目的是生成比用于预训练模型的计算预测更优的结构。嵌入本身是直接的，其中 pLDDT 值首先应用了径向基函数，然后再应用了线性层：

算法 12 径向基

输入： $x \in \mathbb{R}^{\cdots \times L}$, $a \in \mathbb{R}$, $b \in \mathbb{R}$, $n \in \mathbb{Z}^+$

$\Delta = \frac{b-a}{n-1}$ ▷$\mathbb{R}$
$c = [a, a+\Delta, a+2\Delta, \ldots, a+(n-2)\Delta, \ b]$ ▷$\mathbb{R}^n$
$\sigma = \frac{b-a}{n}$ ▷$\mathbb{R}$
$[z]_{\ldots,i,j} = \frac{1}{\sigma}([x]_{\ldots,i} - [c]_j)$ ▷$\mathbb{R}^{\cdots \times L \times n}$
返回 $\exp(-z^2)$ ▷$\mathbb{R}^{\cdots \times L \times n}$

算法 13 pLDI

输入： $x_{\text{plddt}} \in [0,1]^L$, $x_{\text{avgplddt}} \in [0,1]$

$\text{rbf}_{\text{plddt}} = \text{rbf}(x_{\text{plddt}}, 0.0, 1.0, 16)$ ▷$\mathbb{R}^{L \times 16}$
$\text{rbf}_{\text{avgplddt}} = \text{rbf}(x_{\text{avgplddt}}, 0.0, 1.0, 16)$ ▷$\mathbb{R}^{16}$
$z_{\text{avgplddt}} = \text{线性}(\text{径向基函数}_{\text{avgplddt}})$ ▷$\mathbb{R}^d$
$z_{\text{plddt}} = \text{线性}(\text{rbf}_{\text{plddt}})$ ▷$\mathbb{R}^{L \times d}$
$[z_{\text{plddt}}]_{i,:} = [z_{\text{plddt}}]_{i,:} + z_{\text{avgplddt}}$ ▷$\mathbb{R}^{L \times d}$
返回 z_{plddt}

A.1.9.2. 分类法跟踪 98B变体的ESM3在预训练中的最终30,000步包括一个用于处理生物体的分类和物种分类的轨道，该生物体是蛋白质序列的来源。对于每个蛋白质，分类和物种分类被连接以生成完整的分类谱系。然后使用包含训练数据集中前40,000个分类术语的词汇表对术语列表进行标记。在输入时，谱系中每个术语的学习嵌入（维度768）被汇总并通过学习的线性投影投影到一个

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single embedding of d_{model}. This single embedding is then repeated across the length of the sequence and summed with the embeddings of all other tracks. The linear projection is zero-initialized at the start of this stage of training to preserve model behavior, enabling continuation of pretraining with no degradation in performance.

In pretraining we apply random corruption to the taxonomic lineages and train ESM3 to reconstruct the full lineage by predicting dropped terms with a shallow MLP head trained on the final layer's representations. To corrupt the taxonomic lineage, we either drop all terms (probability 25%) or drop a set of the most specific terms of the lineage of size chosen uniformly at random from 1 to the length of the lineage (probability 25%). We also independently drop any taxonomic term with probability 10%. The output head outputs binary classification logits over the full set of 40,000 taxonomic lineage terms, and is trained to predict the missing terms via a binary-cross entropy loss.

A.1.10. ESM3 Inference

Since ESM3 is a bidirectional transformer capable of representing arbitrary factorizations of the joint probability in any order or combination of tracks, we have significant flexibility during inference. We can generate sequence, structure, or function conditioned on any combination of other tracks. We also have a choice of how much compute to apply during generation.

The usual inference strategy is to fix a prompt (which may be a combination of any of the tracks, either fully or partially specified) and choose a track for generation (which may have been partially specified). When predicting the tokens for the generation track, a number of strategies are possible. Here we discuss two notable strategies. Argmax decoding, which predicts all tokens in the generation track in a single forward pass of the model; this computation is $O(L^2)$ in the length of the protein and is extremely efficient. Iterative decoding, on the other hand, samples tokens one position at a time, conditioning subsequent predictions on those already sampled. The runtime for iterative decoding is $O(L^3)$ in the length of the protein.

Additionally, the number of decoding steps can be chosen at runtime. Argmax decoding is equivalent to decoding in one step, while iterative decoding is equivalent to decoding in L steps. It is possible to select any number of decoding steps between these two extremes to find an optimal tradeoff between computation and accuracy for a particular use case. See Appendix A.3.4 for a case study on structure prediction, in which the generation track is the structure tokens track.

When using iterative decoding, ESM3 further allows flexibility in choosing the next position to decode. We choose the position based off of the logits output of ESM3, and for the results of this paper utilize two strategies: entropy decoding, which chooses the position with the lowest entropy after softmax, or max logit decoding, which chooses the position with the maximum logit. To generate k tokens in one pass, we rank by either entropy or max logit and take the top k positions.

In the following algorithm, assume a single forward pass of ESM3 is a function f of a prompt x, and that we can access the logits of a specific token track through a subscript; e.g. sequence logits would be $f_{\text{sequence}}(x) \in \mathbb{R}^{L\times 32}$. Furthermore, denote $\pi(\cdot; z)$ as the probability distribution induced by the logits z, including an implicit softmax, and $T \in \mathbb{R}^L$ a temperature schedule.

Algorithm 14 `generate` from track

Input: $x_{\text{prompt}}, n_{\text{decode}} \in \{1..L\}, T \in \mathbb{R}^{n_{\text{decode}}}$
$k = L/n_{\text{decode}}$ ▷ # steps to decode at each step
for $s \in \{1..n_{\text{decode}}\}$ **do**
  $z_{\text{logits}} =$ `esm3_forward` (x_{prompt}) ▷ $z \in \mathbb{R}^{L\times c_{\text{track}}}$
  $\{p_1, \ldots, p_k\} = \text{ChoosePositions}(z)$
  for $i \in \{p_1, \ldots, p_k\}$ in parallel **do**
    $x_i \sim \pi(x; z_i/T_s)$ ▷Sample i with temp T_s
    $x_{\text{prompt}} = \{x_{\text{prompt}}, x_i\}$ ▷Update prompt
  end for
end for

A.2. Training ESM3

A.2.1. Pretraining Data

A.2.1.1. Sequence Databases

UniRef release 2023_02 is downloaded and parsed from the official UniRef website (77). MGnify90 version 2023_02 is downloaded and parsed from MGnify (37). All non-restricted studies available in JGI on July 31st, 2023 are downloaded and concatenated into the JGI dataset (78). OAS, which includes over a billion antibody sequences from 80 studies, is downloaded and clustered at 95% sequence identity (38).

A.2.1.2. Clustering

In all cases, data is clustered with mmseqs2 (79), with flags `--kmer-per-seq 100 --cluster-mode 2 --cov-mode 1 -c 0.8 --min-seq-id <seqid>`.

During training we re-weight the data distribution using clustering. We first sample a random cluster at the outer level, and then sample a random member of the cluster. To deduplicate cluster members, we perform a first step of clustering at a high level of sequence identity (e.g. 90%), taking one representative per cluster. We then cluster these

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单一的 d_{model} 嵌入被重复应用于序列的长度，并与所有其他轨迹的嵌入相加。在此训练阶段的开始，线性投影被初始化为零，以保持模型行为，从而使预训练的连续进行不会导致性能下降。

在预训练中，我们对分类谱系应用随机腐蚀，训练ESM3 通过预测丢失的术语来重建完整的谱系，使用在最终层表示上训练的浅层多层感知机（MLP）头部。为了腐蚀分类谱系，我们要么丢弃所有术语（概率为25%），要么随机均匀选择一个大小在1到谱系长度之间的特定术语集合（概率为25%）来丢弃。我们还独立地以10%的概率丢弃任何分类术语。输出头部为全套40,000个分类谱系术语输出二分类逻辑值，并通过二元交叉熵损失训练以预测缺失的术语。

A.1.10. ESM3 推理

由于 ESM3 是一种双向变换器，能够以任意顺序或组合表示联合概率的任意分解，因此在推理过程中我们具有显著的灵活性。我们可以基于其他轨迹的任意组合生成序列、结构或功能。我们还可以选择在生成过程中应用多少计算资源。

通常的推理策略是固定一个提示（这可以是任何轨迹的组合，完全或部分指定），并选择一个用于生成的轨迹（该轨迹可能已被部分指定）。在预测生成轨迹的标记时，有多种策略可供选择。这里我们讨论两种显著的策略。Argmax 解码，该策略在模型的单次前向传递中预测生成轨迹中的所有标记；该计算为 $O(L^2)$，以蛋白质的长度为单位，且效率极高。另一方面，迭代解码则一次采样一个位置的标记，并根据已采样的标记条件生成后续预测。迭代解码的运行时间为 $O(L^3)$，以蛋白质的长度为单位。

此外，解码步骤的数量可以在运行时选择。Argmax 解码等同于一步解码，而迭代解码则等同于在 L 步进行解码。 可以在这两种极端之间选择任何数量的解码步骤，以寻找特定用例中计算与准确性之间的最佳权衡。 请参见附录A.3.4 中的结构预测案例研究，其中生成轨迹是结构标记轨迹。

在使用迭代解码时，ESM3 进一步允许在选择下一个解码位置方面具有灵活性。我们根据 ESM3 输出的逻辑值选择该位置，并且对于

本文的结果利用了两种策略：熵解码，选择软最大后熵值最低的位置，或者最大逻辑解码，选择具有最大逻辑值的位置。为了在一次传递中生成 k 标记，我们通过熵或最大逻辑对位置进行排序，并选择前 k 个位置。

在以下算法中，假设 ESM3 的单个前向传递是提示 x 的函数 f，并且我们可以通过下标访问特定标记轨迹的逻辑值；例如，序列逻辑值将为 $f_{\text{sequence}}(x) \in \mathbb{R}^{L\times 32}$。此外，记作 $\pi((

·; z)$是由逻辑值 z 引发的概率分布，包括一个隐式的 Softmax 和 $T \in \mathbb{R}^L$温度调度。

算法 14 从轨迹

输入：$x_{\text{prompt}}, n_{\text{decode}} \in \{1..L\}, T \in \mathbb{R}^{n_{\text{decode}}}$
$k = L/n_{\text{decode}}$ ▷# 每个步骤解码的步骤
对于 $s \in \{1..n_{\text{decode}}\}$do
  $z_{\text{logits}} =$ pt) ▷ $z \in \mathbb{R}^{L\times c_{\text{track}}}$
  $\{p_1, \ldots, p_k\} =$ C选择P位置(z)
  对于 $i \in \{p_1, \ldots, p_k\}$ 并行执行 do
    $x_i \sim \pi(x; z_i/T_s)$ ▷示例 i 与 temp T_s
    $x_{\text{prompt}} = \{x_{\text{prompt}}, x_i\}$ ▷更新提示
  结束
结束

A.2. 训练 ESM3

A.2.1. 预训练数据

A.2.1.1. 序列数据库

UniRef发布2023年02月从官方UniRef网站下载并解析（77）。MGnify90版本2023年02月从MGnify下载并解析（37）。2023年7月31日，在JGI中可用的所有非限制性研究被下载并连接到JGI数据集（78）。OAS包含来自80个研究的超过十亿个抗体序列，下载并以95%的序列同一性进行聚类（38）。

A.2.1.2. C聚类

在所有情况下，数据使用mmseqs2进行聚类（79），使用标志`--kmer-per-seq100 --cluster-mode2 --cov-mode 1 -c 0.8--min-seq-id<seqid>`。在训练过程中，我们通过聚类重新权重数据分布。我们首先在外层随机抽样一个集群，然后随机抽样该集群的一个成员。为了去重集群成员，我们在高序列同一性水平（例如90%）上执行首次聚类步骤，每个集群选择一个代表。然后我们对这些进行聚类

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deduplicated sequences at a lower level of sequence identity (e.g. 70%). We limit the number of samples within a cluster to 20.

A.2.1.3. Inverse Folding

As data augmention we train a 200M parameter inverse folding model and use it to create additional training examples.

The inverse folding model uses geometric attention for structure conditioning and an output projection head for the sequence logits, similar to ESM3. Unlike ESM3, the transformer stack alternates between blocks with geometric attention and standard attention. The model is trained on the sequence and structure pairs in PDB, AlphaFold-DB, and ESMAtlas, with the single training task of (and loss computed on) predicting sequence at the output given structure at the input. Model architecture and training methodology is otherwise substantially similar to ESM3.

This model is used to generate additional sequences corresponding to each structure in the training data for ESM3 (5 sequences per structure for ESMAtlas and AlphaFold-DB, 64 sequences per structure for the PDB). When training ESM3, with 50% probability the original sequence and structure pair is presented to the model as a training example. The other 50% of the time one of the inverse folded sequences is paired with the structure as the training example seen by ESM3.

A.2.1.4. Functional Labels

Functional labels are obtained from InterPro (40) and InterProScan (80), both version 95.0. All annotations for UniProtKB were downloaded from the InterPro website via the 'protein2ipr.dat.gz' file. InterProScan was applied to the entirety of MGnify90 with flags `--goterms --iprlookup --pathways --disable-precalc`. The resultant values are taken as ground truth functional labels for model training.

A.2.1.5. Structural Data

We use all PDB chains, clustered by unique PDB ID and entity ID within the PDB structure. We filter to all structures deposited before May 1, 2020, determined by X-ray crystallography, and better than 9Å resolution (39).

AlphaFoldDB is downloaded as the v4 version specified on their website (4). We notice that structures with high pLDDT are disproportionately alpha helices. Therefore, we ensure globularity by measuring the number of long range (>12 sequence distance) contacts in the chain. If this value is < 0.5L with an L length protein, we omit it from our training set. We also filter out all proteins < 0.7 pLDDT.

ESMAtlas is downloaded as version v0 and v2023_02. Similarly we use a < 0.7 pLDDT filter. We use a 0.7 pTM cutoff as well to enforce globularity. High pTM structures tends to be more compact.

Structural data also includes any functional labels that exist for the corresponding sequence.

A.2.1.6. Solvent Accessible Surface Area and Secondary Structure

For solvent accessibility surface area, we use the Shrake-Rupley rolling probe algorithm as implemented in biotite (81). This generates a set of real numbers, or a nan value when structural coordinates are not provided. Similarly, SS8 labels are generated using the mkdssp tool (82) and taken as ground truth labels. SS8 and SASA tracks were generated for the AlphaFoldDB and ESMAtlas datasets.

A.2.1.7. Purging of Validation Sequences

We keep track of validation set performance on a set of held out sequences from each training set: UniRef, MGnify, and JGI. In order to properly hold out a sufficiently diverse set of validation proteins, we first sample 25000 proteins from each set. Then we use mmseqs easy-search to filter out proteins from this set with a 70% sequence identity threshold. We choose the set of proteins from our training set to be the "query" set, and the set of validation proteins as our "target" set for mmseqs. We use the flags `--alignment-mode 3 -c 0.8 {cov-mode 0 --max-seqs 300 --max-accept 3 --start-sens 2 -s 7 --sens-steps 3.` This query is designed such that early stopping in mmseqs will not affect if we find a hit in the "query" training set.

Train purges are run to generate a list of blacklisted UniRef, MGnify, and JGI IDs, which are removed from the training set.

A.2.1.8. Token counts

The dataset counts in Table S3 are computed after limiting the large clusters to 20. The number of tokens are computed by multiplying the number of sequences with the average length of the dataset.

In order to compute the approximate number of sequences and tokens seen during training, we first compute the number of times the dataset is repeated at the cluster level. Given the the number of repeats, we know the expected number of unique samples seen when sampling with replacement is $n\left(1-\left(1-\frac{1}{n}\right)^k\right)$ with a cluster of size n and k items selected. Computing this on the size of each cluster and number of dataset repeats results in the approximate number of tokens presented in Table S4. Our largest model is trained on all of this data, while our smaller models use a portion

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在较低的序列一致性水平（例如70%）下去重的序列。我们将一个集群内的样本数量限制为20。

A.2.1.3. 反向折叠 作为数据增强，我们训练一个具有200M参数的反向折叠模型，并利用它创建额外的训练示例。该反向折叠模型使用几何注意力进行结构条件化，并采用输出投影头来生成序列逻辑值，类似于ESM3。与ESM3不同的是，该变换器堆在几何注意力块和标准注意力块之间交替。该模型在PDB、AlphaFold-DB和ESMAtlas中的序列与结构对上进行训练，唯一的训练任务是（并计算损失）在输入结构的基础上预测输出的序列。模型的架构和训练方法在其他方面与ESM3基本相似。该模型用于生成与ESM3训练数据中每个结构相对应的额外序列（对于ESMAtlas和AlphaFold-DB，每个结构5个序列，对于PDB，每个结构64个序列）。在训练ESM3时，以50%的概率将原始序列和结构对作为训练示例呈现给模型。而在另50%的情况下，反向折叠得到的序列与结构配对作为ESM3看到的训练示例。

A.2.1.4. FunctionalLabels功能标签来自InterPro（40）和InterProScan（80），均为95.0版本。所有UniProtKB的注释均通过‘protein2ipr.dat.gz’文件从InterPro网站下载。InterProScan被应用于整个MGnify90，附带标志`--goterms --iprlookup--pathways --disable-precalc`。结果值被视为模型训练的真实值功能标签。

A.2.1.5. StructuralData我们使用所有PDB链，根据唯一的PDB ID和PDB结构内的实体ID进行聚类。我们过滤出所有在2020年5月1日之前提交的结构，通过X射线晶体学确定，并且分辨率优于 9 Å（39）。AlphaFoldDB下载为其网站上指定的v4版本（4）。我们注意到具有较高pLDDT 的结构在α-螺旋中所占比例较高。因此，我们通过测量链中长距离（>12 序列距离）接触的数量来确保球状性。如果此值小于 < 0 0.5L（L为序列长度的蛋白质），我们将其从训练集中剔除。我们还过滤掉所有<pLDDT < 0.7 小于0.7。ESMAtlas下载为v0和v2023 02版。Sim-

同样地，我们使用 < 0.7 pLDDT 过滤器。我们也使用 0.7 的 pTM 截止值以强制执行球状性。高 pTM 结构往往更加紧凑。

结构数据还包括与对应序列存在的任何功能标签。

A.2.1.6. 溶剂可接触的表面区域和第二级结构对于溶剂可接触表面面积，我们使用在 biotite 中实现的 Shrake-Rupley 滚动探针算法（81）。这生成一组真实数值，或在未提供结构坐标时生成一个纳值。同样，SS8 标签是使用 mkdssp 工具（82）生成的，并作为真实值标签。为 AlphaFoldDB 和 ESMAtlas 数据集生成了 SS8 和 SASA 轨道。

A.2.1.7. 清除验证序列 我们跟踪来自每个训练集：UniRef、MGnify 和 JGI 的一组排除序列的验证集性能。为了正确保持一个足够多样化的验证蛋白质集，我们首先从每个数据集中抽取 25000 个蛋白质。然后，我们使用 mmseqs easy-search 以 70% 的序列一致性阈值过滤掉该集合中的蛋白质。我们选择来自训练集的蛋白质集合作为“查询”集，将验证蛋白质集合视为 mmseqs 的“目标”集。我们使用标志 --alignment-mode 3 -c 0.8 {cov-mode 0 --max-seqs 300 --max-accept 3 --start-sens 2 -s 7 --sens-steps 3。该查询设计为使早期停止在 mmseqs 中不会影响我们是否能在“查询”训练集中找到匹配。运行训练清除以生成黑名单上的 UniRef、MGnify 和 JGI ID 列表，这些 ID 将从训练集中移除。

A.2.1.8. TokenCounts数据集在表S3 中计算，限制大型集群为20后进行计算。标记的数量通过将序列的数量与数据集的平均长度相乘来计算。为计算训练期间看到的序列和标记的近似数量，我们首先计算数据集在集群级别重复的次数。给定重复的次数，我们知道在替换抽样时，看到的独特样本的预期数量是 $n\left(1-\left(1-\frac{1}{n}\right)^k\right)$，集群的大小为 n，选定的 k 项目数量。对每个集群的大小和数据集重复的数量进行计算，结果在表S4中显示近似标记的数量。我们最大的模型在所有这些数据上进行训练，而我们较小的模型则使用一部分数据。

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of it depending on the model's token budget.

A.2.2. Pretraining Tasks

A.2.2.1. Noise Schedule

In the masked generative framework, corruption is applied to each input to the model. To enable generation, the amount of noise applied to an input is sampled from a distribution with probability mass on all values between 0 and 1.

We select various noise schedules for different tracks with several goals in mind. First, ESM3 should see all combinations of tracks as input and output, enabling it to generate and predict based on arbitrary inputs. Second, ESM3 should maintain a balance of strong representation learning and high quality generations. Third, the type of inputs provided should be representative of what users would like to prompt the model with.

In initial experimentation, we found that a fixed 15% noise schedule led to poor generation results, while a linear noise schedule where probability of each mask rate was constant led to good generation but poor representation learning results. We find a good trade-off between representation learning and generation by sampling the noise schedule from a mixture distribution. 80% of the time, the mask rate is sampled from a $\beta(3, 9)$ distribution with mean mask rate 25%. 20% of the time, the mask rate is sampled from a uniform distribution, resulting in an average overall mask rate of 30%.

The noise schedules applied to each input are listed in Table S6. For the structure coordinate track, we also modify the masking to be applied as span dropping 50% of the time. This ensures that the model sees contiguous regions of masked and provided coordinates, which better mimics the types of inputs users may provide.

A.2.2.2. Track Dropout

Along with applying noise to each track, we want to ensure ESM3 is able to perform well when some tracks are not provided at all (e.g. to perform structure prediction when no structure is provided as input). We enable this by wholly dropping out some tracks with varying probabilities, listed in Table S6.

A.2.2.3. Structure Noise

We apply gaussian noise with standard deviation 0.1 to all coordinates the model takes as input.

A.2.2.4. Tertiary Coordination Sampling

An interesting use case of generative protein models involves conditioning on key structural information, such as an active site, and generating the sequence and structure of a protein that contains this information. It is possible to define an tertiary coordination task as 3 residues which are mutually in contact in structure space ($C\alpha - C\alpha$ distance < 6Å), but are distant in sequence space (≥ 10 positions apart) (26). Training on this conditioning may enable the model to better perform the type of tertiary coordination required for active site sampling.

While this task will be sampled with some probability under the standard noise schedules, we also manually sample the task with 5% probability whenever a structure is available. If the task is sampled and a valid tertiary coordination triplet is found, the structure coordinates for that triplet are shown to the model. For each residue in the triplet, the adjacent residues are also independently shown with 50% probability, which leads to a total size of between 3 and 9 residues. All other structure coordinates are masked. Normal masking is applied to the other tracks.

A.2.2.5. Tertiary Interface Sampling

Predicting and generating binding interfaces is another important task for generative protein models. To help with this capability, we add computational data augmentation that simulates the binding interface task.

We define a tertiary interface as one involving a long range contact ($C\alpha - C\alpha$ distance < 8Å, ≥ 24 sequence positions). When this task is sampled (5% probability whenever a structure is present) and a long range contact is found, then the chain is split into two chains, each containing one side of the contact interface. Suppose the contacting positions are given by the indices i, j. Then the first chain will contain residues between $[\textsc{randint}(1, i-3), \textsc{randint}(i+3, j-15)]$, while the second chain will contain residues between $[\textsc{randint}(i+15, j-3), \textsc{randint}(j+15, L)]$. This ensures there is always a residue gap between the two pseudochains. A chainbreak token "—" is inserted to represent the residue gap.

A.2.2.6. Residue Gap Augmentation

To encourage the model to learn to represent residue gaps using the chainbreak token, we introduce a task which randomly splits a single chain into multiple subchains.

First, a number of chains to sample is sampled from a geometric distribution with probability 0.9, up to a maximum of 9 possible chains. If the number of chains sampled is 1, no additional transformations are applied. A minimum separation of 10 residues between chains is defined. Sequence lengths of the chains along with gaps are sampled from a dirichlet distribution to maintain identically distributed sequence lengths for each chain. This transformation is applied to all samples.

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这取决于模型的标记预算。

A.2.2. 预训练任务

A.2.2.1. 噪声调度

在掩蔽生成框架中，损坏会施加到模型的每个输入上。为了使生成能够进行，施加到输入上的噪声量是从一个概率质量分布中抽样的，该分布的值范围在0到1之间。

我们为不同的轨道选择了各种噪声调度，考虑了多个目标。首先，ESM3应接收所有轨道的组合作为输入和输出，使其能够根据任意输入生成和预测。其次，ESM3应保持强表征学习与高质量生成之间的平衡。第三，提供的输入类型应能够代表用户希望提示模型的内容。

在初步实验中，我们发现固定的15%噪声调度导致生成结果不佳，而线性噪声调度的每个掩蔽率概率保持不变则导致生成良好但表征学习结果不佳。通过从混合分布中抽样噪声调度，我们发现表征学习与生成之间的良好权衡。80%的时间，掩蔽率是从一个 $\beta(3, 9)$分布中抽样的，平均掩蔽率为25%。20%的时间，掩蔽率是从均匀分布中抽样，导致整体平均掩蔽率为30%。

应用于每个输入的噪声调度列于表格S6。对于结构坐标轨道，我们还修改了掩蔽的应用方式，每次以50%的概率进行跨度丢弃。这确保模型看到被掩蔽和提供的坐标的连续区域，这更好地模拟了用户可能提供的输入类型。

A.2.2.2. 轨道丢失

在对每条轨道施加噪声的同时，我们希望确保ESM3在某些轨道完全没有提供的情况下仍能表现良好（例如，在未提供任何结构作为输入时执行结构预测）。我们通过以不同的概率完全丢弃某些轨道来实现这一点，具体见表S6.

A.2.2.3. 结构噪声 我们对模型作为输入的所有坐标施加标准差为0.1的高斯噪声。

A.2.2.4. 三级协调采样 生成蛋白质模型的一个有趣用例涉及对关键结构信息的条件处理，例如一个活性位点，并生成包含此信息的蛋白质的序列和结构。可以将三维协调任务定义为在结构空间中互相接触的3个残基（$C\alpha - C\alpha$距离 < 6 Å），但是在序列空间中相距较远（≥ 10 位置距离）(26)。在这种条件下的训练可能使模型更好地执行活性位点取样所需的三维协调类型。

虽然该任务将在标准噪声调度下以一定概率进行采样，但我们也在结构可用时手动以5%的概率进行任务采样。如果该任务被采样且找到有效的三级配位三元组，则该三元组的结构坐标将展示给模型。对于三元组中的每个残基，相邻残基也以50%的概率独立展示，从而导致总数在3到9个残基之间。所有其他结构坐标均被屏蔽。其他轨迹应用正常屏蔽。

A.2.2.5. 三级界面采样 预测和生成结合界面是生成蛋白质模型的另一个重要任务。为了帮助实现这一能力，我们增加了模拟结合界面任务的计算数据增强。我们将三级界面定义为涉及长距离接触的界面（$C\alpha - C\alpha$ 距离 < 8 Å ≥ 24 序列位置）。当该任务被采样（在结构存在时，概率为5%）并且找到长距离接触时，链将被拆分为两个链，每个链包含接触界面的一侧。假设接触位置由索引 i、 j 给出。那么第一条链将包含残基在 [RANDINT$(1, i-3)$、RANDINT$(i+3, j-15)$], 之间，而第二条链将包含残基在 [RANDINT$(i+15,$ $j-3)$、RANDINT$(j+15, L)$] 之间。这确保了两个伪链之间始终存在残基间隙。插入一个链断符号 "—" 以表示残基间隙。

A.2.2.6. 残基间隙增强 为了鼓励模型学习使用链断符号表示残基间隙，我们引入了一项任务，该任务随机将单个链分割为多个子链。首先，从几何分布中抽样出待抽样的链的数量，概率为0.9，最多可达到9个可能的链。如果抽样的链的数量为1，则不应用额外的变换。定义链之间的最小分离为10个残基。链的序列长度及其间隙从狄利克雷分布中抽样，以维护每个链相同分布的序列长度。此变换适用于所有样本。

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Dataset	Type	Clustering Level	Expansion Level	Tokens	Release
UniRef	Sequence	70% (83M)	90% (156M)	54.6B	2023_02
MGnify	Sequence	70% (372M)	90% (621M)	105.5B	2023_02
JGI	Sequence	70% (2029M)	-	256B	All non-restricted studies available on July 30th, 2023.
OAS	Sequence	95% (1192M)	-	132B	All sequences available on July 30th, 2023.
PDB	Structure	- (203K)	-	0.054B	All chains available on RCSB prior to
PDB Clustered	Structure	70% (46K)	100% (100K)	0.027B	May, 1st, 2020
AlphaFoldDB	Structure	70% (36M)	90% (69M)	40.5B	v4
ESMAtlas	Structure	70% (87M)	90% (179M)	23.5B	v0, v2023_02

Table S3. Pretraining dataset statistics. Includes number of tokens, release, and clustering level. Numbers are derived after dataset filtering.

Dataset Name	Unique Samples (M)	Unique Tokens (M)
UniRef	133	40,177
MGnify	406	65,780
JGI	2,039	265,070
OAS	203	22,363
PDB	0.2	55
AFDB	68	20,510
ESMAtlas	168	38,674
AFDB inverse folded	111	33,300
ESMAtlas inverse folded	251	57,730
Sequence	3,143	484,441
Structure	236	177,710
Annotation	539	105,957
Total unique training tokens		768,109

Table S4. Pretraining unique token statistics. Broken down by token type and dataset type.

Dataset	Inverse Folding	Function Labels	SASA	Secondary Structure
UniRef	✓	✓	-	-
MGnify	✓	✓	-	-
JGI	✗	✗	-	-
OAS	✗	✗	-	-
PDB	✗	✗	✗	✗
AlphaFoldDB	✓	✓	✓	✓
ESMAtlas	✓	✓	✓	✓

Table S5. Data augmentation and conditioning information applied to each dataset.

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数据集	类型	聚类级别	扩展级别	标记	发布
UniRef	序列	70% (83M)	90% (156M)	54.6B	2023_02
MGnify	序列	70% (372M)	90% (621M)	105.5B	2023年02月
J GI	序列	70% (2029M)	-	256B	所有未受限制的研究均可在2023年7月30日。
OAS	序列	95% (1192M)	-	132B	所有序列均可在7月30日获得，2023。
PDB	结构	- (203K)	-	0.054B	在 RCSB 上可用的所有链，截止于
PDB聚类	结构	70% (46K)	100% (100K)	0.027B	2020年5月1日
AlphaFoldDB	结构	70% (36M)	90% (69M)	40.5B	v4
ESMAtlas	结构	70% (87M)	90% (179M)	23.5B	v0，2023年02月

表S3. 预训练数据集统计。包括标记数量、发布和聚类级别。数字是在数据集过滤后得出的。

数据集名称	唯一样本（百万）	唯一标记（百万）
UniRef	133	40,177
MGnify	406	65,780
JGI	2,039	265,070
OAS	203	22,363
PDB	0.2	55
AFDB	68	20,510
ESMAtlas	168	38,674
AFDB 逆折叠	111	33,300
ESMAtlas 逆折叠	251	57,730
序列	3,143	484,441
结构	236	177,710
注释	539	105,957
总独特训练标记		768,109

表S4. 预训练唯一标记统计数据。按标记类型和数据集类型进行分类。

数据集	反向折叠	函数标签	SASA	二级结构
UniRef	✓	✓	-	-
MGnify	✓	✓	-	-
JGI			-	-
OAS			-	-
PDB				
AlphaFoldDB	✓	✓	✓	✓
ESMAtlas	✓	✓	✓	✓

表 S5. 应用到每个数据集的数据增强和条件信息。

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Track	Noise Schedule	Dropout Prob
Sequence	betalinear30	0
Structure Tokens	cosine	0.25
Structure Coordinates	cubic	0.5
Secondary Structure (8-class)	square root	0.9
SASA	square root	0.9
Function Tokens	square root	0.9
Residue Annotations	square root	0.9

Table S6. Noise schedules and dropout probabilities.

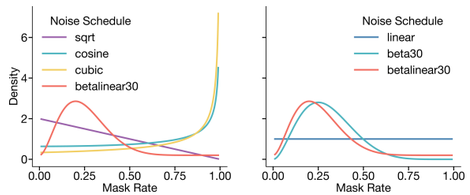

Figure S9. Visualization of noise schedules used. Left shows the probability density function of all noise schedules used. Right shows the betalinear30 distribution (which is drawn from $\beta(3, 9)$ with 80% probability and a linear distribution with 20% probability) against a beta30 distribution (defined by $\beta(3, 7)$) and a linear distribution.

A.2.2.7. Geometric Attention Masking

In the case that multiple chains are provided to the model from either the interface sampling or residue gap augmentation tasks, we mask the geometric attention layer to prevent the model from attending to cross-chain coordinates. This simulates tasks where the structure of individual chains is known, but the interface is unknown.

A.2.3. Training Details

A.2.3.1. Hyperparameters

We train all models using AdamW optimizer (83), with the following hyperparameters: $\beta_1 = 0.9$, $\beta_2 = 0.95$. We use a weight decay of 0.01 and gradient clipping of 1.0. We employ 5K to 20K warmup steps until reaching the maximum learning rate, and utilize a cosine decay scheduler to decay LR to 10% of the maximum learning rate by the end of training.

A.2.3.2. Infrastructure

Our training codebase uses Pytorch. We use Pytorch's FSDP (84) implementation for data parallelism. We also use custom components from the TransformerEngine (85) library.

We have made several optimizations to increase the training speed of our models. For multi-head attention uses, we use the memory efficient implementation from the xformers library (86). We also save activations that are expensive to compute during training when necessary. We employ mixed precision training, utilizing FP8, BF16, and FP32 as needed based on accuracy requirements and kernel availability throughout our network.

A.2.3.3. Stability

Scaling ESM3 to 98 billion parameters with its novel architecture, multi-modal inputs, and low precision computation requirements poses significant training stability challenges. Our model is significantly deeper than its NLP counterparts, and literature has shown that deeper networks are harder to train due to attention collapse (87).

We observed training instability early in the architectural innovation phase, which we addressed through several changes. We apply layer normalization to the query and key vectors within the attention mechanism (88). We observe longer warm up helps (89). Another source of instability is the masking rate in pretraining tasks. We found that a very high masking rate is more likely to cause training divergences than a lower one, especially early in the training. Choosing a masking schedule biased towards lower mask rates improved both performance and training stability. Interestingly, the introduction of conditioning from other modalities also improves training stability, perhaps suggesting that stability is related to the degree of underspecification of a task.

An incorrectly set learning rate is another source of instabil-

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轨迹	噪声调度	丢弃概率
序列	betalinear30	0
结构标记	余弦	0.25
结构坐标	立方	0.5
二级结构（8类）	平方根	0.9
SASA	平方根	0.9
功能令牌	平方根	0.9
残基注释	平方根	0.9

表 S6. 噪声调度和丢弃概率。

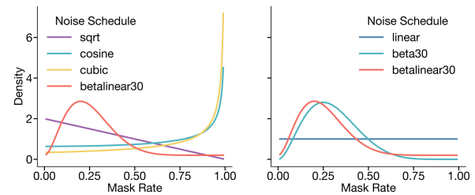

图 S9. 使用的噪声调度的可视化。左侧显示所有使用的噪声调度的概率密度函数。右侧显示betalinear30分布（其以 β(3, 9)的80%概率和线性分布的20%概率绘制）与beta30分布（由 β(3, 7)定义）及线性分布的对比。

A.2.2.7. 几何注意力掩蔽

在从接口采样或残基缺口扩增任务中向模型提供多个链的情况下，我们对几何注意力层进行掩蔽，以防止模型关注跨链坐标。这模拟了已知单个链的结构但未知接口的任务。

A.2.3. 训练细节

A.2.3.1. 超参数

我们使用 AdamW 优化器对所有模型进行训练（83），并采用以下超参数：$\beta_1 = 0.9$, $\beta_2 = 0.95$。我们使用 0.01 的权重衰减和 1.0 的梯度裁剪。我们在达到最大学习率之前，使用 5K 到 20K 的热身步数，并利用余弦衰减调度器在训练结束时将学习率衰减至最大学习率的 10%。

A.2.3.2. 基础设施

我们的训练代码库使用 Pytorch。我们使用 Pytorch 的 FSDP （84）实现数据并行性。我们还使用来自 TransformerEngine （85）库的自定义组件。

我们进行了几项优化，以提高我们模型的训练速度。对于多头注意力的使用，我们采用了来自xformers的内存高效实现。

库 (86)。我们还在必要时保存训练过程中计算成本高昂的激活值。我们采用混合精度训练，根据准确性要求和整个网络中内核的可用性，利用FP8、BF16和FP32。

A.2.3.3. 稳定性

将 ESM3 扩展到 98 十亿个参数，结合其新颖的架构、多模态输入和低精度计算要求，面临着重大的训练稳定性挑战。我们的模型明显比其 NLP 对应物更深，文献表明，较深的网络由于注意力崩溃（87）而更难训练。

我们在架构创新阶段早期观察到了训练不稳定性，我们通过几个变更来解决这一问题。我们在注意力机制中的查询和键向量上应用了层归一化（88）。我们观察到较长的热身有帮助（89）。另一个不稳定的来源是预训练任务中的遮罩率。我们发现，过高的遮罩率比较低的遮罩率更可能导致训练发散，尤其是在训练早期。选择偏向较低遮罩率的遮罩调度改善了性能和训练稳定性。有趣的是，来自其他模态的条件引入也改善了训练稳定性，这或许表明稳定性与任务的不足规范程度相关。

设置不正确的学习率是另一个不稳定的来源-

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ity. To ensure the right balance between learning effectiveness and stability, we optimized the learning rate on smaller models and scaled it according to best practices as outlined in (90, 91). We find empirically that the initialization has a small effect on model stability, and the majority of stabilization can be gained from simply scaling the learning rate at the appropriate rate. By applying the rules in both width-μP and depth-μP, we can simply scale the learning rate inversely proportional to the square root of the number of parameters, and find this results in stable training.

Following these modifications, we successfully trained our 98-billion-parameter model without any issues related to training instability.

A.2.3.4. Staged training

We stage training to alter dataset composition, train on longer contexts that would be too expensive for the entire pretraining, or introduce features such as the taxonomy track (A.1.9.2).

A.3. Model evaluations

ESM3 is both a generative model and a representation learning model that can be adapted for predictive tasks. In this section, we present benchmarking results for both capabilities.

A.3.1. Models

ESM3 models are trained at three scales—1.4B, 7B, and 98B parameters—on approximately 75B, 560B, and 1.8T training tokens, respectively.

The ESM3 1.4B model, trained on 75B tokens and noted for its small size and speed, allows rapid iteration both during training and at inference. Optimal model size and number of training tokens are studied by extrapolating from a series of smaller runs, given a training compute budget, model architecture, and dataset characteristics (22, 24). After determining compute optimality for training, a variety of factors such as release frequency, amount of inference, ease of use, and usage patterns are also taken into account to determine the ideal number of tokens on which to train the model. To enable efficient inference for the benefit of the research community, we have trained two additional versions of ESM3 1.4B, named 1.4B Overtrained and 1.4B Open, which are trained on 300B tokens, far beyond their compute optimality for training.

A.3.2. Data

In the following benchmarks for this section, unless otherwise noted, models are evaluated on a test set of 902 proteins whose structures are temporarily held out from the ESM3 training set. The proteins were sourced from the Continuous Automated Model EvaluatiOn (CAMEO) targets released from May 1, 2020 through Aug 1, 2023 (92).

For contact and structure prediction evaluations, we also evaluate on the CASP14 (71 proteins) and CASP15 (70 proteins) structure prediction benchmarks (93, 94). The CASP14 and CASP15 sets are obtained directly from the organizers.

A.3.3. Representation Learning

The contact prediction model is a multilayer perceptron (MLP) head that operates independently over the representations of each amino acid pair, outputting the probability of contact between them. We use LoRA (95) for finetuning, which is a common alternative to full weight finetuning that uses much less memory while attaining strong performance. LoRA is applied to the base model for finetuning, and the MLP along with the LoRA weights are trained end-to-end using the cross-entropy loss with respect to the ground truth contact prediction map. For the ground truth, all residues at least 6 positions apart in the sequence and within an 8Å Cα-Cα distance are labeled as a contact. All models are trained with LoRA rank 4, batch size 64 and a learning rate of 1e-3 for 10k steps on a mix of sequence and structure data from PDB, AlphaFold-DB, ESMAtlas, and OAS Predicted Structures. Data are sampled in a ratio of 1:3:3:0.03 from these datasets.

Table S7 shows the performance on each structural test set through the metric of precision at L (P@L), which evaluates the precision of the top-L most confident predictions, where L is the length of the protein. The smallest ESM3 model, with 1.4B parameters, achieves a P@L of 0.76 ± 0.02 on the CAMEO test set, which is higher than the 3B parameter ESM2 model (0.75 ± 0.02). Furthermore, it trains on an order of magnitude less compute during pretraining (6.72×10^{20} FLOPs vs. 1.8×10^{22} FLOPs), demonstrating the benefits of multimodal pretraining.

A.3.4. Structure Prediction

ESM3 can directly predict protein structures without additional finetuning by first predicting structure tokens, then decoding these tokens into coordinates. When predicting structure tokens, we follow the strategy outlined in Appendix A.1.10 and test both argmax decoding and full iterative decoding.

For more difficult datasets, such as CASP14 and CASP15, iterative decoding has an outsized impact (see Table S8), whereas for easier datasets like CAMEO, argmax prediction is sufficient. On both the CAMEO and CASP15 datasets, argmax prediction for the 7B model is comparable to ESMFold, and iterative decoding with ESM3 98B surpasses

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为了确保学习有效性与稳定性之间的正确平衡，我们在较小的模型上优化了学习率，并根据（90、91）中概述的最佳实践进行缩放。我们实证发现，初始化对模型稳定性的影响很小，而大多数稳定性来自于在适当的速率下简单缩放学习率。通过在宽度μP和深度μP中应用规则，我们可以简单地将学习率按参数数量的平方根成反比进行缩放，发现这能实现稳定训练。

在进行这些修改后，我们成功训练了我们的980亿参数模型，没有遇到与训练不稳定相关的任何问题。

A.2.3.4. 分阶段训练

我们通过分阶段训练来改变数据集的组成，在较长的上下文上进行训练，该上下文对于整个预训练来说过于昂贵，或引入特征，例如分类法跟踪（A.1.9.2）。

A.3. 模型评估

ESM3既是生成模型也是表示学习模型，可适用于预测任务。在本节中，我们展示了这两种能力的基准测试结果。

A.3.1. 模型

ESM3模型在三个规模上进行训练——1.4B、70亿和98B参数——分别基于约75B、560B和1.8T的训练标记。

ESM3 1.4B模型在75B标记上训练，以其小巧和快速而著称，允许在训练和推理期间进行快速迭代。通过从一系列较小的运行中推断，研究最优模型大小和训练标记数量，考虑到训练计算预算、模型架构和数据集特征（22，24）。在确定训练的计算最优性后，还考虑了多种因素，如发布频率、推理量、易用性和使用模式，以确定训练模型的理想标记数量。为了使研究社区能够进行高效推理，我们额外训练了两个版本的ESM3 1.4B，分别命名为1.4B Overtrained和1.4B Open，这两个版本是在300B标记上训练的，远超其计算训练的最优性。

A.3. 2. 数据

在本节的基准测试中，除非另有说明，否则模型在902种蛋白质的测试集上进行评估，这些蛋白质的结构暂时从ESM3中排除。

训练集。蛋白质来源于2020年5月1日至2023年8月1日发布的连续自动模型评估（CAMEO）目标（92）。

对于接触和结构预测评估，我们还在CASP14（71种蛋白质）和CASP15（70种蛋白质）结构预测基准上进行评估（93， 94）。CASP14和CASP15数据集直接由组织者提供。

A.3. 3. 表征学习

接触预测模型是一个独立操作于每对氨基酸的表征的多层感知机（MLP）头，输出它们之间的接触概率。我们使用 LoRA (95) 进行微调，这是一种常见的全权重微调的替代方案，它在达到较强性能的同时使用更少的内存。LoRA 被应用于基础模型进行微调，同时 MLP 和 LoRA 权重通过交叉熵损失相对于真实值接触预测图进行端到端训练。对于真实值，所有在序列中至少相隔6个位置并且在 8 ˚ACα-Cα 距离内的残基被标记为接触。所有模型均以 LoRA 等级 4、批量大小 64 和学习率 1e-3 在 PDB、AlphaFold-DB、ESMAtlas 和 OAS 预测结构的数据的混合上训练 10k 步。数据按 1:3:3:0.03 的比例从这些数据集中抽样。表S7 显示了每个结构测试集的性能，通过L时的精度指标 (P@L)，该指标评估最有信心的前L个预测的精度，其中L是蛋白质的序列长度。最小的 ESM3 模型，拥有 1.4B 参数，在 CAMEO 测试集上达到 P@L 为 0.76 ± 0.02，超过了 3B 参数的 ESM2 模型 (0.75 ± 0.02)。此外，它在预训练期间计算量少了一个数量级 (6.72×10^{20} FLOPs 对比 1.8×10^{22} FLOPs)，展示了多模态预训练的优势。

A.3. 4. 结构预测

ESM3 可以直接预测蛋白质结构，而无需额外的微调，方法是首先预测结构标记，然后将这些标记解码为坐标。在预测结构标记时，我们遵循附录A.1.10 中概述的策略，并测试 argmax 解码和完全迭代解码。对于更困难的数据集，如 CASP14 和 CASP15，迭代解码具有显著影响（见表S8），而对于较简单的数据集如 CAMEO，argmax 预测是足够的。在 CAMEO 和 CASP15 数据集中，70亿模型的 argmax 预测与 ESM-Fold 相当，而 ESM3 98B 的迭代解码超越了

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Model	CASP14	CASP15	CAMEO
ESM2 3B	0.57 (0.49 - 0.64)	0.57 (0.48 - 0.65)	0.75 (0.73 - 0.77)
ESM3 1.4B	0.56 (0.48 - 0.64)	0.59 (0.50 - 0.66)	0.76 (0.74 - 0.78)
ESM3 7B	0.62 (0.54 - 0.70)	0.64 (0.56 - 0.73)	0.82 (0.80 - 0.84)
ESM3 98B	0.66 (0.57 - 0.74)	0.66 (0.57 - 0.75)	0.85 (0.83 - 0.86)

Table S7. Precision@L results. Measured on CASP14, CASP15 and CAMEO for the ESM3 model family. Intervals represent bootstrapped 95% confidence intervals.

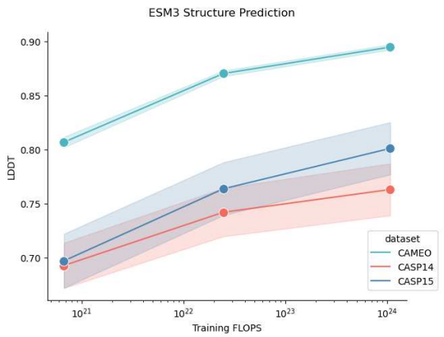

Figure S10. Scaling curves for structure prediction. Error bars are single standard deviations.

ESMFold. Structure prediction scaling curves as a function of training compute, are provided in Fig. S10.

A.3.5. Conditional Likelihood

The conditional likelihood of an output given a prompt serves as a proxy for the generative capabilities of a model. Fig. S11 and Table S9 evaluate the performance of ESM3 as a conditional generative model, using its negative log likelihood (NLL) on the test set. For each track—sequence, structure, function, SASA, and secondary structure—NLL is evaluated both unconditionally and conditioned on each of the other tracks.

Unlike, for example, an autoregressive model, ESM3 is a generative model over masking patterns, so is trained to predict tokens given any masking pattern. The NLL of a sample under ESM3 is given by $\frac{1}{L!}\sum_{o\in\mathbb{O}}\frac{1}{L}\sum_{i=1}^{L}\log p(x_{o_i}|x_{o_1},\ldots,x_{o_{i-1}})$, where O is the set of all decoding orders with normalization constant $Z=\frac{1}{L!}$. This computation is intractable (as the set of all decoding orders is exponential in length of a protein), but can be approximated by sampling a single decoding order o for each x in our dataset. At each step, teacher forcing is used to replace the masked token with the ground truth token and report the mean NLL over the output tokens.

We observe a number of interesting trends in these results. As expected, the unconditional NLL (Fig. S11, black lines) is always higher than conditional, and conditioning on full 3D structure reduces the loss on secondary structure prediction to nearly zero (1.4B: 0.24, 7B: 0.19, 98B: 0.16). Conditioning on sequence results in a lower structure prediction loss than conditioning on secondary structure (98B; sequence: 3.13, secondary structure: 3.37). We observe a clear log-linear relationship between pretraining FLOPs and NLL for the sequence track, under each form of conditioning.

A.3.6. Unconditional Generation

To assess the model's unconditional generation capability, we sampled 100 protein lengths randomly from the PDB and generated 1,024 sequences for each using ESM3 98B with a constant temperature of 0.7. The sampled length distribution is shown in Fig. S13A. Structures for each sequence were predicted using ESM3 7B, and the distribution of pTM and pLDDT are shown in Fig. S13B. ESM3 generates more high-quality structures than ESM2, which was trained using a simple MLM objective over sequence only with a fixed mask rate. Sequence similarity to the training set was computed using mmseqs2 (79) with the following parameters: `--cov-mode 2 -c 0.8 -s 6.0`. Proteins generated unconditionally are similar—but not identical—to proteins found in the training set (Fig. S15) and have high coverage of the training set (Fig. 1E), demonstrating that the model has properly fit the training distribution and does not exhibit mode collapse. We observe a cluster of generations with very high sequence identity to the training set; these correspond to antibody sequences, with the framework regions accounting for the high sequence identity.

We use pTM for evaluating structure predictions from ESM3 instead of pLDDT. This is because pLDDT can be miscalibrated for generated structures and can overestimate the confidence of a prediction. pLDDT is biased towards local structural confidence, which can result in pathologies such as very long alpha helices with high pLDDT at all positions. pTM is a more global measure of structural confidence, and is more robust to these pathologies. Fig. S12 shows that pTM and pLDDT correlation drops for generated sequences (Pearson r: natural = 0.85, generation = 0.79),

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模型	CASP14	CASP15	CAMEO
ESM2 3B	0.57（0.49 - 0.64）	0.57 (0.48 - 0.65)	0.75 (0.73 - 0.77)
ESM3 1.4B	0.56 (0.48 - 0.64)	0.59 (0.50 - 0.66)	0.76 (0.74 - 0.78)
ESM3 7B	0.62 (0.54 - 0.70)	0.64 (0.56 - 0.73)	0.82 (0.80 - 0.84)
ESM3 98B	0.66 (0.57 - 0.74)	0.66 (0.57 - 0.75)	0.85 (0.83 - 0.86)

表 S7. 精确度@L 结果。在 CASP14、CASP15 和 CAMEO 上测量，适用于 ESM3 模型系列。区间表示自助抽样的 95% 置信区间。

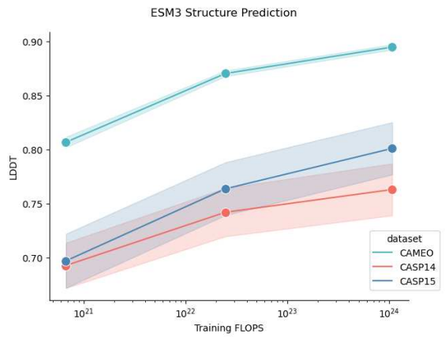

图 S10. 结构预测的缩放曲线。误差条为单个标准差。

ESMFold。作为训练计算的函数，结构预测缩放曲线见图。S10。

A.3.5. 条件似然

给定提示的输出的条件似然作为模型生成能力的代理。图 S11 和表S9 评估了ESM3作为条件生成模型的性能，使用其在测试集上的负对数似然（NLL）。对于每个轨迹——序列、结构、功能、SASA和二级结构——NLL既进行了无条件评估，也在每个其他轨迹的基础上进行了评估。

与自回归模型不同，ESM3是针对掩蔽模式的生成模型，因此其训练是为了在给定任何掩蔽模式的情况下预测标记。样本在ESM3下的NLL由$\frac{1}{L!}\sum_{o\in\mathbb{O}}\frac{1}{L}\sum_{i=1}^{L}\log p(x_{o_i}|x_{o_1},\ldots,x_{o_{i-1}})$给出，其中 O 是包含归一化常数$Z=\frac{1}{L!}$的所有解码顺序的集合。此计算是不可计算的（因为所有解码顺序的集合在蛋白质的长度上是指数级的），但可以通过抽样单个解码顺序来逼近。o 针对我们数据集中每个 x 。在每一步中，教师强制用于用真实值标记替换遮盖标记，并报告输出标记的平均负对数似然。

我们在这些结果中观察到许多有趣的趋势。正如预期的那样，无条件负对数似然（图。S11，黑线）始终高于条件负对数似然，且对完整三维结构的条件化将二级结构预测的损失降低到几乎为零（1.4B：0.24，70亿：0.19，98B：0.16）。与二级结构进行条件化相比，序列条件化产生的结构预测损失较低（98B；序列：3.13，二级结构：3.37）。我们观察到在每种条件化下，预训练FLOPs与序列轨道的负对数似然之间存在明显的对数线性关系。

A.3.6. 无条件生成

为了评估模型的无条件生成能力，我们从PDB中随机抽取了100个蛋白质长度，并使用ESM3 98B以0.7的恒定温度生成了1,024条序列。抽样长度分布如图S13所示。每个序列的结构是使用ESM3 7B预测的，pTM和pLDDT的分布如图S13B所示。与仅使用固定掩码率在序列上进行简单MLM目标训练的ESM2相比，ESM3生成了更多高质量的结构。使用mmseqs2计算与训练集的序列相似性 (79)，其参数如下：`--cov-mode 2 -c 0.8 -s 6.0`。无条件生成的蛋白质与训练集中发现的蛋白质相似——但不完全相同（图S15）并且对训练集具有很高的覆盖率（图1E），这证明模型已经正确拟合训练分布，并未出现模式崩溃。我们观察到有一簇生成序列与训练集具有非常高的序列一致性；这些序列对应于抗体序列，框架区域导致了高序列一致性。

我们使用 pTM 来评估来自 ESM3 的结构预测，而不是 pLDDT。这是因为 pLDDT 可能会对生成的结构进行错误校准，并可能高估预测的信心。pLDDT 对局部结构信心存在偏倚，这可能导致如非常长的 α-螺旋在所有位置具有高 pLDDT 的病理现象。pTM 是对结构信心的更全球性度量，且对这些病理现象更加稳健。图。S12显示生成的序列的 pTM 与 pLDDT 相关性下降（皮尔逊 r：自然 = 0.85，生成 = 0.79），

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	Iterative / $O(L^3)$			Argmax / $O(L^2)$		
Model	**CAMEO**	**CASP14**	**CASP15**	**CAMEO**	**CASP14**	**CASP15**
1.4B Open	0.830	0.705	0.733	0.805	0.640	0.677
1.4B Overtrained	0.846	0.714	0.750	0.825	0.651	0.700
1.4B	0.807	0.693	0.697	0.775	0.608	0.636
7B	0.870	0.742	0.764	0.852	0.607	0.726
98B	0.895	0.763	0.801	0.884	0.719	0.770
ESMFold	0.865	0.728	0.735			
AlphaFold2	0.904	0.846	0.826			

Table S8. Protein structure prediction results. We benchmark ESMFold, ESM3 models, and Alphafold2. Left side: ESM3 iterative inference of structure tokens conditioned on sequence. Because iterative inference is $O(L^3)$ in length of a protein sequence, it is comparable to ESMFold and AF2, both of which share the same runtime complexity. Right panel: Single-pass argmax structure token given sequence. In all cases, the more difficult the dataset, the more iterative decoding appears to help—98B has a +4.4 LDDT boost on CASP14, compared to a +1.0 LDDT boost on CAMEO. Both the Open and Overtrained models are both trained up to 200k steps. The 1.4B model is used for scaling comparisons, and is trained to 50k steps.

		Conditioning				
	Model	Sequence	Structure	Function	SASA	Secondary Structure
Sequence	1.4B	2.31	1.71	2.28	1.81	2.02
	7B	2.04	1.43	2.00	1.47	1.74
	98	1.84	1.21	1.76	1.21	1.50
Structure	1.4B	4.09	4.98	4.93	4.39	4.42
	7B	3.42	4.2	4.18	3.62	3.71
	98	3.13	3.85	3.8	3.24	3.37
Function	1.4B	1.81	1.98	4.52	2.29	2.24
	7B	1.22	1.47	3.75	1.67	1.70
	98	0.93	1.20	3.63	1.41	1.40
SASA	1.4B	1.78	1.81	2.42	2.48	2.10
	7B	1.57	1.66	2.26	2.31	1.92
	98	1.46	1.56	2.15	2.23	1.82
Secondary Structure	1.4B	0.42	0.24	0.70	0.50	0.83
	7B	0.31	0.19	0.57	0.31	0.6
	98	0.26	0.16	0.50	0.25	0.54

Table S9. Negative log-likelihood of each track conditioned on other tracks. Each row is a model size, generating a particular modality. Each column is the conditioning. The diagonal, highlighted with italics, are the unconditional NLL of each track. We observe that indeed adding conditioning improves NLL in all cases.

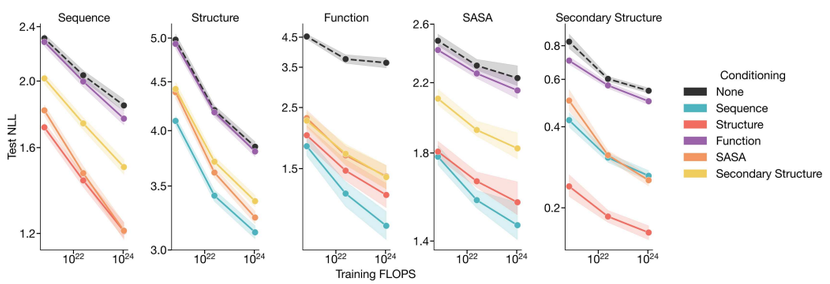

Figure S11. Conditional and unconditional scaling behavior for each track. Loss is shown on the test set (Appendix A.3.2).

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	迭代 / $O(L^3)$			最大化 / $O(L^2)$		
模型	CAMEO	CASP14	CASP15	CAMEO	CASP14	CASP15
1.4B 开放	0.830	0.705	0.733	0.805	0.640	0.677
1.4B 过度训练	0.846	0.714	0.750	0.825	0.651	0.700
1.4B	0.807	0.693	0.697	0.775	0.608	0.636
7B	0.870	0.742	0.764	0.852	0.607	0.726
98B	0.895	0.763	0.801	0.884	0.719	0.770
ESMFold	0.865	0.728	0.735			
AlphaFold2	0.904	0.846	0.826			

表 S8. 蛋白质结构预测结果。我们基准测试 ESMFold、ESM3 模型和 AlphaFold2。左侧：基于序列的 ESM3 迭代推理结构标记。由于迭代推理的长度为 $O(L^3)$ 的蛋白质序列，因此它与 ESMFold 和 AF2 相当，二者的运行时复杂度相同。右侧面板：给定序列的单次最大结构标记。在所有情况下，数据集越困难，迭代解码似乎越能提供帮助——在 CASP14 上，98B 具有 +4.4 的 LDDT 增益，而在 CAMEO 上则为 +1.0 的 LDDT 增益。开放模型和过拟合模型均训练至 200k 步。1.4B 模型用于尺度比较，训练至 50k 步。

		条件化				
	模型	序列	结构	功能	SASA	二级结构
序列	1.4B	2.31	1.71	2.28	1.81	2.02
	7B	2.04	1.43	2.00	1.47	1.74
	98	1.84	1.21	1.76	1.21	1.50
结构	1.4B	4.09	4.98	4.93	4.39	4.42
	7B	3.42	4.2	4.18	3.62	3.71
	98	3.13	3.85	3.8	3.24	3.37
函数	1.4B	1.81	1.98	4.52	2.29	2.24
	7B	1.22	1.47	3.75	1.67	1.70
	98	0.93	1.20	3.63	1.41	1.40
SASA	1.4B	1.78	1.81	2.42	2.48	2.10
	7B	1.57	1.66	2.26	2.31	1.92
	98	1.46	1.56	2.15	2.23	1.82
二级结构	1.4B	0.42	0.24	0.70	0.50	0.83
	7B	0.31	0.19	0.57	0.31	0.6
	98	0.26	0.16	0.50	0.25	0.54

表 S9. 条件于其他轨迹的每个轨迹的负对数似然。每行是一个模型大小，生成特定的模态。每列是条件。对角线部分用斜体突出显示，则是每个轨迹的无条件 NLL。我们观察到，确实添加条件在所有情况下都改善了 NLL。

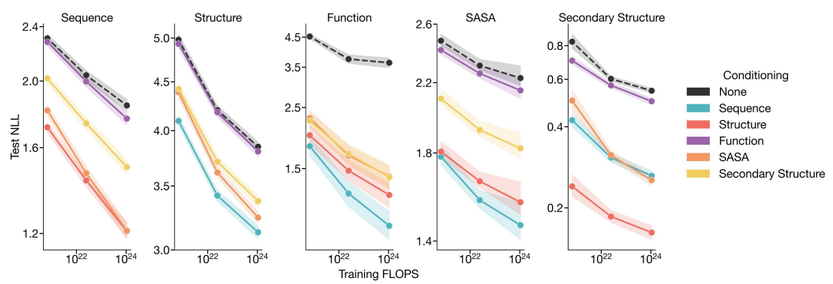

图 S11. 每个轨迹的条件和无条件缩放行为。 损失在测试集上显示（附录 A.3.2）.

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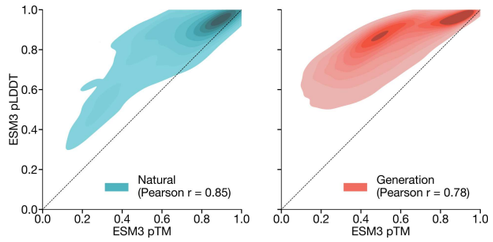

Figure S12. Distribution of pTM and pLDDT. Measured on natural (left) and generated (right) sequences under ESM3 7B structure prediction. Generated sequences show a clearly lower correlation (Pearson r 0.79 vs. 0.85) as well as a mode of sequences with high pLDDT but low pTM. Natural sequences are from the test set (Appendix A.3.2), generations are unconditional generations from ESM3 98B.

and a clear pattern of high pLDDT (> 0.8) but low pTM (< 0.6) emerges.

To visualize the distribution of unconditional generations, we compute sequence embeddings by extracting the final layer outputs produced by running ESM3 7B with sequence inputs only. Protein-level embeddings are computed by averaging over all positions in the sequence to produce a 2560-dim embedding. We then project these embeddings into two dimensions using a UMAP projection (96) fit on a background distribution of 50,000 randomly sampled sequences from UniProt with minimum distance 0.1 and number of neighbors 25. Examples are selected by computing structural clusters with Foldseek-cluster (using default parameters) and sampling the example with highest ESM3 pTM from each cluster. A subset of these cluster representatives are shown in Fig. 1E.

To assess whether ESM3 is biased towards particular secondary structures, we use DSSP to predict the three-class secondary structure of the high-confidence (pTM > 0.8, mean pLDDT > 0.8) generations and measure the percentage of residues that form alpha helices and beta sheets. When compared to a background distribution computed over the PDB, we find that ESM3 closely matches the secondary structure distribution of known proteins (Fig. S13D), unlike other methods which preferentially generate helical structures (15, 26, 28). Finally, to confirm that the structures predicted with high confidence by ESM3 are designable, we inverse folded and re-folded each using ESM3 7B. The majority of generations successfully re-folded with TM-score of greater than 0.8 to the hallucinated structures, demonstrating that ESM3 has high self-consistency for its own high-confidence designs (Fig. S13C).

To explore alternative ways of generating proteins, we assess the quality of proteins generated by a chain-of-thought (CoT) procedure in which ESM3 7B generates the secondary structure (SS8 tokens), then the 3-D backbone coordinates (structure tokens), followed by the amino acid sequence (sequence tokens) (Fig. S14). We compare the quality of amino acid sequences generated from this CoT procedure with the above method of unconditionally directly generating amino acid sequences. We find that the CoT procedure generates sequences that have higher confidence ESM3-predicted structures than the directly-generated sequences as measured by pTM and mean pLDDT (Fig. S14A). Compared to high-confidence (pTM > 0.8, mean pLDDT > 0.8) directly-generated sequences, the high-confidence subset of CoT-generated sequences are also more designable: the CoT-generated sequences have predicted structures whose inverse folded, then re-refolded structures have higher TM-score to the originally predicted structure (Fig. S14C). The CoT-generated sequences show a small bias towards higher alpha and beta proportion compared to those generated directly (Fig. S14D).

A.3.7. Prompt-following Evaluations

To evaluate ESM's ability to follow prompts, we use a set of held-out proteins as described in Appendix A.3.2. The test set is further filtered to remove proteins with length greater than 1024, which removes 7 proteins from the test set. To construct prompts for the structure coordinate, secondary structure, and SASA tracks, we sample a random span of length 15% of the original protein length. The model is then shown the corresponding track for the randomly sampled span, and is tasked with generating the sequence for the entire protein. For example, for the structure track, for a protein of length 100, we may sample a random span of 15 residues from residue 20-35. The model would then have to generate a protein sequence of length 100 conditioned on structure coordinate conditioning from residues 20-35 derived from the original test protein. This same procedure is applied for the secondary structure and SASA tracks. For the function track, we form the prompt by tokenizing the keywords from the InterProScan annotations associated with each sequence. The ESM3 7B model is used for all genera-

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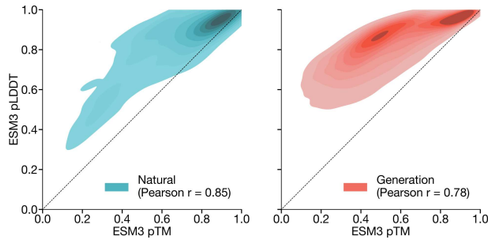

图 S12.pTM 和 pLDDT 的分布.在 ESM3 70亿 结构预测下，对自然（左）和生成（右）序列进行测量。生成序列显示出明显较低的相关性（皮尔逊 r 0.79 对比 0.85），以及存在高 pLDDT 但低 pTM 的序列模式。自然序列来自测试集（附录 A.33 982.），生成是来自 ESM B 的无条件生成。

并且出现高pLDDT（> 0.8）但低pTM（< 0.6）的明显模式。

为了可视化无条件生成的分布，我们通过提取仅由序列输入运行 ESM3 70亿 产生的最后一层输出，计算序列嵌入。蛋白质级嵌入通过对序列中的所有位置取平均来计算，以产生一个2560维的嵌入。然后，我们使用 UMAP 投影将这些嵌入投影到二维空间中（96），其适配于来自 UniProt 的 50,000 个随机抽样序列的背景分布，最小距离为 0.1，邻居数为 25。通过使用 Foldseek-cluster（使用默认参数）计算结构集群并从每个集群中采样具有最高 ESM3 pTM 的示例，选择示例。这些集群代表的子集在图中展示。1E.

为了评估ESM3是否偏向特定的二级结构，我们使用DSSP预测高置信度(pTM > 0.8，平均pLDDT > 0.8)生成的三类二级结构，并测量形成α-螺旋和β-折叠的残基百分比。与在PDB上计算的背景分布相比，我们发现ESM3与已知蛋白质的二级结构分布高度匹配（图S13D），而其他方法则优先生成螺旋结构（15，26，28）。最后，为了确认ESM3预测的高置信度结构是可设计的，我们使用ESM3 7B进行了逆折叠和重新折叠。大多数生成的结构成功重新折叠，TM-score大于0.8，表明ESM3对其自身高置信度设计具有较高的自一致性（图S13C）。

为了探索生成蛋白质的替代方法，我们评估通过链式思维（CoT）程序生成的蛋白质质量，在该程序中，ESM3 7B 生成二级结构（SS8标记），然后是三维骨架坐标（结构标记），最后生成氨基酸序列。

(序列标记) (图。S14). 我们将通过该CoT程序生成的氨基酸序列的质量与上述无条件直接生成氨基酸序列的方法进行比较。我们发现，CoT程序生成的序列在pTM和平均pLDDT（图。S14A）测量下，具有比直接生成的序列更高的信心ESM3预测结构。与高信心（pTM > 0.8，平均pLDDT > 0.8）直接生成的序列相比，CoT生成序列的高信心子集也更易于设计：CoT生成的序列具有其逆折叠然后重新折叠的结构，与最初预测的结构具有更高的TM评分（图。S14C）。与直接生成的序列相比，CoT生成的序列在高α和β比例方面显示出小的偏向（图。S14D).

A.3.7. 响应式评估

为了评估 ESM 跟随提示的能力，我们使用一组如附录中描述的保持不变的蛋白质A.3.2。测试集进一步过滤，以去除长度大于 1024 的蛋白质，这样从测试集中去除了 7 个蛋白质。为了构建结构坐标、二级结构和 SASA 跟踪的提示，我们随机抽取原始蛋白质长度的 15% 的跨度。然后向模型展示相应的随机采样跨度的轨迹，并要求其生成整条蛋白质的序列。例如，对于结构轨迹，对于长度为 100 的蛋白质，我们可以从残基 20-35 中随机抽样 15 个残基的跨度。然后，模型必须生成一个长度为 100 的蛋白质序列，该序列基于来自原始测试蛋白质的残基 20-35 的结构坐标条件。对于二级结构和 SASA 跟踪也应用相同的程序。对于功能跟踪，我们通过对与每个序列相关的 InterProScan 注释中的关键词进行标记化来形成提示。用于所有生成的 ESM3 7B 模型-

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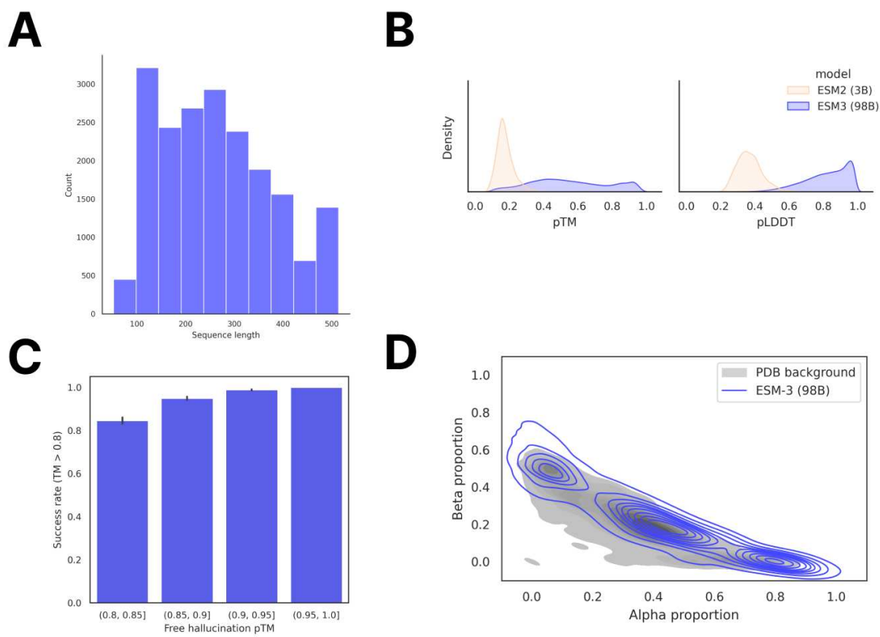

Figure S13. Unconditional generation of high-quality and diverse proteins using ESM3. (A) Distribution of sequence length in the unconditional generation dataset. (B) pLDDT and pTM of unconditional generations from ESM3 compared to sequences designed using the 3B-parameter ESM2 model. (C) Round-trip success rate of high-confidence generations using ESM3. Predicted structures were inverse folded to predict a new sequence and then re-folded to produce a new structure. Success was measured by a TM-score of greater than 0.8 between the original and refolded designs. (D) Secondary structure composition of unconditional generations relative to the distribution of proteins in the PDB, which is shown in gray.

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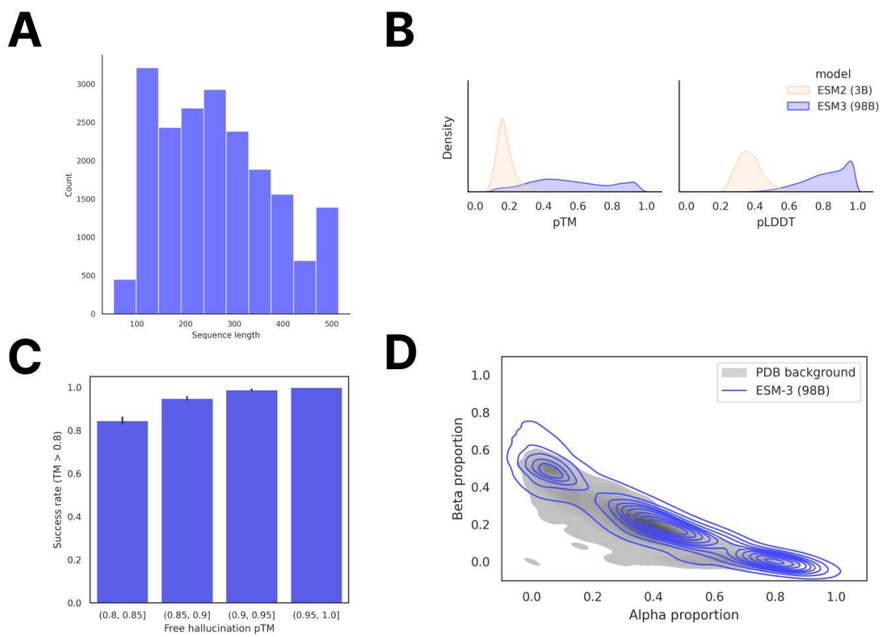

图 S13。使用 ESM3 无条件生成高质量和多样化的蛋白质。(A) 无条件生成数据集中序列长度的分布。(B) ESM3 的无条件生成与使用 3B-参数 ESM2 模型设计的序列相比的 pLDDT 和 pTM。(C) 使用 ESM3 的高可信度生成的往返成功率。预测的结构被逆向折叠以预测新序列，然后重新折叠以生成新结构。成功通过原始设计与重新折叠设计之间的 TM-score 大于 0.8 来衡量。(D) 无条件生成的二级结构组成相对于 PDB 中蛋白质的分布，以灰色显示。

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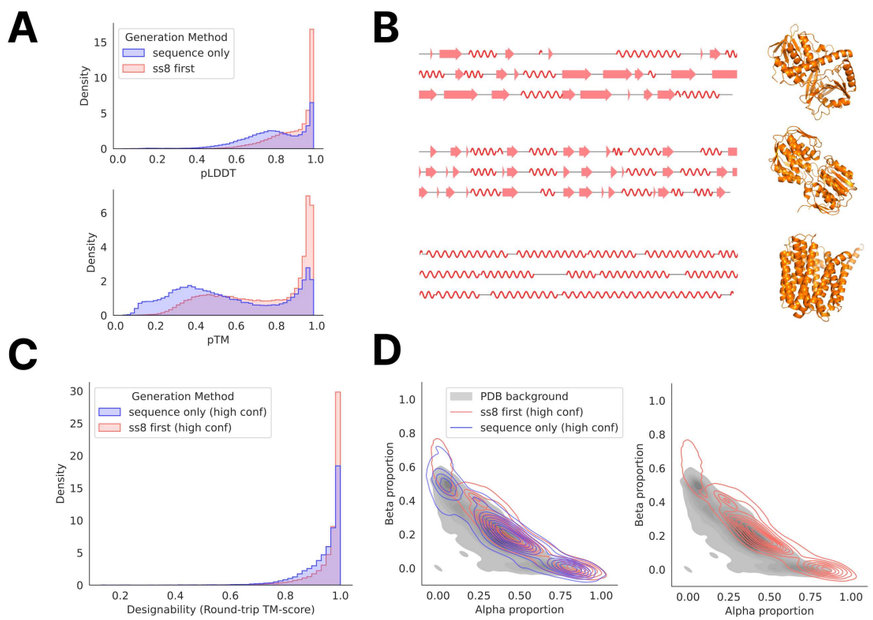

Figure S14. Generation of sequences using chain of thought. SS8 tokens are generated first, followed by structure tokens, then amino acid sequence with the ESM3 7B model. (A) Distribution of mean pLDDT and pTM of sequences generated by chain of thought ("ss8 first") compared to directly generating the sequence ("sequence only"). (B) Sample generations of SS8 tokens and the predicted structure of its corresponding CoT sequence. (C) TM-score between predicted structures of high-confidence (pTM > 0.8, mean pLDDT > 0.8) generated sequences and their corresponding inverse folded, then re-folded structures. (D) Comparison of the secondary structure composition of high-confidence generated sequences to the distribution of proteins in the PDB.

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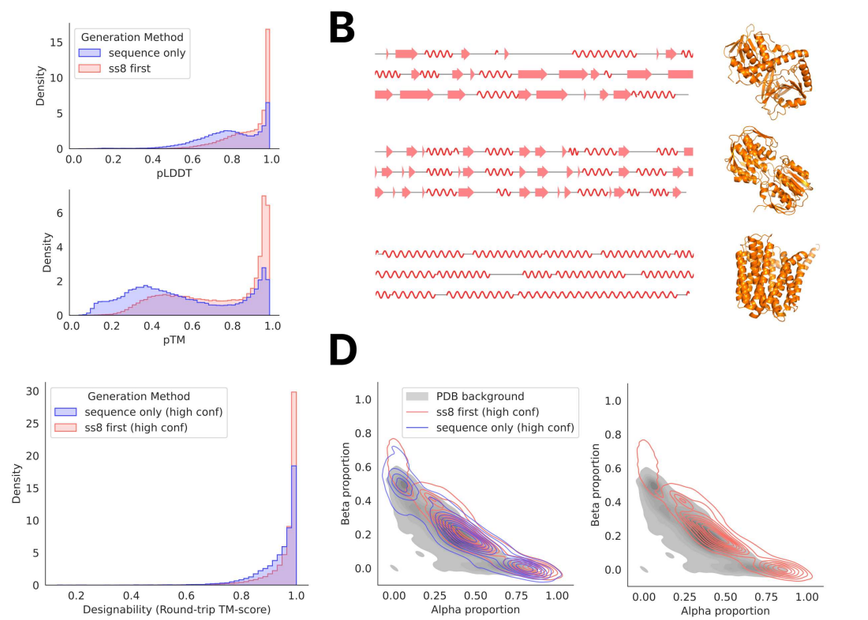

图 S14. 使用思维链生成序列。首先生成 SS8 标记，其次生成结构标记，最后使用 ESM3 7B 模型生成氨基酸序列。(A) 由思维链生成的序列的平均 pLDDT 和 pTM 的分布（“SS8 优先”）与直接生成序列（“仅序列”）相比。(B) SS8 标记的样本生成及其相应 CoT 序列预测结构。(C) 高置信度（pTM > 0.8，平均 pLDDT > 0.8）生成序列的预测结构与其相应的逆折叠再折叠结构之间的 TM-score。(D) 高置信度生成序列的二级结构组成与 PDB 中蛋白质的分布进行比较。

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tions with a temperature of 0.7 and L decoding steps (where L is the length of the sequence). The model generates 64 sequences per prompt, which we use to compute pass@64.

To evaluate the generations, we use ESMFold to fold the sequences generated by ESM3. For the structure coordinate, secondary structure, and SASA tracks, alignment metrics (backbone cRMSD, 3-class secondary structure accuracy, and SASA Spearman ρ) can be calculated using the relevant span in the ESMFold-predicted structure and the original template protein. Continuing the previous example for the structure track, we would compute the RMSD between residues 20-35 in the ESMFold structure predicted of the ESM3-generated sequence and residues 20-35 of the original test protein. For the function annotation track, we run InterProScan (40) on each generated sequence and extract function keywords from the emitted annotations. We report function keyword recovery at the protein level, computing the proportion of all function keywords in the prompt which appear anywhere in the function keywords from the InterProScan annotations of the generation.

A.3.8. Steerable Design

To test the ability of ESM3 to generalize beyond its training distribution under prompting, we evaluate two prompting scenarios. First, we identify proteins which were deposited in the PDB after our training cutoff (December 2020) and choose eight with TM < 0.7 to any structure in our training dataset (PDB IDs: 2JVN chain A, 2KAF chain A, 2L8K chain A, 2MJM chain A, 7ZUO chain A, 8EXF chain B). Using DSSP, we compute the residue-level SS8 and SASA for each of these proteins to prompt ESM3, masking all other tracks. We show in Fig. S15A that the generated proteins are diverse, globular, and closely follow the SS8 and SASA prompts while having no close sequence or structure neighbors in the training set. Interestingly, the original protein sequences are not folded with high confidence or accuracy by ESMFold (mean pTM 0.44, mean TM-score to reference 0.33), suggesting that these are challenging proteins to fold. The ESM3-generated sequences have a similar confidence (mean pTM 0.45) but much higher accuracy (mean TM-score 0.64).

Second, we classify the residue-level secondary structure for a set of eight symmetric protein backbones using DSSP. These proteins were previously designed using ESMFold (5, 97) and have varying secondary structure (alpha and beta) and varying symmetries (5-fold and 8-fold). Again, ESM3 is able to design these proteins successfully with high confidence (pTM > 0.8, pLDDT > 0.8) and low sequence similarity to the training set Fig. S15B. The structural similarity is moderate for these designs due to the high structural conservation of the protomer units in each design. All designs are generated using a constant temperature of 0.7 with L/2 decoding steps, where L is the protein length. We sample 256 sequences for each prompt and filter generations by pTM (> 0.8), pLDDT (> 0.8), and accuracy in satisfying the SS8 prompts (SS8 accuracy > 0.8). Final examples were selected from these filtered designs by visual inspection. Sequence similarity to the training set was computed using the same procedure as the unconditional generations, and structure similarity was computed using Foldseek (41) in TM-score mode (`alignment-type 1`) with sensitivity `-s 7.5`.

A.3.9. Composing Prompts

ESM3 is able to compose multimodal prompts across its input tracks—sequence, structure, SS8, SASA, and function keywords—to generate proteins with novel characteristics. To demonstrate this, we augment the standard functional motif scaffolding task (i.e., partial structure and sequence prompts) with additional conditioning to specify the type of scaffold for ESM3 to design. The functional sites comprise a combination of ligand binding sites coordinated by residues remote in sequence and those defined by short local motifs. For each motif, the coordinates and amino acid identities of all residues from the reference PDB structures are input to the model, with random shuffling and augmentation of the gaps between each active site. See Appendix A.4.5 for a description of this augmentation procedure and the specifications of the ligand-binding sites chosen. In addition to these sites, we also create a set of 12 partial sequence and structure prompts derived from conserved functional motifs (Table S10). These motifs are defined using a combination of the benchmark dataset in Watson et al. (26) and conserved sequence patterns from the Prosite database (98).

The scaffold conditioning is defined using either SS8 tokens (to specify secondary structure composition) or function keywords defined by InterPro accession numbers (to specify a particular fold). For each combination of functional site and scaffold prompt, we sample between 256 and 2048 times to generate proteins with diverse and novel characteristics. All designs were generated with the 7B-parameter model, a constant temperature of 0.7, and $L/2$ decoding steps for a protein of length L.

Secondary structure prompting. We generated proteins under four main classes of secondary structure composition: mostly alpha helices, mostly beta sheets, and mixed alpha-beta proteins (split into alpha/beta, alpha/beta/alpha, and beta/alpha/beta topologies). For each generation, we prompt the model with a random set of SS8 spans up to a total length L, with mask tokens in between. For example, an all-alpha SS8 prompt for a protein of length L=20 might look like `__HHHH____HHHHH_____HH` and a beta-alpha-beta prompt might look like `__EEE____HHHHH_____EE_`, where H is a residue within an alpha helix and E is a residue in a beta

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在温度为0.7和 L 解码步骤下（其中L 为序列长度）。模型为每个提示生成64个序列，我们用其计算pass@64。

为了评估生成结果，我们使用ESMFold对由ESM3生成的序列进行折叠。对于结构坐标、二级结构和SASA轨迹，对齐指标（主链cRMSD、三类二级结构准确性和SASA Spearman ρ）可以通过在ESMFold预测的结构与原始模板蛋白之间的相关跨度计算得到。继续前面的示例，对于结构轨迹，我们将计算ESMFold预测的ESM3生成序列中残基20-35与原始测试蛋白中残基20-35之间的RMSD。对于功能注释轨迹，我们在每个生成的序列上运行InterProScan（40），并从发出的注释中提取功能关键词。我们报告在蛋白质层面上的功能关键词恢复，计算提示中所有功能关键词与生成的InterProScan注释中的功能关键词的出现比例。

A.3.8. 可引导设计

为了测试ESM3在提示下超越其训练分布的泛化能力，我们评估了两种提示场景。首先，我们识别出在我们的训练截止日期（2020年十二月）之后被存储在PDB中的蛋白质，并选择了八个TM < 0.7与我们训练数据集中任意结构的相似性（PDB ID：2JVN链A，2KAF链A，2L8K链A，2MJM链A，7ZUO链A，8EXF链B）。使用DSSP，我们计算了这些蛋白质的残基级SS8和SASA，以提示ESM3，同时屏蔽所有其他轨迹。我们在图S15A中展示生成的蛋白质具有多样性、球状性，并与SS8和SASA提示密切相符，同时在训练集中没有近似的序列或结构邻居。有趣的是，ESMFold对原始蛋白质序列的折叠信心或准确性并不高（平均pTM 0.44，平均TM-score与参考值为0.33），这表明这些蛋白质是折叠的挑战性目标。由ESM3生成的序列具有相似的信心（平均pTM 0.45），但准确性更高（平均TM-score 0.64）。

其次，我们使用 DSSP 对一组八个对称的蛋白质骨架进行残基级别的二级结构分类。这些蛋白质之前是使用 ESMFold 设计的 (5, 97) 并具有不同的二级结构（α 和 β）和不同的对称性（5 重和 8 重）。再一次，ESM3 能够以高置信度成功设计这些蛋白质（pTM > 0.8, pLDDT > 0.8）并与训练集的序列相似性较低，如图。S15B。这些设计的结构相似性适中，因为每个设计中的原型单元具有较高的结构保守性。所有设计均使用恒定的温度 0.7 生成，

L/2 解码步骤，其中 L 是蛋白质长度。我们为每个提示采样 256 条序列，并通过 pTM (> 0.8), pLDDT (> 0.8) 和满足 SS8 提示的准确性（SS8 准确性 > 0.8）来过滤生成。最终示例通过视觉检查从这些过滤设计中选出。使用与无条件生成相同的程序计算与训练集的序列相似性，并使用 Foldseek (41) 在 TM-score 模式下 (对齐类型 1) 计算结构相似性，灵敏度设置为`-s 7.5`.

A.3.9. 编写提示

ESM3能够在其输入轨道上编写多模态提示——序列、结构、SS8、SASA和功能关键词——以生成具有新特征的蛋白质。为了证明这一点，我们通过额外的条件增强标准功能基序支架任务（即部分结构和序列提示），以指定供ESM3设计的支架类型。功能位点包括由序列中远程的残基协调的配体结合位点的组合以及由短局部基序定义的位点。对于每个基序，来自参考PDB结构的所有残基的坐标和氨基酸身份被输入到模型中，同时随机打乱和增强每个活性位点之间的间隙。请参见附录A.4.5以获取此增强过程的描述及所选配体结合位点的规格。除了这些位点外，我们还创建了一组由保守功能基序衍生的12个部分序列和结构提示（表S10）。这些基序是使用Watson et al. (26)中的基准数据集和来自Prosite数据库的保守序列模式（98）的组合来定义的。

支架条件的定义采用SS8标记（用于指定二级结构组成）或由InterPro获取编号定义的函数关键词（用于指定特定的折叠）。对于每个功能位点和支架提示的组合，我们在256到2048次之间进行采样，以生成具有多样化和新颖特征的蛋白质。所有设计均使用7B-参数模型生成，温度保持在0.7，并且 $L/2$ 为长度为 L 的蛋白质进行解码步骤。

二级结构提示。我们生成了四种主要二级结构组成的蛋白质：主要是α-螺旋，主要是β-折叠，以及混合的α-β蛋白质（分为α/β、α/β/α和β/α/β拓扑）。对于每一次生成，我们使用一组随机的SS8范围来提示模型，直到总长度L，中间插入掩码标记。例如，一个针对长度为 L=20 的蛋白质的全α SS8提示可能看起来像`__HHHH____HHHHH_____HH`，而一个β-α-β提示可能看起来像`__EEE____HHHHH_____EE_`，其中H是α-螺旋中的残基，E是β-折叠中的残基。

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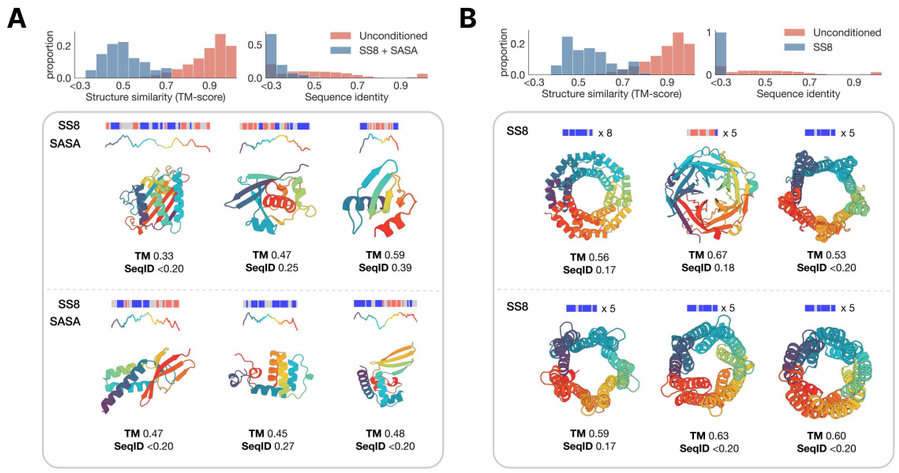

Figure S15. Prompting ESM3 to generalize beyond its training distribution. (A) Proteins designed using SS8 and SASA prompts derived from recent structures in the PDB with low structural similarity to the training set. Prompts along the protein length are visualized above each generation; secondary structure is shown using three-class (alpha = blue, beta = orange, coil = gray) and SASA is shown as a line plot colored by residue index to match the cartoon below. (B) Symmetric proteins designed using SS8 prompting. Histograms show the similarity to the nearest training set protein by structure (TM-score) and sequence (sequence identity) compared to unconditional generation.

Motif	PDB ID	Chain ID	PDB Residue Identifiers
ACE2 binding	6vw1	A	19-89, 319-366
Ferredoxin	6e6r	A	1-44
Barstar binding	7mrx	B	25-47
P53 binding	1ycr	B	19-28
PD-1 binding	5ius	A	63-83, 119-141
DNA-binding helix-turn-helix	1lcc	A	1-52
P-loop	5ze9	A	229-243
Double EF-hand	1a2x	A	103-115, 139-152
Lactate dehydrogenase	1ldb	A	186-206
Renal dipeptidase	1itu	A	124-147
Ubiquitin-activating enzyme E1C binding	1yov	B	213-223
DNA topoisomerase	1a41	A	248-280

Table S10. Functional motif definitions for conserved regions.

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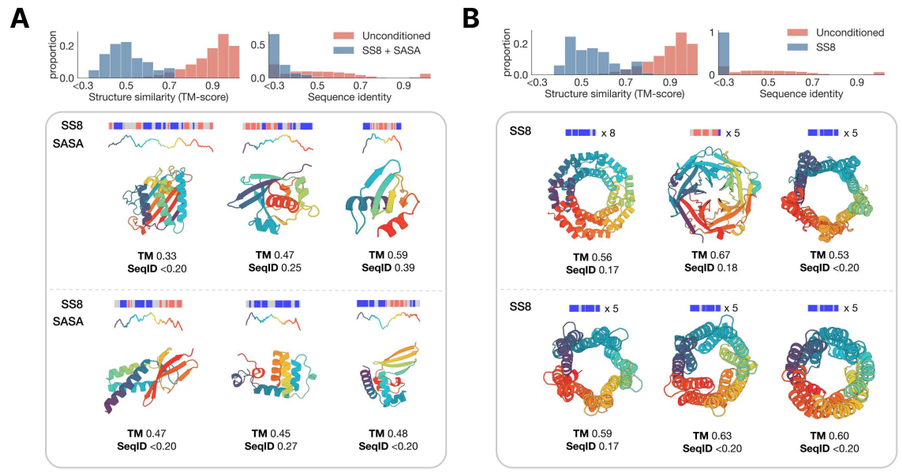

图 S15. 提示 ESM3 超越其训练分布的泛化能力。 (A) 使用来自 PDB 的与训练集结构相似度较低的近期结构所衍生的 SS8 和 SASA 提示设计的蛋白质。沿蛋白质长度的提示在每次生成上方可视化；二级结构使用三类（α = 蓝色，β = 橙色，卷曲 = 灰色）表示，SASA 以线图形式显示，颜色由残基索引匹配下方的漫画图。(B) 使用 SS8 提示设计的对称蛋白质。直方图显示与最近的训练集蛋白质在结构（TM-score）和序列（一致性）上的相似性，相较于无条件生成。

基序	PDB ID	链 ID	PDB 残基标识符
ACE2 结合	6vw1	A	19-89, 319-366
铁氧还蛋白	6e6r	A	1-44
Barstar结合	7mrx	B	25-47
P53结合	1ycr	B	19-28
PD-1结合	5ius	A	63-83, 119-141
DNA结合螺旋转弯螺旋	1lcc	A	1-52
P环	5ze9	A	229-243
双EF手	1a2x	A	103-115, 139-152
乳酸脱氢酶	1ldb	A	186-206
肾脏二肽酶	1itu	A	124-147
泛素激活酶E1C结合	1yov	B	213-223
DNA拓扑异构酶	1a41	A	248-280

表S10. 保守区域的功能基序定义。

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strand. We then combine this with the augmented partial structure and sequence tracks given by a functional site motif. To increase the diversity of the scaffolds and maximize the probability of generating physically realizable prompt combinations, we generate between 256 and 1024 designs for each combination of SS8 and functional site motif. For each generation, we uniformly sample a random length L between 150 and 400. Then, we produce a set of secondary structure spans with length 5-20 residues, each separated by a gap of 3-10 residues, such that the total length adds up to L. Finally, to avoid incompatibility between the partial structure and secondary structure constraints, we also mask the SS8 tokens at positions where structure is specified by the functional site prompt. Secondary structure–prompted designs were assessed by running DSSP on the designed sequence and measuring the fraction of prompted residues which were assigned the correct secondary structure. Success was determined by a pTM > 0.8, all-atom cRMSD $<$ 1.5 for the functional site, and SS8 accuracy > 0.8.

Keyword prompting. To prompt the model to generate proteins with a specific fold, we extracted the set of InterPro tags associated with a set of proteins from the test set for which ESM3 achieved keyword recovery of greater than 80% (Fig. 2A). These tags were then converted into keywords and used to prompt the model in combination with the partial sequence and structure constraints. The list of prompts and function tags is given in Table S11. Keyword-prompted designs were assessed using a self-consistency evaluation, i.e. whether the model successfully predicts any of the prompted InterPro accessions for the designed sequence. Success was determined by a pTM > 0.8, all-atom cRMSD < 2.0, and number of InterPro accessions recovered > 0.

We assess novelty of each motif-scaffold combinations by measuring the TM-score between the generated scaffold and the chain from which the motif is derived (Table S12). This confirms that the model is not retrieving the original motif scaffold, particularly for secondary structure–prompted scaffolds where we do not provide any explicit instructions to produce diverse designs. For the motifs derived from ligand binding residues (magnesium, serotonin, calcium, zinc, protease inhibitor 017, and Mcl-1 inhibitor YLT), we additionally use Foldseek to search the PDB for any other proteins which share that motif (as defined by BioLiP (99)), as a more stringent evaluation of novelty. For all but zinc-binding and magnesium-binding motifs, Foldseek finds no significant hits at an E-value threshold of 1.0. The hits discovered for zinc and magnesium have only modest TM-score (0.76 and 0.64), demonstrating that the model still finds novel scaffolding solutions for these ligands. To assess whether the generated scaffolds are likely to be designable, we measure a self-consistency TM-score (scTM) under orthogonal computational models by inverse-folding the designed structure with ESM-IF1 (42) (using a temperature of 0.5) and re-folding with ESMFold (5). We report the best scTM over 8 inverse folding designs in Table S12.

A.3.10. Multimodal Editing Examples

First, we describe the procedure for generating the protein compression example shown in Fig. 2D. A series of prompts of length 150 were constructed. The sequence and structure of the catalytic triad of trypsin (PDB 1Y3V) (H57, D102, S195) were placed in the prompt using the following procedure: three random residue numbers between 20 and 130 were sampled such that the minimum pairwise difference in position between each of the residues was no less than 20. Then, H57 from the template trypsin was placed at the lowest sampled number, D102 at the second lowest, and S195 at the largest number, thus respecting the left-to-right ordering of the catalytic triad in the template trypsin. 128 prompts were generated by this procedure. Each of these prompts was combined with a function keyword prompt derived from the template protein, specifically InterPro (40) tags IPR001254 (serine proteases, trypsin domain) and IPR009003 (peptidase S1, PA clan), to arrive at a final set of 128 prompts. The base ESM 7B model was then prompted to generate the sequence of the remaining 147 residues of the protein conditioned on the randomly placed catalytic triad sequence and structure coordinates and function keywords. $L = 150$ decoding steps were used with a temperature of 0.7, with 32 generations per prompt. Generations were then filtered by all-atom RMSD at the prompt positions, ESM3 pTM, and InterPro Scan keyword outputs, with the generation shown in Fig. 2D selected finally by visual inspection.

Generation quality was measured using ESMFold (5) pTM of the generated sequence, in addition to self-consistency. For self-consistency, we inverse fold the ESM3-predicted structure of the generation with ESM-IF1 (42) 8 times and re-fold with ESMFold, reporting the mean and std of the TM-scores between the 8 ESMFold-predicted structures and the ESM3-predicted structure. To perform a blast search of the sequence, we use a standard Protein Blast search (54). We set the max target sequences parameter to 5000 and sort results by sequence length and sequence identity, selecting the first sequence that is a serine protease. This yields the reference WP_260327207 which is 164 residues long and shares 33% sequence identity with the generation.

We showcase two further examples of protein editing. First, ESM3 is prompted to bury an exposed helix in a protein with an alternating alpha-beta sandwich fold. The prompt is constructed as follows: the prompt is of the same length as the template protein (PDB 1LBS). We identify a buried helix (mean SASA 0.32 Å^2) between residues 106-116 of the template protein. Structure coordinates from this region are

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链。然后，我们将其与由功能位点模体提供的增强部分结构和序列轨迹相结合。为了提高支架的多样性并最大化生成物理可实现的提示组合的概率，我们为每个SS8和功能位点模体的组合生成256到1024个设计。对于每次生成，我们均匀地从150到400之间随机抽取一个长度 L。接着，我们生成一组长度为5-20个残基的二级结构区间，每个区间之间相隔3-10个残基，使得总长度加起来为 L。最后，为了避免部分结构和二级结构约束之间的不兼容性，我们还会掩蔽在功能位点提示所指定的结构位置处的SS8标记。通过对设计的序列运行DSSP并测量被提示的残基分配到正确的二级结构的比例来评估二级结构提示的设计。成功的判定基于pTM > 0.8、全原子cRMSD <1.5与功能位点的匹配，以及SS8准确率 > 0.8。

关键词提示。为了提示模型生成具有特定折叠的蛋白质，我们提取了与来自测试集的一组蛋白质相关联的InterPro标签，这些蛋白质的ESM3关键字恢复率超过了80%（图。2A）。随后将这些标签转换为关键词，并与部分序列和结构约束结合使用，提示模型。提示和功能标签的列表见表S11。关键词提示的设计通过自我一致性评估来进行评估，即模型是否成功预测为设计序列所提示的任何InterPro访问号。成功的标准是pTM > 0.8，所有原子cRMSD < 2.0，以及恢复的InterPro访问号的数量> 0。

我们通过测量生成的支架与源自该模体的链之间的TM-score来评估每个模体-支架组合的新颖性（表S12）。这确认模型并未检索原始模体支架，特别是在二级结构提示的支架中，我们并未提供任何显式指令以产生多样化的设计。对于源自配体结合残基（镁、血清素、钙、锌、蛋白酶抑制剂017和Mcl-1抑制剂YLT）的模体，我们另外使用Foldseek在PDB中搜索任何共享该模体的其他蛋白质（如BioLiP所定义（99））， 作为对新颖性的更严格评估。除了锌结合和镁结合模体外，Foldseek在1.0的E值阈值下未发现显著的命中。对于锌和镁发现的命中仅有适度的TM-score（0.76和0.64），表明模型仍然为这些配体找到新颖的支架解决方案。为了评估生成的支架是否可能是可设计的，我们通过逆折叠计算模型下测量自一致TM-score（scTM）。

使用 ESM-IF1 的签名结构（42）（使用温度 0。

5) 和使用ESMFold进行重新折叠（5）。我们在表中报告了8个反向折叠设计中的最佳scTMS12。

A.3.10. 多模态编辑示例

首先，我们描述生成图中所示蛋白质压缩示例的过程。2D。构建了一系列长度为150的提示。催化三联体胰蛋白酶（PDB 1Y3V）的序列和结构（H57、D102、S195）通过如下程序放置在提示中：随机抽取20到130之间的三个残基编号，确保每个残基之间的最小位置差异不小于20。然后，将模板胰蛋白酶中的H57放置在最低抽取的编号处，D102放置在第二低的编号处，S195放置在最大编号处，从而遵循模板胰蛋白酶中催化三联体的从左到右的顺序。通过此过程生成了128个提示。每个提示与源自模板蛋白的函数关键词提示相结合，具体为InterPro（40）标签IPR001254（丝氨酸蛋白酶，胰蛋白酶域）和IPR009003（肽酶S1，PA类），以得出最终的128个提示集。然后，基础的ESM 7B模型被提示生成其余147个残基的序列，这些残基的序列是以随机放置的催化三联体序列和结构坐标以及函数关键词为条件生成的。 $L = 150$ 解码步骤的温度设置为0.7，每个提示生成32个样本。随后，通过所有原子RMSD、提示位置、ESM3 pTM和InterPro Scan关键词输出对生成结果进行了过滤，最终通过目视检查选取了图中所示的生成结果。2D。

生成质量是通过使用ESMFold (5) 生成序列的pTM进行测量的，此外还考虑自一致性。对于自一致性，我们使用ESM-IF1反折叠生成的ESM3预测结构（42）8次，并使用ESMFold重新折叠，报告8个ESMFold预测结构与ESM3预测结构之间的TM-score的平均值和标准差。为了执行序列的比对搜索，我们使用标准的蛋白质比对搜索（54）。我们将最大目标序列参数设置为5000，并根据序列长度和序列一致性对结果进行排序，选择第一个为丝氨酸蛋白酶的序列。这得到了参考WP 260327207，其序列长度为164个残基，并与生成物具有33%的序列一致性。

-

我们展示两个进一步的蛋白质编辑示例。首先，ESM3被提示将一个暴露的螺旋埋入一个具有交替的阿尔法-贝塔三明治折叠的蛋白质中。提示的构造如下：提示的长度与模板蛋白（PDB 1LBS）相同。我们识别模板蛋白中残基106-116之间的一个埋藏螺旋（平均SASA 0.32 ˚A^2）。该区域的结构坐标为

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Scaffold	Reference	InterPro tags	Total Length
Beta propeller	8siuA	IPR001680 (1-350) IPR036322 (1-350) IPR015943 (1-350)	353
TIM barrel	7rpnA	IPR000652 (0-248) IPR020861 (164-175) IPR035990 (0-249) IPR013785 (0-251) IPR000652 (2-249) IPR022896 (1-249)	252
MFS transporter	4ikvA	IPR011701 (1-380) IPR020846 (1-380) IPR036259 (1-380)	380
Immunoglobulin	7sbdH	IPR036179 (0-116; 124-199) IPR013783 (0-206) IPR003597 (124-202) IPR007110 (0-115; 121-207) IPR003599 (6-115) IPR013106 (11-114)	209
Histidine kinase	8dvqA	IPR003594 (47-156) IPR003594 (47-158) IPR004358 (118-137) IPR004358 (141-155) IPR004358 (101-112) IPR005467 (0-158) IPR036890 (4-159) IPR036890 (3-156)	166
Alpha/beta hydrolase	7yiiA	IPR029058 (0-274) IPR000073 (26-265)	276

Table S11. InterPro tags extracted from CAMEO test set proteins for prompting with fold specification.

Site	Scaffold	Novelty (TM to original)	Designability (scTM)
017	beta	0.264	0.967
ACE2	alpha	0.606	0.871
CA	Immunoglobulin	0.441	0.781
Double-EF-hand	ab-hydrolase	0.293	0.969
MG	TIM-barrel	0.328	0.980
Renal-dipeptidase	alpha-beta-alpha	0.644	0.933
SRO	mfs-transporter	0.345	0.992
Topoisomerase	histidine-kinase	0.269	0.948
YLT	alpha-beta	0.229	0.899
ZN	alpha	0.567	0.996

Table S12. Novelty and designability metrics. Metrics for motif scaffolds shown in Fig. 2C. Novelty is measured by computing the TM-score to the original scaffold from which the motif is derived. Designability is measured by self-consistency TM-score over eight samples by inverse folding with ESM-IF1 and refolding with ESMFold. All designs are distinct from their original scaffolds while retaining high designability.

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支架	参考	InterPro 标签	序列长度
β-螺旋	8siuA	IPR001680 (1-350) IPR036322 (1-350) IPR015943 (1-350)	353
TIM桶	7rpnA	IPR000652 (0-248) IPR020861 (164-175) IPR035990 (0-249) IPR013785 (0-251) IPR000652 (2-249) IPR022896 (1-249)	252
MFS 转运蛋白	4ikvA	IPR011701 (1-380) IPR020846 (1-380) IPR036259 (1-380)	380
免疫球蛋白	7sbdH	IPR036179 (0-116; 124-199) IPR013783 (0-206) IPR003597 (124-202) IPR007110 (0-115; 121-207) IPR003599 (6-115) IPR013106 (11-114)	209
组氨酸激酶	8dvqA	IPR003594 (47-156) IPR003594 (47-158) IPR004358 (118-137) IPR004358 (141-155) IPR004358 (101-112) IPR005467 (0-158) IPR036890 (4-159) IPR036890 (3-156)	166
α/β水解酶	7yiiA	IPR029058 (0-274) IPR000073 (26-265)	276

表 S11. 从CAMEO测试集蛋白质中提取的InterPro标签，用于折叠规格的提示。

位点	支架	新颖性（TM与原始的比较）	可设计性（scTM）
017	β	0.264	0.967
ACE2	α	0.606	0.871
CA	免疫球蛋白	0.441	0.781
双EF手	ab水解酶	0.293	0.969
MG	TIM桶	0.328	0.980
肾二肽酶	阿尔法-贝塔-阿尔法	0.644	0.933
SRO	mfs转运蛋白	0.345	0.992
拓扑异构酶	组氨酸激酶	0.269	0.948
YLT	阿尔法-贝塔	0.229	0.899
ZN	α	0.567	0.996

表 S12。新颖性和可设计性指标。在图样中显示的支架的指标。2C. 新颖性通过计算与原始支架的 TM-score 来衡量，原始支架是从中派生出该图样的。可设计性根据八个样本的自一致性 TM-score，通过与 ESM-IF1 的反向折叠及与 ESMFold 的重新折叠来衡量。所有设计均与其原始支架不同，同时保持高度的可设计性。

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placed in the prompt at the same residue indices, to prompt ESM3 to generate the same helix. This is composed with a SASA prompt of 40.0 for each of the 11 helix residues, prompting ESM3 to place this helix on the surface of the protein. Finally, we prompt with the secondary structure of 5 central beta strands surrounding the buried helix, residues 33-36, 62-65, 99-103, 125-130, and 179-182. ESM3 7B is then used to generate 512 protein sequences conditioned on this prompt using $\frac{L}{2}$ decoding steps and a temperature of 0.7. Designs are filtered by ESM3 pTM and adherence to the SASA prompt. The final generation is chosen by visual inspection. The generation is evaluated as described above (ESMFold pTM 0.71, scTM mean 0.82, std 0.045). Examining the generation, ESM3 is able to satisfy the input constraints: the generated protein maintains the structure of the helix (backbone cRMSD 0.18 Å) and the alternating alpha-beta fold (both the generation and the template have 7 strands alternating with helices), while exposing the helix motif to the surface (mean SASA 28.35 Å^2). Furthermore, the generation is structurally distinct: a Foldseek search (41) of AlphaFold-DB, ESMAtlas, and PDB in TM-align mode reveals no hit with TM-score greater than 0.76.

We also use ESM3 to generate an idealized TIM Barrel with 11-fold symmetry. This generation is undertaken in two steps. First, we derive a secondary structure and function keyword prompt from a reference TIM Barrel (PDB 5EKY). The secondary structure of the reference protein is computed using DSSP and then idealized to construct a prompt for ESM3. To construct the secondary structure prompt, the length of each helix and strand is fixed at 7 residues. Each helix and strand region is then separated by 3 mask tokens, with a mask token appended to the N and C termini of the prompt as well. This yields a secondary structure prompt of total length 159, which is combined with a function keyword prompt derived from the reference protein: keywords are derived from IPR013785 (aldolase-type TIM barrel) and IPR000887 (KDPG/KHG aldolase). ESM3 7B is then used to generate 256 samples with L decoding steps and a temperature of 0.7. The design shown is chosen by filtering by ESM3 pTM and visual inspection. In the second step, the secondary structure prompt from the first step is expanded to contain 11 helix-strand subunits, for a total prompt length of 225 residues (4 mask tokens are now appended to the N and C termini, rather than just 1). ESM3 7B is then used to generate 256 samples with L decoding steps and a temperature of 0.7, with generations filtered by ESM3 pTM and visual inspection. The generation is evaluated as described above (ESMFold pTM 0.69, scTM mean 0.97, std 0.011). The generation is structurally distinct: a Foldseek search (41) of AlphaFold-DB, ESMAtlas, and PDB in TM-align mode reveals no hit with TM-score greater than .61.

A.4. Alignment

A.4.1. Algorithm

Since the introduction of RLHF (44) there have been a number of algorithms developed to tune large models trained via unsupervised learning to better follow instructions and generally align their generations to user preferences (45, 46, 100, 101). We use IRPO (Iterative Reasoning Preference Optimization) due to its simplicity in implementation and good performance. The IRPO loss combines supervised finetuning with contrastive learning from preference pairs. IRPO operates on a dataset $\mathcal{D} \sim (y_w, y_l, x)$ consisting of prompt x and a pair of completions y_w (preferred) and y_l (not preferred). It also operates on two separate models: the reference model π_{ref} and the current model π_θ. The reference model π_{ref} is the fixed base model of the same scale, and the current model π_θ is the model being optimized.

$$\mathcal{L}_{\text{IRPO}}(\pi_\theta; \pi_{\text{ref}}) = \mathcal{L}_{\text{NLL}} + \alpha\mathcal{L}_{\text{DPO}} = - \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\frac{\log \pi_\theta(y_w|x)}{|y_w|+|x|} + \alpha \log \sigma\left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right] \quad (2)$$

The IRPO loss contains two terms. The $\mathcal{L}_{\text{NLL}}$ term maximizes the log likelihood of the preferred example normalized by the length of the sequence, providing signal to reinforce the good generations from the model. The $\mathcal{L}_{\text{DPO}}$ term is the contrastive preference tuning term, which increases the difference in log likelihoods between the preferred and not preferred examples while staying close to the reference model (45). The use of the reference model serves as a regularizer to prevent overfitting to the preference dataset, which can often be small. There are two hyperparameters, α and β. α weights the relative importance of the supervised with the preference loss and the β parameter controls how close we stay to the reference model: the higher the beta, the closer we stay. We minimize this loss with respect to the current model parameters θ.

ESM3 is a multi-modal model so the prompt can be any combination of the input tracks of (partial) sequence, structure, and function and the generation y can be any of the output tracks. In our experiments we always generate the amino-acid sequence so this will be our running example from now on. Since an amino-acid sequence y can be generated from prompt x in many multi-step ways computing the full likelihood $\pi(y|x)$ would involve integrating over all possible multi-step decoding paths. Since this is intractable, we use a surrogate that mirrors pretraining, shown in Eq. (3) and described below.

$$\log \pi(y|x) \approx \mathbb{E}_m\left[\sum_{i\in m} \log p(y_i|y_{\setminus m}, x)\right] \quad (3)$$

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在提示中放置在相同的残基索引位置，以提示ESM3生成相同的螺旋。这是与每个11个螺旋残基的40.0的SASA提示组合而成的，提示ESM3将此螺旋放置在蛋白质的表面上。最后，我们用围绕埋藏的螺旋（残基33-36、62-65、99-103、125-130和179-182）的5个中心β链的二级结构进行提示。然后使用ESM3 70亿生成基于此提示的512个蛋白质序列，使用 $\frac{L}{2}$ 解码步骤和0.7的温度。设计通过ESM3 pTM和对SASA提示的遵循进行过滤。最终生成通过目视检查进行选择。生成的评估如上所述（ESMFold pTM 0.71，scTM平均值0.82，标准差0.045）。检查生成的结果，ESM3能够满足输入约束：生成的蛋白质保持螺旋的结构（主链cRMSD 0.18 ˚Å）和交替的α-β折叠（生成和模板都有7个链与螺旋交替），同时将螺旋模体暴露于表面（平均SASA 28.35 ˚Å^2）。此外，生成在结构上是独特的：Foldseek搜索（41）在TM-align模式下对AlphaFold-DB、ESMAtlas和PDB进行的检查未发现TM-score大于0.76的命中。

我们还使用ESM3生成具有11重对称性的理想化TIM桶。此生成分为两个步骤进行。首先，我们从参考TIM桶（PDB 5EKY）派生出二级结构和功能关键词提示。参考蛋白质的二级结构使用DSSP计算，然后被理想化以构建ESM3的提示。为了构建二级结构提示，每个螺旋和链的长度固定为7个残基。然后每个螺旋和链区域之间用3个掩膜标记分隔，同时在提示的N端和C端也附加一个掩膜标记。这产生的二级结构提示总长度为159，与从参考蛋白质派生的功能关键词提示结合：关键词来自IPR013785（醛缩酶型TIM桶）和IPR000887（KDPG/KHG醛缩酶）。随后使用ESM3 7B生成256个样本，解码步骤为 L ，温度为0.7。所示设计通过ESM3 pTM筛选和目视检查进行选择。在第二步中，第一步的二级结构提示扩展为包含11个螺旋-链亚单元，提示总长度为225个残基（现在在N端和C端附加了4个掩膜标记，而不仅仅是1个）。随后使用ESM3 7B生成256个样本，解码步骤为 L ，温度为0.7，生成的结果通过ESM3 pTM筛选和目视检查进行筛选。生成的结果如上述所述进行评估（ESMFold pTM 0.69，scTM均值0.97，标准差0.011）。生成的结构上是独特的：Foldseek搜索。

(41) 的 AlphaFold-DB、ESMAtlas 和 PDB 在 TM-align 模式下未能发现 TM-score 大于 .61 的匹配。

A.4. 对齐

A.4.1. 算法

自从引入 RLHF (44) 以来，已经开发出多个算法来调整通过无监督学习训练的大型模型，以更好地遵循指令，并普遍使其生成与用户偏好对齐 (45, 46, 100, 101)。我们使用 IRPO（迭代推理偏好优化），因为其实现简单且表现良好。IRPO 损失将监督微调与来自偏好对的对比学习相结合。IRPO 在一个数据集上操作 $\mathcal{D} \sim (y_w, y_l, x)$ ，该数据集由提示 x 和一对完成 y_w (优选) 和 y_l(非优选) 组成。它还在两个独立的模型上运行：参考模型 π_{ref} 和当前模型 π_θ。参考模型 π_{ref} 是相同规模的固定基本模型，而当前模型 π_θ 是正在优化的模型。

$$\mathcal{L}_{\text{IRPO}}(\pi_\theta; \pi_{\text{ref}}) = \mathcal{L}_{\text{NLL}} + \alpha\mathcal{L}_{\text{DPO}} = - \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\frac{\log \pi_\theta(y_w|x)}{|y_w|+|x|} + \alpha \log \sigma\left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right] \quad (2)$$

IRPO损失包含两个项。 $\mathcal{L}_{\text{NLL}}$ 该项最大化经过序列长度归一化的优选示例的对数似然，为模型提供强化良好生成的信号。该 $\mathcal{L}_{\text{DPO}}$ 项为对比优先调优项，它增加了优选与非优选示例之间对数似然的差异，同时保持与参考模型的接近程度 (45)。使用参考模型作为正则化器，防止对偏好数据集过拟合，而偏好数据集通常较小。有两个超参数， α和 β。 α 的权重控制监督损失与偏好损失的相对重要性，而 β 参数则控制我们与参考模型的接近程度：β越高，我们就越接近。我们通过当前模型参数最小化该损失。 θ.

ESM3是一个多模态模型，因此提示可以是（部分）序列、结构和功能的输入轨道的任意组合，而生成的y可以是任何输出轨道。在我们的实验中，我们始终生成氨基酸序列，因此从现在开始，这将是我们进行的示例。由于氨基酸序列 y 可以通过多种多步骤方式从提示 x 生成，因此计算完整的似然 $\pi(y|x)$ 将涉及对所有可能的多步骤解码路径进行积分。由于这一计算是不可行的，我们使用一个替代方法，反映了预训练，如公式（3）所示，下文进行描述。

$$\log \pi(y|x) \approx \mathbb{E}_m\left[\sum_{i\in m} \log p(y_i|y_{\setminus m}, x)\right] \quad (3)$$

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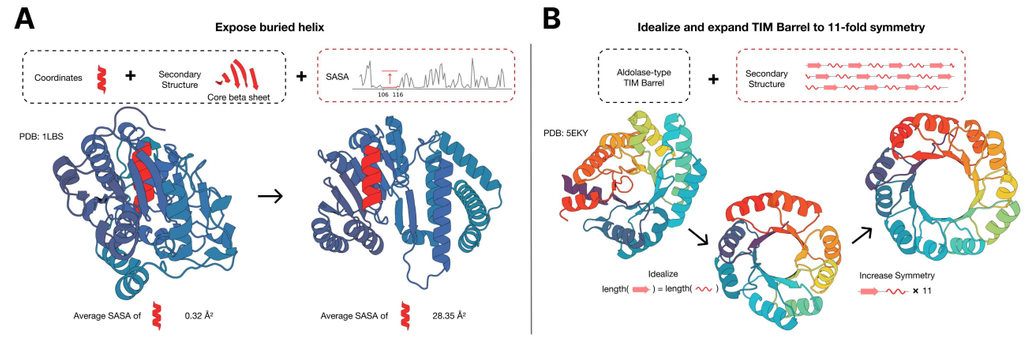

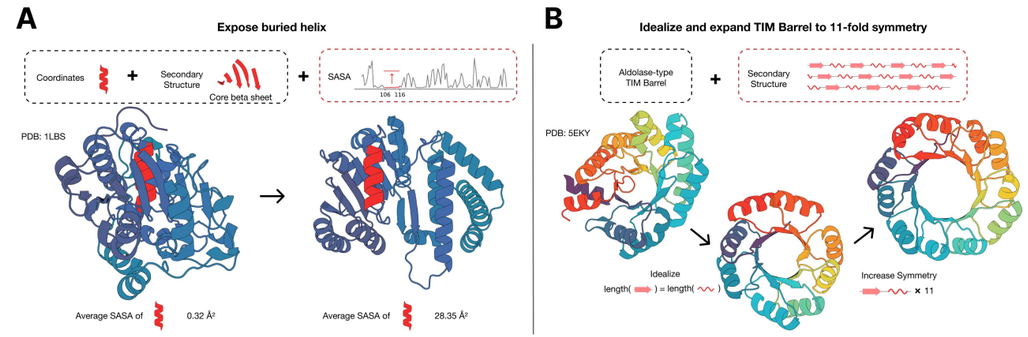

Figure S16. Multimodal protein editing with ESM3. (A) ESM3 exposes a buried helix in an protein while maintaining the alternating alpha-beta sandwich fold of the protein. (B) ESM3 is used in a two-step iterative edit, where first secondary structure prompting and function prompting are used to idealize a reference TIM barrel. Secondary structure prompting is then used to increase the number of subunits in the TIM barrel from 8 to 11.

图 S16. 使用 ESM3 的多模态蛋白质编辑。(A) ESM3 在保持蛋白质交替的阿尔法-贝塔三明治折叠结构的同时，暴露了蛋白质中一个埋藏的螺旋。(B) ESM3 在二步迭代编辑中使用，首先通过二级结构提示和功能提示对参考 TIM 桶进行理想化。然后使用二级结构提示将 TIM 桶中的亚单位数量从 8 增加到 11。

To approximate the likelihood of a generation y from prompt x, we mask y with a mask sampled from a linear noise schedule, prompt ESM3 with $\{y_{\setminus m}, x\}$, and compute the cross-entropy of ESM3 logits with the masked positions of y. During training, the same mask is used to compute the likelihoods for the reference policy vs current policy, as well as for the preferred sample vs non preferred sample.

为近似从提示x生成 y 的可能性，我们使用从线性噪声调度中抽样得到的掩码对 y 进行掩膜处理，使用提示ESM3与$\{y_{\setminus m}, x\}$，并计算掩膜位置的ESM3逻辑值的交叉熵。
y. 在训练期间，使用相同的掩码来计算参考策略与当前策略的可能性，以及首选样本与非首选样本的可能性。

A.4.2. Preference Tuning Intuition

Rearranging the DPO term of the loss function gives some insight into how it finetunes the model for the preference pairs.

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[-\log\sigma\left(-\beta z_\theta(x, y_l, y_w)\right)\right] \quad (4)$$

where

$$z_\theta(x, y_l, y_w) = \log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} - \log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} = \log\frac{\pi_{\text{ref}}(y_w|x)}{\pi_{\text{ref}}(y_l|x)} - \log\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)}$$

The function $f(z) = -\log\sigma(-\beta z) = \log(1+\exp(\beta z))$ is the softplus function, and is an approximation of the hinge function; in other words $f(z) = \beta z$ when $z >> 0$ and $f(z) = 0$ when $z \ll 0$. Because of this property, there are two cases. In the case where

$$\log\frac{\pi_{\text{ref}}(y_w|x)}{\pi_{\text{ref}}(y_l|x)} >> \log\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)} \quad (5)$$

$f(z)$ is in the linear regime, so the loss function is simply maximizing the likelihood ratio $\log\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)}$. In the case where

$$\log\frac{\pi_{\text{ref}}(y_w|x)}{\pi_{\text{ref}}(y_l|x)} \ll \log\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)} \quad (6)$$

the loss has saturated. This ensures that we do not deviate too far from the reference model.

These dynamics also hold true in the case of ESM3 fine-tuning. Although we use a surrogate instead of the true likelihood, the loss will increase the surrogate of the preferred pair over the non preferred pair until the current model deviates too much from the reference model.

A.4.3. Evaluation Metrics

Possibly the most important part of preference tuning is how the generations are separated into preference pairs. The desired objectives for a generation are quality and correctness. Quality refers to the viability of the sequence to be a stable protein. Correctness refers to the extent to which it follows the given prompt; also called prompt consistency. This section only deals with structure coordinate prompts, so prompt consistency can be measured via constrained site RMSD (cRMSD), which is the RMSD between the backbone coordinates of the prompt and the corresponding coordinates in the predicted structure of the generated sequence. Sequence quality can be measured via predicted-TM (pTM) of a structure predictor on the generated sequence.

As with any metric, especially one which is really a surrogate such as a structure predictor, there is a risk of over optimizing: the model keeps improving the specific metric e.g. in our case pTM but the actual property of interest, the viability of the sequence to be a stable protein, stops correlating with the metric (102). Using orthogonal models

A.4.2. 偏好调整直觉

重新排列损失函数中的 DPO 项为模型的偏好对提供了一些见解。

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[-\log\sigma\left(-\beta z_\theta(x, y_l, y_w)\right)\right] \quad (4)$$

其中

$$z_\theta(x, y_l, y_w) = \log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} - \log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} = \log\frac{\pi_{\text{ref}}(y_w|x)}{\pi_{\text{ref}}(y_l|x)} - \log\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)}$$

函数 $f(z) = -\log\sigma(-\beta z) = \log(1+\exp(\beta z))$ 是softplus函数，并且是铰链函数的近似；换句话说 $f(z) = \beta z$ 当 $z >> 0$ 和$f(z) = 0$ 当 $z \ll 0$。由于这一特性，有两种情况。在其中一种情况下

$$\log\frac{\pi_{\text{ref}}(y_w|x)}{\pi_{\text{ref}}(y_l|x)} >> \log\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)} \quad (5)$$

$f(z)$ 位于线性区域，因此损失函数仅仅是最大化似然比 $\log\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)}$。在此情况下

其中

$$\log\frac{\pi_{\text{ref}}(y_w|x)}{\pi_{\text{ref}}(y_l|x)} \ll \log\frac{\pi_\theta(y_w|x)}{\pi_\theta(y_l|x)} \quad (6)$$

损失已经饱和。这确保我们不会偏离参考模型过远。

这些动态在ESM3微调的情况下同样适用。尽管我们使用的是替代品而不是实际的似然，但损失将会增加优选对的替代品，相对于非优选对，直到当前模型过于偏离参考模型。

A.4.3. 评估指标

偏好调优中可能最重要的部分是如何将生成的结果分隔成优选对。生成的目标是质量和正确性。质量是指序列作为稳定蛋白质的可行性。正确性是指它遵循给定提示的程度；也称为提示一致性。本节仅涉及结构坐标提示，因此可以通过约束位点RMSD (cRMSD)来衡量提示一致性，cRMSD是提示的主链坐标与生成序列的预测结构中相应坐标之间的RMSD。序列质量可以通过生成序列的结构预测器的预测-TM (pTM) 来衡量。

与任何度量一样，尤其是作为结构预测器的替代指标，存在过度优化的风险：模型不断提高特定度量，例如在我们的案例中 pTM ，但实际关注的属性，即序列作为稳定蛋白质的可行性，停止与该度量相关 (102)。使用正交模型

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to rank our training dataset vs to perform evaluation helps mitigate this.

To create the training datasets, generations are evaluated according to backbone cRMSD and pTM of ESM3 7B to maintain a consistent structure predictor across all datasets. After the preference tuning phase, the generations from the tuned models are evaluated with ESMFold backbone cRMSD and pTM as an orthogonal model. Training on ESM3-scored preference pairs while evaluating using ESMFold provides a more objective evaluation.

A.4.4. Training Dataset

All ESM3 model scales are trained with the IRPO loss (Eq. (2)) on their respective preconstructed training datasets consisting of structure coordinate prompts and generations. The datasets have 16 generations each for 30,000 prompts from the respective ESM3 model. Preference selection is determined via a threshold of metrics. A sample is considered "good" if it has ESM3 7B $\text{pTM} > 0.8$ and backbone cRMSD to its structure prompt < 1.5Å.

Each "good" sample is paired with a "bad" sample to create a preference pair. We found that enforcing a gap between metrics of paired generations improves results, so to qualify as a "bad" sample the generation must have a delta $\text{pTM} = \text{pTM}_{\text{good}} - \text{pTM}_{\text{bad}} >= 0.2$ and delta backbone $\text{cRMSD} = \text{cRMSD}_{\text{good}} - \text{cRMSD}_{\text{bad}} < -2$Å. Each prompt can have multiple preference pairs, and prompts with no valid preference pair are discarded.

The structure prompts are composed of a variety of proteins adapted from our pretraining pipeline. 50% of the prompts are synthetic active sites, while the other 50% are structure coordinates randomly masked with a noise schedule. All of the structure prompts are derived from PDB structures with a temporal cutoff of before May 1st, 2020.

The synthetic active sites are derived by finding sequences from PDB with coordinating residues. For these structures, the amino acid identities are included in the prompt.

The remaining structure track prompts are masked according to a cosine noise schedule. 50% of the noise scheduled prompts are masked in completely random positions, and the other 50% are masked according to an autocorrelation mechanism that prefers sequentially masked positions.

Each model's training dataset consists of generations of its own reference model. For each prompt, we generate samples from the corresponding ESM3 model scale using iterative decoding with $L/4$ steps, where L is the length of the prompt. We anneal the temperature from 1.0 to 0.5 over the decoding steps.

A.4.5. Evaluation Dataset: Tertiary Coordination

Tertiary coordination tasks require the generation of proteins which satisfy challenging tertiary interaction constraints. The model is prompted with the sequence and coordinates of a set of residues which are near in 3D space, but distant in sequence. To evaluate performance on these tasks, we curate a dataset of 46 proteins with ligand binding sites from the Biolip dataset (99). All selected proteins were deposited in the PDB after the training set cutoff date (2020-12-01). The coordinating residues shown to the model are given by the ligand binding sites defined in the Biolip dataset (Table S13).

ESM3 is prompted with the sequence and coordinates of the residues for a particular ligand binding site. We ask ESM3 to generate novel structures by applying multiple transformations to the prompt. The total sequence length is sampled evenly to be 150, 250, or 350 residues (regardless of the original sequence length). Next, we define a contiguous span of coordinating residues to be prompt residues with fewer than 5 sequence positions between them. The order and the distance between contiguous spans of residues is shuffled. Together, this ensures that, for example, the original protein will no longer satisfy the prompt. We consider a generation a success if backbone $\text{cRMSD} < 1.5$Å and $\text{pTM} > 0.8$.

We construct a total of 1024 prompts for each ligand and generate a completion for each prompt with the model we are evaluating. We report Pass@128, which is an estimate for the fraction of ligands with at least one successful completion after 128 prompts per ligand. We estimate this using an unbiased estimator (Chen et al. (103), Page 3) using the success rate over 1024 prompts. We visualize randomly selected successful generations for both the base model and finetuned model in Fig. S18.

A.4.6. Supervised Finetuning

To judge the value of preference tuning, we also train a supervised finetuning (SFT) baseline where we finetune the model to increase likelihood of the high quality samples without the preference tuning loss. The 1.4B, 7B, and 98B models solve 14.2%, 33.7%, and 44.6% of tertiary coordination tasks at 128 generations, respectively, which improves upon the base models but is much lower than their corresponding preference tuned versions.

A.4.7. Training Hyperparameters

Each IRPO model is trained for 1000 steps using RMSProp. The learning rates are 1e-5, 1e-5, and 5e-6 for the 1.4B, 7B, and 98B, respectively, annealed using a cosine schedule after a 150 step warmup. Gradient norms are clipped to 1.0.

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对我们的训练数据集进行排序与执行评估有助于缓解这一问题。

为了创建训练数据集，生成物根据ESM3 7B的主链cRMSD和pTM进行评估，以维持所有数据集中一致的结构预测器。在偏好调优阶段之后，从调优模型中生成的物使用ESMFold主链cRMSD和pTM作为正交模型进行评估。使用ESM3评分的偏好对进行训练，同时使用ESMFold进行评估，提供了更为客观的评估。

A.4.4. 训练数据集

所有ESM3模型的规模均使用IRPO损失（公式（2））在各自预构建的训练数据集上进行训练，该数据集由结构坐标提示和生成物组成。每个数据集包含来自各自ESM3模型的30,000个提示的16个生成物。通过指标的阈值来确定偏好选择。如果样本的ESM3 7BpTM > 0.8 和主链cRMSD 与其结构提示 < 1.5 ˚A.被认为是“良好”的。

每个“好”样本与一个“坏”样本配对以创建偏好对。我们发现强制在配对生成的度量之间保持间隔可以提高结果，因此，要符合“坏”样本的标准，生成必须具有 delta pTM $= \text{pTM}^{\text{good}} - \text{pTM}_{\text{bad}} >= 0.2$ 和 delta backbone cRMSD $= \text{cRMSD}^{\text{good}} - \text{cRMSD}_{\text{bad}} < -2$ ˚A。每个提示可以具有多个偏好对，而没有有效偏好对的提示将被丢弃。

结构提示由我们预训练管道中改编的各种蛋白质构成。50% 的提示为合成活性位点，而另 50% 的提示为随机掩蔽的结构坐标，掩蔽方式采用噪声调度。所有结构提示均来源于在 2020年5月1日之前的 PDB 结构。

合成活性位点是通过从具有协调残基的 PDB 中找到序列而获得的。对于这些结构，氨基酸的身份也包含在提示中。

剩余的结构轨迹提示根据余弦噪声调度进行掩蔽。50%的噪声调度提示在完全随机的位置被掩蔽，另50%的提示根据优先选择顺序掩蔽的位置的自相关机制进行掩蔽。

每个模型的训练数据集由其自身参考模型的生成组成。对于每个提示，我们使用具有 $L/4$ 步的迭代解码从相应的ESM3模型规模生成样本，其中 L 是提示的长度。我们在解码步骤中将温度从1.0退火到0.5。

A.4. 5. 评估数据集：三级协调三级协调任务需要生成满足具有挑战性的三级相互作用约束的蛋白质。模型根据一组在三维空间中接近但在序列上相距较远的残基的序列和坐标进行提示。为了评估在这些任务上的表现，我们从Biolip数据集中挑选包含46个具有配体结合位点的蛋白质数据集（99）。所有选定的蛋白质在训练集截止日期（2020年12月1日）之后已被存放在PDB中。提供给模型的协调残基来自Biolip数据集中定义的配体结合位点（表S13）。ESM3根据特定配体结合位点的残基的序列和坐标进行提示。我们要求ESM3通过对提示应用多重变换生成新结构。总序列长度均匀地采样为150、250或350个残基（与原序列长度无关）。接下来，我们定义为提示残基的连续协同残基跨度，在其间少于5个序列位置。连续残基跨度的顺序和距离被打乱。这保证了例如，原始蛋白质将不再满足提示。我们认为生成成功如果主链 cRMSD < 1.5 ˚A和pTM > 0.8。我们为每个配体构建总共1024个提示，并使用我们正在评估的模型为每个提示生成一个完成结果。我们报告Pass@128，它是对每个配体在128个提示后至少有一个成功完成的配体比例的估计。我们使用无偏估计器（Chen et al.（103），第3页）根据1024个提示的成功率进行估计。我们在图S18中可视化随机选择的成功生成，适用于基础模型和微调模型。

A.4. 6. 监督微调为了评估偏好调优的价值，我们还训练了一个监督微调（SFT）的基线，在此基线上我们微调模型以提高高质量样本的似然性，而不使用偏好调优损失。1.4B、70亿和98B模型在128次生成中分别解决了14.2%、33.7%和44.6%的第三协调任务，这一结果相比基础模型有所改善，但仍远低于相应的偏好调优版本。

A.4. 7. 训练超参数每个IRPO模型使用RMSProp训练1000步。学习率分别为1e-5、1e-5和5e-6，适用于1.4B、70亿和98B模型，在经过150步预热后使用余弦调度进行逐步降低。梯度范数被剪切到1.0。

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PDB ID	Coordinating Residues	Ligand ID
7map	D25 G27 A28 D29 D30 G48 G49 V50	017
7n3u	I305 F310 V313 A326 K328 N376 C379 G382 D386 F433	05J
7exd	D103 I104 C107 T108 I174 H176 T182 W306 F309 E313 Y337	05X
8gxp	W317 C320 A321 H323 V376 F377 L396 I400 H479 Y502	06L
7n4z	M66 C67 R124 L130 C134 Y135 D152 F155	08N
7vrd	A40 S41 H161 Q169 E170 E213 D248 D324 K349 H377 R378 S379 K400	2PG
7zyk	V53 V66 V116 H160 N161 I174 D175	ADP
6yj7	K23 V24 A25 Y45 T46 A47 F115 I128	AMP
8ppb	H185 F198 K209 Q249 D250 L251 D262 K336 I415 D416	ATP
7knv	E33 F94 E95 D125	CA
7xer	Y466 L505 T525	CLR
7tj6	F366 G367 T378 R418	CMP
6xm7	H167 H218 H284 H476	CO
7bfr	Q62 X126 H248	CO3
6xlr	X272 Y495 H496 H581	CU
6tnh	N40 A41 S127 T128 Q187 L191 C201 T202 V236	DGP
7ndr	F73 S101 F102 D103 R106	EDO
8axy	H68 H109 E144	FE
7o6c	E62 E107 Q141	FE2
8aul	P31 M32 T33 Q106 H185 R237 S319 G320 G321 G342 R343 F369 Y370	FMN
7vcp	N37 D38 Q54 F97 S98 R159 D160 E214 Y276 W297	FRU
7b7f	G167 T168 G189 W195	FUC
8d0w	F73 L136 E137 F329	GAL
7yua	T13 T14 I15 D40 H85 S86 D87 D110 N290	GDP
7w1a	L44 Y88 L91 I212	GMP
7ljn	G71 S72 D91 K236 S253 V254 D309 R310	GTP
6s4f	Y84 N87 K88 V131 Q132 L133 D155 F157 I276 P309 G310 G313 P314 V317	KUN
7mg7	Y12 G98 L99 Y100 A207 D208 G227 R228	MAN
7qow	D12 T118 E268	MG
7dmm	E181 E217 D245 D287	MN
7qoz	G11 G12 I13 Y34 D35 V36 A86 G87 V126 T127 N128 H185 M235	NAD
7v2r	G89 F93 K98 F101 E121 Y204 E209 F229	NAI
7a7b	F51 Y128 K165 N166 S167 Y186 R187 I248 G249 A299	NAP
7pae	M20 L22 L38 V49 I53 C56 K57 R61 Q78 V80 W90 I109 M117 I129 L147 Y149	O7T
8egy	H82 K83 S186 G230 S231 N232 E345 S368 G369	PLP
7qow	S65 R129 D273 H465	PO4
7wmk	E77 L124 R129 S174 T189 Q191 W241 D304 E306 K349 D410 W411 Y486	PQQ
7pl9	D607 A608 Y637 M638 Y705 G706 M735 K736	RET
7yf2	G153 E174 L175 L209 N210 L211 Y295	SAH
7v6j	G207 D230 L231 D250 M251 K264	SAM
7ys6	D106 C110 N288	SRO
6w8m	A22 A23 G70 S110 T111 G112 V113 Y114	TJY
8g27	S258 D294 K435 R717	UDP
7xyk	R24 C170 R190 S191 D193 N201 H231 Y233	UMP
8g3s	H224 F228 V249 M250 V253 R263 T266 L267 F270	YLT
8it9	T92 P93 R96 Y108 L109 K216 V228 S229 H231 H232	ZL6

Table S13. Tertiary coordination dataset. Selected PDBs and coordinating residues (along with binding ligand) for each protein sample in the tertiary coordination dataset.

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PDB ID	协调残基	配体 ID
7map	D25 G27 A28 D29 D30 G48 G49 V50	017
7n3u	I305 F310 V313 A326 K328 N376 C379 G382 D386 F433	05J
7exd	D103 I104 C107 T108 I174 H176 T182 W306 F309 E313 Y337	05X
8gxp	W317 C320 A321 H323 V376 F377 序列长度396 I400 H479 Y502	06L
7n4z	M66 C67 R124 序列长度130 C134 Y135 D152 F155	08N
7vrd	A40 S41 H161 Q169 E170 E213 D248 D324 K349 H377 R378 S379 K400	2PG
7zyk	V53 V66 V116 H160 N161 I174 D175	ADP
6yj7	K23 V24 A25 Y45 T46 A47 F115 I128	AMP
8ppb	H185 F198 K209 Q249 D250 序列长度 L251 D262 K336 I415 D416	ATP
7knv	E33 F94 E95 D125	CA
7xer	Y466 L505 T525	CLR
7tj6	F366 G367 T378 R418	CMP
6xm7	H167 H218 H284 H476	CO
70亿fr	Q62 X126 H248	CO3
6xlr	X272 Y495 H496 H581	CU
6tnh	N40 A41 S127 T128 Q187 序列长度191 C201 T202 V236	DGP
7ndr	F73 S101 F102 D103 R106	EDO
8axy	H68 H109 E144	FE
7o6c	E62 E107 Q141	FE2
8aul	P31 M32 T33 Q106 H185 R237 S319 G320 G321 G342 R343 F369 Y370	FMN
7vcp	N37 D38 Q54 函数97 S98 R159 D160 E214 Y276 W297	FRU
70亿7f	G167 T168 G189 W195	FUC
8d0w	F73 序列长度136 E137 F329	GAL
7yua	T13 T14 I15 D40 H85 S86 D87 D110 N290	GDP
7w1a	L44 Y88 L91 I212	GMP
7ljn	G71 S72 D91 K236 S253 V254 D309 R310	GTP
6s4f	Y84 N87 K88 V131 Q132 序列长度133 D155 f157 I276 P309 G310 G313 P314 V317	KUN
7mg7	Y12 G98 序列长度99 Y100 A207 D208 G227 R228	MAN
7qow	D12 T118 E268	MG
7dmm	E181 E217 D245 D287	MN
7qoz	G11 G12 I13 Y34 D35 V36 A86 G87 V126 T127 N128 H185 M235	NAD
7v2r	G89 F93 K98 F101 E121 Y204 E209 F229	NAI
7a7b	F51 Y128 K165 N166 S167 Y186 R187 I248 G249 A299	NAP
7pae	M20 序列长度22 序列长度38 V49 I53 C56 K57 R61 Q78 V80 W90 I109 M117 I129 序列长度147 Y149	O7T
8egy	H82 K83 S186 G230 S231 N232 E345 S368 G369	PLP
7qow	S65 R129 D273 H465	PO4
7wmk	E77 序列长度124 R129 S174 T189 Q191 W241 D304 E306 K349 D410 W411 Y486	PQQ
7pl9	D607 A608 Y637 M638 Y705 G706 M735 K736	RET
7yf2	G153 E174 序列长度 L209 N210 L211 Y295	SAH
7v6j	G207 D230 序列长度 D250 M251 K264	SAM
7ys6	D106 C110 N288	SRO
6w8m	A22 A23 G70 S110 T111 G112 V113 Y114	TJY
8g27	S258 D294 K435 R717	用户数据报协议
7xyk	R24 C170 R190 S191 D193 N201 H231 Y233	UMP
8g3s	H224 F228 V249 M250 V253 R263 T266 序列长度267 F270	YLT
8it9	T92 P93 R96 Y108 序列长度109 K216 V228 S229 H231 H232	ZL6

表 S13. 三级协调数据集。每个蛋白质样本的已选 PDB 和协调残基（连同结合配体）在三级协调数据集中。

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For all IRPO runs $\beta = 0.05$ and $\alpha = 0.8$. The SFT baseline uses the same hyperparameters, but with $\alpha = 0.0$ to disregard the preference tuning term.

A.5. GFP

ESM3 generates a dim distant GFP, B8, and a bright distant protein, esmGFP. Details are provided below on computational methods, experimental protocols, results, and post-experiment analyses.

A.5.1. Generation and Selection

The base ESM3 7B model generates candidate GFP designs for laboratory testing using a single prompt and a chain of thought over sequence and structure tokens. Candidates are filtered and ranked by metrics at several steps in the process. Experiment 1 tests candidates across a range of sequence identity to a template, yielding multiple GFPs including dim hit B8. Experiment 2 consists of designs starting a chain of thought from the sequence of B8, yielding numerous bright GFPs including C10 which we term esmGFP. This section details the computational protocol that generated and selected candidate GFP designs for Experiments 1 and 2, shown in Fig. 4B. Protocols, metrics, and selection conventions are separately introduced and then synthesized in descriptions of the two experiments, at the end of the section.

A.5.1.1. Model

All candidate GFP designs were created using the base ESM3 7B model with no finetuning. Throughout generation, the model is prevented from decoding cysteine residues.

A.5.1.2. Prompt

All candidate GFP designs in Experiment 1 are produced with a chain of thought beginning from a single prompt. The goal of the prompt is to capture essential residue identities and structural features needed for chromophore formation and fluorescence, leaving other degrees of freedom open for the model to generate diverse designs.

Template To this end, we prompt ESM3 with a minimal set of sequence and structure information from 16 residues near the chromophore formation site from a template protein. We select a pre-cyclized intermediate crystal structure from (53), PDB ID 1QY3, as our template. We reverse the chromophore maturation slowing mutation R96A in 1QY3 so the prompt contains Arg96. We subsequently refer to the full sequence and structure of 1QY3 with mutation A96R as 1QY3 A96R or the template.

Sequence prompt The sequence portion of our prompt consists of 7 template residues: Met1, Thr62, Thr65, Tyr66, Gly67, Arg96, and Glu222. Residues 65-67 form the chromophore. Met1 ensures proper start codon placement. Residues 62, 96, and 222 are described in (53) and other works to have key catalytic roles in chromophore formation.

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对于所有的IRPO运行 $\beta = 0.05$ 和{name}。SFT基线使用相同的超参数，但$\alpha = 0.0$ 以忽略偏好调优项。

A.5. 绿荧光蛋白

ESM3生成了一个暗淡的远程GFP，B8，以及一个明亮的远程蛋白质，esmGFP。以下提供了关于计算方法、实验协议、结果和实验后分析的详细信息。

A.5.1. 生成和选择

基础的ESM3 7B模型利用单个提示及通过序列和结构标记的思维链生成用于实验室测试的候选GFP设计。候选者在多个步骤中通过指标进行筛选和排序。实验 1 测试候选者在与模板的序列一致性范围内，产生多个GFP，包括暗淡的命中B8。实验 2包含以B8的序列开始思维链的设计，产生多个明亮的GFP，包括我们称之为esmGFP的C10。本节详细阐述了生成和选择实验1和实验2候选GFP设计的计算协议，如图。4B. 协议、指标和选择惯例将单独介绍，然后在对两项实验的描述中进行综合，位于本节的末尾。

A.5.1.1. 模型

所有候选 GFP 设计均使用基础的 ESM3 7B 模型创建，无需微调。在生成过程中，模型被防止解码半胱氨酸残基。

A.5.1.2. P提示

在实验 1 中，所有候选 GFP 设计都是以由单一提示开始的思考链产生的。提示的目标是捕捉颜色基团形成和荧光所需的基本残基身份和结构特征，同时为模型生成多样化设计留下其他自由度。模板为此，我们使用从模板蛋白中提取的16个残基在色团形成位点附近的最少序列和结构信息对 ESM3 提出提示。我们选择来自 (53) 的预环化中间晶体结构，PDB ID 1QY3，作为我们的模板。我们在 1QY3 中反转色团成熟减缓突变 R96A，使提示包含 Arg96。随后，我们将带有突变 A96R 的 1QY3 的完整序列和结构称为 1QY3 A96R 或模板。序列提示我们提示的序列部分由7个模板残基组成：Met1、Thr62、Thr65、Tyr66、Gly67、Arg96 和 Glu222。残基 65-67 形成色团。Met1 确保适当的起始密码子位置。残基 62、96 和 222 在 (53) 和其他文献中被描述为在色团形成中发挥关键催化作用。

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Simulating 500 million years of evolution with a language model

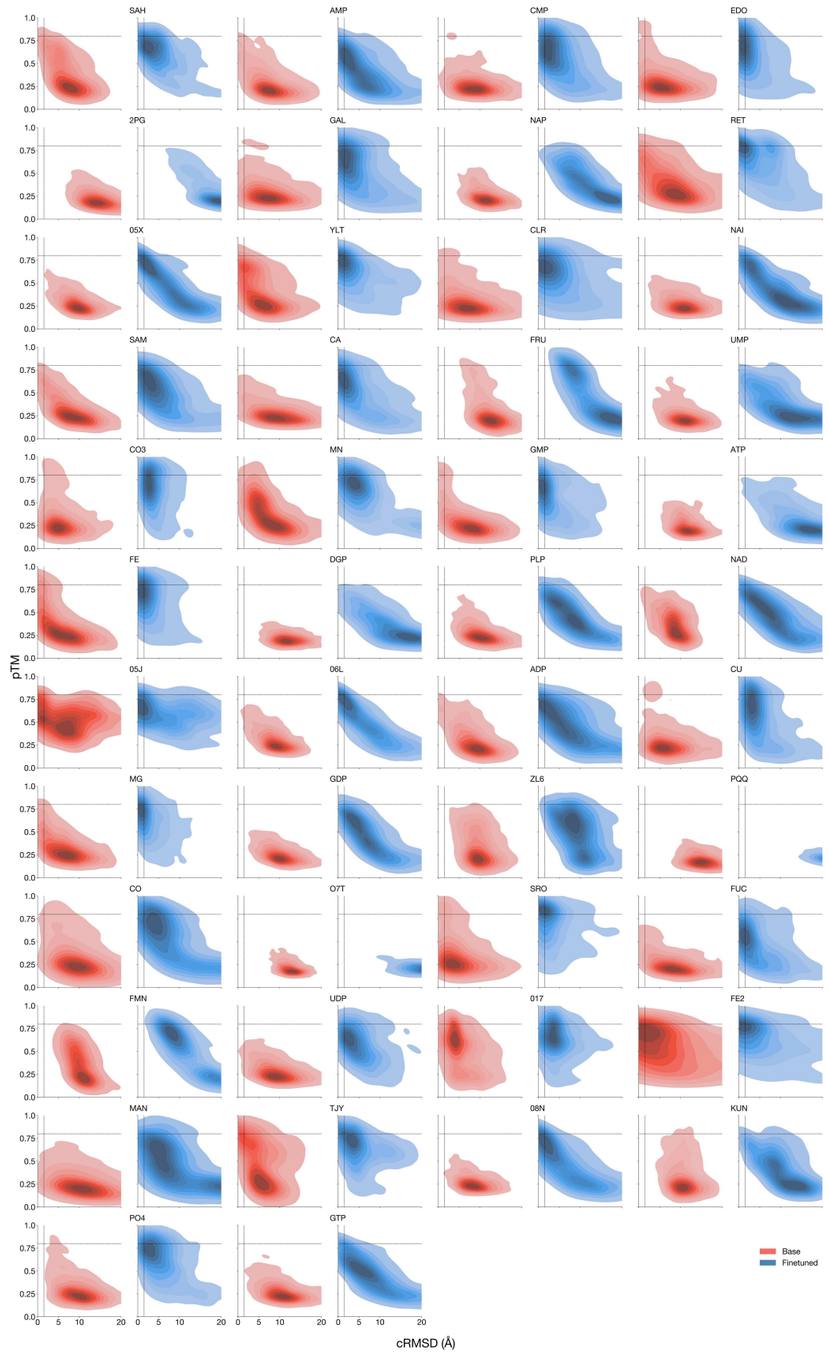

Figure S17. Alignment improves model generations. pTM, backbone cRMSD distributions of generations from the 98B base model and aligned model for all ligands in the tertiary coordination dataset. Each ligand/model pair has 1024 generations.

用语言模型模拟五亿年的进化

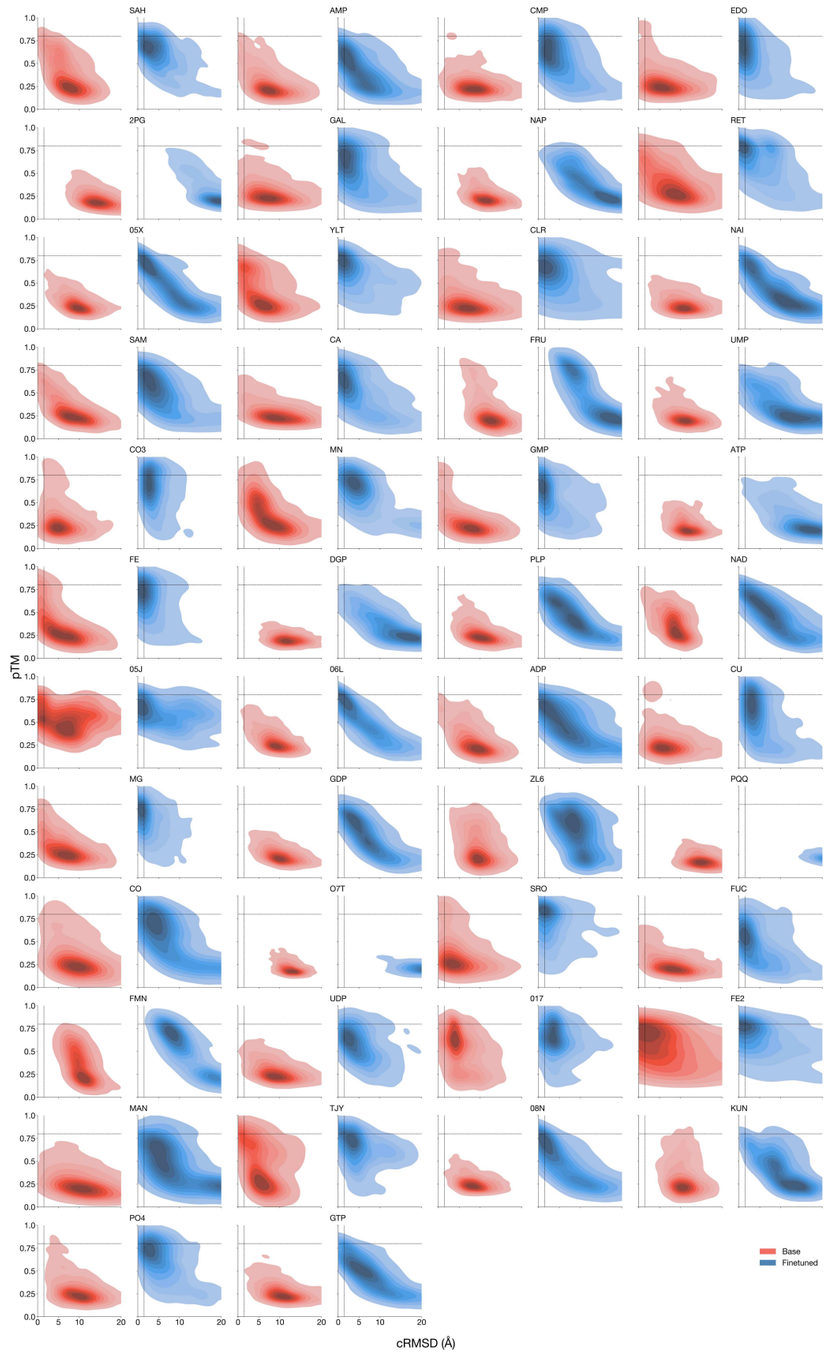

图 S17. 对齐改善模型生成。来自98B基础模型和对齐模型的pTM、主链cRMSD分布，用于三级协调数据集中所有配体的生成。每个配体/模型对拥有1024个生成。

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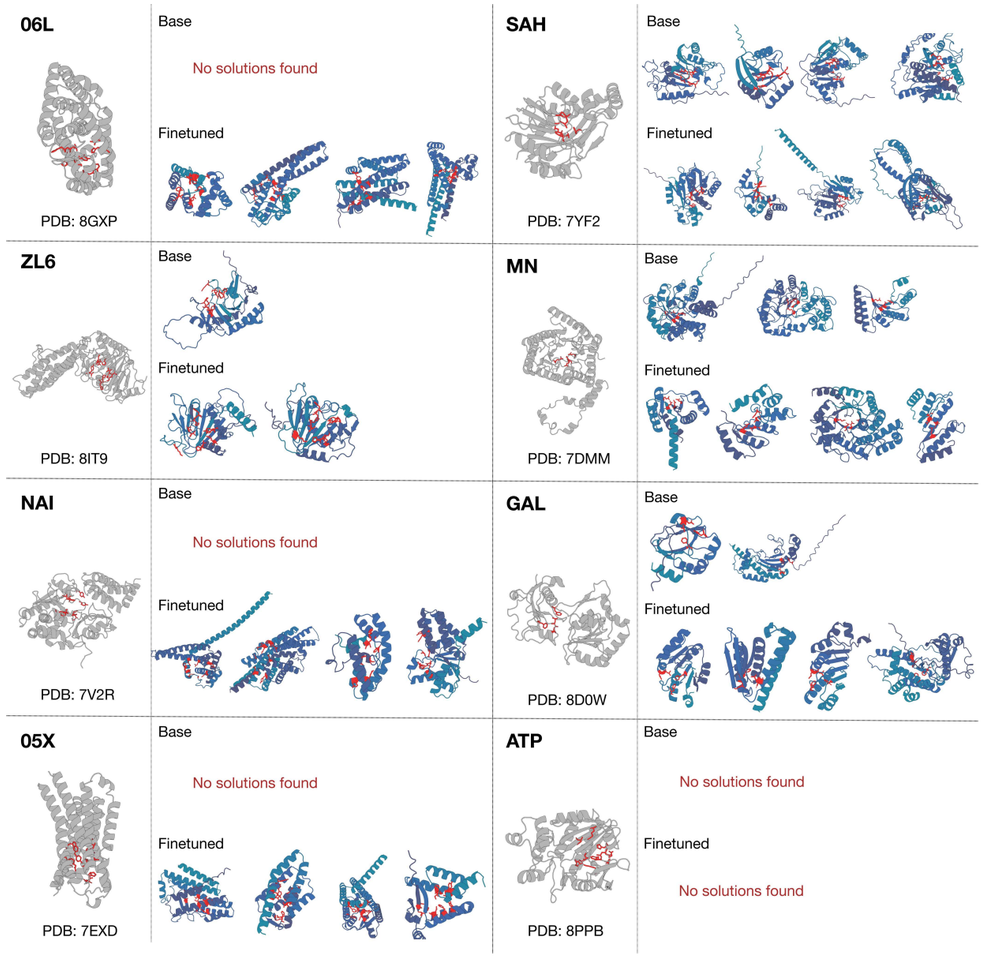

Figure S18. Randomly selected successful generations from the base and finetuned 98B parameter models. A random sample of ligands is selected and visualized with the ground truth PDB chain from which the ligand was taken. Solutions produced by ESM3 are diverse, and the finetuned model gives significantly more successes (out of 1024 total samples).

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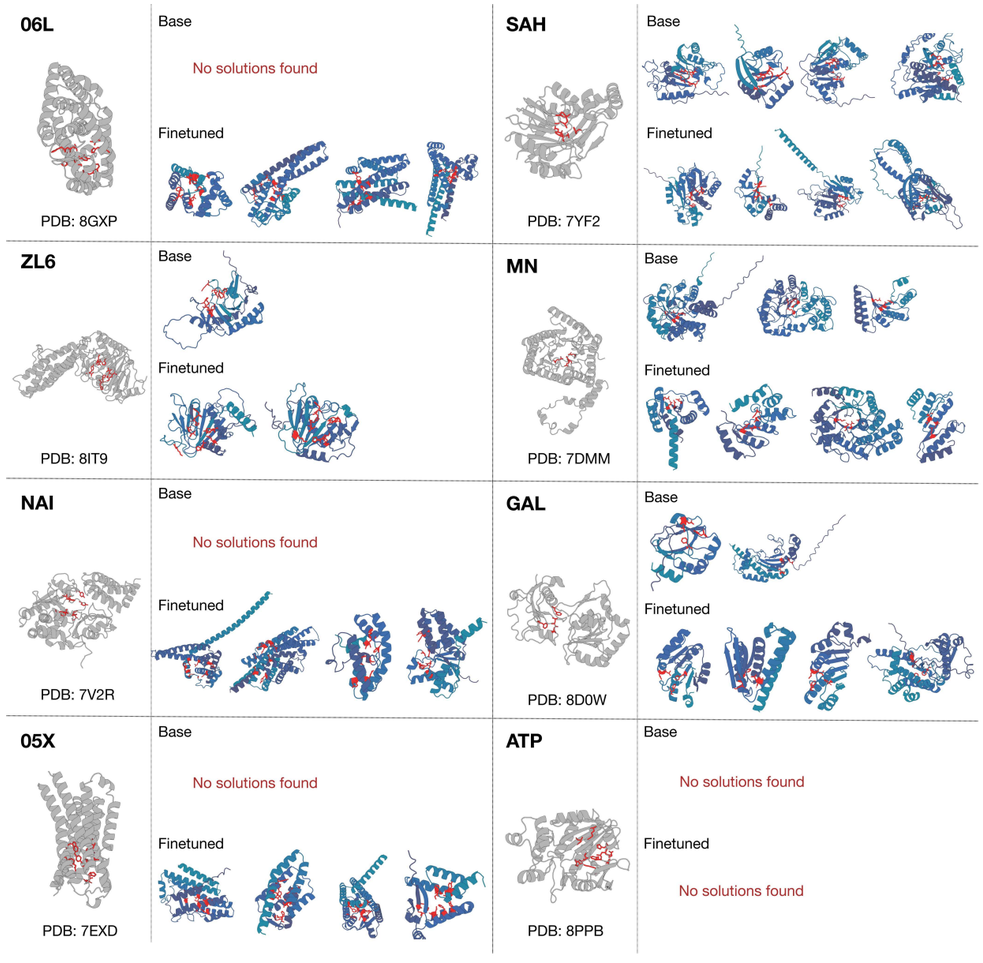

图 S18. 从基础和微调的98B参数模型随机选择的成功生成。选择随机样本的配体，并用真实值PDB链进行可视化，该链是配体所取自的。ESM3产生的解决方案多样化，微调模型在1024个总样本中显著提供了更多成功。

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Structure prompt The structure portion of our prompt consists of structure tokens and backbone atomic coordinates taken from 16 template residues at positions 96, 222, and 58-71 (inclusive) which roughly captures the central alpha helix. The unique geometry of the central alpha helix is known to be crucial for chromophore formation (53).

All other positions and tracks in the prompt are masked. The overall prompt length is 229, matching that of the template. Residue indices are contiguous and begin from 1.

A.5.1.3. Joint Sequence Structure Optimization

We employ the following procedure to jointly optimize the sequence and structure of designs throughout our experiments: While annealing temperature linearly from 1 to 0, we perform multiple iterations of first predicting the structure of a designed sequence and subsequently Gibbs sampling each position in the sequence for that predicted structure. In algorithmic form:

Algorithm 15 `gibbs_seq_given_struct`

Input: ESM3 f, sequence $x \in \{0..20\}^L$, structure y, temperature t

for $i = \text{shuffle}(\{1, ..., L\})$ **do**
  $x_i \sim \exp\left(\log f(x_i \mid x_{\setminus i}, y)/t\right)$
end for
return x

Algorithm 16 `joint_optimize`

Input: ESM3 f, initial sequence x_1, iterations I, initial temperature t_1, final temperature t_f

for $i = 1, \ldots, I$ **do**
  $t_i = (t_f - t_1) \cdot (i/(I-1)) + t_1$
  $y_i =$ `generate`$_{struct}(f, x_i, len(x_i), T = 0)$
  $x_{i+1} =$ `gibbs_seq_given_struct`(f, x_i, y_i, t_i)
end for
return x_{I+1}

Three variants of `gibbs_seq_given_struct` in `joint_optimize` were employed for Experiments 1 and 2. Joint optimization occasionally produces repetitive spans of amino acids when temperature is annealed to low values. Variant 1 and 2 are intended to address this, in differing ways. Variant 3 is an experiment in biasing the logits with a PSSM of known natural GFPs. Half of the candidates in Experiment 2 were produced using Variant 3. This half did not include esmGFP.

1. **Variant 1: Negative Local Sequence Guidance** We bias the logits of the model away from those produced just from a highly local span of the sequence. Specifically, we use classifier-free guidance (104):

$$\text{logits}' = \text{weight} * (\text{logits}_{\text{cond}} - \text{logits}_{\text{uncond}}) + \text{logits}_{\text{uncond}}$$

but push away from the logits produced by inputting just 7 residues centered on the position being sampled, with weight 2 and nothing else. All other sequence positions and all other model inputs are left blank.

$$\text{logits}' = 2 * (\text{logits}_{\text{cond}} - \text{logits}_{\text{local_seq}}) + \text{logits}_{\text{local_seq}}$$

2. **Variant 2: Max Decoding Entropy Threshold** We optionally skip resampling of sequence during Gibbs sampling at positions whose entropy over sequence tokens exceeds a user specified threshold.

3. **Variant 3: PSSM Bias** In Experiment 2 only, we experiment with both including and excluding a PSSM-based bias during Gibbs sequence sampling. Specifically, we add a PSSM constructed from 71 natural GFPs (see Appendix A.5.1.4 for details) directly to the sequence output logits of the model, with a user-specific weight. esmGFP did not use this option; it was produced with weight 0.

A.5.1.4. Metrics

GFP designs are produced and scored by a number of ESM3-derived and independent metrics. Unless otherwise noted, designed structures are predicted using ESM3 with only sequence as input, using iterative decoding of structure tokens with temperature 0 and subsequent decoding of backbone coordinates with an older version of the structure token decoder.

The following is an exhaustive list of metrics used. An exact break down of where and how specific metrics are used can be found in Appendix A.5.1.5, Appendix A.5.1.6 and Appendix A.5.1.7.

Template Chromophore Site RMSD is calculated via an optimal alignment (105) of N, C, CA, and inferred CB atoms at positions 62, 65, 66, 67, 96, and 222 in the predicted structure of a design and the template (crystal) structure.

Template Helix RMSD is calculated in the same way, but for N, C, CA atoms only, at design and template positions 58-71 (inclusive).

1EMA Helix RMSD is a metric proposed in (106). An RMSD is calculated between alpha helix residues in the predicted designed structure and a specific crystal structure of avGFP, PDB ID `1EMA`. Our calculation differs slightly from (106). We calculate RMSD for

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结构提示我们提示的结构部分由结构标记和来自位置96、222以及58-71（包括）的16个模板残基的主链原子坐标组成，这大致捕捉了中心α螺旋。已知中心α螺旋的独特几何形状对色团形成至关重要（53）.

提示中所有其他位置和轨迹均被屏蔽。整体提示长度为229，与模板一致。残基索引连续，并从1开始。

A.5.1.3. 联合序列结构优化

我们采用以下程序在实验过程中联合优化设计的序列和结构：在温度从1线性退火到0的过程中，我们进行多次迭代，首先预测设计序列的结构，然后针对该预测结构对序列中的每个位置进行吉布斯采样。其算法形式为：

算法15吉布斯 `gibbs_seq_given_struct`

输入：ESM3 f，序列 $x \in \{0..20\}^L$，结构 y，温度 t 1: 对 $i = \text{shuffle}(\{1, ..., L\})$ 做 2: $x_i \sim \exp\left(\log f(x_i \mid x_{\setminus i}, y)/t\right)$3: 结束循环 4: 返回 x

算法16联合_1 `joint_optimize`

输入：ESM3 f，初始序列 x_1，迭代 I，初始温度 t_1，最终温度 t_f1: 对于 $i = 1$，…，I做2: $t_i = (t_f - t_1) \cdot (i/(I-1)) + t_1$3: $y_i =$ 生成 $_{struct}(f, x_i, len(x_i)$ $T = 0$)4: $x_{i+1} =$ 吉布斯_序列_给定_结构 (f, x_i, y_i, t_i) 5: 结束循环6: 返回 x_{I+1}

在联合_优化的实验中，采用了吉布斯_序列_给定_结构的三种变体用于实验1和实验2。联合优化在温度回火至低值时，偶尔会产生重复的氨基酸序列。变体1和变体2旨在通过不同方式解决此问题。变体3则是通过已知的自然绿荧光蛋白的PSSM来偏置逻辑值的实验。实验2中的一半候选者是使用变体3生成的，而这一半不包括esmGFP。

1. 变体1：负局部序列引导 我们将模型的逻辑值偏离那些产生的值仅从序列的高度局部范围。具体来说，我们使用无分类器引导（104）：

$$\text{logits}' = \text{weight} * (\text{logits}_{\text{cond}} - \text{logits}_{\text{uncond}}) + \text{logits}_{\text{uncond}}$$

但是，对于通过输入仅以被采样位置为中心的 7 个残基，权重为 2 而不使用其他任何内容所产生的逻辑值，进行推开处理。所有其他序列位置和所有其他模型输入均留为空白。

$$\text{logits}' = 2 * (\text{logits}_{\text{cond}} - \text{logits}_{\text{local_seq}}) + \text{logits}_{\text{local_seq}}$$

2. 变体 2：最大解码熵阈值 我们可选地跳过吉布斯采样中在其序列标记的熵超过用户指定阈值的位置的序列重采样。

3. 变体 3：PSSM 偏置在实验 2 中，我们实验性地包括和排除在吉布斯序列采样期间基于 PSSM 的偏置。具体来说，我们从 71 个自然绿荧光蛋白中构建了一个 PSSM（详见附录 A.5.1.4），并直接将其添加到模型的序列输出逻辑值中，使用用户指定的权重。 esmGFP 不使用此选项；它的权重为 0。

A.5.1.4. 度量

GFP 设计是通过多个 ESM3 派生和独立度量生成和评分的。除非另行说明，否则设计的结构是使用 ESM3 仅以序列作为输入进行预测的，采用温度为 0 的结构标记的迭代解码，以及使用较旧版本的结构标记解码器对主链坐标进行后续解码。

以下是使用的指标的详尽清单。有关特定指标使用的确切细分，可以在附录A.5.1.5、附录 A.5.1.6 和附录 A.5.1.7 中找到。

模板色素位点RMSD 通过对位于预测结构和模板（晶体）结构中的位置62、65、66、67、96和222的N、C、CA和推断的CB原子的最佳对齐（105）进行计算。

模板螺旋RMSD以相同的方式计算，但仅限于设计和模板位置58-71（含）。

1EMA螺旋RMSD 是（106）中提出的一个指标。RMSD是在预测设计结构和avGFP的特定晶体结构之间的α螺旋残基之间计算的，PDB ID`1EMA`。我们的计算与（106）略有不同。我们计算RMSD为

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N, C, CA and inferred O atoms, and consider only positions 60-64 and 68-74 (both ranges inclusive) to exclude chromophore positions 65-67.

Sequence Pseudo-perplexity is calculated as defined in (107). Given a protein sequence, positions are masked one at a time, negative log-likelihoods of input tokens at masked positions are averaged across all positions in the sequence, and the result is exponentiated.

Round-trip Perplexity is calculated for a designed sequence via predicting its structure with ESM3, and then evaluating the perplexity of the sequence given that predicted structure under a single forward pass of ESM3.

N-gram Score is calculated as the E_{ngram} term defined in (11). This score assesses the divergence between the N-gram frequencies of residues in the designed sequence and those found in a background distribution, derived from UniRef50 2018_03. Specifically, for a function ngram_i that takes in a sequence x and an N-gram order i, and a precomputed distribuion of background N-gram frequencies $\text{ngram}_{i,bg}$, the score is calculated as:

$$E_{\text{ngram}} = \sum_{i \in \{1,2,3\}} D_{KL}(\text{ngram}_i(x), \text{ngram}_{i,\text{bg}}) \quad (7)$$

PSSM A position-specific scoring matrix (PSSM) is constructed from a MSA of 71 natural GFPs (108). Specifically, at positions aligned to our template, frequencies for the 20 canonical amino acids (excluding gaps) are transformed to log odds via dividing by the uniform background ($p(aa) = 0.05$), adding an epsilon of 1e-9, and applying log base 2. This produces a matrix of scores of size 229 x 20.

PSSM score We extract from the PSSM values at (position, amino acid) pairs occurring in an input sequence. These are averaged to produce a score.

N-terminus Coil Count is metric intended to measure structural disorder at the N-terminus of a design. We observed that predicted structures have various levels of disorder in this region. To quantify it for possible filtering, we apply mkdssp (82) to the ESM3-predicted structure of a design, and record how many of the first 12 positions are reported as having SS8 labels in {S,T,C}.

A.5.1.5. Selection Criteria

Among Experiment 1 and 2, designs are selected for testing by first applying a set of filters, and then selecting the top-N designs according to a score-based ranking. Scores are calculated by summing the values of several metrics, which are each normalized across designs to have zero mean and unit variance and which are negated when appropriate so that lower values are always better.

Common Filters The following filters are applied in both Experiments 1 and 2.

- Template Chromophore Site RMSD <1.5Å
- Template Helix RMSD <1.5Å
- N-gram Score <5

Common Score Terms The following score terms are used in both Experiments 1 and 2.

- Sequence Pseudo-perplexity
- Round-trip Perplexity
- ESM3 pTM

A.5.1.6. Generation and Selection of Designs for Experiment 1

In this experiment, we generate a set of GFP designs for experimental testing with a range of sequence identities to our template. Designs are generated by a chain of thought: from the prompt, ESM3 decodes all masked structure tokens, then all masked sequence tokens. Lastly, sequence and structure tokens are jointly optimized.

Initial Generation Starting from the prompt, we first generate 38k structures by decoding masked structure tokens one at a time using a fixed temperature sampled uniformly from the range (0, 1.25) for each generation. To focus compute on the most promising structures, we filter according to Template Chromophore Site RMSD <1Å, yielding 24k selected structures. We next generate ≈ 4 sequences for each structure with a temperature uniformly sampled from the range (0, 0.6), yielding 92k total sequences.

Selection We select a subset of promising initial generations for further optimization by applying Common Filters with N-gram score's threshold modified to <5.5, ranking designs according to {Common Score Terms, mean ESM3 pLDDT, mean ESMFold pLDDT, and ESMFold pTM}, and selecting the best 40 designs in each interval of 0.1 sequence identity to the template sequence in [0.2, 1.0], 320 in total.

Joint Sequence Structure Optimization We then jointly optimize the sequence and structure of designs. Using 30 iterations in each case, we run 5 seeds of optimization with max decoding entropy threshold = 1.5 and 2 seeds of optimization with negative local sequence guidance = 2.0, yielding 67k total designs. Designs from every iteration are included in this pool.

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N、C、CA 和推断的 O 原子，仅考虑位置 60-64 和 68-74（两个范围均包括在内），以排除发色团位置 65-67。

序列伪困惑度的计算方法如定义所示

(107)。给定一个蛋白质序列，逐个位置进行掩蔽，掩蔽位置的输入标记的负对数似然值在序列的所有位置上进行平均，结果被指数化。往返困惑度 是通过预测其结构并使用 ESM3 评估该序列在给定预测结构下的困惑度而计算得出的，在 ESM3 的单次前向传递过程中进行评估。N-gram得分 的计算方法为 E_{ngram} 中定义的项

(11)。该得分评估设计序列中残基的 N-gram 频率与背景分布中发现的频率之间的差异，该背景分布源自 UniRef50 2018 03。具体而言，对于函数 ngram_i，该函数接收一个序列 x 和一个 N-gram 顺序 i 以及一个预先计算的背景 N-gram 频率分布 $\text{ngram}_{i,\ bg}$，得分计算如下：

$$E_{\text{ngram}} = \sum_{i \in \{1,2,3\}} D_{KL}(\text{ngram}_i(x), \text{ngram}_{i,\text{bg}}) \quad (7)$$

PSSM 位置特异性得分矩阵（PSSM）是通过对71个自然绿荧光蛋白（108）的MSA构建而成的。具体而言，在与我们的模板对齐的位置上，20种经典氨基酸（不包括缺口）的频率通过除以均匀背景（$p(aa) = 0.05$）转换为对数赔率，并添加1e-9的epsilon，然后应用以2为底的对数。这将产生一个大小为229 x 20的得分矩阵。

PSSM得分 我们提取输入序列中（位置，氨基酸）对的PSSM值。这些值被平均以生成一个得分。

N末端卷曲计数 是一个旨在测量设计N末端结构无序程度的指标。我们观察到预测结构在此区域具有不同程度的无序性。为了对其进行量化以便进行可能的过滤，我们将mkdssp（82）应用于设计的ESM3预测结构，并记录前12个位置中有多少被报告为具有SS8标签在{S,T,C}。

A.5.1.5. 选择标准

在实验 1 和实验 2 之间，通过首先应用一组过滤器来选择设计进行测试，然后根据基于得分的排名选择前 N 个设计。得分通过对几个指标的数值进行求和计算，这些指标在各设计之间进行归一化处理，使其具有零均值和单位方差，并在适当时进行否定，以确保较低的值总是更好。

常见过滤器 在实验1和实验2中应用了以下过滤器。

- 模板色素位点RMSD <1.5 ˚A
- 模板螺旋RMSD <1.5 ˚A
- N-gram得分 <5

常见得分术语 在实验1和实验2中使用了以下得分术语。

- 序列伪困惑度
- 往返困惑度
- ESM3 pTM

A.5.1.6. 实验 1 的设计生成和选择

在本实验中，我们生成了一组针对实验测试的 GFP 设计，其序列与模板的相似度范围不同。这些设计是通过一系列思维过程生成的：从提示开始，ESM3 解码所有被遮蔽的结构标记，然后解码所有被遮蔽的序列标记。最后，序列和结构标记被联合优化。

初始生成 从提示开始，我们首先通过逐一解码遮蔽的结构标记生成 38,000 个结构，使用从 (0, 1.25) 范围内均匀采样的固定温度进行每次生成。为了将计算集中在最有前景的结构上，我们根据模板色素位点RMSD <1 ˚A 进行过滤，得到 24,000 个选定结构。接下来，我们为每个结构生成 ≈ 4 序列，使用从 (0, 0.6) 范围内均匀采样的温度，最终生成 92,000 个总序列。

选择我们选择一组有前景的初始生成进行进一步优化，通过应用常见过滤器，N-gram得分的阈值调整为 <5.5，按照 {常见得分术语、平均ESM3 pLDDT、平均ESMFold pLDDT和ESMFold pTM}进行设计排名，并选择每个0.1序列一致性区间内的最佳40个设计，针对模板序列在 [0.2, 总共1.0], 320个设计。

联合序列结构优化随后我们共同优化设计的序列和结构。在每种情况下使用30次迭代，我们运行5个种子的优化，最大解码熵阈值为 = 1.5，并以负局部序列引导 = 2.0进行2个种子的优化，共生成 67k个设计。每次迭代的设计都被纳入这个池中。

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Selection To select a set of designs for laboratory testing, we apply {Common Filters, N-terminus Coil Count <6}, rank designs according to {Common Score Terms, ESMFold pTM, 15 * PSSM Score}, and select the best 88 designs across 8 buckets of sequence identity to our template among intervals of width 0.1 in range [0.2, 1].

A.5.1.7. Generation and Selection of Designs for Experiment 2

In this experiment, we perform further refinement of the dim, distant GFP found in Experiment 1, B10. To produce a diversity of designs, we sweep over a number of settings: two variations of refinement are performed, and 2 selection protocols are used.

Local Joint Optimization Starting from our dim GFP design, B10, we perform `joint_optimize` using a full grid sweep of the following sets of settings: Initial temperatures {0.001, 0.01, 0.05, 0.1, 0.5}, PSSM bias weights {0, 0.01, 0.05, 0.1, 0.5}, Max decoding entropy thresholds {0.8, 1, 1.25, 1.5, 2.0}. For each unique settings combination, we use 20 iterations of optimization with 3 seeds, continuing the final step of Gibbs sampling until convergence. After accounting for some distributed system machine failures, this yields 6.3k total candidate designs.

Selection We select two sets of 45 designs for laboratory testing via two filters and a shared set of ranking criteria.

1. **Set 1**: We filter according to {PSSM Bias $\neq 0$, Common Filters, RMSD to starting structure <1Å, identity to starting sequence in (0.7, 1.0)}.
2. **Set 2**: We filter according to {PSSM Bias $= 0$ (no bias), Common Filters, RMSD to starting structure <1Å, Identity to starting sequence in (0.9, 1.0)}. esmGFP comes from this pool.

For each set, we rank according to {Common Score Terms, 8 * PSSM Score, 15 * 1EMA Helix RMSD} and select 45 designs each for testing.

A.5.2. Experimental Methods and Data Analysis

A.5.2.1. Strains and plasmids

We designed a custom bacterial expression vector containing an Ampicillin-resistance gene, the BBa_R0040 TetR promoter, the BBa_B0015 terminator, and a Bsa-I golden gate site between the promoter and terminator. GFP designs were codon optimized for *E. coli* expression, compatible golden gate overhangs were added, and sequences were ordered from IDT (Integrated Device Technology Inc.). They were then cloned by golden gate assembly into the vector. We evaluated our GFP designs in the *E. coli* host Mach1.

A.5.2.2. Fluorescence assays of GFP designs

To evaluate the fluorescence of our GFP designs, we transformed our designs into Mach1 cells. For each of two replicates of a design, a colony was seeded into a 1 mL TB culture containing 50 µg/mL carbenicillin. Cultures were grown in 96 deep well blocks at 37 °C in an Infors HT Multitron Shaker with a shaking speed of 1000 RPM for 24 hours. After 24 hours, 1 µL of the cultures were diluted in 200 µl of 0.2 µm filtered DPBS.

Fluorescence intensity of the samples was then quantified at the single cell level using a NovoCyte Quanteon Flow Cytometer (Fig. S19).

The remaining cultures were spun down at 4000 g for 10 minutes, resuspended and lysed with 300 µL lysis buffer (1x bugbuster, 500 mM NaCl, 20 mM Tris-HCl pH 8, 10% glycerol, cOmplete™ , EDTA-free Protease Inhibitor Cocktail), incubated at room temperature on a Belly Dancer Orbital Shaker for 10 minutes, and lysate clarified by centrifugation at 4000 g for 20 minutes. 100-120 µl lysate was transferred to a 96 well black clear-bottom plate, and GFP fluorescence was measured using a Tecan Spark Reader. Fluorescence emission was captured at 515 nm with a 10 nm bandwidth and excited with 485 nm with a 10 nm bandwidth. Absorbance was captured at 280 nm with a 3.5 nm bandwidth to assess total protein content per well. For longer time points, plates containing lysate were sealed and incubated at 37°C for up to 7 days prior to measuring fluorescence. GFP fluorescence values were first ratio normalized within a well by their absorbance at 280 nm, and then further ratio normalized across wells using the measured values from a negative control *E. coli* containing vector without GFP. Data from two replicates was then averaged for (Fig. 4B bottom) and (Fig. 4C).

Overview photos of the plates (Fig. 4B top) were taken with an iPhone 12 mini under blue light illumination from an Invitrogen Safe Imager 2.0 Blue Light Transilluminator.

For excitation spectra, emission was captured at 570 nm with a 50 nm bandwidth, while the excitation wavelength was varied from 350 to 520 nm with a 10 nm bandwidth. For emission spectra, an excitation wavelength of 430 nm was used with a 50 nm bandwidth, while emission was captured at varying wavelengths from 480 to 650 nm with a 10 nm bandwidth. Excitation and emission spectra were normalized by their maximum values (Fig. 4C).

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选择为了选择一组用于实验室测试的设计，我们应用{常见过滤器、N末端螺旋计数 <6}，根据 {常见得分术语、ESMFold pTM、15 * PSSM得分}对设计进行排名，并在范围 [0 中以宽度为 0.1 的区间选择 88 个最佳设计，这些设计在序列相似性方面与我们的模板相符合。2, 1]。

A.5.1.7. 实验 2 的设计生成和选择

在本实验中，我们对在实验 1 中发现的微弱、遥远的 GFP 进行进一步的细化，B10。为了产生多样化的设计，我们遍历多个设置：执行两种细化变体，并使用 2 种选择协议。

局部联合优化从我们的微弱 GFP 设计 B10 开始，我们执行联合_优化，使用以下设置的一整套网格搜索：初始温度 {0.001、0.01、0.05、0.1、0.5}，PSSM 偏置权重 {0, 0.01、0.05、0.1、0.5}，最大解码熵阈值 {0.8、1、1.25、1.5、2.0}。对于每个唯一的设置组合，我们使用 20 次优化迭代和 3 个种子，持续进行吉布斯采样的最终步骤，直到收敛。在考虑到某些分布式系统机器故障后，这产生了 6.3k 个候选设计。

选择我们选择了两个设计集，每个包含45个设计，供实验室测试，利用两个过滤器和一套共享的排序标准。

1. 集 1：我们根据 {PSSM 偏置 $\neq 0$、常见过滤器、与起始结构的 RMSD <1 °A 以及与起始序列的相似度在 (0.7, 1.0) 之间 } 进行过滤。
2. 集 2：我们根据 {PSSM 偏置 $= 0$（无偏置）、常见过滤器、与起始结构的 RMSD <1 °A，以及与起始序列的相似度在 (0.9, 1.0) 之间 } 进行过滤。esmGFP 来自此池。对于每个集合，我们根据 {常见得分术语、8 * PSSM 得分、15 * 1EMA 螺旋 RMSD } 排序，并每个集合选择 45个设计进行测试。

A.5.2. 实验方法与数据分析

A.5.2.1. 菌株与质粒

我们设计了一个定制的细菌表达载体，包含一个氨苄青霉素抗性基因，BBa R0040 TetR 启动子，BBa B0015 终止子，以及一个位于启动子和终止子之间的 Bsa-I 金门位点。GFP 的设计经过了针对 大肠杆菌 表达的密码子优化，兼容

添加了金门重叠序列，并从 IDT（集成设备技术公司）订购了序列。随后，它们通过金门组装克隆到载体中。我们在大肠杆菌宿主 Mach1 中评估了我们的 GFP 设计。

A.5.2.2. GFP设计的荧光测定

为了评估我们的GFP设计的荧光，我们将设计转化为Mach1细胞。对于每个设计的两个重复实验，种植一个菌落于包含50 µg/mL氨苄青霉素的1 mL TB培养基中。在37 °C的Infors HT Multitron摇床中，培养在96孔深孔板中以1000 RPM的摇动速度培养24小时。24小时后，将1 µL的培养物稀释于200 µL经0.2 µm过滤的DPBS中。

样品的荧光强度在单细胞水平上使用NovoCyte Quanteon流式细胞仪定量（图S19）。

剩余的培养物以4000 g离心10分钟，重悬并用300 µL裂解缓冲液（1x bugbuster，500 mM NaCl，20 mM Tris-HCl pH 8，10%甘油，cOmpleteTM，无EDTA蛋白酶抑制剂鸡尾酒）裂解，在室温下在Belly Dancer摇床上孵育10分钟，然后在4000 g下离心20分钟澄清裂解液。100-120 µL裂解液转移至96孔黑色清底板，并使用Tecan Spark读数器测量GFP荧光。在515 nm处捕获荧光发射，带宽为10 nm，并在485 nm处激发，带宽为10 nm。280 nm处的吸光度被捕获，带宽为3.5 nm，以评估每孔的总蛋白质含量。对于较长时间点，含有裂解液的板被封闭并在37°C孵育长达7天，然后测量荧光。GFP荧光值首先根据其在280 nm处的吸光度在孔内进行比率归一化，然后使用来自于不含GFP的负对照大肠杆菌测量的值在孔之间进一步进行比率归一化。来自两个重复实验的数据随后被平均（图4B底部）和（图4C）。

板的概述照片（图.4B上）是在Invitrogen Safe Imager 2.0 蓝光透射仪的蓝光照明下，使用i Phone 12 mini拍摄的。

对于激发光谱，发射在570 nm处捕获，带宽为50 nm，而激发波长在350到520 nm之间变化，带宽为10 nm。对于发射光谱，使用430 nm的激发波长，带宽为50 nm，而发射则在480到650 nm的不同波长下进行捕获，带宽为10 nm。激发和发射光谱通过其最大值进行了归一化（图.4C）。

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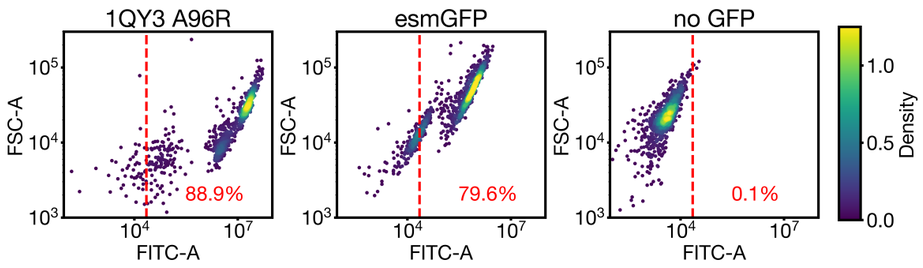

Figure S19. Flow cytometry data confirms cells expressing esmGFP can be detected at the single cell level. Forward Scatter-Area (FSC-A), a measure of cell size vs Fluorescein Isothiocyanate-Area (FITC-A), a measure of GFP-like fluorescent signal, for expressing 1QY3 A96R, esmGFP, and a negative control that does not express any GFP. A gate was set at the 99.9% quantile for the negative control data, and the fraction of cells passing the gate were quantified for each sample.

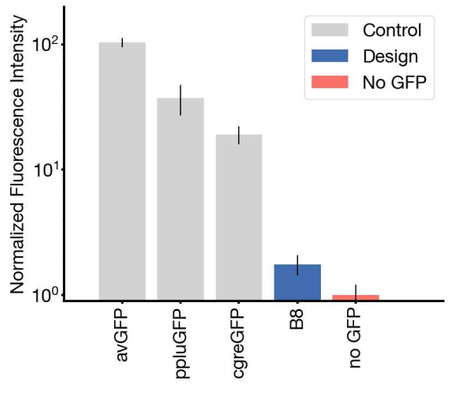

Figure S20. Replication of design B8 and select controls. Results are averages of eight wells across two plates.

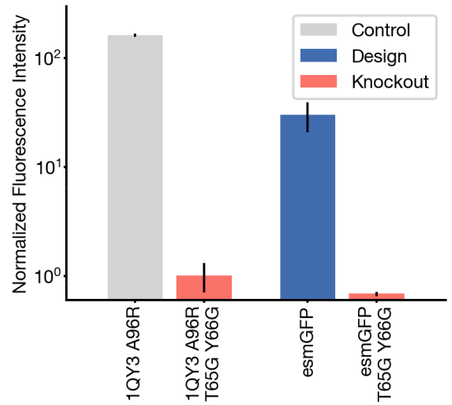

Figure S21. Chromophore knockout mutations. T65G and Y66G reduces fluorescence of both 1QY3 A96R and esmGFP to background levels.

A.5.2.3. Additional GFP experiments

Plate overview photographs (Fig. 4B top) were taken over two weeks since the initial lysate was created and over one week after the final plate reader quantification was done, and so possibly show additional brightness from slow chromophore maturing designs. We observed some low level contamination of wells H11 (vector with no GFP or designs) and H12 (lysis buffer only) in the photograph of Experiment 1 (Fig. 4B top left). Some of this contamination is already visible in well H12 during the initial plate reader quantification (Fig. 4B bottom left). To address potential contamination concerns we performed an additional replication of B8 and observed a similar level of brightness to Experiment 1 (50x less bright than natural GFPs) (Fig. S20).

Chromophore knockout versions of 1QY3 A96R and esmGFP were created through additional T65G and Y66G mutations. These variants, along with 1QY3 and esmGFP, were synthesized and measured as part of an independent replicate performed by Genscript following the *E. Coli* based fluorescent plate reader assay described above. Normalization was performed with an OD600 measurement of the cells prior to lysis. Analysis otherwise proceeded as above. Two replicates were performed for each design and results were averaged. Chromophore knockout reduced fluorescence to background levels (Fig. S21).

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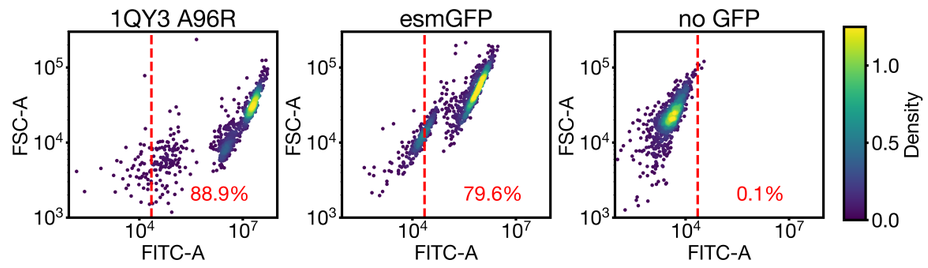

图 S19. 流式细胞术数据确认表达 esmGFP 的细胞可以在单细胞水平上被检测到。正向散射-面积 (FSC-A)，作为细胞大小的测量，与荧光异硫氰酸酯-面积 (FITC-A)，作为 GFP 类荧光信号的测量，针对表达 1QY3 A96R、esmGFP 和不表达任何 GFP 的负对照。为负对照数据设定了 99.9% 分位数的门，量化通过该门的细胞比例，针对每个样本进行评估。

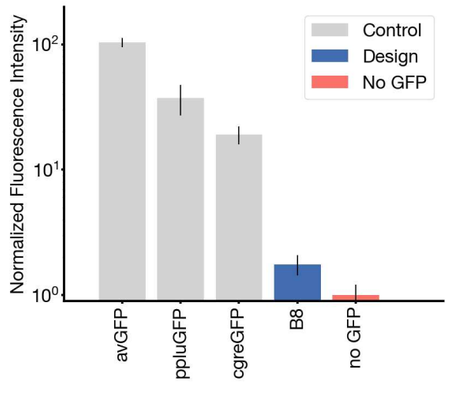

图 S20. 设计 B8 的复制和选择控制。结果是两个板块中八个孔的平均值。

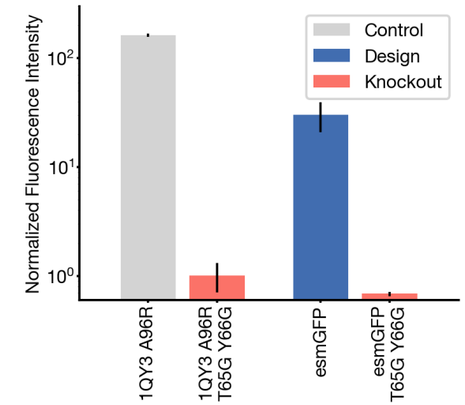

图 S21. 发色团缺失突变。T65G 和 Y66G 将 1QY3 A96R 和 esmGFP 的荧光降低至背景水平。

A.5.2.3. 附加 GFP 实验

板块概述照片 (图。4B顶部) 自初始裂解液生成以来的两周内拍摄，并且在最终板读定量完成后的一周内，因此可能显示出来自缓慢色素成熟设计的额外亮度。我们观察到在实验 1 的照片中，H11井（无 GFP 或设计的载体）和 H12井（仅裂解缓冲液）有一些低水平的污染 (图。4B左上)。在初始板读定量过程中，H12井中的一些污染已经可见 (图。4B左下)。为了解决潜在的污染问题，我们对 B8 进行了额外的重复实验，并观察到与实验 1 相似的亮度水平（比自然 GFPs 亮度低 50 倍） (图。S20)。

发色团缺失版本的 1QY3 A96R 和 esmGFP 通过额外的 T65G 和 Y66G 突变创造。这些变体与 1QY3 和 esmGFP 一起，被合成并作为 Genscript 进行的独立重复实验的一部分进行测量，遵循大肠杆菌 基于荧光平板读数检测的描述。测量细胞在裂解前的 OD600 以进行标准化。其余分析如上所述进行。对每个设计进行了两次重复实验，结果取平均值。发色团缺失使荧光降低到背景水平（图。S21）。

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A.5.3. Sequence searches and comparisons

A.5.3.1. Database searches

BLAST `nr` search: esmGFP's sequence was searched with BLAST's online server using the non-redundant sequences database `nr` with all default settings. tagRFP's sequence was taken from the top hit. The exact top hit found was `TagRFP [Cloning vector pLX-B2-TagRFP-T`, Sequence ID `ASG92118.1` and is shown in its entirety in Table S14.

Train set search: MMseqs2 (79), version 15.6f452, was used to search all datasets that ESM3 was trained on at the maximum available expansion level; for cluster resampling datasets all cluster members are searched, not just cluster centers. The goal is to search against every possible sequence that ESM3 may have seen during pretraining. Settings are selected for conducting a high sensitivity search: `-s 6 -a --max-seqs 10000`.

A.5.3.2. Sequence Identity calculations

To calculate sequence identities involving the two highlighted GFP designs (B8, esmGFP) and select reference proteins, the following procedure is used. MAFFT (109) v7.525 is applied with all default settings to the sequences of B8, esmGFP, the top tagRFP sequence found by BLAST, eqFP578 (from FPBase (110)), the template (PDB ID 1QY3, with mutation A96R), and avGFP (from FPBase). Identities between two sequences are calculated as the number of matching non-gap residues at aligned positions divided by the minimum non-gapped length of the query and target protein. This is the same sequence identity formula used in Appendix A.5.4. Aligned sequences and identities and mutation counts to esmGFP are provided in Table S14.

A.5.3.3. Inner-barrel mutation count

Positions in esmGFP are described as internal if they have SASA $<$ 5 in their predicted structure. SASA is calculated as in Appendix A.2.1.6) from the all-atom structure of esmGFP, predicted with ESM3 7B.

A.5.4. Phylogenetic Analysis

Sequences and metadata of natural and designed fluorescent proteins were obtained from FPBase (110). An initial set of 1000 proteins was filtered according to the following criteria: presence of a specified parent organism, an amino acid sequence between 200 and 300 residues long, a specified emission maximum, and no cofactors. NCBI taxonomy database was used to obtain taxonomic information about each species. These sequences were further filtered according to keep those that had species found by NCBI and were Eukaryotic but not from Chlorophyta (to exclude Channel-

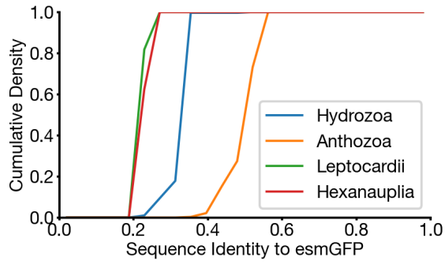

Figure S22. Sequence identity of esmGFP. Comparison with natural and designed GFPs from the four major classes found in nature.

rhodopsin like proteins). The 648 sequences that passed these criteria, along with the sequence for esmGFP, were aligned to a multiple sequence alignement using MAFFT and sequence idenity was computed between each pair of sequences as described above. All pairs within and across taxa were considered for (Fig. 4F). All designed sequences were considered to belong to the species annotated as their parent organism.

All 648 used sequences belonged to the Leptocardii (e.g. laGFP), Hexanauplia (e.g. ppluGFP), Hydrozoa (e.g. avGFP), or Anthrozoa (e.g. efasGFP) classes. The sequence identity of esmGFP was computed to each protein in these classes (Fig. S22). esmGFP was found to be closest to Anthrozoan GFPs (average sequence identity 51.4%) but also shares some sequence identity to Hydrozoan GFPs (average sequence identity 33.4%).

To estimate the millions of years of evolutionary distance by time between esmGFP and known fluorescent proteins we built an estimator to go from sequence identity between pairs of GFPs to millions of years (MY) apart. We used the following six Anthozoan species *Acropora millepora*, *Ricordea florida*, *Montastraea cavernosa*, *Porites porites*, *Discosoma sp.*, *Eusmilia fastigiata* along with the six GFPs amilGFP, rfloGFP, mcavGFP, pporGFP, dis3GFP, efasGFP respectively. These species and GFPs were chosen because they were annotated in both a recent time calibrated phylogenetic analysis of the Anthozoans (57) and a recent study of GFPs (48). Each of these species contains multiple GFP like sequences including red and cyan FPs. These particular GFPs were chosen as they were annotated to be the main GFP in each species. The millions of years between each species was estimated as twice the millions of years to the last common ancestor annotated in the time calibrated phylogenetic analysis. Using statsmodels (111), a line of best fit was fit between MY and sequence identity. The line was required to pass through a sequence identity of 1.0 and 0 MY. The MY to esmGFP was then estimated using this line

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A.5. 3. 序列搜索和比较

A.5.3.1. 数据库搜索

BLAST `nr` 搜索：esmGFP 的序列通过使用非冗余序列数据库`nr` 的 BLAST 在线服务器进行了搜索，采用所有默认设置。tagRFP 的序列取自最顶部的命中。找到的确切顶部命中是`TagRFP [`克隆载体 pLX-B2-TagRFP-T，序列 ID `ASG92118.1`，其完整内容在表S14中显示。

训练集搜索：MMseqs2 (79)，版本 15.6f452，被用于搜索所有在最大可用扩展级别上ESM3训练的数据集；对于聚类重采样数据集，所有聚类成员都被搜索，而不仅仅是聚类中心。目标是搜索ESM3在预训练期间可能见过的每一个序列。选择的设置用于进行高灵敏度搜索：`-s 6-a --max-seqs10000`。

A.5.3.2. Sequence Identitycalculations为了计算涉及两个高亮显示的GFP设计（B8，esmGFP）和选择的参考蛋白质的序列一致性，采用以下程序。MAFFT (109) v7.525在所有默认设置下应用于B8、esmGFP、通过BLAST找到的顶级tagRFP序列、eqFP578（来自FPBase (110)），模板（PDB ID 1QY3，含有突变A96R）以及avGFP（来自FPBase）的序列。在对齐位置上，两个序列之间的一致性通过匹配的非缺口残基数除以查询和目标蛋白质的最小非缺口长度来计算。这与附录 A.5.4中使用的序列一致性公式相同。对齐的序列、一致性和到esmGFP的突变计数在表S14中提供。

A.5.3.3. Inner-barrelmutation count在esmGFP中，若其SASA $<$ 5 小于5，则描述为内部位置。在全原子结构的esmGFP中，SASA的计算方式与附录 A.2.1.6) 一致，该结构是通过ESM3 7B进行预测的。

A.5. 4. 系统发育分析

从 FPBase (110) 获取了天然和设计的荧光蛋白的序列及元数据。根据以下标准筛选出最初的 1000 种蛋白质：存在指定的亲本生物，氨基酸序列长度在 200 到 300 残基之间，指定的发射最大值，以及无辅因子。使用 NCBI 分类数据库获取每个物种的分类信息。这些序列进一步过滤，以保留那些由 NCBI 发现的物种，并且是真核生物但不属于绿藻 （以排除 Channel-

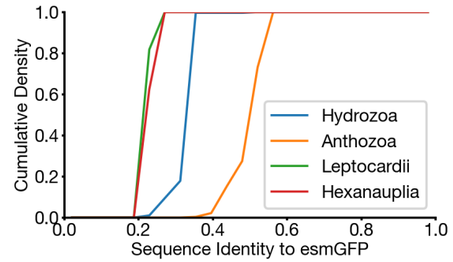

图 S22. esmGFP 的序列一致性。与自然界中发现的四大主要类群中的天然和设计 绿荧光蛋白 进行比较。

视紫红质样蛋白）。通过这些标准筛选出的 648 个序列，以及 esmGFP 的序列，使用 MAFFT 进行了多序列比对，并按照上述描述计算了每对序列之间的序列一致性。考虑了所有类群内外的配对（图。4F）。所有设计的序列被视为属于标注为其亲本生物的物种。

所有648个使用的序列属于侧头目（例如laGFP）、六足幼虫类（例如ppluGFP）、水母（例如avGFP）或珊瑚虫（例如efasGFP）类别。esmGFP与这些类别中的每种蛋白质的序列一致性被计算得出（图。S22）。发现esmGFP与珊瑚虫的绿荧光蛋白具有最高的相似性（平均序列一致性为51.4%），同时也与水母的绿荧光蛋白共享一些序列一致性（平均序列一致性为33.4%）。

为了估算 esmGFP 与已知荧光蛋白之间的百万年 (MY) 进化距离，我们建立了一个估算器，以根据 GFP 之间的序列一致性来计算相距多少百万年。我们使用了以下六种刺胞动物物种米氏珊瑚,佛罗里达海葵,大洞石珊瑚 , 黄条珊瑚 , 盘海葵 , 直立扁海葵，以及六种 GFP amilGFP, rfloGFP, mcavGFP, pporGFP, dis3GFP, efasGFP。选择这些物种和 GFP 是因为它们在最近的时间校准刺胞动物系统发育分析中被注释（57）和在最近的 GFP 研究中被注释（48）。每种物种都包含多个类似 GFP 的序列，包括红色和青色荧光蛋白。这些特定的 GFP 被选择是因为它们被注释为每个物种的主要 GFP。每种物种之间的百万年估计为最后一个共同祖先的百万年数的两倍，该祖先在时间校准的系统发育分析中被注释。使用 statsmodels (111)，在百万年与序列一致性之间拟合了一条最佳拟合线。该线必须通过序列一致性为 1.0 和 0 MY。然后使用这条线估计 esmGFP 的百万年数

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and the sequence identity of esmGFP to the nearest known protein.

A.6. Open model

We are releasing the ESM3 source code and model weights of an open model, ESM3-open. ESM3-open is a 1.4B-parameter model we trained without OAS antibody sequences and with precautionary risk mitigations for release to the academic research community.

As part of this release, we follow guidance from the Principles for the Responsible Development of AI for Biological Design (112). We adopted precautionary risk mitigations, described in Appendix A.6.1, and performed risk evaluations, detailed in Appendix A.6.2. Additionally we conducted a review of the risks and benefits of releasing ESM3-open with experts from the scientific community. We provided reviewers access to ESM3-open, along with a detailed technical report on our risk evaluations. We received unanimous feedback from our reviewers that the benefits of releasing the model greatly outweigh any potential risks.

We see this release as a first step and plan to work with the scientific community to continue to improve processes around responsible development. Open models enable the scientific community to better understand and reduce any potential risks of biological design tools. As our understanding develops alongside the capabilities of future models, we plan to continuously improve our evaluation frameworks, safeguards, and mitigation strategies.

A.6.1. ESM3-open Mitigations

As a precaution, we filtered the training data of ESM3-open to minimize model performance on sequences of potential concern while otherwise maintaining performance. We also removed the capability for the model to follow prompts related to viruses and toxins.

Filtering sequences of potential concern. Previous work has shown that the performance of protein language models is closely related to the number of similar sequences present in the training data (5). We therefore removed sequences aligned to potentially-concerning proteins from the training data in order to reduce the capability of ESM3-open on these sequences.

We identified and removed sequences unique to viruses, as well as viral and non-viral sequences from the Select Agents and Toxins List (113) maintained by the CDC and USDA. The U.S. Department of Health & Human Services recommends filtering based on the Select Agents list as part of their Screening Framework Guidance for Providers and Users of Synthetic Nucleic Acids (114).

To filter data, we create two denylists: the Viral Denylist and the Select Agent Denylist. We then remove all sequences from the training set that are detected to align to those in the denylists by MMseqs2 at or above a given sequence identity threshold.

To create the Viral Denylist, we identify ∼4M sequences that are annotated as viral in UniProt and align almost exclusively to other viral sequences in UniProt. This gives us a procedure that removes viral proteins with both high sensitivity and specificity (as measured by UniProt taxonomic annotations). To create the Select Agents Denylist we identify all sequences in UniProt belonging to organisms on the Select Agents and Toxins List (113). This process gives us 147K non-viral sequences and 40K additional viral sequences.

For each denylist, MMseqs was used to query against the full set of training databases, (including PDB, UniRef, MGnify, and JGI) and all hits were removed from the training set. This filter removes a total of 10.6M sequences across all training sets.

Removal of keywords of concern. There are a number of keyword prompts associated with viruses and toxins that we aim to remove. We first identify a list of harmful keywords with the following steps:

1. We curate a list of filter terms associated with viruses and toxins. The full filter term list is available upon request.
2. We then identify all InterPro tags whose free-text term names contain at least one of the filter terms.
3. We identify keywords that are associated with flagged InterPro tags but that are not associated with non-flagged InterPro tags. We remove those keywords. Keywords which are associated with both flagged and non-flagged InterPro tags (e.g. “extracellular region”) are not removed.
4. We additionally remove all keywords that themselves directly contain one of the filter terms

Of the original 68,103 keywords that ESM3 is trained with, this filter removes a total of 9,462 (14%), creating a new vocabulary of 58,641 keywords.

The function vocabulary is defined via vectors representing Term Frequency Inverse Document Frequency (TF-IDF) which are then tokenized using Locality Sensitive Hashing (LSH), as previously described in Appendix A.1.8. To remove flagged keywords, they are first removed from the TF-IDF vocabulary by removing the entries corresponding to flagged keywords. This reduces the TF-IDF vector size to 58,641. The LSH tokenization is defined by 64 hyperplanes, each defined in the TF-IDF space, i.e. a Euclidean

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以及 esmGFP 与最近已知蛋白质的序列一致性。

A.6. 开放模型

我们正在发布开放模型 ESM3-open 的 ESM3 源代码和模型权重。ESM3-open 是一个 1.4B 参数模型，我们在不使用 OAS 抗体序列的情况下进行了训练，并采取了谨慎的风险缓解措施，以便向学术研究社区发布。

作为此次发布的一部分，我们遵循《人工智能在生物设计中负责任发展的原则》(112)。我们采用了在附录 A.6.1 中描述的谨慎风险缓解措施，并进行了在附录 A.6.2 中详细说明的风险评估。此外，我们与科学界的专家对发布 ESM3-open 的风险和利益进行了审核。我们向评审人员提供了对 ESM3-open 的访问权限，以及有关我们风险评估的详细技术报告。我们收到了评审人员一致反馈，认为发布该模型的好处大大超过了任何潜在风险。

我们将此次发布视为第一步，并计划与科学界合作，继续改善与负责任发展相关的过程。开放模型使科学界能够更好地理解和减少生物设计工具的潜在风险。随着我们理解的不断发展以及未来模型能力的提升，我们计划持续改善我们的评估框架、安全措施和缓解策略。

A.6.1. ESM3开放缓解

作为一种预防措施，我们过滤了ESM3-open的训练数据，以最小化模型在潜在关注序列上的表现，同时保持其他方面的表现。我们还移除了模型跟随与病毒和毒素相关的提示的能力。

过滤潜在关注的序列。先前的研究表明，蛋白质语言模型的性能与训练数据中相似序列的数量密切相关（5）。因此，我们从训练数据中移除了与潜在关注蛋白质对齐的序列，以降低ESM3-open在这些序列上的能力。

我们识别并移除独特于病毒的序列，以及选择性病原体及毒素名单中（113）的病毒和非病毒序列，该名单由CDC和USDA维护。美国卫生与公众服务部建议根据选择性病原体名单进行过滤，以作为其合成核酸提供者和用户筛选框架指导的一部分（114）。

为了筛选数据，我们创建了两个拒绝列表：病毒拒绝列表和选择性病原体拒绝列表。然后，我们移除从训练集中检测到与拒绝列表中序列比对的所有序列，这些序列由mmseqs2在或高于给定的序列一致性阈值所检测。

为了创建病毒拒绝列表，我们识别 ∼4在UniProt中注释为病毒的M序列，并几乎仅与UniProt中的其他病毒序列对齐。这为我们提供了一种程序，能够以高敏感性和特异性（根据UniProt的分类注释进行测量）去除病毒蛋白。为了创建选择性病原体拒绝列表，我们识别所有属于选择性病原体及毒素名单（113）的UniProt中的序列。此过程为我们提供了147K非病毒序列和40K额外的病毒序列。

对于每个拒绝列表，使用MMseqs对完整的训练数据库集进行查询（包括PDB、UniRef、MGnify和JGI），并从训练集中移除所有命中。这一过滤器在所有训练集中总共移除了10.6M序列。

去除相关关键词。我们旨在去除与病毒和毒素相关的多个关键词提示。我们首先通过以下步骤识别有害关键词列表：

1. 我们整理了一份与病毒和毒素相关的过滤词汇列表。完整的过滤词汇列表可根据请求提供。
2. 然后，我们识别所有自由文本术语名称中包含至少一个过滤词的 InterPro 标签。
3. 我们识别与标记的 InterPro 标签相关的关键词，但与未标记的 InterPro 标签无关。我们移除这些关键词。与标记和未标记的 InterPro 标签均相关的关键词（例如“细胞外区域”）不被移除。

4. 我们还移除所有直接包含一个过滤词的关键词。在原本的 68,103 个关键词中，ESM3 训练所用的该过滤去除了 9,462 个（14%），形成了一个新的词汇表，共 58,641 个关键词。函数词汇表是通过代表词频逆文档频率（TF-IDF）的向量定义的，然后使用局部敏感哈希（LSH）进行分词，如附录A.1.8中先前所述。为了移除标记关键词，首先从 TF-IDF 词汇表中去除与标记关键词对应的条目。这将 TF-IDF 向量大小减少到 58,641。LSH 分词由 64 个超平面定义，每个超平面在 TF-IDF 空间中定义，即欧几里得空间。

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Protein	Sequence Identity to esmGFP	Mutations to esmGFP	Aligned Sequence
B8	0.93	15	-MSKVEELIKPEMKMKLEMEGEVNGHKFSIEAEGEGKPYEGKQTIKAWSTT-GKLPFAW DILSTSLTYGFRMFTKYPEGLEEHDYFKQSFPEGYSWERTITYEDGATVKVTSDISLED GVLINKIKFKGTNFPSDGPVM-QKKTTGWEPSTELITPDPATGGLKGEVKMRLKLEGGG HLLADFKTTYRSKKKEK-LPLPGVHYVDHTIRNEKAPHPEGKEYVVQYETAVARLA--- --------
esmGFP	1.0	0	-MSKVEELIKPDMKMKLEMEGEVNGHKFSIEAEGEGKPYEGKQTIKAWSTT-GKLPFAW DILSTSLTYGNRAFTKYPEGLEQHDFFKQSFPEGYSWERTITYEDGATVKVTADISLED GVLINKVKFKGENFPSDGPVM-QKKTTGWEASTELITPDPATGGLKGEVKMRLKLEGGG HLLADFKTTYRSKKKEK-LPLPGVHYVDHRIVNEKATHPEGKEYMIQYEHAVARLA--- --------
tagRFP	0.58	96	MVSKGEELIKENMHMKLYMEGTVNNHHFKCTSEGEGKPYEGTQTMRIKVVEGGPLPFAF DILATSFMYGSRTFINHTQGIP--DFFKQSFPEGFTWERVTTYEDGGVLTATQDTSLQD GCLIYNVKIRGVNFPSNGPVM-QKKTLGWEANTEMLY--PADGGLEGRTDMALKLVGGG HLICNFKTTYRSKKPAKNLKMPGVYYVDHRL--ERIKEADKETYVEQHEVAVARYCDLP SKLGHKLN
eqFP578	0.53	107	----MSELIKENMHMKLYMEGTVNNHHFKCTSEGERKPYEGTQTMKIKVVEGGPLPFAF DILATSFMYGSKTFINHTQGIP--DLFKQSFPEGFTWERITTYEDGGVLTATQDTSLQN GCIIYNVKINGVNFPSNGSVM-QKKTLGWEANTEMLY--PADGGLRGHSQMALKLVGGG YLHCSFKTTYRSKKPAKNLKMPGFHFVDHRL--ERIKEADKETYVEQHEMAVAKYCDLP SKLGHR--
template	0.38	143	-MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTLKFICTT-GKLPVPW PTLVTTLTYGVQCFSRYPDHMKQHDFFKSAMPEGYVQERTISFKDDGNYKTRAEVKFEG DTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYITADKQKNGIKANFKIRHNIEDGS VQLADHYQQNTPIGDGP-VLLPDNHYLSTQSALSKDPN-EKRDHMVLLEFVTAAGI--- --------
avGFP	0.36	146	-MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTLKFICTT-GKLPVPW PTLVTTFSYGVQCFSRYPDHMKQHDFFKSAMPEGYVQERTIFFKDDGNYKTRAEVKFEG DTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGS VQLADHYQQNTPIGDGP-VLLPDNHYLSTQSALSKDPN-EKRDHMVLLEFVTAAGITHG MDELYK--

Table S14. Multiple sequence alignment of select GFP designs (B8, esmGFP) and reference proteins. Template is the full sequence of our template structure (PDB ID 1QY3), with chromophore slowing mutation A96R removed. tagRFP is the full sequence of the top hit returned by BLAST search of the nonredundant database `nr`, avGFP and eqFP578 are from FPBase. Sequence identities for GFP designs are in general calculated as the number of non-gap matches at aligned positions, divided by the minimum length of the query and target ungapped sequences. Here, only sequence identities to esmGFP are shown. Similarly, the number of mutations to esmGFP are calculated as the number of mismatches at aligned positions where esmGFP does not have a gap.

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蛋白质	与esmGFP的序列一致性	与esmGFP的突变	对齐序列
B8	0.93	15	-MSKVEELIKPEMKMKLEMEGEVNGHKFSIEAEGEGKPYEGKQTIKAWSTT-GKLPFAW DILSTSLTYGFRMFTKYPEGLEEHDYFKQSFPEGYSWERTITYEDGATVKVTSDISLED GVLINKIKFKGTNFPSDGPVM-QKKTTGWEPSTELITPDPATGGLKGEVKMRLKLEGGG HLLADFKTTYRSKKKEK-LPLPGVHYVDHTIRNEKAPHPEGKEYVVQYETAVARLA--- --------
esmGFP	1.0	0	-MSKVEELIKPDMKMKLEMEGEVNGHKFSIEAEGEGKPYEGKQTIKAWSTT-GKLPFAW DILSTSLTYGNRAFTKYPEGLEQHDFFKQSFPEGYSWERTITYEDGATVKVTADISLED GVLINKVKFKGENFPSDGPVM-QKKTTGWEASTELITPDPATGGLKGEVKMRLKLEGGG HLLADFKTTYRSKKKEK-LPLPGVHYVDHRIVNEKATHPEGKEYMIQYEHAVARLA--- --------
tagRFP	0.58	96	MVSKGEELIKENMHMKLYMEGTVNNHHFKCTSEGEGKPYEGTQTMRIKVVEGGPLPFAF DILATSFMYGSRTFINHTQGIP--DFFKQSFPEGFTWERVTTYEDGGVLTATQDTSLQD GCLIYNVKIRGVNFPSNGPVM-QKKTLGWEANTEMLY--PADGGLEGRTDMALKLVGGG HLICNFKTTYRSKKPAKNLKMPGVYYVDHRL--ERIKEADKETYVEQHEVAVARYCDLP SKLGHKLN
eqFP578	0.53	107	----MSELIKENMHMKLYMEGTVNNHHFKCTSEGERKPYEGTQTMKIKVVEGGPLPFAF DILATSFMYGSKTFINHTQGIP--DLFKQSFPEGFTWERITTYEDGGVLTATQDTSLQN GCIIYNVKINGVNFPSNGSVM-QKKTLGWEANTEMLY--PADGGLRGHSQMALKLVGGG YLHCSFKTTYRSKKPAKNLKMPGFHFVDHRL--ERIKEADKETYVEQHEMAVAKYCDLP SKLGHR--
模板	0.38	143	-MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTLKFICTT-GKLPVPW PTLVTTLTYGVQCFSRYPDHMKQHDFFKSAMPEGYVQERTISFKDDGNYKTRAEVKFEG DTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYITADKQKNGIKANFKIRHNIEDGS VQLADHYQQNTPIGDGP-VLLPDNHYLSTQSALSKDPN-EKRDHMVLLEFVTAAGI--- --------
avGFP	0.36	146	-MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTLKFICTT-GKLPVPW PTLVTTFSYGVQCFSRYPDHMKQHDFFKSAMPEGYVQERTIFFKDDGNYKTRAEVKFEG DTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGS VQLADHYQQNTPIGDGP-VLLPDNHYLSTQSALSKDPN-EKRDHMVLLEFVTAAGITHG MDELYK--

表 S14. 选定的 GFP 设计（B8, esmGFP）和参考蛋白质的多序列对齐。模板是我们模板结构的完整序列（PDB ID 1QY3），发色团慢化突变 A96R 被去除。tagRFP 是非冗余数据库 nr 中 BLAST 搜索返回的最佳命中序列的完整序列，avGFP 和 eqFP578 来自 FPBase。GFP 设计的序列相似性通常计算为对齐位置上非缺口匹配的数量，除以查询序列和目标无缺口序列的最小长度。在这里，仅显示与 esmGFP 的序列相似性。同样，针对 esmGFP 的突变数量计算为在对齐位置上 esmGFP 没有缺口的错配数量。

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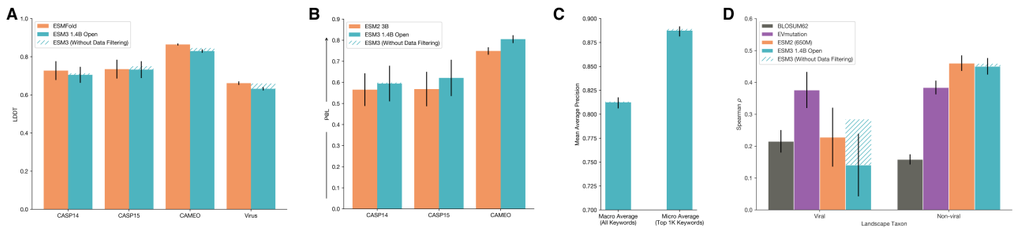
Simulating 500 million years of evolution with a language model

使用语言模型模拟五亿年的进化



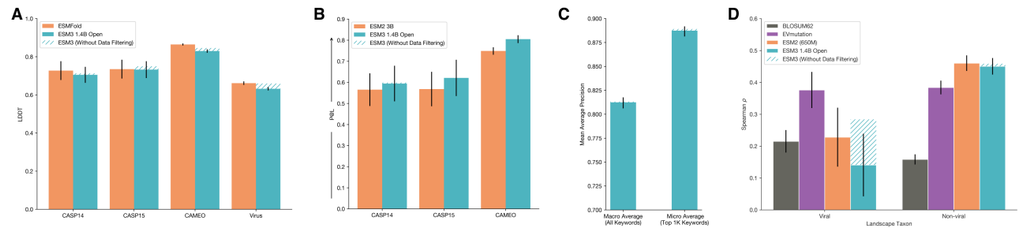


Figure S23. ESM3-open is a powerful predictor of structure and function trained for open release. **A: Structure Prediction** ESM3-open (blue) is competitive with ESMFold (orange) on structure prediction as measured by LDDT on CAMEO and CASP14/15. See Appendix A.3.4 for details on this evaluation. **B: Representation Learning** ESM3-open (blue) is competitive with ESM2-3B (orange) on representation learning as measured by contact prediction P@L for finetuned representations. See Appendix A.3.3 for details on this evaluation. **C: Function Keyword Prediction.** ESM3-open function prediction performance, as measured by Mean Average Precision across function keywords. ESM3-open achieves 0.81 precision across all keywords, and 0.89 for the top 1K most prevalent keywords in the test set (Appendix A.3.2). We use the same evaluation framework as in Appendix A.1.8.2.2. We report both the macro and micro averages as in Fig. S8. In each of the preceding evaluations, the data mitigation minimally impacted performance, as compared to a compute-matched model without data mitigations (hatched blue). **D: Zero-shot Fitness Prediction**. Fitness prediction performance as measured by correlation (Spearman ρ) across 217 Deep Mutational Scanning datasets collated in ProteinGym. Left and right subplots indicate viral (left) and non-viral (right) DMS datasets. The four columns per group indicate different models. ESM3-open performs substantially worse than EVMutation (purple) on viral fitness prediction, while being competitive with ESM2 (orange) on non-viral fitness prediction. Viral fitness prediction was substantially impacted by the data mitigation, while non-viral fitness prediction was not (hatched blue).

图 S23。ESM3-open 是一个强大的结构和功能预测器，经过开放发布的训练。A：结构预测ESM3-open（蓝色）在结构预测方面与 ESMFold（橙色）竞争，就 LDDT 在 CAMEO 和 CASP14/15 的测量结果而言。详见附录A.3.4 中关于此评估的详细信息。B：表征学习ESM3-open（蓝色）在表征学习方面与 ESM2-3B（橙色）竞争，就微调表征的接触预测 P@L 测量而言。详见附录A.3.3 中关于此评估的详细信息。C：功能关键词预测。ESM3-open 在功能关键词上的功能预测性能，按照功能关键词的平均平均精确度进行测量。ESM3-open 在所有关键词上的精度为 0.81，在测试集中最常见的前 1K 关键词上的精度为 0.89（附录A.3.2）。我们使用与附录A.1.8.2.2中相同的评估框架。我们报告了与图S8一致的宏平均和微平均。在上述所有评估中，数据缓解对性能的影响最小，与没有数据缓解的计算匹配模型（带阴影的蓝色）相比。 D：零样本适应性预测。根据 217 个在 ProteinGym 中整理的深度突变扫描数据集的相关性（Spearman ρ）测量的适应性预测性能。左侧和右侧的子图表示病毒（左）和非病毒（右） DMS 数据集。每组的四列表示不同的模型。ESM3-open 在病毒适应性预测方面的表现显著低于 EVMutation（紫色），而在非病毒适应性预测中与 ESM2（橙色）竞争。数据缓解对病毒适应性预测的影响显著，而对非病毒适应性预测的影响并不显著（带阴影的蓝色）。

space with one dimension per keyword. We redefine the hyperplanes to be in the reduced space by removing the dimensions corresponding to the flagged keywords. This permanently removes the information required for tokenization of the flagged keywords. This mitigation is highly selective and does not change the tokenization for any non-flagged keywords.

A.6.2. ESM3-open Evaluations

In the section below, we outline our evaluations of ESM3-open performance. When appropriate, we compare ESM3-open to either existing open models, (e.g. ESM2 or ESMFold), or to a compute-matched version of ESM3-open, trained without any data mitigations.

Structure Prediction In Fig. S23A, we show that ESM3-open achieves competitive performance on structure prediction as measured by LDDT on CASP14, 15 and CAMEO, showing very slight degradation from our compute-matched 1.4B model without data filtering. The evaluation framework is described in Appendix A.3.4.

We also measure the ability of ESM3 to predict the structure of a subset of viral proteins. In Fig. S23A we evaluate structure prediction on a set of structures derived from viruses that were purged from the PDB training set. For the chains in PDB that were $> 70\%$ sequence identity hits to the Viral Denylist, we cluster at 40% sequence identity and then select the longest chain (with length ≤ 1024) from each cluster. ESMfold and ESM3-open achieved an average LDDT of 0.66 and 0.63, respectively, on the viral structures. Without the data mitigation, a compute-matched ESM3-open would have achieved an average LDDT of 0.66. This is substantially worse than the performance on generic structure prediction on CAMEO, and CASP14, where ESMFold achieved an average LDDT of 0.86 and 0.73, and ESM3-open achieved an average of LDDT of 0.83 and 0.70.

Representation Learning. ESM3-open achieves strong performance on representation learning, slightly outperforming ESM2 (3B) on contact prediction as measured by precision at L (P@L) on structures derived from CASP14/15, and CAMEO, see Fig. S23B. The evaluation framework is described in Appendix A.3.3.

Function Keyword Prediction. ESM3-open is able to predict function keywords for proteins in a validation set derived from UniRef and annotated with InterProScan, see Fig. S23C. ESM3-open achieves a Mean Average Precision for all keywords of 0.81 (macro average), and a precision of 0.89 (micro average) for the top 1000 keywords, discarding common terms such as "the". The evaluation framework is the same as that described in Appendix A.1.8.2.2.

Zero-shot Viral Fitness Prediction. We measure the ability of ESM3 to identify viable sequences and understand the effects of mutations on viral proteins. The evaluation consists of the single mutant variants from 217 Deep Mutational Scanning (DMS) datasets collected in ProteinGym (115).

每个关键词对应一个维度的空间。我们通过移除与标记关键词相对应的维度，重新定义超平面在降维空间中的位置。这永久性地去除了标记关键词分词所需的信息。这种缓解措施具有高度选择性，并且不会改变任何未标记关键词的分词。

A.6.2. ESM3开放评估

在下面的章节中，我们概述了ESM3-open的性能评估。在适当情况下，我们将ESM3-open与现有的开放模型（例如ESM2或ESM-Fold）进行比较，或与未经任何数据缓解训练的ESM3-open的计算匹配版本进行对比。

结构预测在图S23A中，我们展示了ESM3-open在结构预测方面的竞争性表现，依据LDDT对CASP14、15和CAMEO的测量结果，显示出与我们无数据过滤的计算匹配1.4B模型相比，仅有非常轻微的退化。评估框架详述于附录A.3.4。

我们还测量了ESM3预测一部分病毒蛋白结构的能力。在图S23A中，我们评估了基于从PDB训练集中移除的病毒衍生结构集的结构预测。对于在PDB中的链，这些链与病毒拒绝列表的 $> 70\%$序列一致性命中，我们在40%序列一致性处进行聚类，然后从每个聚类中选择最长的链（长度为 ≤ 1024）。

ESMfold和ESM3-open在病毒结构上的平均LDDT分别达到了0.66和0.63。如果没有数据减轻，计算匹配的ESM3-open的平均LDDT将为0.66。这显著低于在CAMEO和CASP14上进行一般结构预测的表现，其中ESMFold的平均LDDT为0.86和0.73，而ESM3-open的平均LDDT为0.83和0.70。

表征学习。ESM3-open在表征学习上表现出色，在接触预测方面略微超过了ESM2 (3B)，通过在源自CASP14/15和CAMEO的结构上测得的序列长度L的精度(P@L)进行评估，见图。S23B. 评估框架在附录A.3.3中进行了描述。

功能关键词预测。ESM3-open能够预测来自UniRef的验证集中的蛋白质的功能关键词，这些蛋白质已通过InterProScan进行注释，见图。S23C. ESM3-open对所有关键词的平均平均精确度为0.81（宏观平均），对于前1000个关键词的精度为0.89（微观平均），排除了诸如“the”等常见术语。评估框架与附录A.1.8.2.2中描述的相同。

零-shot病毒适应度预测。我们测量了ESM3识别有效序列的能力，并理解突变对病毒蛋白的影响。评估由来自217个深度突变扫描(DMS)数据集的单突变变体组成，这些数据集在ProteinGym中收集（115）。

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This includes 28 DMS landscapes from viral proteins and 189 from other proteins. We evaluate the correlation (Spearman ρ) between the predicted variant effect and measured variant effect. The predicted variant effect is measured as the difference between the logit value for the variant allele and the logit value of the wildtype allele at a given masked position (18).

First, we compare the performance of ESM3-open to a compute-matched version of ESM3-open which did not undergo any data filtering. Applying data filtering as a mitigation reduces average Spearman ρ performance on viral fitness prediction from 0.28 (ESM3-small) to 0.17 (ESM3-open), while performance on non-viral proteins is not adversely affected, changing from 0.46 (ESM3-small) to 0.45 (ESM3-open). We also compare the performance of ESM3-open to existing open model baselines. Fig. S23D assesses performance relative to the EVMutation (116) baseline. EVMutation is a Markov Random Field model (not deep learning-based) trained on a multiple sequence alignment of the target protein. BLOSUM62 is a baseline based on amino acid substitution frequencies. After mitigations, ESM3-open performance on viral landscapes is low compared to EVMutation and on-par with BLOSUM62.

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这包括来自病毒蛋白的28个DMS景观以及189 从其他蛋白质中。我们评估预测变体效应与测量变体效应之间的相关性（Spear- man ρ）变体效应。预测的变体效应被测量为变体等位基因的logit 值与给定掩蔽下的野生型等位基因的 logit 值之间的差异

位置（18）。首先，我们比较 ESM3-open 的性能与未经过任何数据过滤的计算匹配版本的 ES M3-open 的性能。实施数据过滤作为一种缓解措施，将病毒适应度预测的平均 Spearman ρ 性能从 0.28（ESM3-small）降低到 0.17

(ESM3-open)，而非病毒蛋白的性能并未受到不利影响，从 0.46（ESM3-small）变为 0.45（ESM3-open）。我们还比较了 ESM3-open 的性能与现有开放模型基准的性能。图。S23D

评估相对于 EVMutation （116）基线的性能。EVMutation 是一种马尔可夫随机场模型（非深度学习基础），基于目标蛋白的多序列比对进行训练。BLOSUM62 是基于

关于氨基酸替代频率。在采取缓解措施后，ESM3-open在病毒景观上的表现相较于EVMutation较低，并且与BLOSUM62持平。

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